



NATIONAL DIGITAL TRANSFORMATION STRATEGY (2025-30)



Special Assistant to the Chief Adviser

**Ministry of Posts, Telecommunications
and Information Technology**

It is with a profound sense of responsibility and a clear vision for the future that I present the **National Digital Transformation Strategy (2025–2030)**.

My tenure as the Special Assistant to the Chief Adviser was defined by a singular mission: to dismantle a legacy of systemic fragmentation and siloed decision-making that had long stifled innovation and resulted in redundant infrastructure. This was no easy task and I am greatly thankful to my technical and support team for their hard work as they went above and beyond with minimal support. We prioritized digital sovereignty, data rights, and institutional interoperability. We inherited an ecosystem characterized by fragmented digital assets and overlapping mandates. This strategy represents our collective response to those systemic challenges, providing a roadmap to transition Bangladesh from a consumer of technology into a regional digital leader.

The philosophy underpinning this strategy is anchored in the vision of a "Connected State" where every citizen is empowered. We have enacted a foundational legal triad—the Cyber Safety, Personal Data Protection, and National Data Governance Ordinances of 2025, which anchor the digital ecosystem in a robust legal perimeter. These reforms shift our focus from state control to citizen protection, granting individuals ownership of their data and ensuring digital rights are enforceable. Most significantly, through the National Data Governance Ordinance (NDGO), we have established the National Data Governance Authority (NDGA) as an independent body to oversee data privacy, AI ethics, and interoperability across all government agencies.

We believe that institutional rebirth is the prerequisite for operational excellence. We have initiated a strategic turnaround for our agencies, shifting the focus from "hard concrete" to "soft infrastructure" and business-led facilitation. We have transitioned a2i into a dedicated GovTech Agency and restructured the Bangladesh Computer Council (BCC) to serve as the nation's digital infrastructure backbone. Through the journey from MESH to STAR to DSTAR, we are ensuring that government data is collected once and used many times, eliminating the bureaucratic friction that has long burdened our citizens.

Our vision for technical sovereignty is embodied in our strategy. We have moved decisively to end the era of vendor lock-in by establishing the National Source Code Repository, ensuring the state owns and controls the code essential for public services. By operationalizing the National Fiber Bank and achieving 100% fiber-to-tower connectivity, we are building the "digital rails" required for a truly inclusive economy. This strategy is ultimately about trust. By launching a National Digital Transformation Dashboard for real-time progress tracking and independent third-party audits, we have made the government directly accountable for its digital promises.

To our successor and the dedicated public servants of the division, we leave behind a framework designed to outlast political cycles. The responsibility now lies in the discipline of implementation. The journey from a captured past to a sovereign digital future has begun. Let us ensure that no community is left behind in the march toward a knowledge-based economy.

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Finally, we thank the subject matter experts, stakeholders, and team members who shared their time, knowledge, and ideas that informed this compendium.

Disclaimer

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Executive Summary

National Digital Transformation Strategy (2025-2030)

Bangladesh stands at a defining moment in its digital evolution. The **National Digital Transformation Strategy (2025-2030)** serves as the comprehensive roadmap to transition the nation from a consumer of technology to a regional digital leader. Anchored in the vision of a **"Connected State,"** this strategy addresses systemic fragmentation, strengthens governance, and lays the foundation for a secure, inclusive, and innovation-driven economy.

Strategic Vision & The Nine Pillars

The strategy is built upon **nine core pillars** designed to modernize governance, empower citizens, and catalyze economic growth:

1. **Unified Digital Public Infrastructure (DPI):** Establishing the "National Digital Rail" through the **National Responsible Data Exchange (NRDEX)** to enable secure, interoperable data sharing across all agencies.
2. **Citizen-Centric Governance:** Ensuring **95% of government services** are delivered end-to-end digitally by 2030, enforcing a "Zero-Visit" mandate for service verification.
3. **Data Sovereignty & Legal Certainty:** Anchoring the ecosystem in a robust legal triad—the **National Data Governance Ordinance (NDGO)**, **Personal Data Protection Ordinance (PDPO)**, and **Cyber Safety Ordinance (CSO)**—to protect citizen rights and national data assets.
4. **National Connectivity Backbone:** Guaranteeing high-speed access by operationalizing the **National Fiber Bank** and achieving 100% fiber-to-tower connectivity.
5. **Autonomous Regulatory Oversight:** Empowering the **National Data Governance Authority (NDGA)** as an independent constitutional body to oversee data privacy, AI ethics, and interoperability.
6. **Economic Resilience & Innovation:** Targeting **\$10 billion in ICT exports** by streamlining cross-border digital trade, establishing a **Universal Payment Gateway**, and modernizing tax collection via the **NBR API Connectivity Hub**.
7. **Future-Ready Human Capital:** Building a workforce of **8 million professionals** skilled in Deep Tech (AI, Cybersecurity, Semiconductors) under the **BEAR framework** (Biotech, Electronics, AI, Robotics).
8. **Institutional Interoperability:** Enforcing the **"Collect Once, Use Many Times"** principle to eliminate redundant data collection and bureaucratic friction.
9. **Proactive Security & Resilience:** Transitioning from reactive defense to **Active Threat Intelligence** through the **National Security Operations Center (NSOC)** and the **Cyber Safety Ordinance**.

The Legal & Governance Foundation

The strategy is underpinned by a transformative legal architecture enacted in 2025:

- **The Cyber Safety Ordinance (CSO)** shifts focus from state control to citizen protection, criminalizing modern cybercrimes (e.g., deepfakes, sextortion) while safeguarding freedom of expression.
- **The Personal Data Protection Ordinance (PDPO)** grants citizens ownership of their data, establishing rights to access, correction, and erasure.
- **The National Data Governance Ordinance (NDGO)** mandates interoperability, creates the NDGA, and establishes the **Bangladesh National Digital Architecture (BNDIA)** as the technical standard for all government systems.

DPI as the Delivery Backbone (Interoperability-by-Design)

A central thrust is the shift from point-to-point integration to standardized, governed exchange across agencies—implemented through DPI components and mandatory technical requirements (e.g., API-based exchange, metadata standards, complete audit trails, and zero-trust patterns). The document explicitly enumerates key components in the national data exchange and trust stack (e.g., NRDEX, NSOC, TRUST-CCA) and sets a direction for standardized APIs and interoperable data flows across government services.

Digital Identity as a Foundational Layer

The strategy defines a **Digital Identity Framework (OneID)** to ensure clarity, interoperability, and trust across the identity lifecycle and across agencies. It notes that Bangladesh's identity ecosystem is anchored in the **National Identity Registration Act, 2010**, and highlights legal ambiguity created after the **Unique Identification Number (UIN)** concept introduced in a later Act was repealed—therefore recommending consolidation/clarification through legal review (e.g., amend the 2010 Act or create a new consolidated instrument).

It also specifies the high-level identity design: a **17-digit opaque identifier** designed for governance and scalability, intentionally **not encoding personal attributes** such as DOB/gender to reduce privacy risk.

Sectoral Transformation and Citizen Outcomes

Beyond core platforms, the strategy emphasizes sector-level transformation where DPI and identity enable practical outcomes. Examples include:

- **Digital Social Protection:** addresses fragmentation, duplication/leakage and transparency gaps; proposes a Unified Beneficiary Database under a national authority linkage and digitized G2P payments, plus analytics-driven targeting and

ministry dashboards, with explicit 2030 outcomes such as full digitization and leakage reduction targets.

- **Justice & Judiciary digitization:** targets the large case backlog with e-courts and end-to-end digital case management, with outcomes including digitization of new filings and backlog reduction targets by 2030.
- **Resilience & Public Safety:** integrates AI/IoT/satellite/drone-enabled early warning, pipeline/grid monitoring, emergency alerting and intelligent transport safety, backed by a KPI framework over 2025–2030 (coverage expansion and response-time reduction goals).

Human Capital: A Multi-Track National Skills Program

The strategy treats capacity building as a “make-or-break” constraint: growth in ICT exports and freelancing is acknowledged, but current training programs are described as fragmented, mismatched with industry demand, and unevenly distributed geographically—requiring institutionalized, outcome-focused reform.

It then proposes a **lifecycle-based, multi-track skills framework**, including foundational digital skills through schooling, plus scaled infrastructure such as digital classrooms and a national education cloud, supported by KPI tables that define annual scale-up targets through 2030 (e.g., coverage of digital classrooms, student e-learning usage, teacher ICT training volumes).

Institutional and Governance Reforms

The document includes a substantial reform agenda across institutions and governance mechanisms, emphasizing that digital transformation requires enforceable coordination, standards, procurement reform, cybersecurity capability, and data-center/cloud operating reforms. The table of contents indicates detailed reform sections for multiple institutions and cross-cutting governance topics (coordination, PPP, procurement, funding mechanisms, M&E).

Monitoring, Evaluation, and Accountability

A defined M&E framework is included with KPIs, real-time dashboarding, independent audits/third-party evaluation, and feedback loops—positioning performance management as an integral control layer, not an afterthought.

Phased Roadmap to 2030 and Beyond

The strategy uses phased implementation planning through 2030 and identifies some initiatives with longer time horizons beyond 2030. The roadmap section explicitly lays out multi-phase sequencing (foundation, scaling/optimization, consolidation/global

integration) and includes “long-term vision” items beyond 2030 such as semiconductor industry development and national drone coverage, as well as a deep-tech identity framing (BEAR: Biotech, Electronics, AI, Robotics).

Conclusion

The **National Digital Transformation Strategy 2025-2030** is not just a plan for digitization but a blueprint for a **unified, citizen-centric Bangladesh**. By harmonizing legal mandates with technical infrastructure and human capital development, it aims to build a state that is **efficient, transparent, and equitable**—where technology serves the citizen, and the digital economy drives national prosperity.

1. Strategic Vision & Context

1.1 Background

The National Digital Transformation Strategy provides a comprehensive framework for 2025-2030, addressing systemic challenges while fostering socio-economic growth and building a secure, inclusive, and technologically literate society.

The strategy is anchored in **nine core pillars**:

1. Unified Digital Public Infrastructure
2. Citizen-Centric Governance
3. Data Sovereignty & Legal Certainty
4. National Connectivity Backbone
5. Autonomous Regulatory Oversight Over Data and Privacy
6. Economic Resilience & Innovation
7. Future-Ready Human Capital
8. Institutional Interoperability
9. Proactive Security & Resilience

The overarching vision is to secure a resilient, sustainable, and future-ready economy that is technologically advanced yet equitable and human-centric.

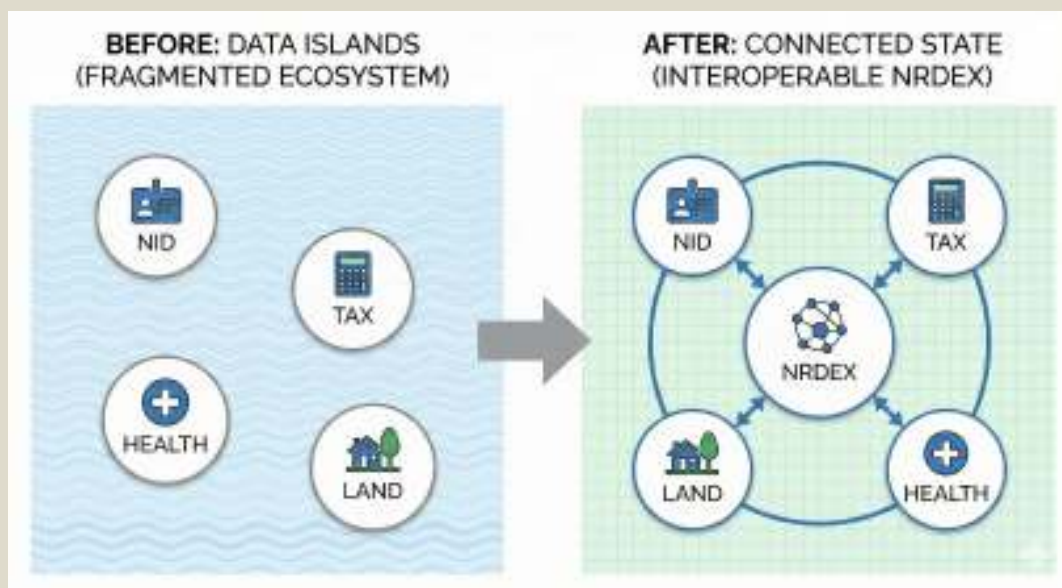
With over 120 million internet users and a rapidly growing digital economy, Bangladesh must act decisively. Outdated legal frameworks, siloed systems, and vendor lock-ins risk impeding innovation and increasing inequality. This strategy provides the roadmap for swift reforms that will protect sovereignty, empower citizens, and establish Bangladesh as one of South Asia's premier digital hubs.

Comprehensive Digital Governance and Reform for Bangladesh

Bangladesh faces enormous opportunities as well as significant challenges at this critical juncture in its digital transformation journey. The government has implemented a thorough digital transformation plan with a planned timeline from 2025 to 2030 in order to traverse this terrain. The goal of this roadmap is to go beyond simple digitization and create a strong, secure, and inclusive digital ecosystem that will promote economic expansion, improve the effectiveness of governance, and guarantee sustainable development. Building a robust, just, and future-ready economy that can take advantage of cutting-edge technologies like blockchain and artificial intelligence for the good of all citizens is the fundamental tenet of the strategy.

The following major goals form the basis of the strategy, which aims to modernize the country's digital policies and infrastructure:

1. Create Interoperable Digital Infrastructure: Dismantling the institutional silos that presently impede digital advancement is one of the main objectives. The nation's data systems are disjointed and do not have secure interoperability, including those for tax records, health information, and National ID (NID). The roadmap suggests establishing a National Data Exchange (NRDEX) and a national digital architecture to enable safe, instantaneous data sharing between government organizations. The quality and security of citizen data will be enhanced, operational expenses will be decreased, and redundant data collection will be eliminated. More effective and citizen-focused services will result from the guiding principle of "collect once, use many times."



2. Strengthen Cybersecurity & Data Governance: To guard against the increasing risk of cyberattacks and data breaches, the strategy gives cybersecurity top priority. This entails implementing the Personal Data Protection Ordinance, 2025 and its 2026 Amendment, which strengthens data sovereignty. In support of these legislative efforts, Bangladesh has set ambitious targets to train 20,000 cybersecurity experts by 2027 and increase this to 50,000 by 2030. Building a robust national defense against cyber threats and creating a workforce with the skills to meet demands worldwide depend heavily on this massive upskilling program.

3. Reform the ICT Division for Efficient Service Delivery: The ICT Division will experience substantial reform in order to guarantee the strategy's successful execution. The objective is to make it a more effective and flexible organization that can spearhead digital transformation in all government agencies. This will entail improving technical capability, simplifying procedures, and revising the guidelines for dealing with suppliers and other governmental bodies.

4. Expand Digital Services for Businesses & Citizens: The roadmap calls for a comprehensive transformation of the way government services are provided. Moving past partial digitization to a point where all services can be accessed and completed fully online is the goal of expanding end-to-end digital services. By eliminating the need

for in-person meetings and bureaucracy, this will significantly increase efficiency and convenience for citizens.

5. Facilitate a thriving innovation ecosystem and digital economy: The strategy is intended to stimulate economic expansion. This includes modernizing tax collection via the NBR API Connectivity Hub, establishing a Universal Payment Gateway to unify disparate payment systems, and improving the regulatory climate to attract foreign direct investment and startups. Fostering an innovation-driven ecosystem that can advance the country's digital economy is the aim.

6. By 2030, increase the number of ICT professionals to 7-8 million by upskilling them: Investing in human capital is a fundamental component of the strategy. The government has set the ambitious goal of increasing the number of ICT professionals from the current number to 7-8 million by 2030. This will be accomplished by a number of measures, such as updated university curricula and collaborations with foreign organizations to offer top-notch instruction and certifications.

7. Citizen-Centric "Paperless" Services: The roadmap calls for a comprehensive transformation toward 95% End-to-End Digital Services. The goal is to enforce a "Zero-Visit" mandate, where citizens can Apply, Pay, Verify, and Receive services (such as Land Mutation or Passport renewal) fully online via mobile devices, eliminating the need for physical lobbying and intermediary bureaucracy.

7. Future-Ready Human Capital (The BEAR Framework): Investing in human capital is a fundamental component. The government has set the ambitious goal of building a workforce of 8 million professionals by 2030. The focus shifts from basic literacy to Deep-Tech mastery under the BEAR framework (Biotech, Electronics, AI, and Robotics), supported by updated university curricula and international certification partnerships.

8. AI-Resilient Labor Market : The strategy incorporates a proactive plan to upskill blue collar workers and migrant workforce. Recognizing the threat of automation, the government aims to preserve jobs by implementing **"Human-in-the-Loop"** training programs, and just transition for sustainable job creation, ensuring workers are equipped with digital skills to manage AI-assisted manufacturing rather than being replaced by it.

8. Create NDGA: The roadmap suggests creating an independent Data & AI Authority to guarantee strong and objective governance. Operating under the Supreme Court's supervision, this body will be immune to administrative or governmental meddling. In order to uphold national data sovereignty and foster public trust, its mandate will include monitoring AI policies, enforcing data protection laws, and serving as an impartial arbiter in disputes involving data.

To sum up, the National Digital Transformation Strategy is a thorough and progressive road map for Bangladesh's future. By giving institutional reform, cybersecurity, workforce development, and ethical governance top priority, it tackles fundamental issues head-on. In addition to updating the nation's digital infrastructure, this roadmap

aims to empower its people, spur economic expansion, and establish Bangladesh as a leader in the world's digital arena.

1.2 Current Digital Landscape & Challenges

Bangladesh's ICT ecosystem faces significant systemic challenges that this strategy aims to address comprehensively:

1.2.1 Fragmented Digital Ecosystem

Major data repositories including the National ID (NID), Birth and Death Registration System (BDRIS), Civil Registration and Vital Statistics (CRVS), Bangladesh Bank, Bangladesh Bureau of Statistics (BBS), National Board of Revenue (NBR), Health MIS, and BANBAIS operate in isolation without secure interoperable mechanisms. This fragmentation results in:

- Redundant infrastructure and increased operational costs across ministries
- Lack of coordination among government agencies leading to duplicated efforts
- Citizen data vulnerabilities from inconsistent security protocols
- Ministries procuring standalone assets that fail to serve end-users effectively
- Data-informed policymaking and decision making is difficult when data is siloed

1.2.2 Data Governance & Security Risks

Bangladesh has experienced serious data breaches, with sensitive citizen information including NID data, birth and death registrations, financial transactions, and mobile banking details exposed on the dark web. The absence of a structured data protection policy and robust cybersecurity framework has led to:

- Unauthorized access and misuse of citizen data undermining public trust
- Cybersecurity vulnerabilities including weak firewalls and absence of a fully operational National CERT
- Mismanagement of government digital infrastructure leading to systemic inefficiencies
- Citizens more vulnerable to cyber-threats etc. due to data leakage

1.2.3 Incomplete Digital Service Delivery

While over 350 services have been digitized, most remain partially digital, requiring physical visits for verification and approvals. The MyGov and D-Nothi platforms have not achieved full adoption across ministries. Key gaps include:

- Continued dependency on manual verification that bottlenecks service delivery
- Inconsistent user experience across different government portals

- Poor service and interface designs which lead to increased reliance on middlemen for service delivery.
- Limited mobile optimization despite a comparatively higher number of mobile users.

1.2.4 Digital Economy Bottlenecks

Despite hosting over 100 million digital financial accounts, Bangladesh lacks:

- A universal payment ecosystem integrating banks, Mobile Financial Services (MFS), and government transactions
- Interoperable taxation, VAT, and financial reporting mechanisms
- A startup-friendly regulatory environment, limiting FDI inflows

1.2.5 Skills Gap

While Bangladesh has a large youth demographic, the workforce suffers from a critical "skills mismatch" that threatens future competitiveness. There is a profound gap between the output of the traditional education system and the demands of the global digital economy.

- There is currently no master plan identifying high-demand competencies. Training programs are often static, failing to adapt to rapid shifts in technology (e.g., Generative AI, Quantum Computing).
- While entry-level freelancers exist, there is a severe shortage of mid-level technical managers and solution architects capable of leading complex projects.
- Current initiatives struggle to reach key demographics—specifically rural women, madrasa students, and marginalized communities—leaving a significant portion of the potential workforce untapped.
- While basic digital literacy is improving, deep expertise in cybersecurity, cloud architecture, and data science remains scarce, forcing reliance on foreign consultants for critical infrastructure.

1.2.6 Policy Paralysis & Regulatory Gaps

The current legal and regulatory framework is reactive rather than proactive, creating friction that stifles innovation and export growth.

- Outdated foreign exchange regulations make it difficult for IT firms to pay for essential global services (cloud hosting, software licenses), forcing many to operate through informal channels or register offshore.
- The absence of mandated data protection laws leaves citizen data exposed to abuse by both state and non-state actors. There is no legal recourse for data breaches or unauthorized surveillance.
- Without a mandatory "Once-Only" data policy, government agencies continue to operate in silos, leading to massive duplication of efforts and citizen harassment.

- Policies fail to address emerging technologies. There is no clear framework for AI ethics, blockchain assets, or drone usage, creating uncertainty that deters foreign direct investment (FDI).
- The lack of a robust Cyber Safety Ordinance has left citizens vulnerable to cyber-harassment and financial fraud, eroding trust in digital platforms.

1.3 Vision Statement

"Establishing Bangladesh as a trusted, interoperable, and resilient nation by leveraging Digital Public Infrastructure (DPI) to drive inclusive growth, ensure data sovereignty, and empower every citizen through a connected governance ecosystem."

This vision captures Bangladesh's aspiration to leverage information and communication technology as the cornerstone of national development. It recognizes that digital transformation must serve all citizens equitably, driving economic prosperity while ensuring no community is left behind in the march toward a knowledge-based economy.

1.4 Mission Statement

"To build a secure and unified digital backbone by operationalizing the Bangladesh National Digital Architecture (BNDIA), mandating interoperable data exchange, and deploying citizen-centric DPI that eliminates fragmentation, ensures transparency, and catalyzes a future-ready digital economy."

The mission translates the vision into actionable mandates, emphasizing three interconnected objectives: modernizing governance structures, revolutionizing public service delivery, and catalyzing economic growth through targeted ICT investments. This tripartite approach ensures that digital transformation benefits both the state apparatus and the citizens it serves.

1.5 Guiding Principles

The strategy is built upon foundational principles that guide all reform initiatives and policy decisions:



Principle	Description
Citizen-Centric Design	All digital services and infrastructure are designed with the end-user in mind, prioritizing accessibility, simplicity, and inclusivity across all demographics and regions.
Interoperability First	Breaking down institutional silos through mandatory data exchange standards, unified APIs, and the "collect once, use many times" philosophy to eliminate redundancy.
Digital Sovereignty	Ensuring government ownership of critical source code, preventing vendor lock-in through open standards, and maintaining control over national data assets.
Transparency & Accountability	Publishing service metrics in real-time, conducting public audits of digital systems, and establishing clear KPIs for all reform initiatives.
AI-Driven Efficiency	Leveraging artificial intelligence and data analytics for evidence-based policymaking, predictive governance, and automated service delivery where appropriate.
Collaborative Ecosystem	Fostering partnerships between government, industry, academia, and civil society to drive innovation and ensure sustainable implementation.

Localized Innovation	Adopting global best practices while tailoring solutions to Bangladesh's unique socioeconomic context, with Bangla-first interfaces and local language accessibility.
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1.6 Nine Strategic Pillars

The roadmap is structured around nine interconnected pillars that collectively address Bangladesh's digital transformation needs:

1. **Unified Digital Public Infrastructure (DPI):** Establishing the Digital Public Infrastructure. This pillar shifts focus from individual systems to a unified framework for Digital Identity, Payment Gateways, and Secure Data Exchange.
2. **Citizen-Centric Governance:** Operationalizing the "Digital by Default" and "Bangla First" mandates. It ensures that all public services are inclusive, mobile-optimized, and end-to-end digital by 2030.
3. **Data Sovereignty & Legal Certainty:** Anchoring the digital ecosystem in a robust legal perimeter. This pillar ensures that national data assets are protected, individual privacy is sacrosanct, and digital rights are enforceable through the PDPO, CSO, and NDGO.



4. **National Connectivity Backbone:** Treating high-speed connectivity as a fundamental right. This involves transitioning to a National Fiber Bank and ensuring 100% fiber-to-tower connectivity to eliminate the rural-urban digital divide.

5. **Autonomous Regulatory Oversight:** Empowering the National Data Governance Authority (NDGA) as an independent constitutional body. The NDGA serves as the guardian of the national digital architecture, ensuring interoperability across all 100+ ministries.
6. **Economic Resilience & Innovation:** Fostering a \$10 billion digital economy. This pillar streamlines the startup ecosystem, enables cross-border digital trade, and modernizes fiscal systems like the Tax-VAT API Hub.
7. **Future-Ready Human Capital:** Moving from general literacy to Deep-Tech mastery. The goal is to build a workforce of 8 million professionals skilled in AI, Cybersecurity, and the BEAR framework (Biotech, Electronics, AI, Robotics).
8. **Institutional Interoperability:** Enforcing the "Collect Once, Use Many Times" principle. This eliminates bureaucratic friction by mandating that government agencies share data through secure, real-time APIs instead of manual transfers.
9. **Proactive Security & Resilience:** Transitioning from passive defense to Active Threat Intelligence. This pillar operationalizes the National Security Operations Center (NSOC) to protect critical information infrastructure in real-time.

1.7 Global Benchmarks

Digital Public Infrastructure (DPI) models pioneered by leading digital nations provide valuable reference points for Bangladesh's transformation:

Country/Region	Key Achievements	Lessons for Bangladesh
Estonia	99% of services online; X-Road data exchange; blockchain-secured identity; digital residency program	Federated architecture model; decentralized yet secure systems; annual savings equivalent to 2% of GDP
Singapore	GovTech Agency centralizes 1,300+ applications; rapid prototyping; citizen-focused innovation	Centralized tech procurement; service design standards; AI integration in healthcare and governance
India	Aadhaar digital identity for 1.4B citizens; UPI payment platform processing billions monthly; clear regulatory frameworks	Scale-appropriate DPI; economic inclusion through digital payments; population-scale identity management
South Korea	Data-driven policymaking; advanced 5G infrastructure; comprehensive e-government services	Analytics-based governance; high-speed connectivity as enabler; private sector collaboration models

Denmark/EU	GDPR compliance; green ICT practices; citizen privacy protection; DANIDA funding model	Strong privacy frameworks; sustainable digital practices; milestone-based funding for talent development
UK	Government Digital Service (GDS); agile procurement; open standards; user-centric design methodology	Skill-based recruitment; agile project management; value-driven procurement over lowest cost



Bangladesh's roadmap aligns with these established international models while placing strong emphasis on **affordability, inclusivity, and regional leadership** to ensure that digital transformation reflects the nation's socioeconomic priorities while maintaining global competitiveness.

1.8 Reform Urgency

With more than **120 million internet users** and a rapidly expanding digital economy, Bangladesh stands at a critical juncture. The urgency for reform stems from multiple converging factors:

Risk of Missed Opportunities: Global industries are rapidly adopting IoT, AI-driven automation, and green ICT practices. Without swift reforms, Bangladesh risks losing opportunities to attract foreign investment and create millions of high-skilled jobs.

Competitive Pressure: Regional competitors are advancing rapidly. Bangladesh must act decisively to establish itself as one of South Asia's premier digital hubs.

Security Imperatives: Past data breaches and security vulnerabilities demand immediate strengthening of cybersecurity frameworks and data protection mechanisms.

Citizen Expectations: A digitally aware population increasingly expects seamless, efficient, and transparent government services.

Bridging the Digital Divide: Despite high internet penetration, a stark disparity remains between urban centers and rural peripheries. Services such as telemedicine, e-learning, and digital commerce have to be scaled up to reach marginalized communities. Failing to bridge this gap risks creating a "two-tier" society where digital exclusion perpetuates economic poverty.

The Youth Employment Crisis & Skills Mismatch: Bangladesh faces a paradox of high graduate unemployment alongside a shortage of skilled ICT professionals. The traditional education system is struggling to keep pace with industry demands. Immediate reforms are essential to convert the country's "demographic dividend" into a "digital dividend" by upskilling millions of unemployed graduates in deep-tech fields (AI, Cybersecurity, Blockchain), ensuring they are employable in the global market.

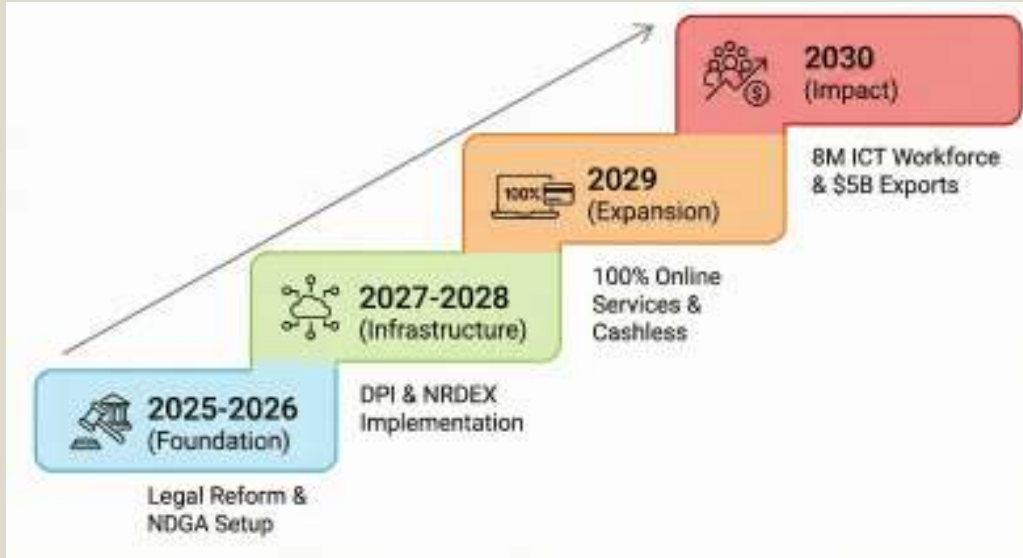
Safeguarding Civil Liberties & Data Privacy: As the government collects unprecedented amounts of personal data (NID, Health, Financial), the risk of misuse, whether by state actors or private corporations has grown exponentially. Reform is urgent to establish a rights-based framework (via the PDPO) that protects citizens from mass surveillance, unauthorized profiling, and data abuse, thereby restoring public trust in digital governance.

Mitigating AI-Induced Labor Displacement: The unregulated adoption of Artificial Intelligence and automation poses a significant threat to labor-intensive sectors like the Ready-Made Garments (RMG) industry and traditional BPOs. There is a critical need for proactive policy planning to address potential mass layoffs. The strategy must prioritize "Human-in-the-Loop" economic models and safety nets to ensure that AI serves as a tool for augmentation and productivity rather than a driver of sudden unemployment.

This urgency necessitates quick policy updates, robust regulatory organizations like NDGA, and strategic investments in digital infrastructure. Prompt reforms will **protect sovereignty, empower citizens, and position Bangladesh for sustainable digital growth.**

1.9 Strategic Objectives (2025-2030)

The following major objectives form the basis of the strategy, aimed at modernizing the country's digital policies and infrastructure:



Strategic Objective	Description & Target
Unified Digital Public Infrastructure (DPI)	Operationalize the "National Digital Rail" via the NRDEX platform. Mandate the "Collect Once, Use Many Times" principle to eliminate data silos and ensure seamless, secure API-based exchange across all agencies.
Proactive Cyber Resilience	Operationalize the Cyber Safety Ordinance (CSO). Transition from reactive defense to Active Threat Intelligence via the NSOC. Train 50,000 cybersecurity professionals to protect Critical Information Infrastructure (CII).
Institutional Restructuring	Redefine ICT Division as the "Policy Architect" and DoICT as the "Grassroots Bridge." Enforce Vendor Neutrality and open standards in procurement to break vendor lock-in and ensure competitive pricing.
Citizen-Centric "Paperless" Services	Achieve 95% End-to-End Digital Services by 2030. Mandate a "Zero-Visit" policy for verification, ensuring all services (Apply, Pay, Verify, Receive) are accessible via mobile without physical office visits.
Digital Economy & Fiscal Integration	Deploy the NBR API Connectivity Hub for real-time tax transparency and the Universal Payment Gateway for cashless transactions. Foster a startup-friendly climate to

	achieve \$10 Billion in ICT exports by 2030.
Future-Ready Human Capital	Build a workforce of 8 million professionals focused on Deep-Tech mastery. Shift curriculum focus to the BEAR Framework (Biotech, Electronics, AI, Robotics) to move beyond basic data entry to high-value engineering.
AI-Resilient Labor Market	Proactively upskill the RMG and Migrant workforce to mitigate automation risks. Implement "Human-in-the-Loop" training programs to ensure technology augments rather than replaces labor during the Industry 4.0 transition.
Autonomous Regulatory Oversight	Operationalize the National Data Governance Authority (NDGA) as an independent constitutional body. Empower it to enforce the PDPO, audit algorithms for bias, and serve as the supreme guardian of citizen data rights.
Data Sovereignty & Legal Certainty	Enforce the "Public Money, Public Code" policy to ensure state ownership of source code. Establish a robust legal perimeter that guarantees Data Sovereignty, protecting national data assets from unauthorized foreign jurisdiction.

End of Chapter 1

2. Legal Landscape

2.1 Why Legal Reform Matters

Bangladesh's digital transformation agenda requires more than new platforms and digitized services; it requires legal certainty that enables trust, safeguards citizen rights, and creates enforceable rules for data and cyber risk management. The strategy therefore treats the legal landscape as a foundational layer for the entire reform program, particularly for:

- **Trust and legitimacy:** citizen confidence in digital services depends on privacy protection, transparent enforcement, and meaningful redress.
- **Interoperability at scale:** secure data exchange across agencies requires clear legal authority, consistent standards, and defined accountability.
- **Future-readiness:** emerging technologies (AI, data exchange infrastructure, identity systems, cloud) require modern terminology, governance models, and compliance duties.

In 2025, the Government enacted three landmark ordinances that collectively establish the legal foundation for a secure, interoperable, and citizen-centric digital ecosystem:

1. **Cyber Safety Ordinance, 2025 (CSO)** – Addressing cybercrime, online safety, and digital security
2. **Personal Data Protection Ordinance, 2025 (PDPO)** – Protecting citizens' privacy and data rights
3. **National Data Governance Ordinance, 2025 (NDGO)** – Enabling secure data sharing and interoperability



These three ordinances work in concert to create a comprehensive legal architecture. The CSO establishes the security perimeter, the PDPO protects individual rights within that perimeter, and the NDGO enables productive and secure use of data across government and the private sector. Together, they align Bangladesh with international standards such as GDPR while addressing the nation's unique socioeconomic context.

2.2 Cyber Safety Ordinance, 2025 (CSO)

2.2.1 Background and Purpose

The Cyber Safety Ordinance, 2025 has been enacted to update and replace previous legislation, including the Digital Security Act, 2018 and the Cyber Security Act, 2023. This new legal framework modernizes terminology, protects citizens' rights, and addresses emerging cyber threats while balancing security needs with freedom of expression.

Addressing Past Misuse

A core driver for this reform was the documented misuse of previous legislation for **politically motivated harassment**. For example, under the DSA and CSA, hundreds of cases were filed against journalists, activists, and ordinary citizens for "tarnishing the image of the state" or "hurting religious sentiments". Notable instances of misuse included:

- **Arbitrary Arrests:** Numerous individuals were detained for social media posts, often without clear evidence of a crime, leading to prolonged pretrial detentions.
- **Effect on Expression:** Broadly defined sections regarding "defamation" and "propaganda" were used to suppress legitimate dissent and investigative journalism.

The CSO decisively breaks this cycle by **repealing the CSA, 2023** and **annulling all pending cases** under its most controversial sections (including sections 21, 25, 28, and 29).

A Shield for the Citizens, not Government

The new ordinance pivots focus toward protecting citizens—especially **women and children**—from modern, non-political cybercrimes that were previously neglected. It introduces explicit protections against:

- **Gender-Based Crimes:** Criminalizing Non-Consensual Sexual Imagery (NCSI), "revenge porn," AI-generated deepfake pornography, and sextortion.
- **Protection of Minors:** Dedicated provisions for Child Sexual Abuse Material (CSAM).

- **Procedural Guardrails:** To prevent harassment, only **the police, the victim, or their legal representatives** are now permitted to file a case, effectively ending the practice of third-party politically motivated suits.

By including **domain experts and civil society** in the judiciary process and the National Cyber Security Council, the CSO ensures that technical and human rights perspectives guide the law's application, moving Bangladesh toward a trusted and secure digital perimeter.

2.2.2 Key Features and Provisions

Recognition of Rights

For the first time in Bangladesh's legal framework, the ordinance recognizes citizens' right to continuous internet access as part of their cyber protection. This landmark provision acknowledges the internet as essential infrastructure for modern life and economic participation.

Modernized Definitions and Terminology

Section 2 introduces and modernizes several critical definitions to address contemporary digital challenges:

- **Digital Identity** – Formal recognition of citizens' online identities
- **National Security Operations Center (NSOC)** – Central hub for cyber threat monitoring
- **Cyber Terrorism** – Attacks on digital infrastructure with intent to cause mass harm, loss of life
- **Pornography Harassment** – Non-consensual sharing of intimate content
- **Child Sexual Abuse Material** – Specific provisions for protecting minors
- **Disinformation Campaigns** – Coordinated efforts to spread false information

Proactive Threat Intelligence and Response

The ordinance establishes a robust institutional framework for real-time cyber defense, transitioning from reactive recovery to proactive, intelligence-led protection.

- **NCERT and NSOC Operations:** It mandates the creation of the National Cyber Emergency Response Team (NCERT) to coordinate national incident response and the National Security Operations Center (NSOC) to serve as the apex monitoring hub for the state's digital perimeter.
- **Mandatory Reporting for CII:** Entities designated as Critical Information Infrastructure (CII)—including banks, power grids, and telecommunication hubs—are now legally required to report cyber incidents to the NCERT and maintain secure connectivity with the NSOC for real-time threat monitoring.
- **Active Intelligence Sharing:** The framework moves beyond passive defense to active intelligence sharing. The National Cyber Security Agency (NCSA) is

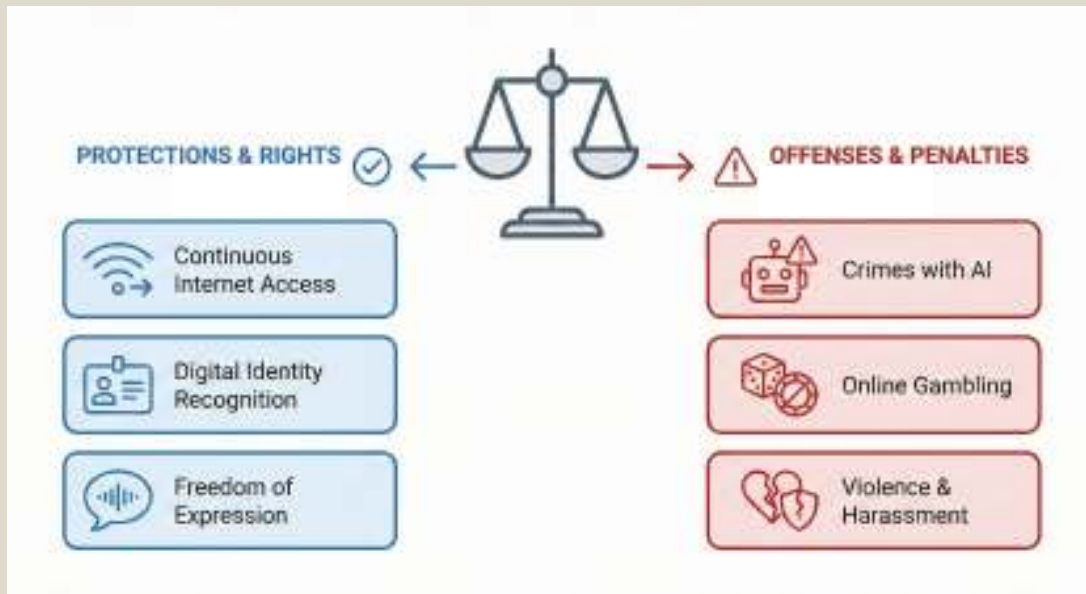
empowered to analyze telemetry data and disseminate actionable threat intelligence across domestic sectors.

- **International Cooperation and Threat Intel:** Recognizing that cyber threats are borderless, the CSO specifically mandates International Threat Intelligence Sharing. The NCSA is authorized to act as the national point of contact for global cybersecurity alliances, enabling Bangladesh to exchange real-time threat data, malware signatures, and vulnerability research with international CERTs, foreign security agencies, and global technology partners. This ensures the national defense is fortified by global early-warning systems against emerging zero-day exploits and state-sponsored actors.

Criminal Offenses

The ordinance establishes clear categories of punishable offenses:

Offense Category	Description & Provisions
Crimes with AI	Crimes committed using artificial intelligence or AI-powered tools are explicitly punishable, addressing emerging technological threats.
Online Gambling	Section 20 addresses and criminalizes cyber gambling operations.
Violence & Harassment	Section 25 addresses pornography harassment, blackmailing, and sextortion. Section 25(3) includes special provisions for offenses against women and children. "Revenge porn" is defined as non-consensual sharing of intimate images.
Identity Forgery	Section 23(4) categorizes forgery of citizen's national identity as a punishable crime.
Instigating Violence	Section 26 addresses publication of content that clearly instigates religious or communal hatred and violence, while preserving freedom of expression.



Content Blocking and Removal

- **Section 8(1)** grants the National Cybersecurity Agency (NCSA) power to remove or block digital content posing threats to national security or public safety
- **Section 8(3)** mandates the government to maintain full transparency regarding all blocked or removed digital content. Any content removed must be unblocked if tribunal permission is not obtained within three days of the action. Furthermore, the relevant authority is obligated to periodically release a comprehensive list of all blocked and removed content to ensure public transparency and institutional accountability.

Justification: The three-day tribunal requirement balances security needs with due process, preventing arbitrary censorship while allowing rapid response to genuine threats.

National Council Restructuring

- Section 12 details the restructuring of the National Cyber Security Council
- Section 12(1) provides for expert individuals or civil society representatives as council members
- The Chief Adviser or Prime Minister serves as Chairman of the Council

Victim Protection and Discharge

- **Section 30** provides for monetary compensation to victims, ordered by the tribunal
- **Section 31(4)** allows immediate release of individuals if no evidence is found for offenses like cyber terrorism, pornography harassment, and blackmailing

Repeal of Previous Laws

The ordinance repeals the Cyber Security Act, 2023 and annuls all pending cases under specific sections of that act, including sections 21, 24, 25, 26, 27, 28, 29, 31, and 34.

Section 46 specifies which offenses are bailable, non-bailable, or compoundable, providing clarity for law enforcement and judiciary.

2.2.3 Implementation Framework

To ensure effective implementation of the CSO, the following supporting measures are being developed:

- **Bangladesh Cybersecurity Strategy 2026-2030:** National-level framework for building a resilient and secure digital ecosystem
- **Cyber Safety Policy:** Supplementary guidance for public and private sectors on secure digital operations
- **NSOC Establishment:** \$5 million World Bank proposal through EDGE project for National Security Operations Center
- **Specialized Training:** Programs for judiciary and law enforcement on cybercrime investigation and digital forensics
- **Public Awareness Campaigns:** Multimedia content in Bengali and English focusing on digital hygiene and social engineering awareness

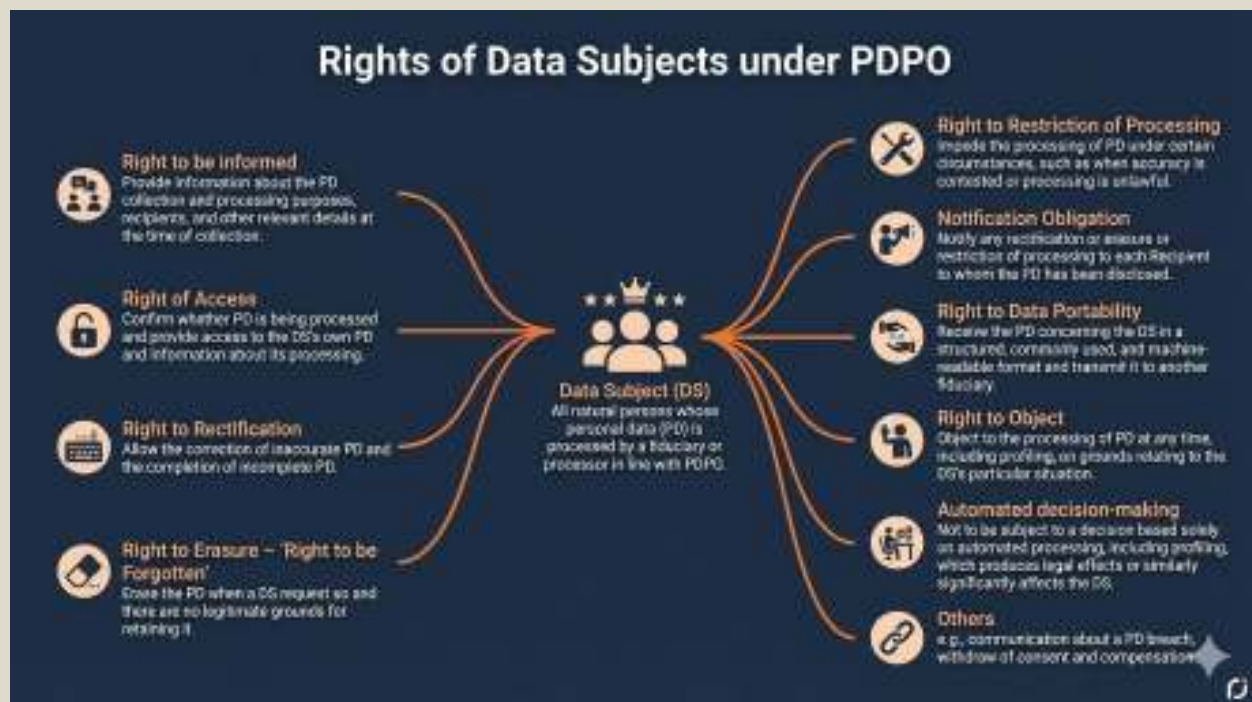
2.3 Personal Data Protection Ordinance, 2025 (PDPO)

2.3.1 Background and Purpose

The Personal Data Protection Ordinance, 2025 establishes a comprehensive legal framework for protecting personal data and privacy rights of citizens in Bangladesh. This ordinance introduces modern data protection standards aligned with international best practices, including GDPR principles, while accounting for Bangladesh's developmental context.

2.3.2 Recognition of Data Subject Rights

The ordinance recognizes citizens' fundamental rights regarding their personal data, establishing unprecedented control over how personal information is collected, processed, and shared:



Right	Description (Section)
Right to Access	Data subjects can access their personal data held by data-fiduciaries (Section 10)
Right to Data Portability	Right to receive personal data in structured, commonly used format and transfer to another data-fiduciary through Federated Interoperable Ecosystems (Section 11)
Right to Correction & Erasure	Data subjects can request correction of inaccurate data and erasure under specified conditions (Section 12)

Right to Consent Withdrawal	Data subjects can withdraw consent and request deletion of their personal data (Section 13)
System-wide Propagation	Updates, corrections, or erasures must be propagated across all systems where the data is stored (Section 14)

2.3.3 Data Processing Requirements

Lawful Basis for Processing

Section 5 requires data-fiduciaries to obtain explicit consent from data subjects before processing personal data, with specific conditions for lawful processing. This consent-first approach ensures citizens maintain control over their information.

Sensitive Personal Data

Section 6 provides special protections for sensitive categories including:

- Genetic data and biometric data
- Health data and medical records
- Sexual orientation information
- Financial data
- Children's data (requiring verifiable parental consent under Section 8 for individuals under 18)

2.3.4 Data-Fiduciary Obligations

- **Transparency (Section 15):** Data-fiduciaries must maintain transparent practices and inform data subjects about data processing
- **Security Measures (Section 17):** Implementation of technical and organizational security measures to prevent data breaches
- **Data Breach Notification (Section 20):** Mandatory notification to the Authority and affected data subjects in case of personal data breaches
- **Data Protection Impact Assessment (Section 21):** Required for significant data-fiduciaries processing sensitive data
- **Chief Data Officer (Section 23):** Significant data-fiduciaries must appoint a Chief Data Officer

2.3.5 Cross-Border Data Transfer

- **Section 29** establishes conditions for transferring personal data outside Bangladesh, requiring adequate protection levels in recipient jurisdictions. Pursuant to the 2026 Amendment to Section 29, any cloud-based processing of Critical Information Infrastructure (CII) or Restricted Personal Data requires the maintenance of at least one synchronized real-time copy within Bangladesh.

- **Section 30** provides for anonymization, pseudonymization, and de-identification of data for cross-border transfers

2.3.6 Offenses and Penalties

Offense	Penalty
Non-compliance with consent requirements (Section 36)	Up to 3 years imprisonment or fine up to 10 lakh taka
Unauthorized processing of sensitive personal data (Section 37)	Up to 5 years imprisonment or fine up to 20 lakh taka
Unauthorized disclosure of children's personal data (Section 38)	Up to 3 years imprisonment or fine up to 5 lakh taka
Unauthorized interception or extraction of personal data (Section 39)	Up to 5 years imprisonment or fine up to 10 lakh taka
Unauthorized disclosure of personally identifiable information (Section 41)	Up to 5 years imprisonment or fine up to 15 lakh taka
Corporate liability for company violations (Section 44)	Officers and company held jointly liable
Violations by government officials (Section 47)	Government departments and officers held accountable

2.4 National Data Governance Ordinance, 2025 (NDGO)

2.4.1 Background and Purpose

The National Data Governance Ordinance, 2025 establishes a comprehensive framework for data governance, interoperability, and management across Bangladesh. This ordinance creates the National Data Governance Authority (NDGA) and sets standards for data sharing, protection, and utilization in both public and private sectors. It represents the enabling infrastructure for Bangladesh's transition from fragmented "data islands" to an integrated, trusted digital state.

2.4.2 Key Definitions and Concepts

Section 2 introduces comprehensive definitions critical to understanding the framework:

- **Interoperability:** The ability of different systems to exchange and use information through technological, semantic, legal, and organizational dimensions
- **Zero-Trust Architecture:** Security model requiring verification for every access request regardless of network location
- **Digital Public Infrastructure (DPI):** Shared, secure, interoperable digital systems including identity, payments, and data exchange
- **NRDEX Platform:** National Responsible Data Exchange platform for trusted API-based data sharing
- **BNDIA:** Bangladesh National Data Governance and Interoperability Architecture

2.4.3 National Data Governance Policy Board

Establishment and Structure

- **Section 5** creates the National Data Governance Policy Board with 15 members including government officials, technical experts, private sector representatives, and civil society members
- **Section 8** establishes the Chief Adviser as Chair
- **Section 10** establishes a Secretariat for administrative support
- The NDGA is an independent statutory body with perpetual succession and its own fund



2.4.4 Organizational Structure of the NDGA

NDGA will be headed by a Chairperson and up to six Members, supported by six specialized divisions:

1. **Data Protection & Management Division** – data classification, privacy, and consent management
2. **Technical Standards & Interoperability Division** – API certification, metadata standards, and NRDEX operations
3. **Legal & Compliance Division** – PDPO, cybersecurity, and enforcement
4. **Administration & Finance Division** – budgeting, staffing, and fund oversight
5. **Coordination & Capacity Building Division** – sectoral data offices, training, and awareness
6. **Citizen Data & AI Governance Division** – unified digital ID, AI registry, algorithmic ethics, and innovation sandbox

2.4.5 Mandate and Core Functions of the NDGA

Domain	Key Responsibilities
Data Governance & Interoperability	Implement BNDIA; operate NRDEX; standardize APIs; ensure metadata and audit trails; maintain the National Citizen Data Repository
Privacy & Security	Enforce PDPO and Cybersecurity Ordinance; monitor breaches; accredit data centers; maintain encryption and TRUST-CCA standards

AI Ethics & Model Oversight	Certify AI algorithms used by public agencies; manage national AI registry; audit datasets for bias and transparency; maintain AI innovation sandbox
Compliance & Enforcement	Conduct inspections and VAPT audits; impose penalties for misuse; coordinate with the Grievance Redress Council
Capacity Building	Train 5,000 officials in data stewardship and AI ethics; publish open data handbooks and interoperability toolkits

Powers and Functions (Section 23)

- Enforce ordinance provisions and ensure compliance across government
- Manage BNDIA and NRDEX Platform operations
- Ensure data interoperability across government and private sectors
- Facilitate data sharing through Data Resharing Agreements (DRA)
- Oversee digital infrastructure development
- Manage electronic identity (e-ID) systems with unified identity management
- Establish National Datacentre Ecosystem (NDE) with disaster recovery
- Provide technical certifications and compliance oversight
- Issue sector-specific guidelines for banking, MFS, and digital payments

2.4.6 Chief Data Officers (CDOs)

Each ministry will appoint a **Chief Data Officer (CDO)** responsible for data inventory, interoperability compliance, and coordination with NDGA. This ensures institutional accountability and creates a network of data stewards across government.

2.4.7 Data Classification Framework

Section 31 establishes a four-tier classification system for data repositories:

NDGO Data Classification Hierarchy



Data Category	Description
Core Citizen Data	11-digit citizen identification, Tax Identification Number (TIN), birth/death registration, land records, judicial system repositories
Government Policy & Administration	Internal government data used for policy formulation and administrative decisions
Public Sector Service Delivery	Data generated through public service delivery requiring interoperability across agencies
Private Sector Service Delivery	Data from private sector entities regulated under NDGA guidelines for banking, MFS, and digital payments

2.4.8 Organizational Responsibilities (Section 34)

All organizations must comply with the following requirements:

- Compliance with ordinance and BNDIA guidelines
- Purpose-based data processing and interoperability
- Use of NRDEX Platform for all data exchange
- Technology infrastructure modernization with zero-trust architecture
- Cloud-first policies and API-based integration
- Open-source platform adoption where applicable

- Support for Digital Public Infrastructure development

2.4.9 Cross-Border Data Transfer and Penalties

Section 43 requires Authority approval for international data transfers. Violations are punishable by up to **25 years imprisonment or fine up to 25 lakh taka**, reflecting the seriousness with which Bangladesh views data sovereignty.

Section 60 establishes a three-year transitional period for organizations to achieve full compliance.

2.4.10 Dispute Resolution Mechanism

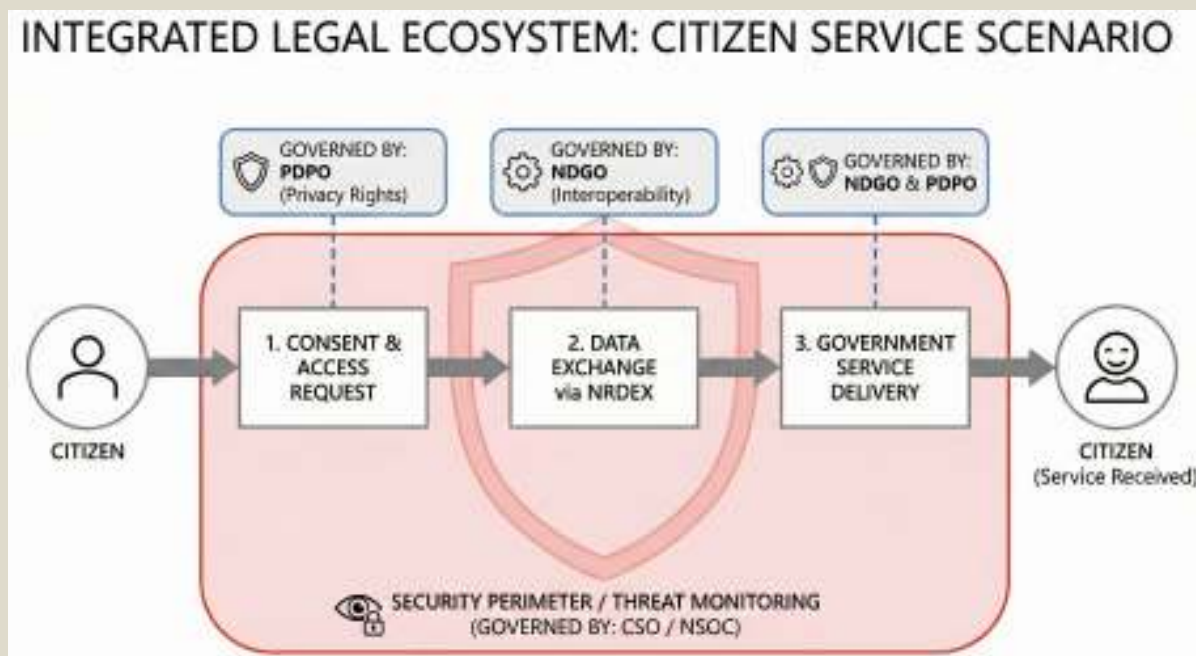
- **Grievance Redressal Committee (Sections 46-48):** Handles citizen complaints with decisions within 30 days (extendable to 45 days)
- **Appeals (Section 49):** Appeals to designated tribunal within 30 days
- **Tribunal Powers (Section 50):** Can summon witnesses and examine evidence

2.5 Legal Framework Integration

2.5.1 Complementary Roles

The three ordinances form an integrated legal architecture where each addresses a distinct but interconnected domain:

Aspect	CSO	PDPO	NDGO
Primary Focus	Cybersecurity & Crime Prevention	Individual Privacy Rights	Data Sharing & Interoperability
Key Institution	NCSA, NSOC	Data Protection Authority	NDGA
Primary Beneficiary	Society & State	Individual Citizens	Government & Economy
Core Platform	NSOC	Consent Management	NRDEX, BNDIA



2.5.2 Cross-References and Dependencies

The ordinances explicitly reference each other to ensure coherent implementation:

- **NDGO Section 42** requires organizations to establish consent management systems per the PDPO
- **NDGA's Legal & Compliance Division** is responsible for enforcing both PDPO and the Cybersecurity Ordinance
- **Data breach notifications under PDPO** are coordinated with NCSA under CSO provisions

- **AI governance under the National AI Policy** is supported by the data governance framework of PDPO and NDGO

2.6 Strategic Impact of Legal Reforms

Strategic Impact of the Legal Trinity

The simultaneous enactment of the Cyber Safety Ordinance (CSO), Personal Data Protection Ordinance (PDPO), and National Data Governance Ordinance (NDGO) creates a synergistic "Legal Trinity" that fundamentally reshapes Bangladesh's digital ecosystem. This framework moves beyond isolated regulation to create a cohesive environment for trust, innovation, and security.

Impact on Citizen Trust & Rights

- **From Fear to Empowerment:** By repealing the Digital Security Act and introducing the CSO, the state shifts from a posture of surveillance to one of protection. Citizens regain the confidence to participate in the digital economy without fear of arbitrary harassment, fostering a vibrant online civil society.
- **Data Ownership:** The PDPO grants citizens legal ownership of their digital footprint. The "Right to Forget" and "Consent Management" provisions empower individuals to control how their personal data is used by both state and private entities, building profound trust in digital services.

Impact on Digital Economy & Innovation

- **Market Certainty for Investors:** The clear data classification and cross-border transfer rules in the PDPO and NDGO provide international investors with regulatory certainty. This predictability is crucial for attracting FDI into the high-tech sector, as global firms require GDPR-aligned frameworks to operate.
- **Unlocking the API Economy:** The NDGO's mandate for interoperability and the establishment of the **NRDEX** platform eliminates the "data silos" that stifle innovation. Startups can now build value-added services on top of government data (e.g., credit scoring using utility payment history), catalysing a new wave of fintech and govtech solutions.

Impact on National Security & Sovereignty

- **Proactive Cyber Resilience:** The CSO's institutionalization of the **NSOC** and **NCERT** transitions national defense from reactive firefighting to proactive threat intelligence. Mandatory reporting for Critical Information Infrastructure (CII) ensures that the state has real-time visibility into systemic risks.

- **Data Sovereignty:** By enforcing data localization for critical sectors while allowing controlled cross-border flows for commerce, Bangladesh asserts sovereign control over its most valuable national asset—its data—without isolating itself from the global internet.

2.7 Key Highlights Summary

4. **Centralized Governance:** NDGA created with broad regulatory powers over data, privacy, and AI
5. **Interoperability Platform:** NRDEX and BNDIA ensure seamless data exchange across sectors
6. **Digital Infrastructure:** Framework for secure, interoperable systems including unified identity management
7. **Rights Protection:** Comprehensive data subject rights aligned with GDPR principles
8. **Strict Cross-border Controls:** Authority approval required for international data transfers
9. **Significant Penalties:** Heavy fines and imprisonment for violations ensure compliance
10. **Three-Year Transition:** Phased implementation approach allowing organizations to adapt

End of Chapter 2

3. Digital Public Infrastructure (DPI)

3.1 Introduction and Definition

Digital Public Infrastructure (DPI) refers to the set of shared, secure, and interoperable digital systems that form the foundation for delivering public and private services. These core systems typically include national digital identity platforms, payment gateways, data exchange frameworks, and the technical standards that allow different services to work together seamlessly.

A clear and strategic definition of DPI is essential for guiding Bangladesh's digital transformation. It provides a common framework for ministries, agencies, and stakeholders; ensures investments are inclusive, citizen-focused, and secure; and supports alignment with international standards. Without a shared definition, efforts risk becoming fragmented, leading to duplication, inefficiency, and weaker governance.

3.1.1 Bangladesh DPI Definition

As defined in the National Data Governance Ordinance 2025 (NDGO), Digital Public Infrastructure encompasses:

- **National Digital Identity Platforms:** Systems enabling citizen identification, verification, and authentication
- **Payment Gateways:** Universal payment systems integrating banks, MFS, and government platforms
- **Data Exchange Frameworks:** Secure, API-based platforms for cross-agency data sharing (NRDEX, BNDIA)
- **Technical Standards:** Zero-trust architecture, pseudonymized identity tokens, data minimization principles, Data Sovereignty & Mirroring

3.1.2 Implementation Approach

Bangladesh will formally adopt a national DPI definition through policy that:

1. **Clearly defines the scope** — including identity, payments, data exchange, and interoperability tools
2. **Establishes governance principles** — covering data privacy, cybersecurity, accessibility, and accountability
3. **Adopts a hybrid model** — with government managing core infrastructure while enabling private sector, academia, and civil society to build interoperable services
4. **Integrates the DPI definition** into the ICT Strategic Roadmap and relevant sectoral strategies

3.2 The Interoperability Journey: MESH → STAR → DSTAR

To move toward an integrated, trusted, and intelligent digital state, the Government of Bangladesh has adopted a phased interoperability roadmap—**MESH** → **STAR** → **DSTAR**—anchored in the National Data Governance Ordinance 2025 (NDGO). This ordinance legally establishes an autonomous National Data Governance & AI Authority (NDGA) under the Prime Minister's Office to regulate, facilitate, and secure cross-agency data exchange and responsible AI use.

The journey represents a systemic transition from isolated data systems to a cohesive digital public infrastructure. Each stage builds upon the previous, progressively introducing new architectural components, governance mechanisms, and technical capabilities that transform how government agencies interact with each other and serve citizens.

3.3 Stage 0: MESH to STAR Transition

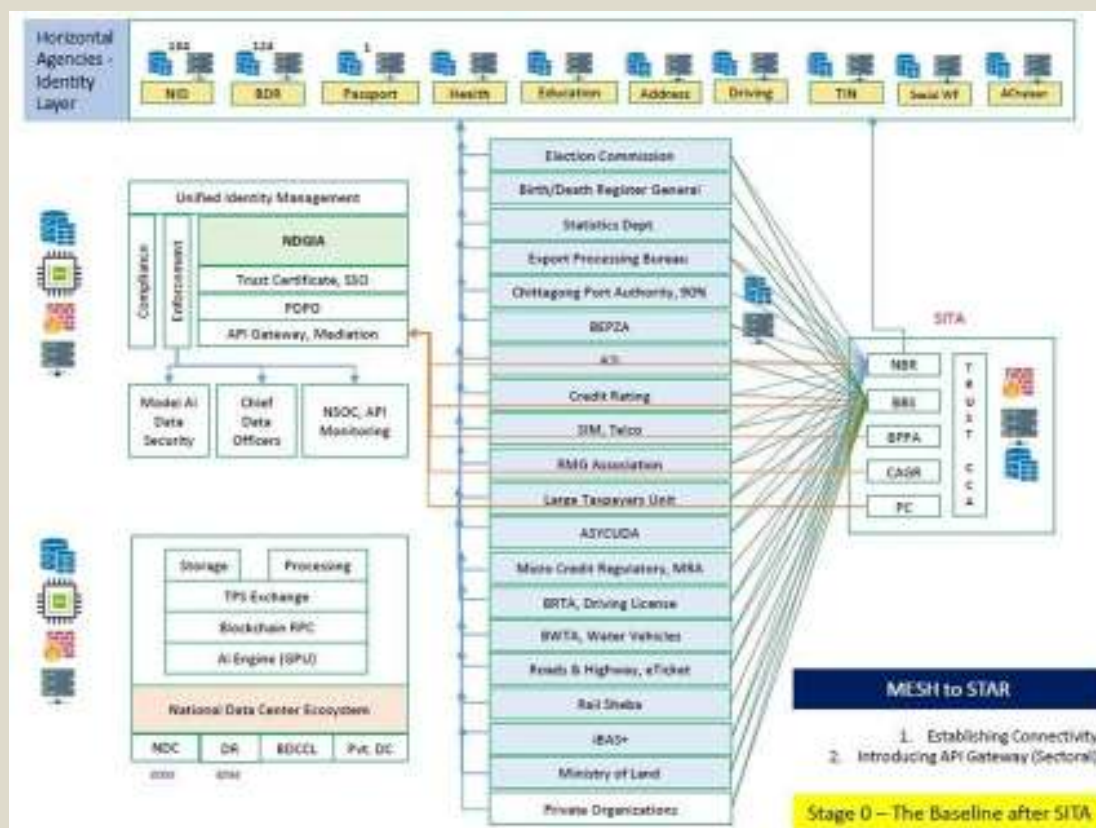


Figure 3.1: Stage 0 – MESH to STAR

Architectural Overview

Stage 0 represents the foundational architecture implementation. At this baseline, the system exhibits the following characteristics:

Horizontal Identity Layer: At the top of the architecture, identity registries remain siloed. The National ID (NID) database contains 184 million records, the Birth and Death Registration (BDR) system holds 124 million records, and Passport maintains 1 million records. Additional identity sources include Health, Education, Address, Driving, TIN, Social Welfare, and AChalaan—each operating independently without a unified identity management framework.

SITA: The SITA (Strengthening Institutions for Transparency and Accountability), a World Bank project which aims to connect key agencies including NBR (National Board of Revenue), BBS (Bangladesh Bureau of Statistics), BPPA (Bangladesh Public Procurement Authority), CAGR (Comptroller and Auditor General), and PC (Planning Commission) through an integration point. This gateway represents the first attempt at cross-agency data exchange but operates with limited scope.

Fragmented Service Connections: Multiple service agencies are connected through point-to-point integrations rather than through a centralized gateway. These include the Election Commission, Birth/Death Register General, Statistics Department, Export Processing Bureau, Chittagong Port Authority, BEPZA, A2i, Credit Rating agencies, SIM/Telco providers, RMG Association, Large Taxpayers Unit, ASYCUDA, Micro Credit Regulatory (MRA), BRTA (Driving License), BWTa (Water Vehicles), Roads & Highway (eTicket), Rail Sheba, iBAS+, Ministry of Land, and Private Organizations. Each connection creates a mesh-like network with inherent inefficiencies.

Trust Framework (TRUST-CCA): A preliminary trust and certificate authority framework exists to ensure authenticity and integrity of data transactions, but it is not yet universally enforced across all connected systems.

Key Actions in MESH to STAR Transition

1. **Establishing Connectivity:** Creating the initial network links between previously isolated systems
2. **Introducing API Gateway (Sectoral):** Deploying sector-specific API gateways to enable structured data exchange

Challenges Addressed

The MESH stage presented significant challenges: duplicated citizen data across registries, inconsistent data formats, no standardized APIs, security vulnerabilities from multiple point-to-point connections, and the inability to implement cohesive governance policies. Stage 0 marks the beginning of addressing these issues through the SITA project while acknowledging that substantial architectural improvements are still required.

3.4 Stage 1: The Foundation of NDGO (STAR to DSTAR Transition)

Stage 1 represents a transformational leap in Bangladesh's digital architecture. This phase introduces the foundational elements required for the National Data Governance Ordinance (NDGO) implementation, establishing the critical infrastructure components that will enable true interoperability.

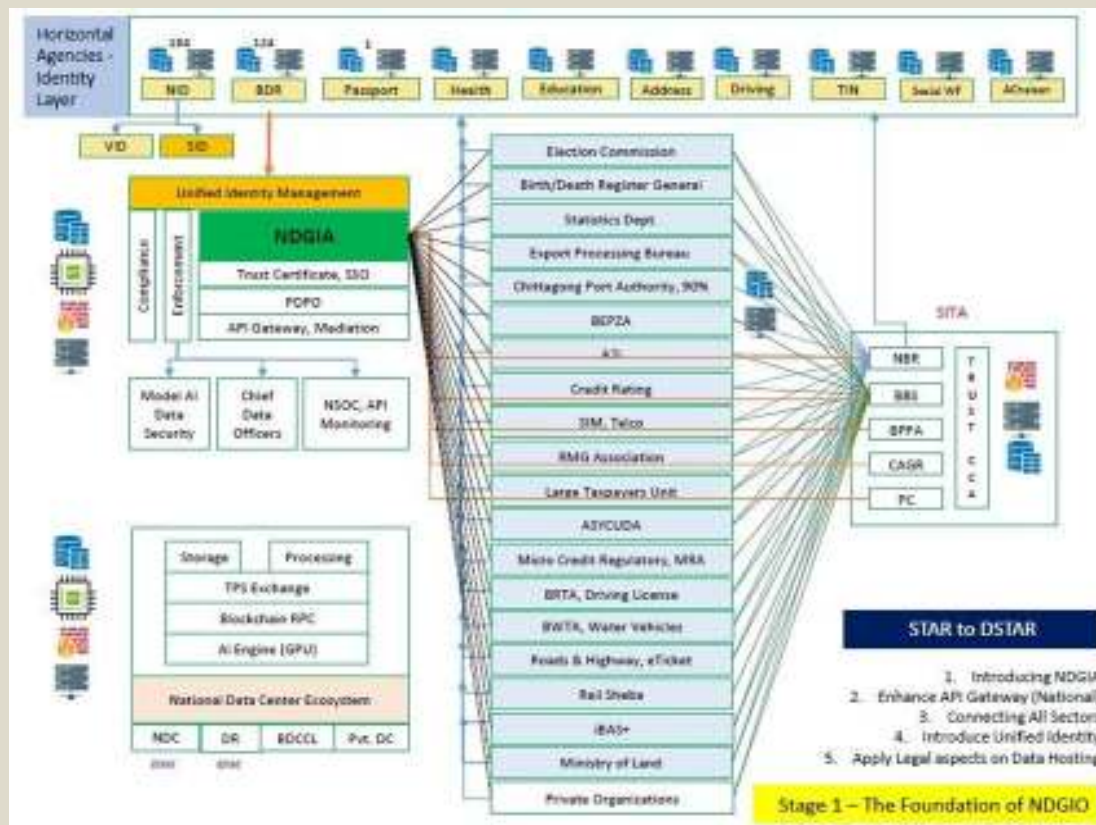


Figure 3.2: Stage 1 – Foundation of NDGO (Initial Phase)

Key Architectural Enhancements

1. Introduction of Virtual ID (VID) and Sectoral ID (SID): A critical enhancement at this stage is the introduction of derived identity tokens. The Virtual ID (VID) provides citizens with a privacy-preserving identifier that can be used instead of their actual NID for service transactions, reducing the risk of identity theft. The Sectoral ID (SID) enables sector-specific identification without exposing the core national identity, allowing for controlled data access based on purpose limitation principles enshrined in the Personal Data Protection Ordinance (PDPO).

2. NDGA (National Data Governance Authority): The establishment of NDGA marks the institutional foundation for data governance. Operating under the Unified Identity Management framework, NDGA serves as the central authority responsible for

implementing the Bangladesh National Data Governance and Interoperability Architecture (BNDIA), operating the National Responsible Data Exchange (NRDEX) platform, certifying and monitoring digital platforms handling citizen data, and coordinating Chief Data Officers across ministries.

3. Enhanced Compliance and Enforcement Layer: The architecture introduces a robust compliance framework including Trust Certificate and Single Sign-On (SSO) capabilities for secure authentication, PDPO enforcement mechanisms ensuring privacy protection, and API Gateway with mediation services for standardized data exchange.

4. Supporting Infrastructure Components: New components include Model AI Data Security frameworks, Chief Data Officers at institutional level, and NSOC (National Security Operations Center) with API Monitoring capabilities for real-time surveillance of data anomalies.

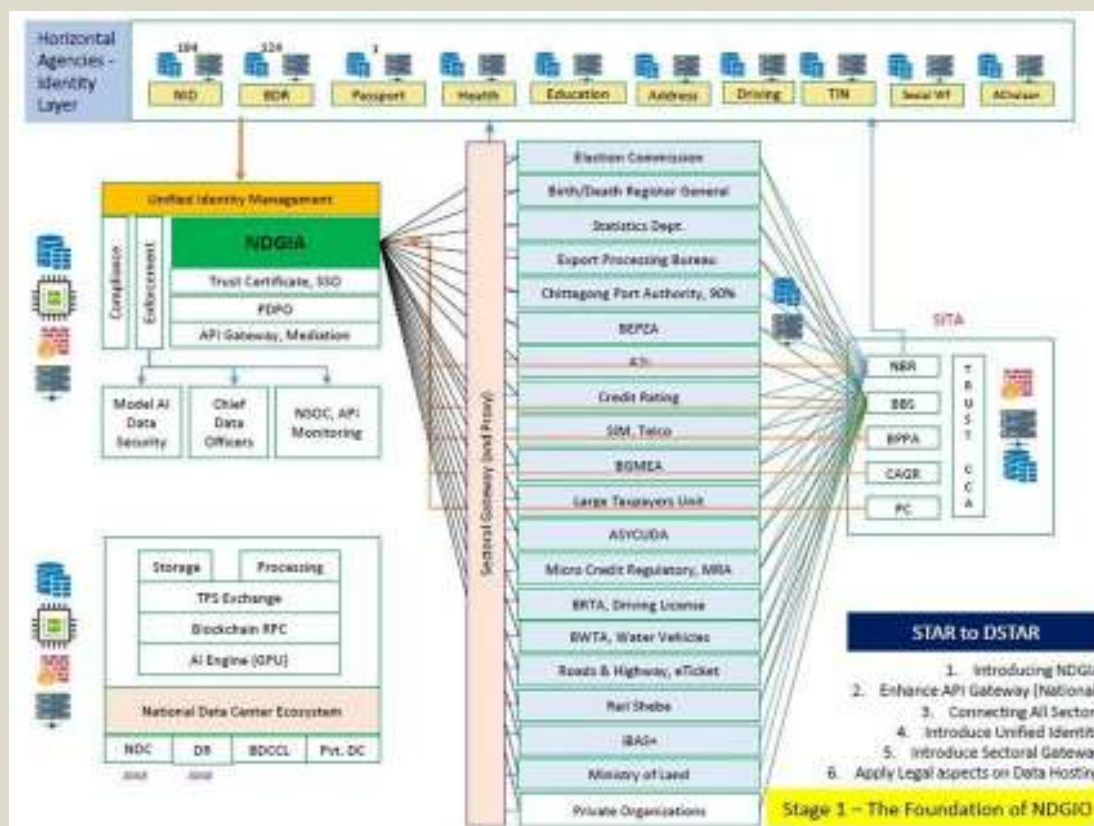


Figure 3.3: Stage 1 – Introduction of Sectoral Gateway and Proxy Layer

Sectoral Gateway Architecture

The progression within Stage 1 introduces the **Sectoral Gateway (and Proxy)** layer—a critical architectural component that fundamentally changes how services connect to the national data infrastructure. This gateway serves as an intermediary layer between individual service agencies and the core NDGIA infrastructure.

The Sectoral Gateway provides protocol translation and format standardization across different legacy systems, request routing and load balancing for efficient resource utilization, security proxy services including authentication, authorization, and audit logging, and data transformation services to ensure compatibility with BNDIA standards.

STAR to DSTAR Transition Actions

3. **Introducing NDGIA:** Establishing the central governance authority for data and AI
4. **Enhanced API Gateway (National):** Scaling from sectoral to national-level API management
5. **Connecting All Sectors:** Bringing all government agencies under unified architecture
6. **Introduce Unified Identity:** Implementing VID and SID for privacy-preserving identity management
7. **Introduce Sectoral Gateway:** Deploying proxy and mediation services
8. **Apply Legal Aspects on Data Hosting:** Enforcing PDPO and requirements for mandatory domestic mirroring of CII data and other data localization requirements.

3.5 Stage 2: Shaping the DPI (Full DSTAR Implementation)

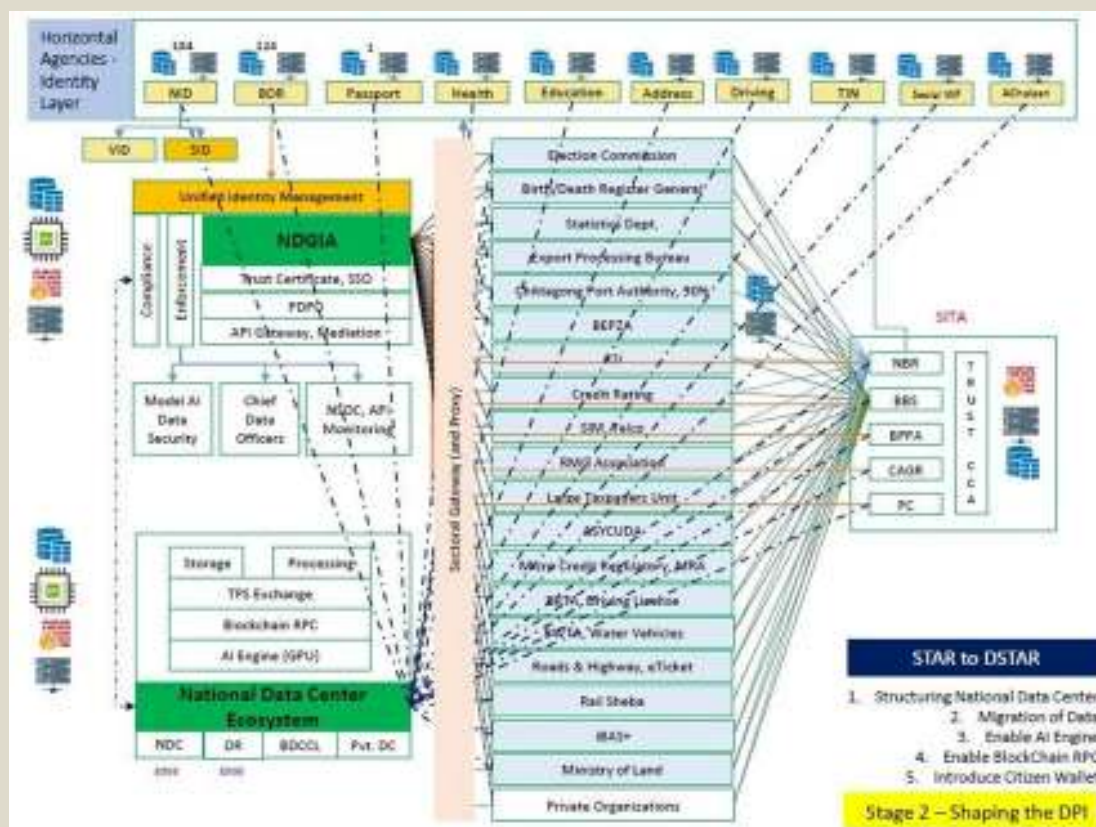


Figure 3.4: Stage 2 – Shaping the DPI (Full DSTAR Architecture)

Stage 2 represents the culmination of Bangladesh's DPI vision—a fully integrated, intelligent, and citizen-centric digital public infrastructure. This stage introduces advanced technological capabilities while consolidating the governance frameworks established in earlier phases.

National Data Center Ecosystem

The most significant architectural addition in Stage 2 is the fully operationalized **National Data Center Ecosystem**. This ecosystem comprises four integrated components:

NDC (National Data Center): The primary sovereign infrastructure for hosting critical government data and services. The NDC serves as the anchor for all government data operations, ensuring data sovereignty and security compliance.

DR (Disaster Recovery): Geographically distributed disaster recovery facilities ensuring business continuity and data resilience. This component safeguards against data loss and ensures service availability during emergencies.

BDCCL (Bangladesh Data Centre Company Limited): The commercial arm providing scalable cloud services for government and enterprise clients. BDCCL enables value-added services for freelancers, the gaming industry, and the broader digital services ecosystem.

Pvt. DC (Private Data Centers): Accredited private sector facilities operating under NDGA certification, expanding capacity while maintaining compliance with national standards.

Advanced Processing Capabilities

Stage 2 introduces sophisticated processing infrastructure that enables Bangladesh to leverage emerging technologies:

TPS Exchange: High-throughput transaction processing services capable of handling the scale required for national-level digital services, from tax settlements to citizen service requests.

Blockchain RPC (Remote Procedure Call): Distributed ledger capabilities enabling transparent, immutable record-keeping for critical government transactions. This technology supports applications ranging from land registration to educational credential verification.

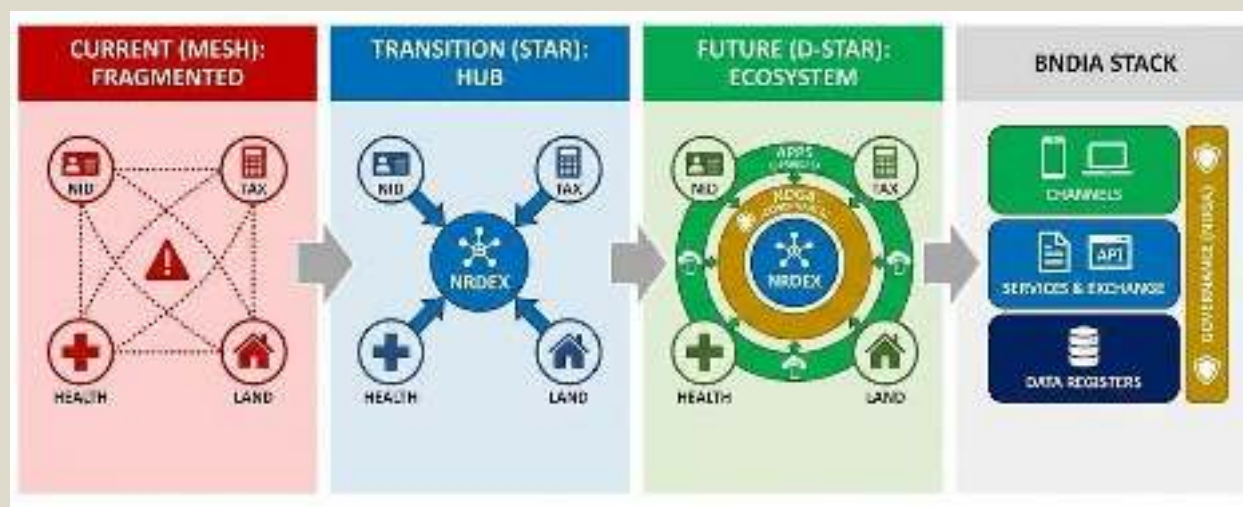
AI Engine (GPU): Dedicated GPU-powered artificial intelligence processing capabilities supporting predictive analytics for governance, AI-based anomaly detection for security, natural language processing for citizen services, and machine learning models for policy optimization. This infrastructure positions Bangladesh to deploy AI-powered predictive governance for real-time decision-making.

STAR to DSTAR Final Transition Actions

9. **Structuring National Data Center:** Integrating NDC, DR, BDCCL, and private facilities into a cohesive ecosystem
10. **Migration of Data:** Systematically moving legacy data to sovereign infrastructure
11. **Enable AI Engine:** Deploying GPU-powered processing for intelligent governance
12. **Enable Blockchain RPC:** Implementing distributed ledger for transparency and trust
13. **Introduce Citizen Wallet:** Launching the National Document Wallet for secure citizen credential management

3.5.1 Interoperability Phases

Phase	Characteristics	Key Challenges/Actions
MESH (Current State)	Fragmented "data islands"—isolated registries for NID, taxation, land, education, health without real-time coordination	• Duplication of digital assets • Increased operational costs • Data breaches exposing citizen information • Lack of end-to-end digital services
STAR (2025-2026)	Secure hub-and-spoke model connecting priority ministries through SITA project and standardized APIs	• Deploy SITA project across 5-10 ministries • Appoint Chief Data Officers • Begin PDPO compliance audits • Establish NDGA Secretariat
DSTAR (2027-2030)	Data-driven STAR: Full interoperability with AI-enabled analytics, unified identity, and zero manual data exchange	• Operationalize NRDEX nationwide • Integrate 100+ ministries and agencies • Launch AI registry and certification • Zero manual data transfers



3.5.2 Justification for Phased Approach

The phased MESH → STAR → DSTAR approach is justified by:

- **Risk Mitigation:** Gradual deployment allows identification and correction of issues before nationwide rollout
- **Capacity Building:** Time for training officials, appointing CDOs, and developing institutional capability

- **Legal Compliance:** Three-year PDPO transitional period allows organizations to adapt to new requirements
- **Stakeholder Buy-in:** Phased integration demonstrates value to hesitant ministries through early wins

3.6 Bangladesh National Data Governance and Interoperability Architecture (BNDIA)

The **Bangladesh National Data Governance and Interoperability Architecture (BNDIA)** is the comprehensive technical and governance framework that enables secure, standardized data exchange across all government entities. Legally established under NDGO Section 21, BNDIA provides the architectural backbone for Bangladesh's DPI.

3.6.1 Core Components

Component	Description
NRDEX Platform	National Responsible Data Exchange—secure API-based platform for trusted cross-agency data sharing with purpose-based access controls and audit trails
SITA Project	Strengthening Institutions for Transparency and Accountability—secure API exchange connecting NBR, BBS, BPPA, Planning Commission, and OCAG during STAR phase
Unified Identity Layer	Interoperability between NID, Birth Registration, Passport, and Business ID with system-wide propagation of updates
NSOC	National Security Operations Center—real-time monitoring of network and data anomalies across all DPI components
TRUST-CCA Chain	Trust and Certificate Chain Authority—ensures authenticity and integrity of data transactions through PKI infrastructure
National Data Exchange Layer (NDEL)	Scalable platform for all cross-agency data flows with standardized APIs, metadata management, and audit capabilities

3.6.2 Technical Standards and Requirements

BNDIA mandates the following technical requirements for all connected systems:

- **Zero-Trust Architecture:** Every access request must be verified regardless of network location
- **API-Based Integration:** All data exchange through standardized RESTful APIs with proper authentication
- **Cloud-First Policies:** Government agencies required to adopt cloud-first but hybrid-friendly architecture

- **Open-Source Adoption:** Preference for open-source platforms where applicable to prevent vendor lock-in
- **Metadata Standards:** Consistent metadata tagging for all datasets to enable discovery and governance
- **Audit Trails:** Complete logging of all data access and modifications for accountability

3.7 Digital Identity Framework (OneID)

To ensure clarity, interoperability, and trust, Bangladesh requires a structured framework for digital identity covering legal foundations, lifecycle inclusion, standards, and governance. The **Unified Identity Layer (oneID)** concept represents Bangladesh's approach to creating a single, authoritative identity that follows citizens throughout their lives.

3.7.1 Legal Basis

Currently, Bangladesh's digital identity ecosystem is anchored in the **National Identity Registration Act, 2010**. The later 2023 Act introduced the Unique Identification Number (UIN) but has since been repealed, creating ambiguity about the legal foundation for UIN as a universal identifier.

- The strategy recommends a policy review to clarify and consolidate the legal framework for digital identity
- Options include amending the 2010 Act or considering a new consolidated legal instrument
- Interoperability and trust will be guided by existing legal provisions, supplemented by PDPO and NDGA oversight

3.7.2 The 17-Digit OneID Architecture



OneID introduces a standardized **17-digit decimal identifier** designed for governance, provenance, and scalability:

- **Position 1 (Fixed Prefix):** '1' identifies the Bangladesh national identity space.
- **Positions 2-3 (Reserved):** '00' reserved for future expansion.
- **Position 4 (Producer Agency):** Identifies the issuing authority (e.g., '1' for Election Commission, '2' for Registrar General).
- **Positions 5-6 (Sub-Agency):** Identifies the specific service point type (e.g., Union Parishad, Hospital).
- **Positions 7-17 (Citizen Number):** Unique 11-digit random number assigned to the individual.

Note: The identifier is opaque and non-predictable; it does not encode personal attributes like date of birth or gender to protect privacy.

3.7.3 Lifecycle Coverage (0-18 Years)

The strategy emphasizes a life-cycle approach to ensure no citizen is excluded:

Life Stage	Identity Foundation	Associated Credentials
Birth (0 years)	Birth Registration as the root identity	Birth Registration Number (BRN)

Childhood (0-17)	Progressive Digital Credentials linked to BRN	Health records, education credentials, social protection IDs
Adulthood (18+)	Automatic transition to full NID-based digital identity	NID, Passport, TIN, Business ID, driving license

Justification: *This lifecycle approach prevents fragmentation (separate education IDs, health IDs, etc.) and ensures continuity. Citizens maintain a single identity thread from birth through adulthood.*

3.7.4 Standards and Interoperability

Bangladesh will adopt internationally recognized standards to guarantee interoperability and vendor neutrality:

- **ISO/IEC 19794 series** for biometric formats
- **ICAO standards** for travel and identity credentials
- **NIST Digital Identity Guidelines** for assurance levels
- **FIDO2/WebAuthn** for passwordless authentication
- **GDPR/eIDAS principles** for alignment with global data protection norms

3.7.5 PKI, Digital Signatures, and Trust Framework

A robust Public Key Infrastructure (PKI) is the bedrock of secure digital transactions. The digital identity system will integrate with an enhanced PKI framework to enable:

- **Secure Authentication & Legally Binding E-Signatures:** Ensuring non-repudiation and legal validity for all digital government and financial transactions.
- **Consent-Based Data Sharing:** Facilitating auditable, citizen-controlled data exchange.
- **Offline Verification:** Supporting QR codes and credential wallets for low-connectivity areas.

Reform of the Controller of Certifying Authorities (CCA): To operationalize this trust framework, the CCA is undergoing strategic reform to address current limitations in adoption, legal recognition, and international interoperability. [See chapter 6.6 for further details.](#)

- **Legal Harmonization:** The CCA will lead the amendment of the Information and Communication Technology Act, 2006 to modernize provisions for digital signatures. This includes removing requirements for "blue-ink" signatures and physical presence in service delivery, ensuring that electronic records and signatures have absolute legal equivalence in courts.

- **Mandatory E-Signatures in Government:** A circular from the Cabinet Division will mandate the use of digital signatures in all critical G2G and G2C services (e.g., e-Nothi, e-GP, iBAS++), moving away from unsecured "system-generated" documents.
- **Financial Integration:** The Ministry of Finance will introduce specific **economic codes** to allocate budgets for digital signature procurement across government offices, removing a key barrier to adoption.
- **Global Trust & Recognition:** The CCA will pursue WebTrust audits to include Bangladesh's Root CA certificate in international trusted stores (Microsoft, Mozilla, Adobe). Furthermore, Mutual Recognition Arrangements (MRA) will be established to facilitate cross-border digital trade and legally binding international contracts.
- **Future-Ready PKI:** Investment in research for Post-Quantum Cryptography to future-proof national encryption standards against emerging threats.

These components will evolve into a **National Trust Framework**, drawing from global models such as EU eIDAS 2.0 and NIST standards, ensuring that Bangladesh's digital ecosystem is secure, legally robust, and globally connected.

3.7.6 Identity Wallet Governance

Bangladesh already has an **NID Wallet** operated by the Election Commission. The strategy clarifies that:

- The wallet should be consolidated and expanded, not duplicated
- Its custodian authority (EC or another designated body) should be clearly defined through policy review
- Core functions should include credential storage, e-signatures, consent management, offline verification, and regulated integration with private-sector apps (banks, fintechs, telcos)

3.7.7 Privacy & Trust Architecture

OneID operates under the strict legal regime of the **National Data Governance Ordinance (NDGO)** and **Personal Data Protection Ordinance (PDPO)**, implementing a multi-layered trust framework that balances verification utility with citizen privacy.

- **Sectoral Identifiers (SIDs):** To mitigate the risk of mass surveillance and cross-sector profiling, OneID employs a system of Sectoral Identifiers. Relying parties in different domains (e.g., a healthcare provider vs. a financial institution) receive distinct, derived identifiers for the same individual. These SIDs cannot be correlated without specific legal authorization or the citizen's explicit consent, effectively isolating data silos and preserving privacy.
- **Tokenization for Verification:** To minimize data exposure, verification services return cryptographically signed **"Yes/No"** tokens or attribute assertions (e.g., "Is Age > 18?" or "Address Match: True") rather than revealing raw personal data.

This "Zero-Knowledge" approach allows service providers to verify eligibility without accessing sensitive underlying records.

- **Citizen Wallet & Selective Disclosure:** The architecture introduces a secure **Citizen Wallet** mobile application that empowers individuals to hold and manage their own verifiable credentials (e.g., Driving License, Education Certificates, Professional Licenses). The wallet supports **Selective Disclosure**, enabling citizens to share only the specific data points required for a transaction (e.g., proving age without revealing name or address) without handing over physical documents.
- **Trust & Certificate Chain Authority (TRUST-CCA):** All credential issuance, digital signatures, and authentication events are anchored in the national PKI framework administered by TRUST-CCA. This ensures that every digital interaction is cryptographically verifiable, non-repudiable, and tamper-evident.

3.7.8 Integration & Interoperability

OneID serves as the foundational "Trust Anchor" for the entire Digital Public Infrastructure (DPI) ecosystem, facilitating seamless integration across disparate sectors through the **National Responsible Data Exchange (NRDEX)**.

- **Digital Social Protection:** OneID integration eliminates "ghost beneficiaries" and duplicate payments by linking beneficiary registries directly to the unique biometric identity. It enables the **Integrated Beneficiary Registry (IBR)** to perform cross-program deduplication, ensuring that social safety net funds reach only eligible, unique individuals.
- **National Address Management System (NAMS):** OneID integrates with NAMS using a "Pointer Model," where the identity record stores a reference (National Address Identifier - NAID) rather than raw address text. This ensures that when an address is updated in NAMS, the change is automatically reflected for all services relying on that identity, enabling reliable service delivery and geographic targeting.
- **Payments Ecosystem:** The framework supports secure **Account Binding**, linking a beneficiary's OneID to their preferred financial address (Bank Account or MFS Wallet). This enables frictionless Government-to-Person (G2P) payments and supports **FIDO-based biometric authentication** at agent points, allowing secure cash-out even for citizens without smartphones.
- **Health Sector:** OneID enables the creation of a portable **Longitudinal Health Record**. By uniquely identifying patients across different facilities (hospitals, clinics, diagnostic centers), it ensures continuity of care, prevents duplicate medical records, and supports national immunization registries.
- **CRVS Integration:** The system maintains a real-time synchronization link with the Civil Registration and Vital Statistics (CRVS) system. Birth registration automatically triggers OneID creation, while death registration triggers immediate identity deactivation and event propagation to subscribed systems (e.g., stopping

pension payments), ensuring the national identity database remains perpetually accurate and current.

3.7.9 Implementation Governance

- **Regulator: National Data Governance Authority (NDGA)** oversees compliance, sets standards, and accredits enrolling agencies.
- **Producer Agencies:** Designated bodies like the **Election Commission (NID Wing)** and **Office of the Registrar General (ORG)** responsible for data capture.
- **Technology:** Built on **Open Standards** (MOSIP-aligned principles) to avoid vendor lock-in, with data hosted exclusively in sovereign government data centers.

3.7.10 Expected Outcomes by 2030

- **100%** of citizens (birth to death) covered by OneID.
- **Zero** manual data entry for government service applications (Once-Only Principle).
- **99.9%** accuracy in beneficiary targeting for social safety nets.
- **Frictionless** eKYC for banking and mobile payments nationwide.

3.8 National Address Management System (NAMS)

The National Address Management System (NAMS) establishes the definitive technical and governance framework for a unified, geocoded national address registry. It solves the critical "Address Ambiguity" problem that currently hampers service delivery, e-commerce logistics, and emergency response in Bangladesh. NAMS transforms addresses from unstructured text into precise, machine-readable geospatial coordinates linked to a unique **National Address Identifier (NAID)**.

Vision and Core Objectives

Vision: To provide every physical location in Bangladesh—residential, commercial, or institutional—with a unique, verified digital address that enables precision service delivery, seamless logistics, and effective emergency response.

Strategic Objectives:

- **Standardization:** Replace unstructured address text with a standardized schema aligned with **ISO 19160** and **UPU S42**.
- **Geocoding:** Link every address to precise geospatial coordinates (Latitude/Longitude), enabling "Dot-on-Map" navigation.
- **Uniqueness:** Assign a persistent, immutable **National Address Identifier (NAID)** to every addressable object.
- **Interoperability:** Serve as the "Single Source of Truth" for address data across all government agencies via **OneID** and **NRDEX** integration.

The National Address Identifier (NAID) Architecture

NAMS introduces the **NAID**, a 12-character alphanumeric code that serves as the permanent digital key for any location.

- **Structure:** The NAID is opaque and non-semantic (e.g., **BD-7834-K92M**), designed to remain valid even if administrative boundaries (Ward, Upazila) change.
- **Persistence:** Unlike street names or holding numbers which may change, the NAID remains constant for the lifecycle of the building or property.
- **Pointer Model:** Government databases (NID, Passport, Utility) will store the **NAID** as a "Pointer" rather than duplicating full address text. This ensures that when an address is updated in NAMS (e.g., a street is renamed), all linked services automatically reflect the change without manual data entry.

Core Components and Integration

NAMS operates as a foundational layer of the Digital Public Infrastructure (DPI):

1. **Central Address Registry (CAR):** The authoritative database hosting verified address records, geocodes, and NAID mappings.
2. **Digital Addressing Portal:** A citizen-facing platform for self-service address registration, verification, and NAID generation.
3. **OneID Integration:** NAMS provides the "Where" to OneID's "Who."
 - **Address Verification:** OneID verification services check the NAID against the NAMS registry to confirm address validity during eKYC.
 - **Service Delivery:** Social protection benefits and emergency services are routed using the precise geocodes linked to the beneficiary's NAID.
4. **Logistics & E-Commerce Integration:** NAMS provides open APIs for logistics companies to resolve NAIDs into precise delivery coordinates, reducing "Last Mile" failure rates and delivery costs.

Implementation and Governance

- **Custodian Agency: Bangladesh Post** serves as the primary custodian of the NAMS registry, leveraging its nationwide field force for verification and data maintenance.
- **Technical Operator: ICT Division / BCC** manages the digital infrastructure and API gateway.
- **Local Government Role:** City Corporations and Union Parishads are responsible for assigning holding numbers and validating new address creation requests at the source.

Expected Outcomes by 2030

- **100%** of households and businesses assigned a NAID.
- **Zero** ambiguity in emergency response routing (Ambulance/Fire/Police) due to precise geocoding.
- **30% reduction** in e-commerce logistics costs due to elimination of address verification calls and failed deliveries.
- **Seamless** utility connection applications using NAID for instant premise verification.

3.9 Digitally Integrated Tax-VAT Ecosystem: NBR API Connectivity Hub

To modernize fiscal administration and strengthen domestic revenue collection, a national NBR API Connectivity Hub will be deployed—enabling secure, real-time, system-to-system linkage between the National Board of Revenue (NBR) and public/private ERP systems, payment service providers, banks, and digital financial platforms.

This reform shifts NBR from relying on delayed, secondary reporting toward automated access to primary financial transactions—transforming Bangladesh's tax lifecycle into a real-time, transparent, and data-driven system anchored in Digital Public Infrastructure (DPI).

3.9.1 Four Pillars of Reform

Pillar 1: ERP Integration & Instant (T+0) Tax Settlement

All major public and private ERP platforms—handling salary disbursement, procurement, contractor payments, vendor management, etc.—will be connected to the NBR hub. Once a payment is initiated within an ERP, a digital payment slip will automatically trigger advance income tax/withholding tax deduction and reporting to NBR. This realizes a T+0 tax settlement model, minimizes leakage, and eliminates manual compliance burdens.

Pillar 2: VAT Net Expansion via Payment & ERP Connectivity

Simultaneously, all payment gateways, MFS/PSP platforms, wallets, and VAT-enabled ERPs will be integrated through a dedicated VAT API, enabling automated VAT capture at the exact point of transaction. This approach expands the VAT base without increasing rates—particularly across growing segments such as e-commerce, services, logistics, and SME supply chains.

Pillar 3: Secure API-Based Architecture & Cyber Protection

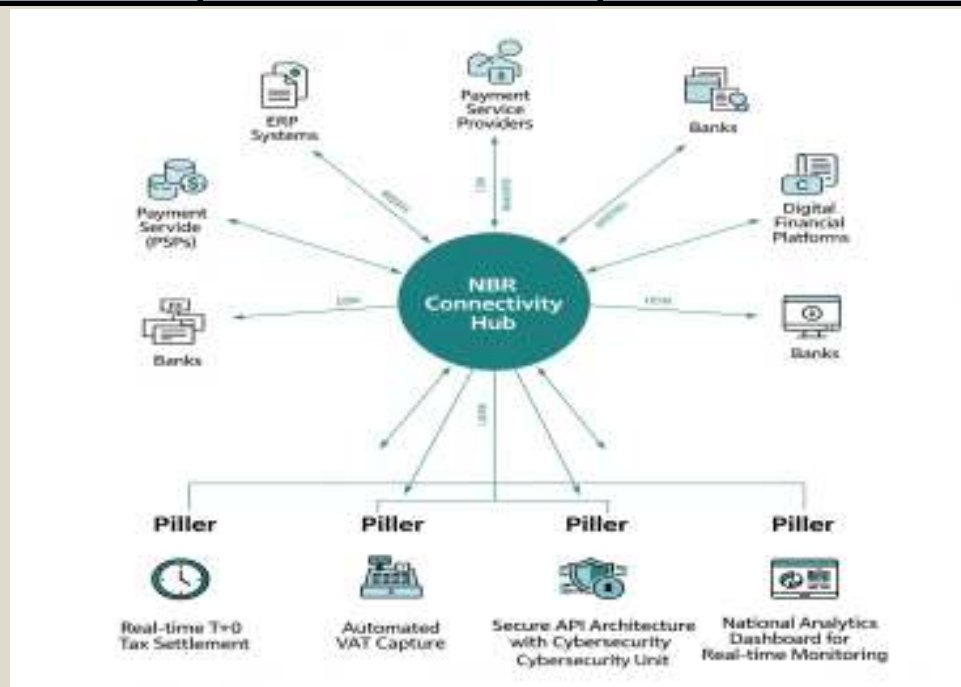
NBR will establish a cyber-secure API Gateway & Connectivity Hub to govern all integrations. Embedded within this will be a new Tax-VAT Cybersecurity Operations Unit, responsible for endpoint protection, encrypted data pipelines, zero-trust access control, monitoring, and regulatory coordination with N-CERT and the proposed National Data Governance & Interoperability Authority (NDGA).

Pillar 4: Real-Time Reporting & Predictive Monitoring

A national analytics dashboard will offer MoF and NBR live insight into revenue flow, taxpayer behaviour, risk patterns, and compliance hotspots. AI-based anomaly detection systems will guide enforcement and forecasting. This same data backbone will feed into planning, budgeting, and public service delivery reforms across government.

3.9.2 Technical Architecture

Component	Function	Integration Points
NBR API Gateway	Central hub for all tax/VAT data exchange	ERPs, banks, MFS, PSPs, customs
Tax-VAT Cybersecurity Unit	Endpoint protection, zero-trust access	N-CERT, NDGA, NSOC
Real-Time Analytics Dashboard	Live revenue monitoring, anomaly detection	MoF, Planning Commission, BBS
ERP Integration Layer	Automated tax deduction at source	Public/private ERP platforms
VAT API Module	Point-of-transaction VAT capture	Payment gateways, wallets, e-commerce



3.9.3 Expected Outcomes

- Transparent, corruption-free tax administration through automated compliance
- Real-time revenue visibility for fiscal planning and policy decisions
- Expanded VAT base without rate increases through comprehensive transaction capture
- Reduced compliance burden for businesses through automated reporting

- Enhanced enforcement capabilities through AI-powered anomaly detection
- Integration with national DPI backbone for seamless cross-government data sharing

3.10 Expanded Role of BBS and Cross-Government Data Ecosystem

A key design principle of the DPI framework is "collect once, use many times"—especially for the integration of Bangladesh Bureau of Statistics (BBS) into a dynamic, real-time data ecosystem. This represents a fundamental transformation in how national statistics are generated and utilized for evidence-based governance.

3.10.1 From Periodic Surveys to Real-Time Intelligence

Instead of receiving periodic, siloed, survey-based inputs, BBS will be granted API-level access to anonymized primary transactional data (e.g., salaries, business payments, sector-wise VAT/tax flows, digital commerce activity, service-based revenue streams).

This will allow BBS to:

Generate High-Frequency Economic Indicators:

Produce employment, consumption, and informal-to-formal transition indicators without waiting for quarterly or annual surveys.

Improve National Accounts:

Enhance price indices, input-output tables, and GDP forecasting models using real-time datasets instead of historical survey data.

Provide Dynamic Sectoral Statistics:

Deliver timely statistics to ministries (e.g., commerce, planning, industries, agriculture) for evidence-based policymaking.

3.10.2 Institutional Transformation

This transition will also reduce reliance on manual Excel/email reporting, enabling BBS to evolve from being a census-survey heavy institution toward a digitally enabled national data intelligence agency. Key transformations include:

- Automated data pipelines replacing manual data collection and entry
- Real-time dashboards for policymakers instead of periodic reports
- API-based data sharing with authorized government agencies and researchers
- Integration with international statistical frameworks for comparability

3.10.3 Cross-Government Integration

Beyond BBS, live data streams from the NBR hub will automatically connect to:

iBAS++ (Integrated Budget and Accounting System):

Real-time budget execution monitoring and fiscal management

Planning Commission's MIS:

Project monitoring and development planning informed by actual economic data

Budgetary Systems:

- Revenue forecasting and expenditure planning based on real transaction flows
- This integration removes the operational burden of data submission by ministries, officials, and SOEs—thereby dramatically improving public financial management and planning efficiency.



3.10.4 Data Governance and Privacy

- All cross-government data sharing will operate under strict governance frameworks:
- Anonymization and pseudonymization of individual-level data
- Purpose and role based access controls aligned with PDPO requirements
- Audit trails for all data access and usage
- NDGA oversight and certification of data-sharing arrangements

3.11 Future-Ready Payment Ecosystem

Bangladesh's payment ecosystem must evolve into a seamless, interoperable, and inclusive infrastructure to support the nation's transition toward a digital economy. The existing system is fragmented, dominated by a few players, and constrained by regulatory bottlenecks. Reform is essential to unlock the potential of Payment System Operators (PSOs), Payment Service Providers (PSPs), FinTech innovators, and digital commerce actors.

A future-ready payments framework must not only enable secure digital financial services for citizens but also integrate with national transport, utility, and commerce systems, ensuring inclusivity, accessibility, and competitiveness.

3.11.1 Breaking Silos and Ensuring Security

The Challenge: The "Silo over Silo" Landscape

The current payment ecosystem in Bangladesh is characterized by fragmented operating silos that hinder true financial inclusion and efficiency.

- **Banking Silos:** The VISA card network is currently limited effectively to only four major banks, creating a restricted "Silo Banking Network."
- **MFS Fragmentation:** Major Mobile Financial Service (MFS) providers—including bKash, Nagad, Sonali Pay, and Upay—have constructed proprietary networks, creating a "silo over silo" situation across the merchant and payment layers.
- **Capacity Constraints:** The limitations of independent infrastructure were highlighted when Nagad, operating on its own system, faced significant challenges handling approximately 80 million transactions during the Eid festival, exposing critical gaps in Transaction per Second (TPS) capacity planning.
- **Interoperability Gaps:** The National Payment Switch Bangladesh (NPSB) has historically been commercial bank-centric. Consequently, effective MFS-to-MFS interoperability remains unrealized, and MFS providers continue to operate within closed loops for merchant and government services.

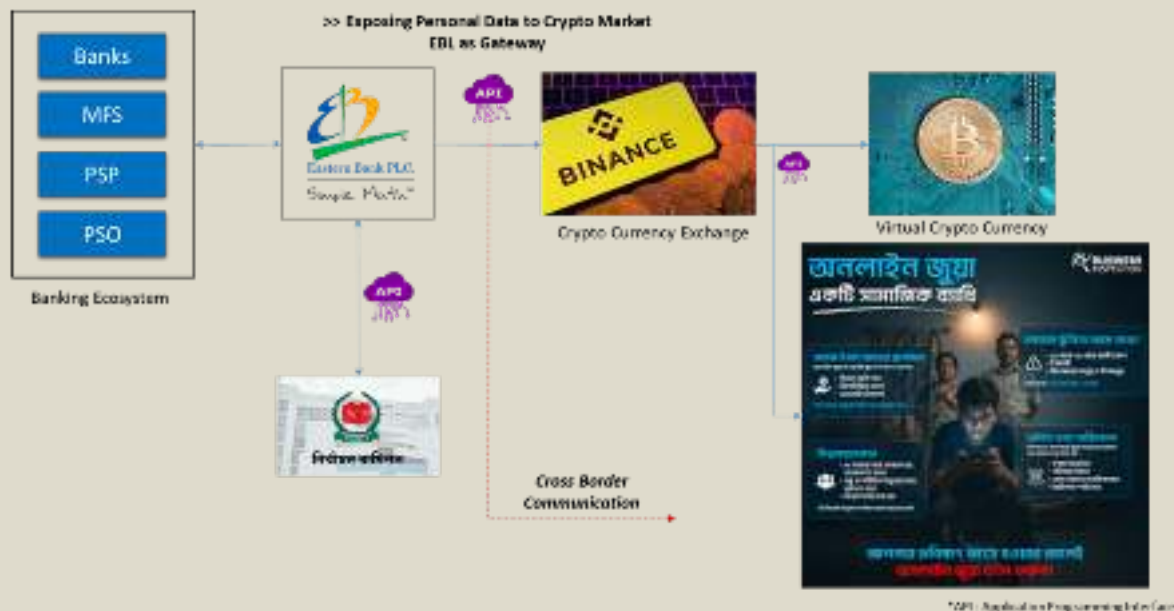


The Solution: A Universal Interoperable Framework

To transition from fragmentation to a unified digital economy, the following structural changes are mandated:

- **Universal Payment Gateway:** A transition from mandatory NPSB connectivity to a truly **Interoperable Universal Payment Gateway**. This will enable critical national functions including Automated e-Invoicing, Pattern Recognition, Capacity Dimensioning, and Financial Intelligence.
- **Central Clearing & NRDEx:** The NPSB must evolve into a Central Clearing System under Bangladesh Bank. High-capacity transactions will be routed through the **National Responsible Data Exchange (NRDEx)** to ensure API-based end-to-end connectivity.
- **Marketplace Guidelines:** New guidelines will enforce that MFS operators cannot directly enter marketplaces without routing through the national switching infrastructure (NPSB/Universal Gateway) to ensure transparency.

Exposure of Online Scams and Crypto Currency Exchange Network with Banking System of Bangladesh



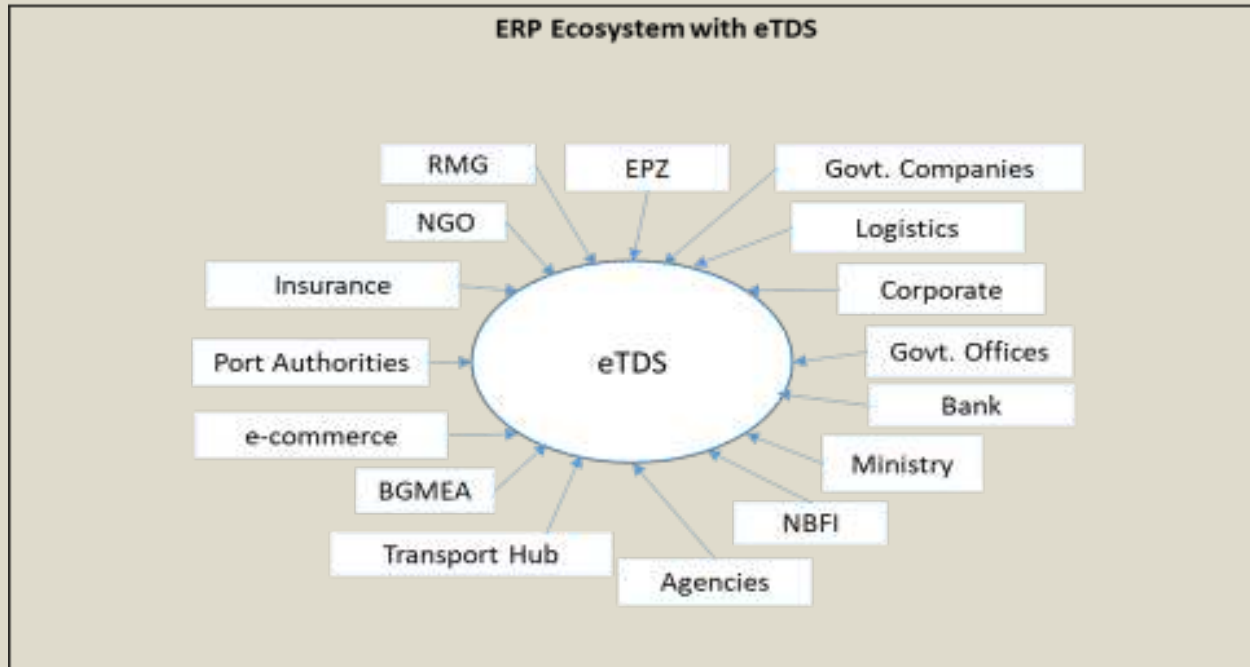
Cross-Border Security and API Monitoring

The digital payment landscape faces emerging threats from unregulated cross-border flows:

- **Crypto-Exchange Risks:** Despite cryptocurrency being unauthorized, platforms like Binance have previously connected via API to local banks for NID verification. This allows external applications to penetrate the internal banking system, posing severe security risks. Strict monitoring of all Cross-Border API communications is required to prevent unauthorized financial flows.
- **Gambling Control via DPI:** To curb online gambling, the Bangladesh Telecommunication Regulatory Commission (BTRC) will utilize **Deep Packet Inspection (DPI)** technology to identify gambling app traffic and introduce latency, effectively slowing down and discouraging illicit usage.

Integration with Taxation and Data Governance

- **Automated Revenue Collection:** By integrating Enterprise Resource Planning (ERP) systems (e.g., corporate salary disbursements) directly with the National Board of Revenue (NBR) via APIs, the ecosystem will enable automatic tax calculation, collection, and the generation of e-Invoices.
- **Data Protection:** Recognizing payment data as sensitive personal information, all stakeholders must strictly adhere to the **Personal Data Protection Ordinance 2025**.

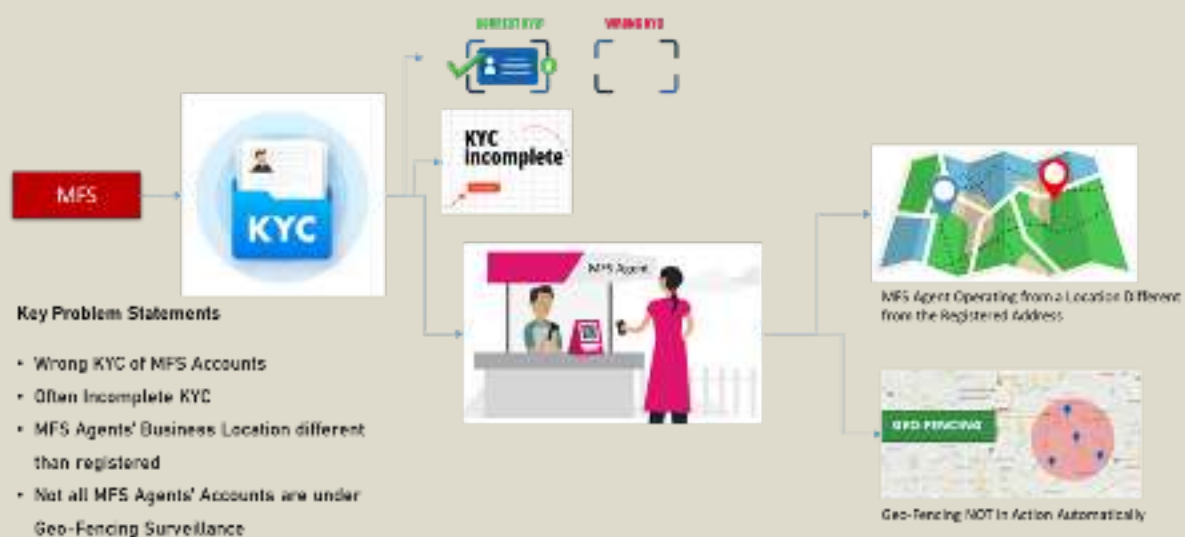


Advanced Intelligence: AI, NEIR, and BFIU

A robust ecosystem requires advanced surveillance and intelligence capabilities:

- **SIM-Device Tagging:** Leveraging BTRC's **National Equipment Identity Register (NEIR)**, MFS accounts will be strictly tagged to a specific SIM and Device. This creates a "Mobile-SIM-Device" triangle, ensuring a complete and accurate e-KYC process.
- **Geo-Fencing:** MFS Agent accounts will be subject to SIM-based geo-fencing to prevent unauthorized remote transactions.
- **Centralized Intelligence (BFIU):** Moving beyond partial local processing, data processing must be centralized through the Bangladesh Financial Intelligence Unit (BFIU).
- **AI & Machine Learning:** Centralized data will be analyzed using AI/ML to enable:
 - Real-time Financial Modeling and Fraud Detection.
 - Automated Suspicious Transaction Reporting (STR).
 - Digital Transaction Forensics.
 - Revenue collection from global platforms (e.g., Meta, Google).

Misleading Metadata Resulting in Incorrect Identification of MFS Accounts



User Experience and Security Enhancements

- **Biometric Authentication:** Reducing reliance on OTPs by shifting toward app-based biometric verification.
- **Incentives:** Providing incentives for digital payments in public services.
- **Charge Control:** Rationalizing and controlling additional charges at Points of Sale (POS).

By implementing this interoperable network and establishing effective coordination between agencies (Bangladesh Bank, BTRC, NBR, BFIU), Bangladesh will build a secure, efficient, and data-driven national payment ecosystem.

Interoperable Framework :: Solution to all of these problems 2/2



3.11.2 Ensuring Market Competitiveness & Fair Access

- Declare bKash as Significant Market Player (SMP) to prevent monopolistic dominance and create a level playing field
- Expand PSO licensing scope to allow QR merchant acquisition, PSP top-ups, account-to-account (A2A) transfers, and settlement for Personal Retail Accounts (PRAs)
- Mandate connectivity to the National Payments Switch of Bangladesh (NPSB) for all PSOs to enable interoperability across banks, MFS, and PSPs

3.11.3 Financial Inclusion & CMSME Empowerment

- Leverage PSOs to onboard micro, small, and ultra-micro entrepreneurs under the PRA framework in partnership with banks, MFS, and PSPs
- Recognize transactional data as collateral to facilitate CMSME access to credit
- Enable crowdfunding and peer-to-peer lending through regulated PSOs to diversify financing options for entrepreneurs

3.11.4 Remittance & Cross-Border Transactions

- Empower PSOs to enhance remittances by enabling real-time transfers, account validation, and standardized APIs
- Reduce remittance costs through competition and interoperability while ensuring AML/CTF compliance

- Permit PSOs to facilitate small-value cross-border transactions, particularly for SMEs in regional trade corridors (e.g., China–Bangladesh, SAARC)

3.11.5 Government-Led Innovation & Sovereign Investment

- Instruct Startup Bangladesh Ltd (SBL) to co-invest in FinTech, InsurTech, and PSO-led innovations that directly support financial inclusion and digital commerce
- Establish a National Data & Payment Analytics Platform to provide regulators and businesses with insights into digital commerce flows, risks, and opportunities
- Position PSOs as true FinTechs by granting them authority to provide embedded financial services for CMSMEs

3.11.6 Unified Ticketing & Transport Payment Integration

- Introduce a Metro Travel Pass / Unified Transport Pass (single card/app) that enables seamless payment across metro rail, buses, tolls, ferries, and future transport systems
- Ensure interoperability with the national payments backbone, allowing users to fund transport passes via banks, MFS, and PSOs
- Integrate with a Super App to unify transport payments, bill pay, fuel charges, plane ticketing, and government fees
- Develop a real-time transaction backbone for instant settlement across all transport operators and financial intermediaries

3.11.7 Policy Actions

Area	Action
Regulatory Reform	Amend PSO/PSP guidelines to expand permissible services (QR, A2A, PRA integration, PSP top-ups)
Competition Framework	Introduce anti-trust regulations for dominant MFS players to encourage fair pricing and service innovation
Financial Inclusion	Expand PRA framework to authorize PSOs as enablers of CMSME merchant settlement and financing
Cross-Sector Integration	Build interoperability between payment systems and public infrastructure services
Innovation & VC	Allocate a FinTech Innovation Fund (via SBL) to invest in PSOs, PSPs, and transport-integrated payment startups
Internationalization	Support PSOs in enabling remittance modernization and cross-border small-value trade payments

3.11.8 Expected Outcomes by 2030

- Inclusive digital finance: 20M+ PRA accounts linked with PSO-enabled settlements
- Expanded IT & transport payments: Single pass usable across metro, bus, ferry, and toll payments nationwide
- Remittance efficiency: >70% of inward remittance via digital channels with near-real-time settlement
- CMSME financing: At least \$1B in credit facilitated via transactional data-driven lending
- Global competitiveness: Interoperable PSO ecosystem aligned with regional trade and global standards

3.12 National Fiber Bank

Bangladesh currently has approximately 78,400 km of government-owned fiber-optic networks across agencies like BTCL, BCC, Power Grid (PGC), and Bangladesh Railway; however, about 40% of this fiber remains unused. Further, only approximately 25% of mobile towers are currently connected via fiber—limiting the ability to roll out high-speed services like 5G and causing duplication of new deployments.

3.12.1 Current State Analysis

Underutilized fiber cores represent a significant national asset:

BCC using just 2 cores out of 48 in some segments (46 unused) for services like Summit and Fiber@Home

BTCL has approximately 12 fiber cores unused nationwide

These inefficiencies lead to higher costs, slower project delivery, and under-realized potential for national backbone capacity, redundancy, and broadband expansion.

3.12.2 Strategic Objective

To optimize the use of existing fiber infrastructure across government agencies, accelerate high-speed broadband deployment, reduce rollout costs, and enable resilient digital services including 5G, IoT, and data interconnection.

3.12.3 Key Initiatives & Implementation Priorities

Governance & Institutional Setup:

- Establish a BTCL-led National Fiber Management Team responsible for oversight, leasing policies, pricing, and technical monitoring
- Define stakeholder roles: ICT Division, BCC, BTCL, PGC, Railway, BTRC, and private sector operators

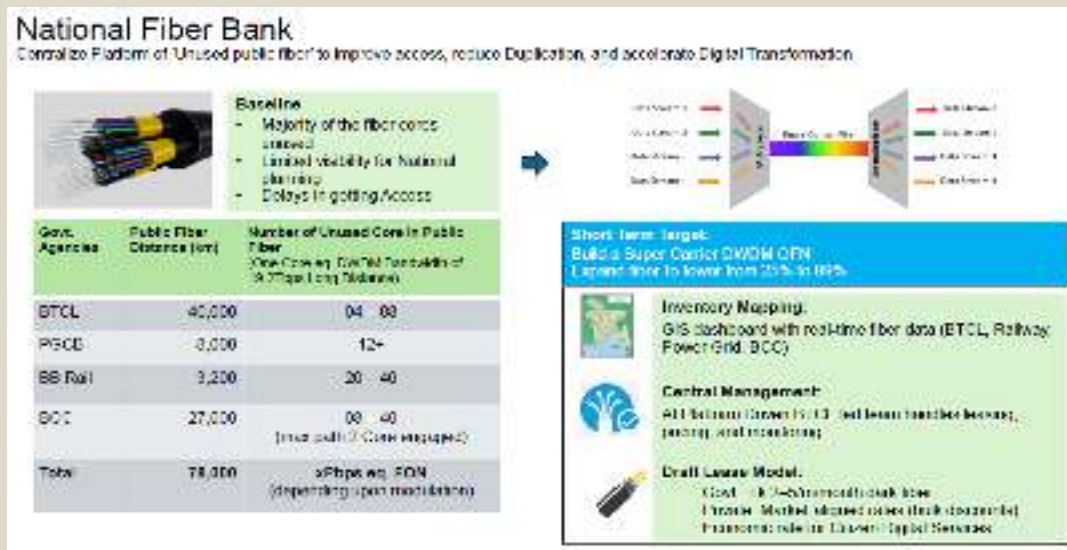
Centralized Inventory & Digital Platform:

- Develop a GIS-based platform cataloguing all government fiber routes: location, number of fiber cores, usage status, underground/overhead, condition, etc.
- Enable online lease applications, monitoring, and reporting

Policy & Regulatory Measures:

- Mandate inclusion of unused government fiber in the Fiber Bank
- Implement fair, subsidised leasing for government/public service use; competitive leasing rates for private operators

- Use DWDM (Dense Wavelength Division Multiplexing) on leased fibers to enhance capacity. BTRC has recently lifted restrictions on mobile operators using DWDM on leased fibers



Pilot Implementation (2025):

By end-2025: complete planning, data collection, platform development, pilot leasing initiatives, and test deployment in a few divisions or regions.

Integration & Upgrades:

- Use underutilized fiber cores to connect more mobile towers (increase fiber-to-tower coverage)
- Encourage public/private collaboration in network sharing and maintenance

3.12.4 KPI Framework (2025-2030)

Year	Fiber Inventory (%)	Towers Fiber-Connected (%)	Unused Cores Utilized (%)	Divisions Operational	Leasing Revenue (BDT M)
2025	25%	35%	5%	3	500
2026	50%	50%	15%	6	2,000
2027	70%	65%	30%	8	4,500
2028	85%	80%	50%	10	8,000
2029	95%	90%	70%	All	15,000

2030	100%	100%	90%	All + cross-border	25,000+
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Note: "Unused Fiber Cores Utilized" refers to proportion of total available cores being leased or deployed. "Leasing Revenue" refers to gross amount collected from leasing to public/private users under Fiber Bank.

3.12.5 Expected Outcomes by 2030

- Full inventory of fiber assets across government with GIS visibility
- All mobile towers served by fiber, enabling significantly higher capacity, lower latency services
- Major reduction in redundant new fiber deployments; costs and time for broadband rollout lowered substantially
- Strengthened backbone for 5G, IoT, data centres, and other high-capacity services
- Transparent, fair leasing model generating measurable revenue to maintain fiber infrastructure

3.13 Implementation Roadmap (2025-2030)

Phase	Timeline	Key Actions
Phase 1: STAR Foundation	2025-2026	• Enact NDGA Rules & Regulations • Establish NDGA Secretariat and Policy Board • Deploy SITA project across 5-10 ministries • Appoint Chief Data Officers • Begin PDPO compliance audits
Phase 2: DSTAR Integration	2027-2028	• Operationalize NRDEX nationwide • Integrate 20+ ministries and regulators • Launch AI registry and certification regime • Conduct first cross-sector data-exchange pilot • Introduce national interoperability dashboard
Phase 3: DPI Consolidation	2029-2030	• Full NDGA autonomy and independent funding • Integrate private-sector APIs (banks, telcos, logistics) • Establish AI innovation sandbox and cross-border pilots

		• Achieve zero manual data exchange in government
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3.14 KPI Framework (2025-2030)

Indicator	2025	2026	2027	2028	2029	2030
Ministries/Agencies integrated via NDGA	5	10	25	40	60	100+
Chief Data Officers appointed	10	20	40	60	80	100+
AI systems certified	0	10	25	50	75	100+
Cross-sector APIs live under NRDEX	1	3	5	8	12	15+
Data breach incidents (government)	<20	<15	<10	<5	<2	0
Officials trained in data/AI governance	500	1,000	2,000	3,500	4,500	5,000+

3.15 Expected Outcomes by 2030

- Full legal and operational establishment of an independent NDGA regulating data, privacy, and AI
- 100+ ministries and agencies interoperable via NRDEX and unified identity
- Zero manual data transfers; all exchanges through secure APIs
- AI governance regime ensuring fairness, transparency, and accountability in public-sector AI
- National Data Exchange Layer recognized as a regional model for secure and ethical digital governance
- Enhanced citizen trust through robust privacy protection, grievance redress, and transparent public-service delivery

End of Chapter 3

4. Sectoral Impacts Outside ICTD

Foundational Digital Systems, Service Improvement, AI Integration, Citizen-Centric Design & Nagorik Sheba

4.1 Introduction: Digital Transformation Beyond ICT Division

While the ICT Division serves as the anchor institution for Bangladesh's digital transformation, the true measure of success lies in the transformative impact across all sectors of government and society. This chapter examines how the Digital Public Infrastructure (DPI) established through the MESH → STAR → DSTAR journey translates into tangible improvements across foundational digital systems, service delivery, AI-enabled governance, and the flagship Nagorik Sheba initiative.

The sectoral impacts documented in this chapter demonstrate that digital transformation is not merely about technology adoption—it is fundamentally about improving citizens' lives through better education, healthcare, agriculture, land administration, law enforcement, justice delivery, social protection, and fiscal administration. Each sector benefits from the interoperability frameworks, unified identity systems, and secure data exchange platforms that form the core of Bangladesh's DPI.

4.2 Citizen-Centric Service Design Framework

Citizen-centric digital service delivery will be at the heart of Bangladesh's digital transformation. All public services will follow a common national standard that prioritises simplicity, accessibility and inclusion, ensuring that citizens—regardless of location, gender, language, or digital ability—can complete essential services with confidence.

Government agencies will adopt user-journey mapping, service-design guidelines, mobile-first interfaces and accessibility requirements so that services reflect real needs and reduce the time, cost and number of visits required. To uphold the **Once-Only Principle**, agencies will reuse verified data from trusted government sources instead of repeatedly asking citizens for the same information. Digital services will progressively move toward end-to-end delivery, supported by secure data exchange, interoperable systems, and reusable building blocks of a future **BanglaGovStack**.

Nine Strategic Actions for the Digital Operating System

#	Strategic Action	Description & Rationale
1	National Definition of Digital Public Service	Adopt definition aligned with IMAPS/European Interoperability Framework: service produces clear administrative outcome, has designated owner, primary end-user group, and digital interface

2	Annual Digital Maturity Assessment	0-4 scale: Analogue → Informational → Semi-digital → Digital with incomplete integration → End-to-End Digital. Ministries submit annual reports with bottlenecks and required support
3	National Service Design & Portal Standards	Mobile-first design, plain-language labels, accessibility for PWDs, multilingual readiness, status tracking, common authentication, uniform UI/UX components, registry of all services
4	Once-Only Principle for Data Reuse	Agencies must reuse verified data from authoritative registers (NID, Birth, Address, Land, Tax, Marriage). Citizens cannot be required to re-submit existing data. Requires secure, logged exchange mechanisms
5	Procurement Reform for Outcomes	Mandatory user testing with women, rural residents, PWDs. Service-level KPIs (completion rate, satisfaction, processing time). Reusable components over custom builds. Government ownership of data/systems
6	Digital-First Legislative Reform	Conduct a comprehensive legal review led by the CCA to amend the ICT Act 2006 and repeal analog provisions requiring physical presence or "blue-ink" signatures. Introduce mandatory Digital Signatures for G2G/G2C services, economic codes for PKI procurement, and legal recognition for Cross-Border Mutual Recognition Arrangements (MRA) to facilitate digital trade.
7	Digital Service Workforce Framework	Specialized roles: Service owners, UX designers, Product managers, Data engineers, Cybersecurity specialists. Competitive pay, career progression, rotational assignments. Training for frontline staff
8	BanglaGovStack: Reusable Building Blocks	Centrally maintained modules: Authentication, Payments, Notifications (SMS/email/WhatsApp), Document storage, Case management, Grievance handling, Multilingual content, Accessibility—aligning with India's GovStack, Estonia's X-Road, UK GDS
9	Public Digital Literacy Programmes	Ensure both service providers and users are equipped to benefit from digital systems. Capacity building for civil servants, clear procurement standards requiring user testing, equitable access initiatives

Why Citizen-Centric Digital Services should be prioritised in the Strategy

Digital transformation succeeds only when services are designed around the real needs, constraints, and expectations of citizens. Despite significant progress, evidence from field consultations and service audits shows that many services remain difficult to navigate, inconsistently designed, or reliant on manual steps. This increases the burden on both citizens and frontline officials and weakens public trust in digital government.

A citizen-centric approach ensures that every public service—digital or analogue—delivers value through simplicity, dignity, accessibility, and predictable

outcomes. This approach reduces administrative workload, improves compliance, and strengthens accountability across the public sector.

The need for a clear national definition of a Public Digital Service

At present there is no agreed national definition of what counts as a “digital public service” (DPS) or a “citizen-centric service.” This leads to inconsistent design, mislabeling of portals as digital services, and difficulty for ministries in planning or prioritising improvements.

Recommendation:

The Strategy should adopt a national definition aligned with international standards (e.g., IMAPS, European Interoperability Framework):

A digital public service is a public service that:

1. Produces a clear administrative outcome (certificate, approval, registration, entitlement).
2. Has a designated service owner within the government.
3. Has a primary end-user group (citizens, businesses, agencies).
4. Includes a user centric digital interface or machine-to-machine process through which the outcome is delivered.

This will allow government to clearly distinguish:

- End-to-end digital services,
- Semi-digital services,
- Application builders,
- Informational portals.

Such classification is essential for prioritisation, budgeting, and accountability.

Annual Digital Maturity Assessment of all public services

Without a systematic way to measure service maturity, it is impossible to understand progress, identify bottlenecks, or direct resources where they are most needed.

Recommendation

Introduce an **Annual Digital Maturity Assessment** for all public services in Bangladesh, using a simple scale (e.g., 0 to 4):

Level	Classification	Description
0	Analogue Service	No digital pathway exists
1	Informational Presence	Instructions available online, but no digital transaction capability
2	Semi-Digital	Some steps online, but manual verification still required
3	Digital with Incomplete Integration	Online application with partial back-office automation
4	End-to-End Digital Public Service	Full digital lifecycle, automated data verification, no manual steps

Each ministry would submit an annual maturity report with:

- Current level of each service,
- Bottlenecks,
- Planned improvements,
- Required support (skills, budget, data access).

This would provide the Strategy with a baseline and a tool for monitoring implementation.

National Standards for Service Design and Government Portals

At present, different ministries and vendors use different layouts, labels, authentication methods, languages, and processes. Citizens must learn each service from scratch, leading to confusion and dependency on intermediaries.

Recommendation

Adopt national Service Design Standards covering:

- Mobile-first design
- Clear process steps and plain-language labels
- Accessibility for persons with disabilities
- Prioritize ease of access for users instead of highlighting ministry officials or activities
- Multilingual readiness
- Status tracking and notifications
- Integration of feedback and grievance redress
- Mandatory user testing with target groups

- Privacy, data minimisation, and transparency principles

Adopt national Portal (there could be thematic, geographic or other portals) Standards covering:

- Single search interface
- Uniform navigation patterns
- Common authentication
- Common UI/UX components
- Consistent accessibility rules
- A maintained registry of all government services

These standards will reduce duplication, improve usability, and create a predictable user experience no matter which service a citizen accesses.

Once-Only Principle and Data Reuse Across the Government

Citizens repeatedly enter information that government already holds—NID details, birth date, parent names, land records, addresses, tax IDs, etc. This violates the Once-Only Principle, increases errors, and creates unnecessary burden on both citizens and staff.

Recommendation

Codify the **Once-Only Principle**:

- Public agencies must reuse verified data held by any other government agency.
- Citizens cannot be required to re-submit or re-enter information already present in a trusted registry.
- Any exception must be justified on legal or security grounds.

This requires:

- A catalogue of authoritative registers (NID, Birth Registration, Address, Land, Tax, Marriage, etc.).
- Secure, logged data exchange mechanisms.
- Clear institutional agreements between agencies.

The impact will be immediate: shorter applications, fewer errors, faster decisions, and reduced reliance on middlemen.

Reforming Procurement for Citizen-Centric Outcomes

Most tenders today focus on software delivery, infrastructure, and the number of services digitised—not on usability, adoption, or outcomes. Vendors are not required to conduct user testing, support inclusive design, or demonstrate service impact.

Recommendation

Revise procurement rules to require:

- Mandatory user testing with specific groups (women, rural residents, PWDs).
- Service-level KPIs (completion rate, drop-off rate, user satisfaction, processing time).
- Compliance with national service design and interoperability standards.
- Reusable components rather than custom builds.
- Clear ownership of data and systems by the government.

This aligns incentives with real-world outcomes rather than technical outputs.

Digital-First Legislative and Regulatory Reform

Many laws still require physical signatures, stamps, or in-person verification, which blocks end-to-end digitisation.

Recommendation

Many laws still require physical signatures, stamps, or in-person verification, which blocks end-to-end digitization. Furthermore, a prevailing misconception that "system-generated documents do not require signatures" undermines legal validity in courts. To address this, the **Office of the Controller of Certifying Authorities (CCA)** must play a central role in harmonization.

Recommendation Conduct a Digital-First Legal Review, led by the CCA and ICT Division, across ministries to:

- **Amend the Information and Communication Technology Act, 2006:** Update the act to reflect modern digital realities, ensuring the legal equivalence of electronic records and signatures is absolute.
- **Identify and Repeal Analog Provisions:** Remove requirements for "blue-ink" signatures, physical stamps, and mandatory physical presence in service delivery.
- **Establish Cross-Border Recognition:** Empower the CCA to facilitate Mutual Recognition Arrangements (MRA) for digital signatures, enabling cross-border digital trade and legally binding international electronic contracts.
- **Global Trust Integration:** Mandate necessary audits (e.g., WebTrust) to ensure Bangladesh's Root CA certificate is included in international trusted stores

(Microsoft, Mozilla, Adobe), preventing browser security warnings that erode user trust.

Introduce:

- **Legally Recognized Digital Signatures:** Enforce the use of digital signatures in all G2G and G2C documents, moving away from unsecured "system-generated" footers.
- **Economic Codes for PKI:** The Ministry of Finance must introduce specific economic codes to allocate budgets for digital signature expenses in government offices.
- **Evidence-Based Risk Assessments:** Guidelines for when physical verification is strictly necessary versus when e-KYC suffices.
- **Future-Ready Clauses:** Enabling legal provisions for Post-Quantum Cryptography, digital wallets, and verifiable credentials.

This will allow services like birth registration, licensing, pensions, and trade certificates to become fully digital, legally admissible, and globally recognized.

Capacity and Staffing for Digital Service Delivery

Digital services cannot be delivered or maintained without the right skills. Today's public service is structurally designed for analogue processes, not continuous delivery of digital products.

Recommendation

Introduce a **Digital Service Workforce Framework:**

Create specialised roles inside ministries:

- Service owners
- UX designers (staff, not contractors)
- Product managers
- Data engineers
- Cybersecurity specialists

Professional pathways:

- Competitive pay scales
- Career progression for digital roles
- Rotational assignments between ministries

Frontline support:

- Training programmes for registrars, UDC operators, hospital staff, and others who deliver digital services daily.

This strengthens sustainability and reduces reliance on vendors.

Reusable Building Blocks: Toward a BanglaGovStack

Almost all ministry and divisions and agencies under them builds their own login system, payment module, document upload tool, notification system, and ERP solutions. This leads to the "reinventing the wheel" syndrome, where public money is wasted procuring the same software repeatedly (e.g., Inventory Management, Payroll, or File Tracking systems) for different departments. It also increases the burden on vendors to build basic infrastructure from scratch for every tender.

Recommendation: Develop **BanglaGovStack**, a set of centrally maintained, open-standard digital building blocks that agencies can combine to build services instantly. This includes:

- **National Source Code Repository (NSCR):** A centralized, secure repository (based on the National Source Code Policy) hosting the source code of all government-owned software.
 - **Mandate for Reuse:** Agencies must check the NSCR for existing modules (e.g., an ERP module used by the Roads Department) before tendering for a new one.
 - **Public Money, Public Code:** Software developed with public funds must be treated as a national asset, available for reuse and customization by other agencies, drastically reducing procurement costs and time.
- **Authentication and Identity (OneID):** Single sign-on for all government portals.
- **Payments:** Unified gateway integration.
- **Notifications:** Centralized SMS, email, and WhatsApp API gateways.
- **Document Storage:** A secure digital locker for citizen credentials.
- **Case Management & Grievance Redress:** Standardized ticketing systems.
- **Multilingual Content Libraries:** Pre-approved Bangla terminology and UI kits.

Government would maintain these centrally, and agencies would assemble them when creating new services. This reduces cost, ensures interoperability, reduces vendor burden, and improves the user experience. This approach aligns with global models such as India's GovStack, Estonia's X-Road, and Singapore's CODEX, tailored to Bangladesh's context.

How These Measures Support Officials and Frontline Staff

This policy note is not about imposing additional burdens on ministries. Rather, it aims to:

- Reduce workload by eliminating repetitive manual verification.
- Improve reliability so frontline staff face fewer system failures.
- Simplify responsibilities through clear standards and shared infrastructure.
- Strengthen trust in public institutions.
- Direct funding to practical reforms with measurable impact.

The Strategy can position these measures as tools that empower ministries, not constrain them.

4.3 Sector-Wide Digital Transformation Impacts

4.3.1 Integrated E-Commerce and Logistics Supply Chain Modernization:

E-commerce in Bangladesh has grown rapidly in recent years but remains heavily **concentrated in metropolitan centers**. Rural producers and entrepreneurs—particularly women and small enterprises—have limited access to digital marketplaces, restricting inclusive growth. At the same time, logistics networks suffer from **addressing gaps, inefficiencies, and fragmented last-mile delivery systems**.

A next-generation strategy is required to **bridge the rural–urban divide, modernize logistics infrastructure, and enhance Bangladesh’s competitiveness in regional and global trade**.

Strategic Objectives

1. Expand **nationwide access** to e-commerce platforms, enabling rural entrepreneurs to directly reach urban and international markets.
2. Establish a **National Address Management System** to standardize delivery locations.
3. Modernize **logistics infrastructure** with digital, geotagged, and interoperable systems.
4. Leverage **AI-driven forecasting** to improve national logistics performance.

Strategic Interventions

1. Nationwide E-Commerce Access & Rural Integration

- Onboard **rural producers, women entrepreneurs, and marginalized groups** onto national e-commerce platforms.
- Create **local digital commerce hubs** offering training, mobile-first inventory tools, and financial literacy.

2. National Address Management System (NAMS)

- Divide Bangladesh into structured **micro-regions with unique identifiers**.
- Introduce **GPS-based geofencing** for accurate deliveries.
- Align with international postal standards to **facilitate cross-border trade**.

3. Smarter Logistics Infrastructure

- Integrate **geotagged address systems** into delivery operations for last-mile optimization.
- Ensure interoperability between logistics firms, platforms, and payment systems under the **BNDA framework**.

4. AI-Driven National E-Logistics Dashboard

- Monitor **real-time flows of goods** across national networks.
- Use predictive analytics for **demand forecasting, route planning, and bottleneck identification**.
- Provide government and private operators with **decision-support tools** for efficiency and cost reduction.

KPI Framework (2025–2030)

Year	Rural Entrepreneurs Onboarded	% Villages with Digital Commerce Hubs	Deliveries Using GPS/Geotag Addresses	AI-Enabled E-Logistics Dashboard Coverage	Cross-Border Readiness (International Standard Addresses)
2025	50,000	5%	10%	Pilot in Dhaka + Chattogram	Pilot
2026	150,000	15%	25%	Expanded to 5 divisions	Initial adoption
2027	400,000	30%	45%	Nationwide rollout (50% coverage)	50% adoption

2028	800,000	50%	70%	75% coverage	National framework aligned with UPU
2029	1.2M	75%	85%	90% coverage	Adopted by leading exporters
2030	1.5M+	100%	100%	100% coverage	Full compliance with international trade standards

Expected Outcomes by 2030

- **1.5 million rural entrepreneurs** directly engaged in digital commerce, with women playing a central role.
- **100% standardized addressing system** nationwide, ensuring delivery reliability and global trade compatibility.
- **Seamless integration** of logistics, marketplaces, and payment systems.
- **AI-driven national dashboard** providing real-time visibility of supply chains.
- Bangladesh positioned as a **regional leader in inclusive e-commerce and smart logistics**.

4.3.2 National Land Automation Initiatives:

The transformation of Bangladesh's land administration is pivoting from fragmented digital services to a unified, fully interoperable ecosystem. While previous initiatives laid the groundwork, the current strategy focuses on eliminating the persistent "hybrid" processing gaps that force citizens into physical bureaucracy.

- **Vumisheba and Service Gaps:** The Ministry of Land launched the **Vumisheba** platform as a single-window entry point for digital services. However, critical assessments reveal that the system is **not yet truly end-to-end digital or interoperable**. Citizens often face a disjointed experience where they can apply online but must still navigate highly complicated manual processes and make time-consuming physical visits to local land offices for final verifications, corrections, and hard-copy submissions.
- **Integrated Land Management Software (land.gov.bd):** To rectify these inefficiencies, the strategy mandates the full deployment and integration of the **Automated Land Management Integrated Software** hosted at land.gov.bd.

This centralized platform will replace standalone "silos" with a unified system that seamlessly connects:

- **E-Mutation:** Fully automated mutation workflows that do not require physical hearings for standard cases.
- **Land Development Tax (LD Tax):** A cashless, transparent tax collection system integrated with the national treasury.
- **Digital Record Room (DRR):** A repository for certified copies of Khatians and Mouza maps, accessible remotely.
- **Interoperability & Data Integrity:** A core objective is to enforce the **"Once-Only" principle** by linking land records with the National ID (NID) database and the Registration Directorate. This ensures that a single change in ownership is automatically reflected across all government registries—preventing the forgery and dual-ownership disputes that currently clog the judicial system.

4.3.3 Police Automation Program:

The Police Automation Program is a cornerstone of the country's move toward smart Policing, transitioning law enforcement from manual, station-centric operations to a data-driven, citizen-first service model.

- **Nationwide Online GD Adoption:** Building on successful metropolitan pilots, the strategy enforces the **nationwide adoption of the Online General Diary (GD) system**. This platform allows citizens across all 64 districts to file non-emergency reports (such as for lost documents, phones, or vehicles) via a dedicated mobile app or web portal. This initiative eliminates the harassment and delays often associated with physically visiting a police station for routine matters.
- **Integrated Case Management System (CDMS):** Beyond simple reporting, the program is digitizing the core investigative lifecycle. This includes:
 - **Crime Data Management System (CDMS):** A centralized database for tracking **First Information Reports (FIRs)**, case evidence, and investigation updates in real-time.
 - **Forensic & Criminal Records:** Digitizing criminal profiles to enable instant biometric verification and background checks nationwide.
- **Transparency & Accountability:** The platform introduces a "Track My Case" feature, providing complainants with real-time SMS and dashboard updates on their case status. By removing the opacity of manual filing, the system fosters greater accountability and public trust in the police force.
- **Predictive Policing Analytics:** The integration of traffic, crime, and complaint data into a central **Personnel Information Management System** enables the use of AI-driven analytics. This allows for predictive resource allocation—deploying patrols to high-risk zones before crimes occur—and ensuring data-backed decision-making for public safety.

4.3.4 Strengthening National Helplines 333 and 999:

The National Helpline **333** serves as Bangladesh's inclusive, voice-based citizen service platform, providing information on government services, complaint management, and non-emergency support across ministries and agencies. In parallel, the National Emergency Service **999** addresses critical, life-threatening situations by connecting citizens to the Police, Fire Service, and Ambulance. However, a considerable number of non-emergency calls are directed to 999 every day, creating operational pressure and reducing efficiency in handling genuine emergencies. Furthermore, public awareness of the distinct role of 333 remains low, limiting its potential as a primary service channel.

To address these challenges, the **technical and operational integration of 333 and 999** is proposed. Through this integration, non-emergency calls received by 999 will be automatically redirected to 333, while genuine emergencies reported to 333 can be seamlessly escalated to 999. This model will reduce congestion on emergency lines while strengthening 333 as the central hub for citizen service information, complaints, and election support.

Objectives

- **Centralized Platform:** Establish 333 as the primary gateway for citizen information and complaint management.
- **Seamless Integration:** Implement technical call integration for automatic redirection between 999 (emergency) and 333 (service).
- **Enhanced Awareness:** Launch targeted campaigns to educate citizens on the specific utility of 333 versus 999.
- **Election Support:** Provide special election-period support for voter information and reporting irregularities.
- **24/7 Trust:** Ensure a reliable, inclusive service platform that enhances public trust in government institutions.

Proposed Scope

1. Integration & Smart Redirection

- **Automatic Redirection:** Implement protocols to forward non-emergency calls from 999 to 333.
- **Emergency Escalation:** Enable immediate transfer of critical cases from 333 to 999 dispatchers.
- **AI-Powered Efficiency:** Introduce **AI-driven voice assistants and analytics** to record caller details, categorize complaints automatically, and transcribe calls in real-time. This will significantly reduce agent workload and enable faster response times.
- **Shared Analytics:** Develop a unified dashboard for call data, reporting, and performance metrics across both services.

2. Government Service Information & Complaints

- **Comprehensive Database:** Maintain up-to-date information on services from 50+ ministries and agencies.
 - **Process Guidance:** Guide citizens through complex processes, documentation requirements, and service delivery steps.
 - **Complaint Escalation:** Register grievances and escalate them to the relevant authorities with tracking numbers.
3. **Strategic Awareness Campaign**
- **Public Education:** Launch nationwide awareness initiatives to clarify the distinction between 999 (Emergencies) and 333 (Services/Information), targeting rural and less digitally literate populations.
4. **Election Period Support**
- **Voter Assistance:** Provide information on polling stations, schedules, and voting rules.
 - **Complaint Reporting:** Receive reports of election irregularities, violence, or harassment and coordinate with the Election Commission and law enforcement.
 - **Inclusive Support:** Offer dedicated assistance for vulnerable groups, including women, the elderly, and persons with disabilities.

Expected Outcomes

- **Optimized Emergency Response:** Reduced congestion at 999 allows faster response to life-threatening situations.
- **Improved Service Access:** Faster complaint handling and transparent escalation for non-emergency issues.
- **Operational Efficiency:** AI integration reduces manual data entry and speeds up call processing.
- **Increased Utilization:** Higher public awareness leads to correct usage of 333 for administrative needs.
- **Institutional Trust:** A responsive, integrated platform fosters greater citizen satisfaction and trust in public services.



4.3.5 Disaster Recovery and Smart National Infrastructure Safety:

As Bangladesh accelerates its digital transformation, **resilience and public safety** must remain at the core of national planning. With rising climate risks, rapid urbanization, and expanding critical infrastructure, a forward-looking framework is essential to:

- Protect citizens during emergencies.
- Safeguard national pipelines, grids, and highways.
- Enable rapid response through data-driven technologies.

This strategy integrates **AI, IoT, satellite, and drone systems** to ensure resilience, early warning, and coordinated national responses.

Strategic Interventions

1. AI-Powered Public Safety Surveillance

- Upgrade urban CCTV systems with **AI-based hazard detection** (gas leaks, smoke, structural collapse, suspicious activity). Proper standards and regulations will be introduced to ensure PDPO and NDGO compliant data collection and AI training from CCTV footage.
- Establish a **centralized Emergency Alert Platform**, linked to 999 and other national call centers.
- Enable **automated alerts** from surveillance feeds for faster dispatch of responders.

2. National Pipeline and Grid Safety Monitoring

- Deploy **sensor networks** across gas and electricity pipelines to monitor leakage, pressure, vibration, and flow.
- Build a **central Infrastructure Safety Monitoring System** to aggregate data, allow predictive maintenance, and enable rapid fault isolation.
- Ensure **coordination with BAPEX, PGCB, DESCO, DPDC, Titas Gas** and other relevant agencies.

3. Satellite and Drone-Based Disaster Response

- Use **satellite imaging** for large-scale disaster forecasting (cyclones, floods, landslides).
- Deploy **drones for post-disaster mapping**, enabling targeted relief, damage assessment, and infrastructure checks.
- Archive and analyze all imagery within the **National Image Repository** (under NDGA).

4. Smart Highway and Transport Infrastructure

- Introduce **embedded sensors and AI systems** on highways to detect accidents, congestion, and environmental risks.
- Integrate toll automation, BRTA vehicle data, and emergency services into a **National Intelligent Transport Safety Platform**.
- Apply GPU-powered data analytics for **real-time traffic and hazard management**.

KPI Framework (2025–2030)

Year	AI-Enabled CCTV in Major Cities	Pipeline/Power Grid Segments with IoT Sensors	Satellite/Drone Coverage for Disaster Zones	Smart Highway Segments Operational	Avg. Response Time Reduction
2025	Dhaka pilot (2,000 cameras)	10%	10 districts	50 km	5%
2026	Dhaka + Chattogram	25%	20 districts	150 km	10%
2027	4 metro cities	40%	40 districts	300 km	20%

2028	8 major cities	60%	60 districts	600 km	30%
2029	Nationwide urban rollout	80%	80 districts	1,000 km	40%
2030	100% major cities	100%	All districts	1,500 km	50%

Expected Outcomes by 2030

- **50% faster response times** during national emergencies.
- **Nationwide AI-enabled surveillance**, integrated with 999 and emergency services.
- **Full pipeline and grid coverage** with predictive maintenance capability.
- **National archive of disaster imagery** (satellite + drone).
- **Smart transport network** covering 1,500 km of highways with real-time safety monitoring.



4.3.6 Digital Accessibility for Persons with Disabilities:

To ensure **inclusive, barrier-free access** to all public and private digital services for persons with disabilities (PwDs), enabling **equal participation** in Bangladesh's digital economy, education, and public service ecosystem.

Why This Is Needed

- According to a2i estimates and national data, **over 9% of Bangladesh's population** (~15 million people) live with some form of disability.
- Yet fewer than **20% of government websites** are fully accessible.
- Accessibility is not just a social obligation — it is a **driver of economic inclusion**, as digitally empowered PwDs can participate in freelancing, e-commerce, and digital jobs.

Strategic Components

1. Mandatory Digital Accessibility Standards

- National guidelines aligned with **W3C Web Content Accessibility Guidelines (WCAG 2.1)**.
- Enforcement for all government websites, apps, and digital public services by 2030.

2. Inclusive Public Platforms (e.g., Nagorik Seba)

- **Assistive Technology Integration:** Screen-reader compatibility, audio navigation, voice guidance, subtitles, and sign language in video content. Simplified layouts for cognitive, motor, and visual impairments.
- **Human-Assisted Digital Access:** Recognizing that not all citizens, particularly those with disabilities or low literacy, can navigate digital interfaces independently, Nagorik Seba centers operate as human-agent-based digital service hubs. Trained entrepreneurs at these centers provide direct assistance, helping PwDs and illiterate citizens apply for services, access benefits, and complete transactions, ensuring no one is left behind due to a lack of digital skills.

3. Accessible Social Security & Benefits Platforms

- Voice-assisted navigation.
- Alternative authentication (fingerprint, OTP, face ID).
- Accessible tracking of social protection payments.

4. Accessible Education Tools

- Multimedia Talking Books, e-learning with Bangla screen readers.
- Text-to-speech integration across education platforms.
- Interactive content for visually and hearing-impaired learners.

5. Braille-Compatible Services

- E-documents, digital libraries, device integration for blind and visually impaired users.

6. Accessible Emergency & Health Services

- 999 and 333 helplines with text-based chat, video relay for the hearing impaired.
- Priority tagging for persons with disabilities.

7. Training & Capacity Building

- Specialized training for **5,000+ developers, designers, and service staff** on inclusive digital design.

8. Assistive Technology Innovation

- Incentives for local startups to build **AI-powered Bangla screen readers, voice-command systems, and affordable assistive devices.**

KPI Framework (2025–2030)

Year	% Govt Websites & Apps Accessible	Accessible Social Security Users	PwDs Using Digital Learning Tools	PwDs with Access to Helplines (999/333)	Developers Trained in Accessibility
2025	20%	100k	50k	50k	1,000
2026	35%	300k	150k	150k	2,000
2027	50%	600k	400k	300k	3,000
2028	70%	1M	800k	600k	4,000

2029	85%	1.5M	1.2M	1M	4,500
2030	100%	2M+	2M+	1.5M+	5,000+

Expected Outcomes by 2030

- **100% accessibility compliance** across all government websites and major private digital platforms.
- **2M+ persons with disabilities** directly benefiting from accessible social security, education, and health platforms.
- **5,000 developers trained** to design inclusive systems.
- A vibrant **assistive technology ecosystem**, creating export-ready products and services.
- Bangladesh recognized as a **regional leader in digital inclusion**.

4.3.7 Digital Social Protection Framework:

Bangladesh has a wide range of **social safety net programs (SSNPs)** — including allowances for the elderly, persons with disabilities, widows, and marginalized groups — serving over **9 million beneficiaries**. However, program delivery remains fragmented, with challenges such as duplication, leakage, and lack of transparency.

A **Digital Social Protection Framework** will ensure targeted, efficient, and inclusive benefits delivery using ICT tools.

Key Provisions

- **Unified Beneficiary Database (UBD):** Consolidate all SSNP data under NDGA, linked to NID and Digital ID Wallet.
- **Targeted E-Cash Transfers:** Social protection payments delivered directly into mobile wallets and bank accounts, reducing leakage.
- **Smart Eligibility & Monitoring:** Use AI/data analytics to eliminate duplication, track poverty dynamics, and identify new beneficiaries.
- **Accessible Platforms:** Enable voice-assisted and disability-friendly systems for citizens to check benefits status.
- **Cross-Agency Integration:** Connect MoSW, MoWCA, and Finance Division through APIs for coordinated benefits delivery.

Expected Outcomes by 2030

- 100% of social protection payments digitized.
- Leakage reduced by 50% through data-driven targeting.
- Real-time monitoring dashboards operational at ministry level.
- Enhanced financial inclusion of marginalized groups (women, PwDs, rural poor).

4.3.8 ICT in Justice & Judiciary:

Bangladesh's judiciary faces a **backlog of over 3.7 million pending cases**, leading to delays in justice delivery. Manual processes, paper records, and lack of integrated case management slow proceedings. To address this, the government is scaling up **a2i's E-Court Initiative** (labeled *MyCourt* or *Amar Adalat* in pilot phases) to introduce a fully digital judicial ecosystem.

Strategic Objective The primary goal is to transform the traditional courtroom into a "Virtual Court" environment that ensures speedy trials, transparent case tracking, and remote accessibility for all citizens, reducing the time and cost of litigation.

Key Provisions

- **Virtual Courtrooms:** Leveraging the success of virtual courts during the pandemic, the initiative standardizes remote hearings for bail, witness testimonies, and civil motions. This reduces the need for physical transport of prisoners and witnesses, enhancing safety and efficiency.
- **Judicial Dashboard & Cause List:** A unified digital platform where lawyers and litigants can check the Daily Cause List, case status, and next hearing dates in real-time via mobile apps, eliminating the need for physical lobbying.
- **Digital Evidence Management:** Implementing a secure repository for uploading and storing digital evidence (audio, video, documents), ensuring chain-of-custody integrity and preventing evidence tampering.
- **e-Filing & e-Summons:** Enabling lawyers to file cases electronically and receive digitally signed summons/notices via SMS and email, drastically reducing procedural delays caused by manual dispatch.
- **AI-Assisted Judgment:** Piloting Artificial Intelligence tools to assist judges in legal research, case summarization, and translating judgments from English to Bangla (and vice versa) to make legal outcomes accessible to the general public.

Expected Outcomes by 2030

- Case backlog reduced by 30% through faster e-filing and hearings.
- 100% digitization of new case filings.
- Establishment of specialized e-courts for commercial disputes.
- Increased citizen trust through transparent, trackable legal processes.

4.4 National E-SERVICE Guideline

Bangladesh has made significant progress in digitalization, with over 350+ services digitized, MyGov platform expansion, and Union Digital Centers providing access across rural areas. However, most services remain **partially digital**, often requiring manual verification and physical presence. To achieve a fully inclusive and efficient digital governance system by 2030, a structured e-Service Guideline is essential to:

- Mandate end-to-end digital services (apply, pay, verify, receive online).
- Ensure equity and inclusion, with Bangla-first interfaces and accessibility standards.
- Build citizen trust through secure, transparent, and efficient service delivery.
- Drive cost savings and efficiency in government operations.
- Align Bangladesh with global digital governance benchmarks (e.g., Estonia, Singapore, India).

Guiding Principles

1. **Digital by Default** – Every new government service must be designed as fully digital from inception.
2. **Bangla First** – All services must provide Bangla-enabled interfaces, ensuring inclusivity.
3. **Mobile-Centric Access** – Given Bangladesh’s high mobile penetration, services must be optimized for mobile devices.
4. **End-to-End Digitalization** – No mandatory physical visits; all steps (application, payment, verification, delivery) must be digital.
5. **Interoperability** – All e-services must ensure interoperability following the NDGA guidelines.
6. **Trust and Security** – Services must comply with **PDPO 2025 (as amended 2026)** and NDGA data protection standards.
7. **Citizen-Centric Design** – Services should be user-tested for accessibility, clarity, and speed.
8. **Accessibility & Inclusion** – Services must meet **digital accessibility standards** for persons with disabilities.
9. **Transparency & Accountability** – Service metrics (processing time, approvals, complaints) must be published in real time.
10. **Continuous Improvement** – Feedback loops and user surveys should inform service redesigns.

Institutional Roles

- **ICT Division (Lead)**: Policy oversight, digital standards, interoperability enforcement.
- **DoICT**: Capacity building, grassroots training, district and upazila service hubs.
- **BCC**: Certification of e-services for compliance, infrastructure hosting.

- **NDGA:** Data protection, interoperability, trust compliance.
- **Line Ministries:** Service design, ownership, and integration with core platforms.
- **a2i:** Innovation, rapid prototyping, and user-centered design pilots.



Implementation Framework

Phase 1 (2025–2026): Foundation

- Digitize **800+ services end-to-end**.
- Enforce common design standards (Bangla, mobile, PDPA-aligned).
- Launch **National e-Service Dashboard** to track KPIs.

Phase 2 (2027–2028): Expansion

- Achieve **75% of all services fully digital**.
- Integrate all services into **MyGov super-app**.
- Introduce **real-time grievance redressal** across ministries.
- Implement AI-based chatbots for citizen interaction.
- Expand the **Nagorik Sheba agent network** to 2,000 centers, ensuring that every Union has a physical touchpoint where illiterate citizens can receive digital support for land, passport, and NID services.

Phase 3 (2029–2030): Smart Service Delivery

- Reach **95% end-to-end digital service coverage**.
- Enable **predictive governance**: AI models anticipate citizen needs (e.g., automatic benefit eligibility).
- Integrate digital ID & document wallet for instant verification.
- Bangladesh enters **UN E-Gov Index Top 15**.

KPI Framework (2025-2030)

Year	% Services End-to-End Digital	Services in Bangla	Active Users (millions)	Citizen Satisfaction (%)	Cost Savings (US\$)
2025	30%	20	15M	60%	\$50M
2026	50%	40	30M	65%	\$120M
2027	65%	60	50M	70%	\$250M
2028	75%	80	75M	75%	\$400M
2029	85%	100	95M	80%	\$600M
2030	95%	120	110M	85%	\$850M

Expected Outcomes by 2030

- **95% of government services delivered digitally end-to-end.**
- **110 million citizens** actively using digital services.
- **\$850M in cost savings** from reduced paper, travel, and manual processes.
- Universal **Bangla digital access** with inclusive service design.
- Enhanced **citizen trust and transparency** in governance.

Bangladesh recognized as a **regional leader in e-governance**.

4.5 Flagship Initiative: Nagorik Sheba Bangladesh

Nagorik Sheba Bangladesh is a nationwide initiative to bring government e-services closer to citizens through community-based service centers operated by trained local entrepreneurs. Unused government premises such as post offices and union facilities are being transformed into modern co-working service hubs, providing citizens with convenient access to services like NID, passport, tax filing, and more. Nagorik Sheba is fundamentally a Human-Assisted Digital Service model. It bridges the digital divide not just by providing internet access, but by providing a trained human intermediary who navigates the complex digital ecosystem on behalf of citizens with low literacy, disabilities, or limited technical skills.

Zero-Cost Interoperability Hub Model

A critical innovation of Nagorik Sheba is its financial model—it operates as an interoperability hub built with **zero cost to public money**. The public-private-community partnership model ensures centers are managed by licensed entrepreneurs under a sustainable social business framework, with private financing from banks, NGOs, and donors replacing direct government subsidies.

Impact on Interoperability and Digital Public Infrastructure (DPI): Nagorik Sheba serves as the "last-mile" physical manifestation of the Bangladesh National Digital Architecture (BNDIA). By acting as a decentralized interoperability hub, it directly supports the transition from fragmented "data islands" to the DSTAR (Distributed Secure Trusted Architecture for Resilience) ecosystem.

Current Status (2025 Pilot)

- Three pilot centers launched in Gulshan, Uttara, and Nilkhet
- 60 entrepreneurs trained and certified (25 women, 35 men), with 6 operating the first centers
- Each center delivers 500+ government services digitally
- Officially inaugurated on 26 May 2025 by Chief Advisor Dr. Muhammad Yunus

KPIs for Nagorik Sheba (2025-2030)

Year	Centers	Entrepreneurs	Women %	Services	Active Users
2025	3	60	42%	500+	50,000
2026	100	500	45%	600+	1M
2027	500	2,000	45%	800+	10M
2028	2,000	5,000	50%	1,000+	25M
2029	5,000	12,000	50%	1,200+	50M

2030	8,000+	20,000+	50%	1,500+	80M+
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Strategic Role in e-Service Ecosystem

- **Accessibility:** Ensures "walk-to-distance" availability of digital services nationwide
- **Inclusion:** Creates jobs for youth and women while bridging the rural-urban divide
- **Trust & Transparency:** Centers use standardized SOPs and digital monitoring for accountability
- **Scalability:** Provides a replicable model to expand e-services sustainably without heavy fiscal burden

Integration with the National e-Service Ecosystem

Nagorik Sheba is not a stand-alone program but a practical expression of the National e-Service Guideline. Its rollout directly operationalizes the core principles of Bangla-first service delivery, citizen-centric design, inclusion, and end-to-end digitalization. Each Nagorik Sheba center acts as a local access point to the government's wider digital ecosystem, ensuring that citizens can reach services without long travel or administrative barriers.

The platform will be fully integrated with MyGov to allow seamless digital applications and service tracking. It will also complement Union Digital Centers (UDCs) by standardizing service quality and expanding coverage beyond existing rural hubs. In addition, the National Portal will serve as the central directory of all services available at Nagorik Sheba centers. The initiative is designed to interlink with helpline services (333 for citizen queries and 999 for emergencies), ensuring citizens have a consistent, trusted, and accessible gateway for all government services.

End of Chapter 4

5. Strategic Policies for Digital Transformation

This chapter outlines the strategic policies essential for Bangladesh's digital transformation journey. These policies establish the regulatory frameworks, governance structures, and operational guidelines necessary to ensure secure, ethical, and sustainable digital development aligned with the UN Sustainable Development Goals.

5.1 Guiding Policy Principles

Before detailing individual policies, this section establishes the cross-cutting principles that govern all policy formulation and implementation within the digital transformation agenda. These principles, while not formally codified in separate policy documents, represent binding guidance that all agencies and implementing bodies must observe.

5.1.1 Vendor Neutrality and Digital Sovereignty

A critical pillar of Bangladesh's digital transformation is the assurance that public digital infrastructure and services remain under national control, free from dependency on any single vendor, technology provider, or proprietary system. Over the past decade, certain government ministries, agencies, and state-owned enterprises have procured "vendor-locked" or "closed" software and hardware systems under the banner of automation. While these systems have enabled short-term operational gains, many have created long-term strategic vulnerabilities.

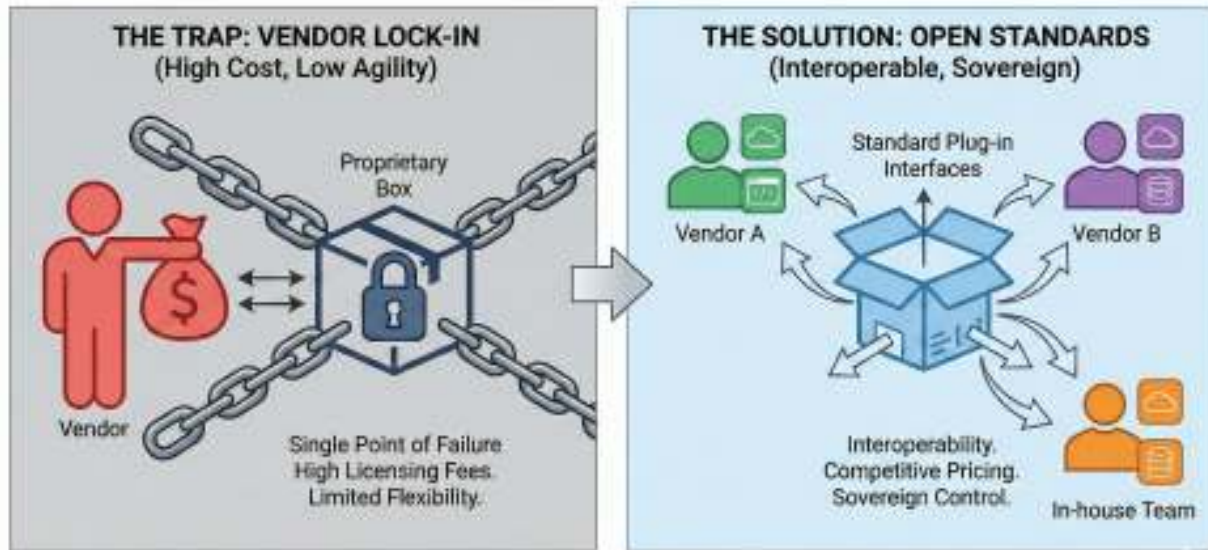
Such systems often come with restrictive licensing models, lack of interoperability, and prohibitively high renewal or change fees. Even minor modifications—such as profile changes, configuration updates, or integration of new features—may require substantial additional payments to vendors, particularly foreign suppliers. In some cases, critical public sector platforms are contractually bound to a vendor's proprietary ecosystem until as far ahead as 2030 or beyond. This dependency limits adaptability, delays service improvements, and imposes excessive financial burdens on the state.

Policy Directives

Ban on Vendor-Locked Systems:

Public funds, whether from national revenue or foreign loans, must not be used to procure software or hardware that creates dependency on a single vendor for its operation, modification, or maintenance. All newly procured systems must support open standards, interoperability, and integration with the National Digital Architecture.

VENDOR LOCK-IN TRAP VS. OPEN STANDARDS (PROBLEM/SOLUTION)



Mandatory Source Code Ownership:

Government agencies must ensure full ownership of the source code for all software they procure or develop. Contracts must include provisions granting the government the legal right to modify, update, and redeploy the software without additional vendor authorization.

Institutional Capacity for Code Maintenance:

Ministries and agencies procuring software must either maintain their own secure Git-based code repository or partner with another government entity that has such capacity (e.g., National Source Code Repository under ICT Division). No automation project should proceed without an identified technical team capable of maintaining and evolving the software post-deployment.

Interoperability and Security by Design:

All new systems must be designed with interoperability, API integration, redundancy, and cybersecurity considerations from inception. Regular code audits and testing must be conducted through the Software and Hardware Quality Testing and Certification (SHQTC) Center.

Phased Transition Away from Existing Vendor-Locked Systems:

For current locked systems, agencies must develop phased migration plans toward open, interoperable solutions. Where legal or contractual barriers exist, renegotiations should be initiated to recover control over system architecture and code.

Strategic Outcomes

Implementing this policy will reduce long-term operational costs by eliminating excessive licensing and change fees, increase agility in adapting systems to evolving

national needs, strengthen cybersecurity by allowing local oversight of all source code, prevent strategic infrastructure from being controlled by a small number of vendors, and promote innovation by enabling multiple domestic and international suppliers to compete on a level playing field.

5.1.2 Open Source for Trust and Transparency

Bangladesh's digital transformation depends on sovereign, interoperable, and cost-effective digital infrastructure. Reliance on vendor-locked proprietary systems creates financial burdens, dependency, and security risks. Inspired by the UN Open Source Principles (endorsed by France in 2025), Bangladesh will adopt an "Open Source by Default" posture for all digital policies to ensure resilience, innovation, and citizen trust.

Strategic Objectives

Mandate open by default for public sector digital systems. Strengthen sovereignty by retaining full source code ownership. Enhance trust and transparency via open repositories and public audits. Promote reuse and interoperability across ministries through the National Source Code Policy. Foster a domestic open source innovation ecosystem (universities, startups, freelancers). Support Bangla-first tools to ensure local language access and inclusivity.

Strategic Pillars (aligned with UN Open Source Principles)

Open by Default: All government-funded software developed under open licenses unless exceptions approved.

Contribute Upstream: Public agencies must contribute fixes, translations, and enhancements to global projects.

Trust and Security: Code published in auditable repositories; regular vulnerability assessments.

Bangla Inclusivity: Prioritize open source platforms that support Bangla interfaces, fonts, and NLP datasets.

Reuse and Interoperability: Mandatory use of the National Source Code Repository for all agencies.

Recognition and Capacity: Civil service engineers' contributions count as KPIs; community developers incentivized.

Sustainability: Dedicated annual budget line for open source maintenance and scaling.

Open Source Implementation Targets (2025-2030)

Year	% New Gov Systems Open Source	Developers Trained	Repositories	Cost Savings (USD)	Bangla Tools
2025	25%	5,000	200	\$20M	5
2026	40%	10,000	400	\$50M	10
2027	55%	20,000	800	\$90M	20
2028	70%	30,000	1,200	\$130M	30
2029	75%	40,000	1,600	\$170M	40
2030	80%	50,000	2,000	\$200M	50

Expected Outcomes by 2030

80% of new government software will be open source. 50,000 developers will be certified in open source skills. 2,000+ public repositories will be in the National Source Code Repository. \$200M in annual savings will be realized from reduced vendor dependency. 50 Bangla-enabled open source platforms will support e-gov and citizen services. Bangladesh will be recognized as a regional leader in open government technology.

5.2 Draft Technology Policies

5.2.1 National AI Policy

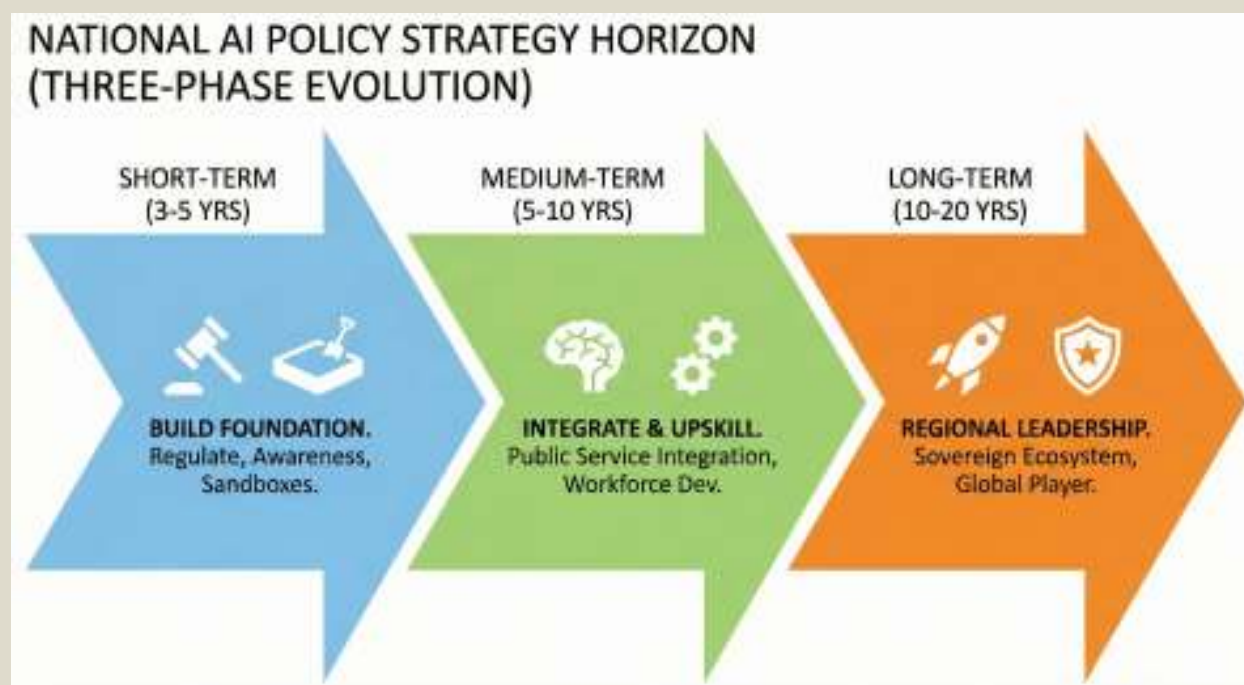
Bangladesh has developed a draft National AI Policy (Draft V1.1: 2026-2030) which establishes a strategic roadmap to leverage Artificial Intelligence as a fundamental driver for sustainable economic growth, governance modernization, and regional leadership. The policy recognizes that Bangladesh, like most of the world, is currently a consumer of AI products rather than a producer, with production concentrated primarily in the US, China, and some European and Asian countries. With over half the population having internet access through smartphones and strengthening connectivity infrastructure, people in Bangladesh are already exposed to and using AI products, mainly Generative AI chatbots like ChatGPT.

The policy is designed to be flexible and adaptable to the accelerated growth and unknowns of AI, prioritizing the regulation of AI consumption, public safety from potential harms, and the creation of an ethical foundation for future domestic innovation.

Core Objectives and Vision

Vision: The development of an ethical and innovative environment for the use and development of AI to accelerate Bangladesh's path towards a middle-income country and aligned with the UN's Sustainable Development Goals (SDGs).

Timeline	Objectives
Short-Term (3-5 Years)	Establish core government offices for AI implementation and monitoring under NDGA; regulate ethical use through risk-based classification; build public awareness; develop regulatory sandboxes for safe experimentation
Medium-Term (5-10 Years)	Fully incorporate AI into public service delivery; foster local AI entrepreneurship through targeted incentives; develop a sustainable, skilled national workforce; establish AI Data Exchange platform
Long-Term (10-20 Years)	Achieve regional leadership in South Asia by building a sovereign AI ecosystem encompassing quality data, compute infrastructure, and talent; position Bangladesh as a competitive player in global AI governance



Bangladesh AI Principles

The Principles of AI in Bangladesh are rooted in the national and cultural context, in alignment with commitments to international norms and standards including UN, OECD, and UNESCO frameworks:

Ethical and Safe: The government shall govern the use and development of AI to improve quality of life and distribute benefits equitably across all segments of society, ensuring AI consumption and production are aligned with ethical standards, human rights, rule of law, and democratic values.

Pro-Innovation and Sustainable Development: The government shall promote innovation through an enabling legal, regulatory, and fiscal environment that supports research and development, startups, and responsible experimentation, including targeted tax incentives, public funding mechanisms, and pilot and sandbox frameworks.

Human-Centric: The government shall prioritize AI systems aligned with human rights, rule of law, and democratic values, requiring mandatory explainability and accountability mechanisms for high-impact decisions.

Inclusive, Equitable, and Non-Discriminatory: The government shall govern AI use to avoid discrimination that can emerge through algorithmic bias, flawed data structures, and human errors.

Subject to Oversight and Accountability: The government shall enforce strict human accountability measures for AI benefits and harms, establishing continuous evaluation, feedback, and learning mechanisms to adapt AI governance frameworks over time.

Transparent and Responsible: The government shall establish transparency measures in AI use, decision-making processes, and mechanisms to build public trust in AI consumption.

Regulatory Framework and Risk Classification

The government shall adopt a risk-based classification framework for AI systems, with regulatory obligations calibrated to the level of risk posed to individuals, society, and the State. The framework is supported by the Personal Data Protection Ordinance (PDPO) 2025 (as amended 2026), the National Data Governance Ordinance (NDGO) 2025, and the Cyber Safety Ordinance 2025.

Risk Category	Description	Requirements
Prohibited (Unacceptable)	AI-enabled social scoring, indiscriminate biometric mass surveillance, manipulative systems exploiting vulnerabilities	Banned entirely; explicit legal prohibition
High-Risk	Law enforcement, biometric ID, creditworthiness, welfare allocation, employment, healthcare decisions	Prior approval, documentation, oversight, mandatory Algorithmic Impact Assessments (AIA)

Limited-Risk	Content generation, chatbots, customer service AI, recommendation systems	Transparency requirements, user disclosure, content labeling
Low-Risk	Spam filters, inventory management, general productivity tools	Minimal compliance; baseline safety standards

Institutional Framework: National Data Governance Authority (NDGA)

The government shall designate the National Data Governance Authority (NDGA) as the central coordinating and regulatory body for AI governance across government. Key responsibilities include:

Issue Technical Standards: Develop and maintain technical standards for AI systems, data formats, and interoperability requirements.

Certify Compliance: Establish certification programs for high-risk AI systems and AI vendors, incorporating both technical standards and ethical considerations.

Oversee Algorithmic Impact Assessments: Mandate and review AIAs for all significant AI systems deployed by public authorities and for high-risk AI systems deployed by private actors.

Manage AI Data Exchange: Establish and manage a dedicated AI Data Exchange to facilitate responsible access to high-quality datasets for AI research, innovation, and public-interest applications.

Data Governance and Sovereignty

Without data, there is no AI. The National Data Governance Ordinance provides a landmark foundation for AI development in Bangladesh. Key provisions include:

Data Sovereignty: Sensitive and critical datasets—including health, biometric, financial, children's, and national security-related data—shall be stored and processed within Bangladesh or in jurisdictions explicitly designated as trusted. For Critical Information Infrastructure (CII), any offshore processing is strictly conditional upon the maintenance of a synchronized real-time domestic copy.

Right to Erasure: Extends to AI systems trained on personal data. Where an AI system has been trained using data that was unlawfully obtained or processed without valid consent, the deploying entity shall implement algorithmic disgorgement measures.

Open Standards: All public-sector AI systems shall adopt open standards, interoperable data formats, and standardized APIs to prevent vendor lock-in and strengthen digital sovereignty.

Children's Data Protection: Prohibit AI companies from collecting data of children under 16 without parental consent. AI-generated targeted advertising directed at minors shall be prohibited.

UNESCO AI Readiness Assessment Methodology (RAM)

Bangladesh has undertaken a comprehensive AI readiness assessment in partnership with UNESCO to evaluate its preparedness for ethical and inclusive AI adoption. This assessment follows UNESCO's Readiness Assessment Methodology (RAM), a systematic framework designed to help countries assess their capacity to develop, deploy, and govern AI systems in alignment with UNESCO's Recommendation on the Ethics of Artificial Intelligence.

Assessment Governance Structure

The ICT Division, under the Ministry of Posts, Telecommunications and Information Technology, leads Bangladesh's AI governance efforts. A National Steering Committee, co-chaired by the ICT Division and UNESCO, has been established to oversee the assessment process. Key collaborating ministries include the Cabinet Division, Law Ministry, Ministry of Public Administration, and Home Ministry. Aspire to Innovate (a2i), launched in 2007 with support from UNDP, the Gates Foundation, and the Government of Bangladesh, provides technical expertise and coordinates innovation initiatives.

Key Assessment Dimensions and Findings

Dimension	Current Status	Key Gaps Identified
Legal & Regulatory	Emerging framework with PDPO 2025, NDGO 2025, Cyber Safety Ordinance 2025	National AI Policy pending finalization; AI-specific procurement policies needed
Institutional Capacity	ICT Division leads; a2i provides innovation support; NDGA proposed	Specialized AI institutions needed; multi-stakeholder governance body required
Infrastructure	National Data Center operational; 54.8% household internet access	GPU compute resources limited; cloud infrastructure fragmented; urban-rural digital divide
Data Ecosystem	NRDEX platform under development; data sharing frameworks emerging	Bengali language datasets inadequate; data governance frameworks need strengthening

Human Capital	Limited AI awareness; skills concentrated in urban areas	Comprehensive AI literacy programs needed; workforce reskilling urgent
Research & Innovation	University research ongoing; startup ecosystem nascent	R&D funding inadequate; industry-academia collaboration limited

Strategic Opportunities

The assessment identified significant opportunities for Bangladesh to leverage AI for inclusive growth:

Ethical Framework Alignment: Establish robust AI frameworks aligned with UNESCO's ethical AI principles, building public trust and international credibility.

Institutional Development: Create specialized AI institutions including research centers, certification bodies, and a multi-stakeholder AI governance body under NDGA coordination.

Language Localization: Build comprehensive Bengali and indigenous language datasets to localize AI applications, ensuring accessibility for all citizens.

Talent Development: Invest in AI research infrastructure and talent development through curriculum updates, vocational training, and international partnerships.

Inclusive Access: Focus on gender equity and digital inclusion to ensure equitable access to AI benefits, particularly for women, rural populations, and marginalized communities.

5.2.2 National Source Code Policy

The National Source Code Policy 2025 ("Public Money, Public Code") establishes a unified, transparent, and secure framework for the development, management, reuse, and governance of all government-funded or government-controlled software in Bangladesh. The policy recognizes source code as a national asset and positions it at the heart of secure, interoperable, and sustainable digital transformation. This policy is issued under the National Data Governance and Interoperability Ordinance 2025, Section 8, and administered by Bangladesh Computer Council (BCC) under NDGA oversight.

Purpose and Vision

Strengthen Digital Sovereignty: All government-funded software is public property unless exceptions apply. Centralized governance ensures long-term control, security, and sustainability.

Reduce Vendor Lock-in: By mandating open-source licensing and reuse-first approaches, agencies are protected from dependency on single vendors.

Increase Software Reuse: A "Reuse First" mandate ensures ministries must search and evaluate existing government-developed software before starting new development.

Improve Transparency and Security: All source code, artifacts, licensing, and metadata must be stored in a national repository with full audit logs, version history, and CI/CD integrity.

Modernize Development Practices: Promote the adoption of modern software development lifecycles (SDLC) and modular architectures (e.g., microservices) to enhance reusability, accelerate deployment, and facilitate continuous upgradation of government digital services.

Promote innovation, skill development, and local industry growth: Open-source code and transparent standards help build local capacity, strengthen trust, and encourage industry–government collaboration.

Scope of the Policy

The policy applies to all ministries, divisions, departments, attached/special/autonomous bodies, and all software, APIs, and digital services developed through national budget, foreign loans or grants, and government-led or government-funded projects. It also applies to public–private projects using government funds, vendor-developed software delivered to the government, and mixed financing models where the government has partial ownership. No software developed with public funds may be deployed to production without being registered in the National Source Code Repository (NSCR).

Key Provisions and Mechanisms

A. National Source Code Repository (NSCR):

A centralized repository operated under BCC/BGD e-Gov (KoUdiL) and governed by the National Interoperability and Asset Management Authority. Every project must upload source code, documentation, build instructions, release tags and commit history, metadata, and SBOM (Software Bill of Materials). No deployment is allowed until the repository entry is complete.

B. Reuse-First Mandate:

Before starting new development, agencies must search the national repository, evaluate existing software or modules, document findings, justify non-reuse with strong technical/legal reasons, and obtain approval from the authority before proceeding. This reduces duplication, cost, and development time.

C. Open Licensing:

Government software must use approved open-source licenses. Non-open or ambiguous licenses are prohibited. Any third-party intellectual property issues must be cleared before publication. The default principle is: "Public money, public code." Exceptions (with strict approval) include national security, confidentiality, and sensitive IP restrictions.

D. Secure Development and Deployment:

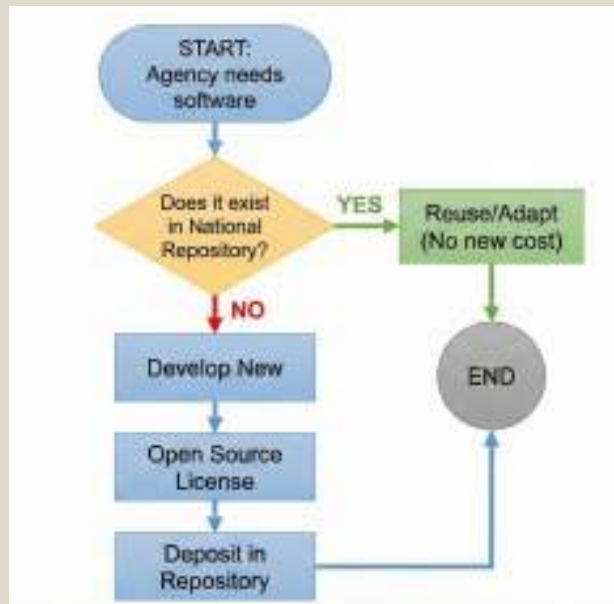
A Coding Standards Committee serves as a national body that maintains coding standards, reviews critical components, issues security guidance, and produces boilerplate and reusable templates. A mandatory CI/CD pipeline requires every government software system to use an approved pipeline that performs automated testing, vulnerability scanning, license verification, and manual approval before production deployment. Mandatory security patching ensures deployed software receives regular security updates, vulnerability fixes, and patch release logs.

E. Role-Based Access Governance:

The policy mandates strict access control in the repository with roles including Viewer (can view code), Contributor (can submit code), Maintainer (manages codebase), Approver (authorizes deployments), and Auditor (performs reviews and compliance checks). All contributors must sign NDAs.

F. Data and Metadata Governance:

All software-related datasets are classified as Open, Restricted, or Regulated. Metadata must be recorded in the national data catalog. Sensitive datasets require NDAs, legal compliance, and access controls.



Compliance, Monitoring and Audit

The policy requires independent and automated audits, self-assessment by each agency, regular compliance reporting, and mandatory focal points in each ministry. Enforcement actions include administrative measures, contract penalties, and legal consequences. The policy must be reviewed every three years to remain aligned with evolving technology and national priorities.

Additional Recommendations

Government Code Exchange Portal: A public portal showcasing government open-source projects, similar to India's OpenForge and UK's GitHub Government initiative.

Mandatory SBOM for all vendors: This improves supply-chain security and traceability.

National Training Program on Secure DevOps: To build capacity across ministries.

Mandatory Open API Registry: Ensures interoperability and prevents API fragmentation.

Integration with National Cybersecurity Strategy: Shared logs, SIEM integration, and secure build pipelines.

Role-Based Access Control

Role	Permissions
Viewer	View repository contents and documentation
Contributor	Push code to repository; NDA signature required

Maintainer	Manage codebase, handle merge requests, manage CI/CD integration; NDA required
Approver	Authorize production deployments; NDA required
Auditor	Analyze logs, conduct compliance reviews, generate audit reports; NDA required

Each agency must designate at least one Maintainer, Approver, and Auditor. Vendors are not granted independent repository access; agencies work with their vendors to manage source code. All user activities are logged for audit purposes, and NDAs remain effective even after role termination.

NDGA and BCC jointly monitor compliance through repository analytics, random audits, and annual agency reports. Each agency must designate a Focal Point for policy compliance. The policy is reviewed every three years to ensure alignment with evolving legal, technical, and administrative requirements.

5.2.3 National Drone Policy

Bangladesh's National Drone Policy treats drones (UAS) as shared public infrastructure to accelerate digital transformation across land, river, environment, agriculture, cities, infrastructure, disaster response, and public safety. The policy's core principle is "collect once, use many times": any publicly funded drone data is standardized and deposited into a National Aerial Imagery Repository (NAIR) within the National Spatial Data Infrastructure (NSDI), and all publicly procured drones are tracked in a National Drone Inventory (NDI).

Governance Framework

Institution	Responsibilities
NDGA	Lead coordinating body for drone data governance; data standards, interoperability, sharing protocols, repository management, and compliance audits
CAAB	Regulate flight safety, licensing, registration, geozones, and operational authorizations; implement UTM system
Survey of Bangladesh	Receive and validate authoritative survey deliverables; maintain mapping standards
BCC	Operate secure cloud infrastructure, APIs, and host UTM systems at National Data Center
SPARRSO	Technical advisor on sensors, remote sensing methods, and data analysis techniques

Strategic Sectors for Drone Use

Agriculture: Crop health monitoring using RGB and multispectral imagery, soil moisture assessment, targeted input application, damage assessment, and crop spraying pilots.

Urban Planning: 3D city models, traffic monitoring, infrastructure inspection, and informal settlement mapping.

Land and Rivers: Cadastral mapping with centimeter-level accuracy, erosion monitoring, char mapping, and identification of illegal structures.

Environment: Forest patrols with infrared cameras for nighttime surveillance, biodiversity monitoring, pollution monitoring with mounted sensors.

Infrastructure: Bridge, rail, and runway inspections; construction progress monitoring; earthwork volume calculations.

Disaster Response: Rapid damage assessment maps, search and rescue with thermal cameras, live feeds to emergency coordination centers.

Law Enforcement: Crowd monitoring, crime scene mapping with 3D models, search and rescue operations, border patrol, and traffic enforcement.

Regulatory and Safety Framework

Category	Description	Requirements
Open	Low-risk operations in designated green zones, below certain altitudes, away from sensitive areas	Minimal paperwork; auto-approval or blanket permission; flight logging required
Specific	Operations in yellow zones; flights near critical infrastructure, at night, or BVLOS	CAAB permission required; SORA-based risk assessment; additional safety measures
Certified	High-risk operations in red zones (airports, military, sensitive facilities)	Both CAAB and High Security clearances required; case-by-case approval

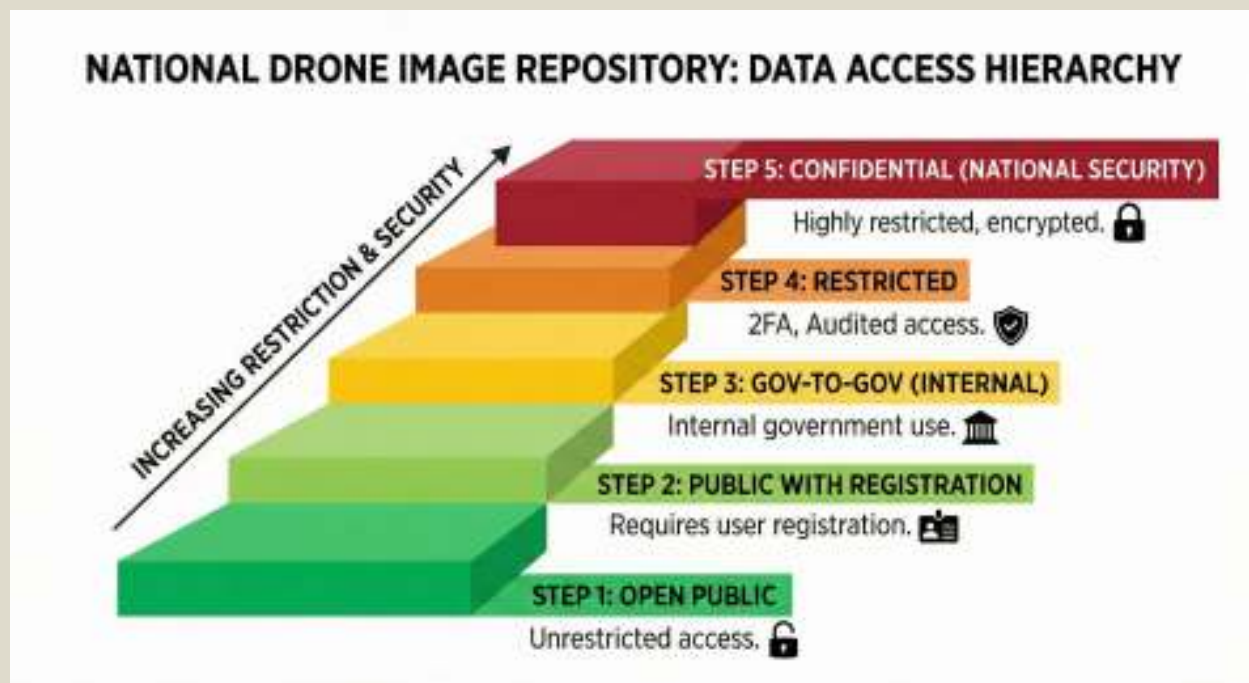


Data Sharing and Repository Access Tiers

The National Drone Image Repository operates under NSDI with a five-tier access model aligned with PDPO data protection requirements:

Access Tier	Description and Examples
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Open Public	Unrestricted access; generalized land cover maps, agricultural surveys, post-disaster damage maps; published under open licenses
Public with Registration	Requires user registration; detailed environmental and infrastructure data; academic research applications
Gov-to-Gov	Internal government data; high-resolution urban imagery, preliminary survey data; authenticated government users only
Restricted	Tightly controlled access; detailed critical infrastructure imagery, law enforcement footage; 2FA required, usage audited
Confidential	National security data; encrypted storage only; secure offline access; metadata visible but files in separate secure systems



Satellite-Drone Complementarity:

The repository includes satellite imagery for comprehensive coverage. However, satellite imagery is significantly more expensive. Drone images compensate for satellite costs while offering superior precision for detailed applications - maximizing cost-effectiveness while ensuring comprehensive spatial coverage at multiple resolution levels.

Implementation KPI Framework (2025-2030)

Year	Agencies	Pilots	Coverage	Datasets	Outcomes
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2025	3	200	5% Upazilas	100	Pilots completed
2026	5+	500	20% Upazilas	500	Maps in 48 hrs
2027	8+	1,000	50% Upazilas	1,500	Land mapping -50%
2028	10+	2,000	75% Upazilas	3,000	Response 24 hrs
2029	12+	3,000	90% Upazilas	5,000	Routine scans
2030	All ministries	5,000	100% districts	10,000+	3-yr ortho cycle

5.2.4 National Cloud Policy

Strategic Context and Rationale

The growth of e-services, digital identity systems, financial technology, and artificial intelligence demands secure, sovereign, and scalable data storage capabilities. Bangladesh's cloud adoption has historically been fragmented, governed by ad-hoc guidelines that created risks of data breaches, vendor lock-in, and non-compliance with emerging global standards. The National Cloud Policy addresses these challenges by establishing clear rules for cloud adoption across government while enabling the innovation necessary for digital transformation.

The policy recognizes that cloud computing is not merely a technology choice but a strategic capability that underpins virtually every digital initiative. From citizen-facing services to internal government operations, from AI-powered analytics to disaster recovery, cloud infrastructure provides the scalable, flexible foundation that modern digital government requires.

Vision and Guiding Principles

Vision: Establish a secure, sovereign, and scalable Government Cloud ecosystem that accelerates digital transformation while maintaining the trust and protection that citizens deserve.

- **Sovereign-by-Design:** Sensitive government data and critical infrastructure remain under Bangladesh legal control at all times.
- **Risk-Based Placement:** Workload placement decisions balance innovation needs with appropriate protection based on data sensitivity.
- **Security Baseline Compliance:** All cloud implementations meet mandatory security, privacy, and resilience controls aligned with PDPO, NDGO, and CSO.

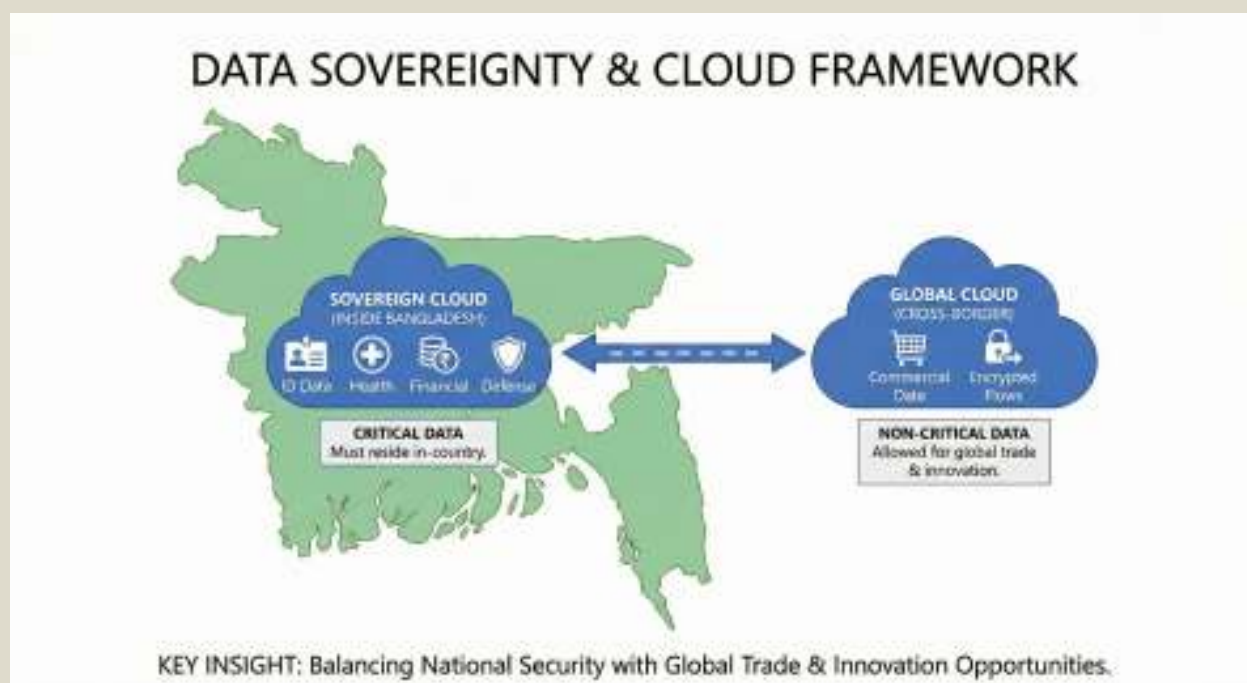
- **Vendor Neutrality:** Procurement ensures interoperability and prevents single-provider dependency through open standards.
- **FinOps Governance:** Cost optimization through centralized financial operations practices including automated anomaly detection and transparent chargeback models.

Data Classification Framework

The policy establishes a five-tier data classification framework that determines permissible cloud deployment models, security controls, and residency requirements:

Level	Category	Description	Residency Requirement
D0	Open	Publicly available information	No restriction
D1	Internal	Internal government use, limited harm if disclosed	Preferred domestic
D2	Confidential	Personal data, operational data requiring protection	Mandatory domestic
D3	Restricted	Sensitive personal data, law enforcement, financial	Sovereign zone only
D4	National Critical	National security, CII-related, diplomatic	NDC sovereign zone only

This classification system directly aligns with NDGO data governance requirements and determines workload placement: D4 workloads are hosted exclusively in National Data Center sovereign environments; D3 workloads require accredited Tier 4 sovereign cloud zones; D2 workloads require domestic data residency with Tier 3+ accredited providers.



Hybrid Multi-Provider Approach

The policy mandates a hybrid multi-provider approach that balances sovereignty with flexibility. Rather than mandating a single cloud provider or model, the policy enables agencies to select the most appropriate deployment pattern for each workload based on data classification, performance requirements, and cost considerations.

The National Data Center (NDC) serves as the primary Cloud Governance and Implementation Authority, exercising regulatory oversight and maintaining the Government Cloud Technical Baseline. Bangladesh Data Centre Company Limited (BDCCL) operates as the primary Government Cloud service provider, offering IaaS and PaaS capabilities with sovereignty guarantees.

Accreditation Framework

Any Cloud Service Provider supporting Government workloads must be accredited against the Government Cloud Control Baseline:

Tier	Permitted Data Classes	Accreditation Requirements
Tier 1	D0 (Open) only	Self-attestation + registry listing
Tier 2	D0-D1 (Open, Internal)	Documented controls, NDC review
Tier 3	D0-D2 (including Confidential)	Independent audit (ISO 27001), penetration testing, DPIA

Tier 4	D0-D4 (including Restricted, Critical)	Enhanced audit, sovereign controls, NCSA approval, CII compliance
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Zero Trust Architecture

The policy mandates Zero Trust Architecture for all cloud implementations, implementing the principle that network location shall not confer trust:

- **No Implicit Trust:** All access requests authenticated and authorized regardless of origin
- **Continuous Verification:** Context-aware access policies evaluating risk signals for each decision
- **Least Privilege:** Elimination of standing privileged access in favor of just-in-time elevation
- **Micro-Segmentation:** Networks segmented to limit lateral movement with deny-by-default policies
- **Assume Breach:** Security architecture designed for detection, containment, and recovery

Implementation KPIs

KPI Metric	2026	2028	2030
Critical data in sovereign cloud	40%	80%	100%
National green data centers	1	2	3-4
ISO/IEC 27001 compliance	50%	80%	100%
Agency cloud migration	30%	60%	100%

5.2.5 National Blockchain Policy

Introduction and Strategic Rationale

As Bangladesh advances toward a fully digital economy, the need for a trusted, immutable, and transparent data layer has become critical. The National Blockchain Policy 2026 establishes the foundational framework for adopting Distributed Ledger Technology (DLT) not merely as a financial instrument, but as a critical component of the nation's Digital Public Infrastructure (DPI).

This policy addresses the "Trust Deficit" in digital governance by replacing manual verification with cryptographic certainty. It aligns strictly with the legal triad of 2025: the **National Data Governance Ordinance (NDGO)**, the **Personal Data Protection Ordinance (PDPO)**, and the **Cyber Safety Ordinance (CSO)**. The policy explicitly pivots away from speculative cryptocurrency trading, focusing instead on sovereign, permissioned blockchain networks that enhance public service delivery, secure land records, and validate citizen identity.

Vision and Mission

Vision: To transform Bangladesh into a blockchain-enabled nation where DLT serves as a sovereign, trusted infrastructure layer enhancing government transparency, citizen empowerment, and economic efficiency while strictly adhering to data sovereignty and privacy rights.

Mission: To establish a sovereign, permissioned blockchain ecosystem that integrates seamlessly with the Bangladesh National Digital Architecture (BNDIA) and the National Responsible Data Exchange (NRDEX), enabling secure and verifiable data sharing across government agencies while protecting citizen rights.

Core Policy Provisions

A) Priority application domains (illustrative, policy-specified examples)

The Policy specifies concrete public-sector and regulated-industry use cases and assigns lead stakeholders. Examples include:

- **Public finance & procurement:** recording tender announcements, bids, evaluation, awards; anchoring budget and expenditure records for real-time tracking; integrating subsidy distribution with **MFS** for last-mile transparency (with Finance Division, iBAS++/CGA, CPTU e-GP, OCAG roles).
- **Supply chain traceability:** pharmaceuticals "lab-to-patient," agriculture "farm-to-fork," RMG traceability for buyer compliance, and export documentation as **Verifiable Credentials** with **TradeTrust** compatibility.
- **Healthcare data exchange:** verifiable vaccination/fitness/screening credentials, consent-based sharing with on-chain consent records, and PDPO compliance for sensitive personal data (MoH/DGHS and private ecosystem roles).

B) Legal and regulatory positioning

The Policy establishes how blockchain artifacts interact with Bangladesh's legal system:

- **Evidentiary status:** blockchain records are treated as electronic records under the **ICT Act 2006** and **Evidence Act 1872 (as amended)**; cryptographically

secure chains may receive a rebuttable presumption of integrity; timestamps evidence existence at a point in time.

- **Smart contracts:** recognized as evidence of contract formation when standard elements exist; **natural language generally prevails** if code conflicts; courts retain authority for corrective actions when code behaves unintentionally.
- **Crypto-asset regulatory perimeter:**
 - **Bangladesh Bank** retains authority over cryptocurrency trading, VASPs, payment services, CBDC, and FX matters;
 - **BSEC** governs security tokens and related market infrastructure;
 - **Ministry of Commerce** governs consumer protection and trade/e-commerce aspects for blockchain applications.
- **Liability and market integrity controls:** validator operators are accountable for secure, honest participation; the policy prohibits predatory **MEV** behaviors (front-running, sandwich attacks, arbitrary reordering) with revocation and penalties; smart contract deployers carry code-quality, upgrade governance, and PDPO “data fiduciary” obligations; NDGA is not liable for third-party application losses on the NBI.
- **Dispute resolution:** may use on-chain mechanisms, ADR, or courts; government blockchain services must maintain grievance redress with defined timelines.

The "Immutability Paradox" and Privacy Solutions

A critical challenge in blockchain adoption is the tension between the technology's inherent immutability (once written, data cannot be deleted) and the **Right to Erasure** mandated by the Personal Data Protection Ordinance (PDPO). To resolve this "Immutability Paradox," the policy mandates a **Privacy by Design** architecture for all public sector blockchain implementations:

- **Off-Chain Data Storage:** Personal or sensitive data shall **never** be stored directly on-chain. Instead, data resides in secured, off-chain encrypted vaults (e.g., within the National Data Center), while only the cryptographic hash (a digital fingerprint) of the data is recorded on the blockchain.
- **Crypto-Shredding for Right to Erasure:** When a citizen exercises their Right to Erasure, the system does not attempt to edit the immutable ledger. Instead, it destroys the specific encryption keys associated with that user's off-chain data. This renders the data mathematically inaccessible and unrecoverable, effectively satisfying the legal requirement for deletion without compromising the blockchain's integrity.

- **Pseudonymization:** To prevent user profiling, the system mandates the use of rotating **Decentralized Identifiers (DIDs)**. This ensures that a user's transaction history cannot be easily correlated to their real-world identity by unauthorized parties.

Technical Architecture: National Blockchain Infrastructure (NBI)

The policy establishes the **National Blockchain Infrastructure (NBI)** as the single, sovereign platform for all government blockchain applications. This approach avoids the risks associated with public, permissionless networks (like Ethereum or Bitcoin) where data governance is impossible.

1. Infrastructure Layer (Sovereignty & Localization)

- **Hosting:** The NBI will be hosted exclusively within Bangladesh's sovereign territory, primarily at the **National Data Center (NDC)** in Kaliakair and **Bangladesh Data Center Company Limited (BDCCL)** facilities. This ensures compliance with Data Localization laws.
- **Disaster Recovery:** A geographically separated Disaster Recovery (DR) site (minimum 200km distance) will maintain a real-time synchronized copy of the ledger with a Recovery Time Objective (RTO) of 4 hours.

2. Network and Consensus Layer

- **Permissioned Network:** The NBI will operate as a consortium blockchain where only authorized government entities can validate transactions.
- **Validator Nodes:** Validator nodes will be operated by trusted ministries including the **ICT Division**, **National Board of Revenue (NBR)**, **Ministry of Land**, **Bangladesh Bank**, and **Ministry of Health**.
- **Consensus Mechanism:** To ensure environmental sustainability, the network will use energy-efficient consensus protocols such as **Proof of Authority (PoA)** or **Practical Byzantine Fault Tolerance (PBFT)**. Energy-intensive Proof of Work (PoW) mechanisms are explicitly prohibited for government deployments.

3. Interoperability & Integration

- **NRDEX as Oracle:** The National Responsible Data Exchange (NRDEX) will serve as the trusted "Oracle," feeding authenticated off-chain data (e.g., NID verification results) into smart contracts.
- **BNDIA Alignment:** All blockchain applications must expose data through standardized RESTful APIs compliant with the **Bangladesh National Digital Architecture (BNDIA)**, ensuring legacy systems can interact with the ledger without needing specialized blockchain knowledge.

Priority Application Domains

The policy identifies six high-impact domains where blockchain will be deployed immediately to solve systemic governance challenges:

A. Digital Identity & Credential Verification (The "OneID" Trust Anchor)

- **Objective:** To eliminate fake certificates and identity fraud.
- **Mechanism:** A national DID registry on the NBI will issue Verifiable Credentials (VCs). Citizens can hold these credentials (birth certificates, driving licenses) in a mobile wallet. Verifiers (e.g., banks, employers) can instantly check the validity of a credential against the blockchain without contacting the issuing authority.

B. Land Records Management (The "Property Chain")

- **Objective:** To stop land forgery and reduce litigation.
- **Mechanism:** The "Property Chain" will record the complete history of a land parcel—from original survey to current mutation. Every transfer of ownership, mortgage, or lien will be an immutable transaction. This creates a single source of truth that prevents the "double-selling" of land.

C. Educational Credential Verification (The "Academic Chain")

- **Objective:** To automate degree verification and combat counterfeit diplomas.
- **Mechanism:** Universities and education boards will issue degrees as digital assets on the NBI. Employers can verify a candidate's degree in milliseconds by scanning a QR code, which checks the cryptographic signature on the blockchain.

D. Public Finance & Procurement (Smart Audit)

- **Objective:** To prevent corruption in public spending.
- **Mechanism:** The **e-GP (Electronic Government Procurement)** system will anchor tender opening logs and bid evaluation scores on the blockchain. This ensures that bid documents cannot be tampered with retroactively. Budget releases and expenditure tracking will also be tokenized to ensure funds reach their intended destination.

E. Supply Chain Traceability

- **Objective:** To ensure food and drug safety.
- **Mechanism:** "Farm-to-Fork" and "Lab-to-Patient" traceability systems will track products through the supply chain. Consumers can scan a medicine packet to verify its origin and ensure it is not counterfeit.

F. Healthcare Data Exchange

- **Objective:** Secure, patient-controlled medical records.

- **Mechanism:** Patients will grant doctors temporary access to their medical history via smart contracts. The blockchain records the "consent" transaction, ensuring an immutable audit trail of who accessed the patient's data and when.

Regulatory and Governance Framework

To ensure this powerful technology is managed responsibly, a robust governance structure is established:

1. National Data Governance Authority (NDGA)

- **Role:** The primary regulator for the NBI. The NDGA is responsible for approving blockchain platforms, designating node operators, and enforcing data classification rules.
- **Oversight:** It will operate a dedicated Blockchain Division to monitor compliance with the NDGO and PDPO.

2. National Blockchain Technical Committee (NBTC)

- **Role:** A specialized technical body comprising experts from academia, industry, and government.
- **Functions:** The NBTC will define technical standards (e.g., cryptographic algorithms), conduct smart contract security audits, and approve upgrades to the NBI protocol.

3. National Cybersecurity Agency (NCSA)

- **Role:** Responsible for the security of the NBI as a Critical Information Infrastructure (CII). The NCSA will monitor the network via the **National Security Operations Center (NSOC)** and coordinate incident response.

4. Regulatory Sandbox

- The NDGA will administer a **Blockchain Regulatory Sandbox**. This allows startups and enterprises to test innovative blockchain solutions (e.g., tokenized assets, smart insurance) in a controlled environment with relaxed regulatory requirements for a limited period (12-18 months). This fosters innovation while protecting consumers.

Legal Position on Crypto-Assets

It is crucial to clarify that this policy **does not** legalize or regulate speculative cryptocurrencies (like Bitcoin) or Virtual Asset Service Providers (VASPs).

- **Bangladesh Bank Jurisdiction:** The regulation of virtual currencies, payments, and forex trading remains the exclusive domain of Bangladesh Bank.

- **BSEC Jurisdiction:** The Bangladesh Securities and Exchange Commission (BSEC) retains authority over security tokens and digital asset investment schemes.
- **Focus:** This policy is strictly focused on **enterprise and government blockchain applications** for service delivery, data integrity, and administrative efficiency.

Implementation Roadmap (2026-2030)

- **Phase 1 (Foundation - 2026):** Establish the NDGA and NBTC. Deploy the initial NBI validator nodes at the National Data Center. Launch pilot projects for Digital Identity and Academic Credentials.
- **Phase 2 (Scale - 2027-2028):** Expand the validator network to key ministries (Land, Health, Education). Integrate NBI with NRDEX. Launch the "Property Chain" pilot. Open the Regulatory Sandbox.
- **Phase 3 (Consolidation - 2029-2030):** Achieve national coverage for land records and educational verification. Implement cross-chain interoperability with international trade networks (e.g., for cross-border logistics).

Conclusion

The National Blockchain Policy 2026 represents a paradigm shift from "trust in people" to "trust in code." By building a sovereign, privacy-preserving, and immutable infrastructure, Bangladesh is laying the rails for a transparent, corruption-free, and highly efficient digital future. This initiative will not only secure citizen data but also significantly reduce the cost of doing business, enhancing Bangladesh's position in the global digital economy.

5.2.6 National Green ICT Policy

The National Green ICT Policy 2026 establishes the governance framework to ensure that Bangladesh's digital transformation advances economic development without compromising environmental sustainability. As the government expands its digital footprint—through data centers, cloud infrastructure, and millions of connected devices—energy consumption and electronic waste are projected to rise exponentially. This policy operationalizes a "Sustainability-by-Design" approach, mandating energy efficiency, renewable integration, and circular economy practices across the public sector.

Vision and Strategic Objectives

Vision: A sustainable, secure, and interoperable Bangladesh where ICT systems deliver public value with minimal environmental harm, and where electronic waste is managed through circular economy principles.

Strategic Objectives (2026–2030):

1. **Decarbonize Digital Infrastructure:** Achieve 30% renewable energy sourcing for public ICT infrastructure by 2028, scaling to 50% by 2041.
2. **Institutionalize Green Procurement:** Transition from "lowest price" to **Lifecycle Costing (LCC)** for all major ICT procurements, ensuring durability and energy efficiency are valued over cheap, short-lived hardware.
3. **Build National E-waste Infrastructure:** Establish a formal Take-back and Treatment system that integrates the informal sector while enforcing environmental standards.
4. **Secure IT Asset Disposition (ITAD):** Ensure that end-of-life government devices are sanitized of sensitive citizen data before disposal, aligning environmental goals with data privacy (PDPO) requirements.

Key Policy Requirements

A. Green Data Centers & Cloud Infrastructure Government data centers are major energy consumers. To curb this, the policy mandates:

- **PUE Limits & Reporting:** All government data centers must establish baseline **Power Usage Effectiveness (PUE)** measurements by 2026 and achieve a 15% efficiency improvement by 2030.
- **Thermal Optimization:** Adoption of **ASHRAE TC 9.9** thermal guidelines, allowing wider temperature ranges to reduce cooling loads (which often account for 40% of total energy).
- **Renewable Integration:** Mandatory disclosure of renewable energy procurement methods. The National Data Center (NDC) aims for **ISO 50001** energy management certification by 2028.



B. Universal Charging Standard (The "One Cable" Rule) To tackle the 3,000-4,000 tonnes of e-waste generated annually from discarded chargers and cables (from ~35 million mobile imports), Bangladesh will align with international standards (IEC 62680 / ITU-T L.1000).

- **Mandate:** All government-procured mobile devices must use **USB Type-C** with USB Power Delivery (USB-PD) by **2027**. Laptops must transition by 2028 where technically feasible.
- **Unbundling:** Procurement preferences will be given to "Charger Unbundled" options where compatible chargers are already available, reducing redundant accessory acquisition.

C. Green Public Procurement (GPP) The government will use its purchasing power to drive market transformation.

- **Lifecycle Costing (LCC):** For ICT procurements exceeding **BDT 50 Lakh**, agencies must use LCC methodology, evaluating not just the purchase price but also operating energy costs and end-of-life disposal costs.
- **Sustainability Weighting:** Tender evaluations shall allocate a minimum **10% weighting** to environmental criteria (e.g., ENERGY STAR/EPEAT certification, repairability, extended warranty).

D. Secure IT Asset Disposition (ITAD) Bridging the gap between Green ICT and Data Privacy:

- **Data Sanitization:** Before any government device (laptop, server, drive) enters the e-waste stream, it must undergo data sanitization aligned with **NIST SP 800-88** guidelines.
- **Verification:** Agencies must maintain a "Chain of Custody" document for retired assets. Devices that stored **Restricted/Confidential** data (under NDGO) require enhanced sanitization verification or physical destruction of storage media to prevent data leakage.

Institutional Governance

Implementation is overseen by a multi-tier governance model to ensure cross-sector coordination:

1. **National Green ICT Steering Committee (NGICESC):** The apex body chaired by the Secretary, ICT Division, with co-chair from the Ministry of Environment (MoEFCC). It sets targets and resolves jurisdictional issues.
2. **Green ICT Implementation Unit (GIU):** A specialized unit within the ICT Division responsible for technical guidelines, the National Green ICT Dashboard, and monitoring PUE compliance.

3. **E-waste Regulatory Directorate:** A designated wing within the Department of Environment (DoE) responsible for licensing treatment facilities and enforcing Extended Producer Responsibility (EPR).

E-Waste Management & Just Transition

Bangladesh generates ~500,000 tonnes of e-waste annually, handled largely by the informal sector under hazardous conditions. The policy promotes a **"Just Transition"**:

- **Integration of Informal Sector:** Rather than banning informal collectors, the policy establishes "Authorized Aggregator" models. Informal workers are trained, equipped with safety gear, and licensed to feed collected waste into formal recycling centers.
- **Target:** Integrate 500 informal workers into formal operations by 2030, ensuring decent work standards and safety.

Implementation Roadmap (2026-2030)

Phase	Timeline	Key Actions
Foundation	2026	Publish Green ICT Procurement Guidelines; Establish GIU; Baseline PUE for NDC.
Integration	2027	Mandatory USB-C for Govt Mobile Devices; Launch National Green ICT Dashboard; Enforce ITAD certification.
Scale	2028	30% Renewable Energy in Public ICT; Mandatory LCC for tenders >50 Lakh; Formalize 200 informal e-waste collectors.
Maturity	2030	15% PUE improvement across all govt data centers; Full circular economy integration for ICT assets.

Conclusion

The National Green ICT Policy ensures that Bangladesh's digital growth is decoupled from environmental degradation. By standardizing ports, enforcing energy efficiency, and securing the disposal of digital assets, the government sets a precedent for the private sector, driving the nation toward a low-carbon, circular digital economy.

5.2.7 National Semiconductor Policy

Bangladesh's Semiconductor Policy (2026–2030) establishes a national framework of binding directives, minimum technical requirements, and institutional coordination mechanisms to develop a credible semiconductor ecosystem—prioritising fabless

design, verification/DFT services, packaging & reliability engineering, and OSAT as realistic entry points for 2026–2030, while deferring wafer fabrication until ecosystem maturity is proven.

Vision and Policy Mandate

Vision: To establish Bangladesh as a competitive, trusted, and sustainable participant in the global semiconductor value chain by 2030, leveraging **Fabless Design**, **OSAT**, and **Trusted Electronics** as primary growth engines.

Strategic Objectives: The policy mandates the achievement of the following national targets by 2030:

- **Export Sovereignty:** Generate **\$1 Billion+** in annual export revenue.
- **Human Capital:** Train and certify **10,000+** engineers under the BASICS framework.
- **Industrial Capability:** Operationalize at least **5 Design Centers** and **2 OSAT facilities**.
- **Sustainability:** Enforce **30% renewable energy** adoption across all semiconductor facilities.

Institutional Governance Architecture

To ensure coordinated implementation, the policy establishes a robust regulatory structure:

1. **National Semiconductor Council:** The apex decision-making body chaired by the Secretary, ICT Division. The Council is empowered to approve incentives, designate semiconductor zones, and coordinate cross-agency security protocols.
2. **Semiconductor Intellectual Property Office (SIPO):** A specialized regulatory office established to protect Industrial IP (Topographies, Layout Designs). SIPO provides the legal certainty required for multinational corporations to offshore sensitive R&D to Bangladesh.
3. **Green ICT Implementation Unit (GIU):** Mandated to enforce **Green Foundry Guidelines**, ensuring rigorous water stewardship and chemical management compliance.



Core Policy Directives

A. The "Fabless First" Industrial Strategy The policy explicitly prioritizes high-value, lower-capital entry points over immediate wafer fabrication.

- **Fabless Design:** Entities focusing on RTL design, Verification, and IP Block creation shall receive priority fiscal support.
- **OSAT (Outsourced Semiconductor Assembly & Test):** The policy designates OSAT as a "High-Priority Industry," enabling duty-free import of testing equipment and specialized machinery.
- **Trusted Electronics:** Manufacturing of secure modules (SIMs, Smart Cards, Biometric Sensors) for the National Digital Public Infrastructure (DPI) is mandated to reduce reliance on non-sovereign supply chains.

B. Workforce Development: The BASICS Mandate The **Bangladesh Skills Initiative for Chip Systems (BASICS)** is established as the national workforce development program.

- **Curriculum Reform:** Universities receiving public grants must modernize curricula to include VLSI, microelectronics, and material science.
- **Shared EDA Infrastructure:** The policy mandates the creation of a "National EDA Cloud," providing students and startups with subsidized access to industry-standard tools (Cadence, Synopsys, Siemens).

C. Investment Framework: The 50:50 Co-Investment Model To de-risk the sector and ensure market discipline, the policy enforces a strict co-investment structure:

- **Public Capital:** Allocated strictly for "Common Goods"—workforce training, shared labs (Clean Rooms), and R&D grants.

- **Private/FDI Capital:** Required for commercial operations, equipment procurement, and scaling. This ensures that public funds are leveraged to catalyze, not replace, private investment.

Strategic Pillars and Security Protocols

1. Intellectual Property Protection Recognizing IP as the currency of the semiconductor industry, the policy extends legal protection to **Layout Designs (Topographies)** and trade secrets. SIPO is authorized to prosecute IP infringement with expedited judicial processes.

2. Security by Design All semiconductor facilities designated as **Critical Information Infrastructure (CII)** under the CSO must implement:

- **Zero-Trust Architecture:** Network segmentation to protect design IP.
- **Secure Access Controls:** Biometric authentication for sensitive zones, compliant with PDPO privacy standards.

3. Sustainability Directives Facilities must adhere to the **National Green ICT Policy:**

- **Renewable Energy:** A roadmap to 100% renewable energy adoption by 2040.
- **Circular Economy:** Mandatory recycling protocols for e-waste and chemical byproducts.

Implementation Roadmap

Phase	Timeline	Strategic Directive
Foundation	2026	Establish National Semiconductor Council and SIPO . Launch BASICS pilot.
Activation	2027	Operationalize SHQTC for hardware testing. Secure first anchor investment for Design Centers.
Acceleration	2028-29	Commission first OSAT facility. Achieve international certification for national labs.
Consolidation	2030	Full ecosystem operational. Feasibility study for Wafer Fabrication initiated.

Conclusion

The National Semiconductor Policy 2026 transforms the semiconductor sector from an aspirational goal into a regulated, strategic industry. By enforcing the **BEAR framework**

(Biotech, Electronics, AI, Robotics) and adhering to strict governance standards, Bangladesh secures its technological sovereignty and economic resilience for the decades ahead.

5.2.8 Cross-Border Digital Trade Policy

The Cross-Border Digital Trade Policy aims to simplify and secure the import and export of goods and services through digital platforms while aligning Bangladesh's trade practices with global standards (WTO, WCO, UNESCAP). It emphasizes creating opportunities for small and medium entrepreneurs, expanding international market access, and ensuring safe, transparent, and efficient cross-border trade.

Industry Context and Challenges

Recent findings from the software and IT-enabled services (ITES) industry highlight critical bottlenecks that this policy seeks to address. Currently, many export-oriented software firms face significant hurdles in paying for essential foreign services—such as cloud hosting (AWS, Azure), software licenses, and digital marketing—due to restrictive outward remittance caps.

- **Operational Expense Gap:** Industry data suggests that a growing software firm often needs to reinvest 30-40% of its revenue into foreign services to maintain operations and growth. However, current regulations limit outward remittance for such expenses (e.g., often capped at \$30,000-\$40,000 annually per card), which is insufficient for scaling companies.
- **Revenue Retention Abroad:** To bypass these limits and avoid service disruptions, many companies are forced to retain a portion of their export earnings in foreign jurisdictions or use informal channels, leading to a discrepancy between actual export performance and officially recorded revenue repatriated to Bangladesh.
- **Double Taxation Issues:** The lack of streamlined tax treaties often results in double taxation for digital services, further disincentivizing formal cross-border transactions.

Key Priorities

- **Trade Facilitation:** Streamlined rules for cross-border B2B and B2C trade, simplified customs processes, dedicated logistics tracking, and digital trade zones near ports.
- **Payments and Regulation:** Internationally connected payment systems, escrow services, and policy support for secure settlements under the Foreign Exchange Control Act. Specifically, this includes revising outward remittance limits for bona fide digital exporters to cover legitimate business expenses like cloud infrastructure and marketing.

- **Tax and Compliance:** Clear VAT, tariff, and reporting guidelines, with APIs linking platforms and customs. Mechanisms to avoid double taxation on digital services will be prioritized.
- **Consumer Protection:** Strengthened dispute resolution, after-sales service requirements, and strict enforcement of intellectual property laws.
- **Capacity Building:** Training programs, institutional support for entrepreneurs, and promotional initiatives to brand Bangladeshi products globally.
- **Data and Monitoring:** Systems for trade data collection, risk analysis, and interoperability to inform policy.

Vision and Strategic Objectives

Vision: To position Bangladesh as a competitive, trusted, and resilient hub for cross-border digital trade by enabling seamless payments, frictionless trade documentation, and secure data flows.

Strategic Objectives (2026–2030):

- **Export Growth:** Increase IT/ITES exports from USD 1.5 billion to **USD 5 billion** by 2030.
- **Paperless Trade:** Reduce average trade documentation processing time by **50%**.
- **Financial Integration:** Enable international payment gateway access for qualified exporters and freelancers.
- **Trust & Compliance:** Achieve **Trust Mark** certification for at least 500 digital exporters.

Core Policy Pillars

A. Modernizing Foreign Exchange Regulations Current foreign exchange rules, designed for traditional goods trade, stifle digital exporters. The policy introduces three critical reforms:

1. **Revenue-Linked Foreign Expense Caps:** Replacing the flat annual remittance cap (e.g., \$40,000) with **tiered limits based on export turnover**. This allows growing firms to pay for essential cloud infrastructure, software licenses, and digital marketing required to scale global operations.
2. **Foreign-Currency Retention Windows:** Permitting exporters to hold a portion of earnings in foreign currency accounts (or offshore for defined periods) to hedge against exchange rate volatility and meet operational expenses without forced conversion losses.
3. **Simplified Freelancer Settlement:** Streamlining compliance for low-value export receipts, allowing platform-verified invoices as sufficient proof for inward remittance, reducing the administrative burden on micro-exporters.

B. Trade Facilitation & Paperless Customs To accelerate cross-border e-commerce, the policy mandates:

- **Paperless Trade:** Recognition of secure electronic signatures and digital records for customs processing, aligning with the UNESCAP Cross-Border Paperless Trade Agreement.
- **Digital Trade Lanes:** Fast-track customs clearance for low-risk, low-value e-commerce shipments, supporting the growth of direct-to-consumer (D2C) exports.
- **API Integration:** Direct integration between major e-commerce platforms and customs systems for automated duty calculation and declaration.

C. Cross-Border Data Flows & Safe Corridors Recognizing that data is the lifeblood of digital trade, the policy establishes a balanced approach:

- **"Data Free Flow with Trust":** Allowing cross-border data transfer for legitimate commercial purposes (e.g., BPO operations, SaaS delivery) while enforcing privacy safeguards.
- **Safe Corridors:** Designating approved data transfer channels for priority markets (e.g., EU, USA) where data protection standards are equivalent, facilitating BPO and ITES contracts without complex individual approvals.

D. Trust Services & Certification To build the "Brand Bangladesh" in the global digital market:

- **Trust Mark Certification:** A national certification scheme signaling that an exporter meets international standards for privacy (PDPO), security (ISO 27001), and service quality. This tiered system (Baseline, Assured, High Assurance) helps Bangladeshi firms prove their credibility to foreign clients.
- **Platform Accountability:** Clear guidelines for cross-border e-commerce platforms regarding consumer protection, dispute resolution, and returns management.

Addressing Structural Constraints

The policy directly tackles long-standing industry grievances:

- **Overseas Subsidiaries:** Extending the right to establish overseas subsidiaries to **all IT/ITES exporters** meeting revenue thresholds, removing the arbitrary age limit that previously restricted established firms from expanding globally.
- **Incentive Rationalization:** Simplifying the documentation required to claim the **IT/ITES export cash incentive**, ensuring that genuine digital exports are not misclassified as general remittances due to bureaucratic hurdles.

Institutional Governance

Implementation is coordinated by the **Ministry of Commerce**, supported by:

- **Bangladesh Bank:** For payment and foreign exchange reforms.
- **NBR/Customs:** For paperless trade and incentive administration.
- **ICT Division:** For technical standards, Trust Mark certification, and exporter capacity building.

Implementation Roadmap (2026-2030)

Phase	Timeline	Key Reform Actions
Foundation	2026	Issue circulars for Revenue-Linked Expense Caps and Freelancer Settlement . Launch Trust Mark consultation.
Expansion	2027-28	Deploy National Digital Trade Gateway . Establish Safe Corridors for key markets. Full e-KYC integration.
Maturity	2029-30	Achieve mutual recognition of electronic trade documents with trading partners. Full paperless trade capability.

Conclusion

The Cross-Border Digital Trade Policy transforms the regulatory environment from a barrier into an enabler. By aligning foreign exchange rules with the realities of the digital economy and digitizing trade facilitation, Bangladesh empowers its entrepreneurs to compete on a level playing field in the global market.

The policy introduces a central coordination committee and expert bodies to oversee implementation, risk management, and global alignment, ensuring Bangladesh becomes a competitive player in cross-border digital commerce.

5.2.9 Digital Skills and Workforce Development Policy

Bangladesh's digital economy is expanding rapidly, but workforce skills remain unevenly distributed across regions, education levels, and sectors. Existing initiatives are fragmented and lack a legal framework to ensure continuity, accountability, and alignment with market demand. The National Digital Skills and Workforce Development Policy 2026 establishes a unified national framework to build the human capital required for Bangladesh's digital transformation. It addresses the critical "Skills Mismatch" that currently constrains the growth of the ICT sector, ensuring that workforce development is demand-driven, inclusive, and aligned with global standards. The policy shifts from fragmented, project-based training to a permanent, institutionally embedded capacity-building model.

Vision and Strategic Objectives

Vision: A digitally capable Bangladesh where every citizen possesses the skills to participate safely and productively in the digital economy, and where a world-class ICT workforce drives innovation, exports, and public service excellence.

Strategic Objectives (2026–2030):

- **Universal Literacy:** Train **20 million students** in foundational digital literacy and coding.
- **Workforce Readiness:** Certify **1 million youth** in market-ready freelancing and vocational digital skills.
- **Advanced Specialization:** Produce **100,000 graduates** with advanced competencies in AI, Data Science, Cybersecurity, and Cloud Computing.
- **Public Sector Capacity:** Train **50,000 government officials** in digital governance and AI ethics.
- **Inclusion:** Ensure at least **40% female participation** in all publicly funded training programs.

The 5-Track Skills Highway

The policy organizes skills development into five distinct tracks targeting specific population segments:

Track 1: Foundational Skills (Schools & Early Learning)

- **Target:** Primary and secondary students (ages 6–16).
- **Focus:** Integrating digital literacy, online safety, and basic coding/computational thinking into the national school curriculum. Content will be **Bangla-first** to ensure accessibility.

Track 2: Vocational & Youth Employment (TVET & Freelancers)

- **Target:** School leavers, unemployed youth (16–25), and madrasa students.

- **Focus:** Market-ready skills for immediate employment (graphic design, web dev, digital marketing).
- **Madrasa Track:** Specialized training in Arabic/Farsi/Bangla localization and content moderation to target the GCC outsourcing market.

Track 3: Higher Education & Advanced Tech (Deep Tech)

- **Target:** University students.
- **Focus:** "Deep Tech" competencies—AI, Blockchain, Semiconductors, Cybersecurity.
- **Mechanism:** Establishing **Innovation Accelerators** at major universities (BUET, DU, etc.) and mandating 6-month industry internships.

Track 4: Lifelong Learning & Workforce Reskilling

- **Target:** Existing workforce, SME owners, returning migrants, women at home.
- **Focus:** Reskilling for the digital economy—e-commerce management for SMEs, digital remittance tools for migrants, and micro-work platforms for home-based female workers.

Track 5: Public Sector Leadership

- **Target:** Civil servants, judiciary, law enforcement.
- **Focus:** Competencies in **Digital Governance, Data Privacy (PDPO), Cybersecurity Hygiene, and AI Ethics.**

Institutional Architecture & Reform

The policy mandates a clear division of labor to prevent duplication:

1. **Policy Authority: ICT Division** sets the strategic direction and funding priorities.
2. **Standards Body: National Skills Development Authority (NSDA)** is the sole authority for setting competency standards, accrediting training providers, and managing the **National Skills Registry**.
3. **Grassroots Delivery: Department of ICT (DoICT)** is reformed into the primary training delivery arm, utilizing its network of Upazila and District offices to deliver foundational and intermediate training nationwide.
4. **Advanced Certification: Bangladesh Computer Council (BCC) and BKIICT** focus on advanced, specialized certifications and industry partnerships.

Key Enablers

- **National Skills Registry:** A centralized database tracking all verified certifications linked to the **OneID** system, allowing employers to instantly verify a candidate's credentials.

- **Freelancer.gov.bd:** A unified platform for freelancer registration, training enrollment, and the issuance of the **Freelancer ID Card**, which facilitates banking and social protection access.
- **Green Digital Skills:** Embedding sustainability competencies (energy efficiency, e-waste management) across all tracks to support the Green ICT Policy.

Financing & Accountability

- **Outcome-Based Funding:** Public funding for training providers will be linked to verified outcomes (certification rates, employment placement) rather than just enrollment numbers.
- **Industry Co-Investment:** Encouraging private sector participation in curriculum design and funding for specialized tracks (e.g., vendor-specific certifications like AWS, Cisco).

Implementation Roadmap (2026-2030)

Phase	Timeline	Key Actions
Foundation	2026	Publish National Digital Skills Taxonomy; Launch National Skills Registry ; Pilot Madrasa Track.
Scale	2027-28	Roll out training via DoICT Upazila Hubs ; Integration with Freelancer.gov.bd ; Green Skills modules mandatory.
Maturity	2029-30	Full nationwide coverage; International mutual recognition of NSDA certifications; 1M freelancers active.

Conclusion

The Digital Skills Framework moves Bangladesh from ad-hoc training to a structured, lifecycle-based human capital development model. By aligning training with industry needs and global standards, it ensures that the "Demographic Dividend" is translated into a "Digital Dividend," powering the nation's economic growth.

5.3 Regulatory Frameworks

Beyond strategic policies, Bangladesh's digital transformation requires sector-specific regulatory frameworks addressing emerging digital applications. The following frameworks establish governance structures for sectors with significant growth potential and consumer protection requirements.

5.3.1 National Digital Social Protection Framework

The Digital Social Protection Framework establishes systems for implementing Digital Social Protection Delivery Systems (DSPDS) in Bangladesh, transforming how social safety net benefits reach vulnerable populations.

Vision: Ensure that every eligible Bangladeshi can access social protection benefits with dignity, efficiency, and respect for their rights and well-being.

Key Components: Dynamic Social Registry (dSR) enabling on-demand registration; Beneficiary Operations Management Systems (BOMS) for programme-specific operations; Integrated Beneficiary Registry (IBR) for cross-programme coordination; Multi-Programme Multi-Provider Payments Platform progressing toward G2P 4.0 architecture.

Guiding Principles: Human-centred design; dynamic inclusion enabling registration when needed; interoperability by design; collect-once, use-many with governance; privacy by design; explainability and due process; resilience and continuity.

Indicator	2026 Target	2030 Target
Beneficiaries in dynamic registry	10 million	40 million
Programmes on unified payment platform	20	100+
Digital payment coverage	60%	95%
Grievance resolution within 30 days	70%	90%
Exclusion error rate	< 10%	< 5%

5.3.2 National Electronic Games Industry Regulatory Framework

The Electronic Games Industry Framework establishes comprehensive governance for Bangladesh to participate competitively in the global games ecosystem while protecting citizens from potential harms. The global games market exceeds USD 200 billion annually, presenting significant export and employment opportunities.

Vision: Position Bangladesh as a competitive participant in the global electronic games ecosystem through domestic development, export-oriented production, and robust consumer protection.

Key Provisions: Bangladesh Game Certification Board (BGCB) for age rating and content classification; mandatory probability disclosure for randomised monetisation (loot boxes); dark pattern prohibitions protecting consumers from manipulative design; parental controls and spending limits for minor protection; Game Process Outsourcing (GPO) export incentives.

Consumer Protection: Fundamental consumer rights including informed purchase decisions, access to support, and data portability. Prohibition of predatory monetisation targeting minors. Age assurance requirements proportionate to content risk.

Indicator	2026 Target	2030 Target
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Game development employment	5,000	25,000
Game export revenue (USD)	20 million	100 million
Games with age rating compliance	50%	95%
Consumer complaints resolved within 30 days	60%	80%
Local game studios	50	200

5.3.3 National E-Waste Management Framework

The E-Waste Management Framework establishes comprehensive regulatory, operational, and technical requirements for managing electronic waste in Bangladesh, implementing Extended Producer Responsibility (EPR) principles that align incentives with environmental outcomes.

Strategic Linkage: This Framework operates as a companion document to the National Green ICT Policy, together forming an integrated approach to lifecycle sustainability in the ICT sector.

Extended Producer Responsibility: Producers bear financial responsibility for end-of-life management of products they place on market, implemented through Producer Responsibility Organisations (PROs) pooling resources for collection and treatment infrastructure.

IT Asset Disposition (ITAD): Government organisations shall dispose of IT assets only through NDGA-accredited ITAD service providers meeting both data security (aligned with CSO/PDPO) and environmental standards. Data sanitisation must achieve NIST 800-88 equivalent standards.

Just Transition: Formalisation of the informal e-waste sector protecting existing livelihoods while improving working conditions, targeting integration of 500 informal workers into formal operations by 2030.

Indicator	2026 Target	2028 Target	2030 Target
Collection rate (ICT equipment)	15%	35%	65%
Licensed treatment facilities	5	15	25
Informal workers formalised	100	300	500
Material recovery rate	70%	80%	90%
PROs operational	2	5	10

5.13 Chapter Summary

This chapter outlined nine strategic policies and three regulatory frameworks forming the foundation of Bangladesh's digital sovereignty and innovation ecosystem. Together, these instruments:

- **Establish Ethical Principles:** Ensure digital sovereignty through vendor neutrality, source code ownership, and AI ethics with risk-based oversight.
- **Modernize Infrastructure & Trade:** Enable drone usage as a shared public resource, facilitate cross-border digital trade aligned with international standards, and promote cloud sovereignty.
- **Build Trust & Security:** Mandate open source for transparency, enforce secure coding practices, and implement blockchain for immutable public records.
- **Drive Sustainability & Skills:** Institutionalize green ICT practices to reduce carbon footprints and build a future-ready workforce through inclusive digital skills development.
- **Foster Industrial Growth:** Create a strategic roadmap for the semiconductor industry, positioning Bangladesh in the global deep-tech value chain.

These policies collectively ensure Bangladesh's digital transformation is technologically advanced, secure, sustainable, inclusive, and aligned with global best practices while preserving national interests.

End of Chapter 5

6. Institutional reform

6.0 Introduction

Bangladesh's digital transformation cannot succeed without fundamental reform of the institutions responsible for driving and sustaining change. The current institutional landscape within the ICT Division and its affiliated agencies reflects legacy structures that were designed for an earlier era of standalone projects and siloed operations. As Bangladesh moves toward an integrated Digital Public Infrastructure (DPI) ecosystem aligned with the Bangladesh National Digital Architecture (BNDA), institutional mandates, capacities, and coordination mechanisms must be modernized.

This chapter outlines comprehensive reforms across five key institutions: the Bangladesh Computer Council (BCC), the Department of ICT (DoICT), the National Cybersecurity Agency (NCSA), Access to Information (a2i), and the flagship ICT projects requiring redesign. Each reform agenda is justified by identified gaps, aligned with the overall strategy vision, and supported by clear Key Performance Indicators (KPIs) to measure progress.

6.1 Bangladesh Computer Council (BCC) Restructuring

6.1.1 Rationale for Reform

The Bangladesh Computer Council serves as the apex technical body for government ICT initiatives. However, its current operational model is characterized by fragmented project execution, inadequate human resource development programs, and limited regional presence. As Bangladesh transitions toward a knowledge-driven digital economy, BCC must evolve from a project-executing agency into a strategic enabler of national digital infrastructure, workforce development, and data governance.

6.1.2 Digital Infrastructure Backbone

National Service Bus (NSB) Integration: BCC will strengthen the National Service Bus in full integration with the Bangladesh National Digital Architecture (BNDA). This ensures interoperability across all government systems, compliance with DPI standards, and secure integration of Digital Identity systems. The NSB will serve as the middleware backbone enabling ministries and agencies to share data through standardized APIs without duplicating infrastructure.

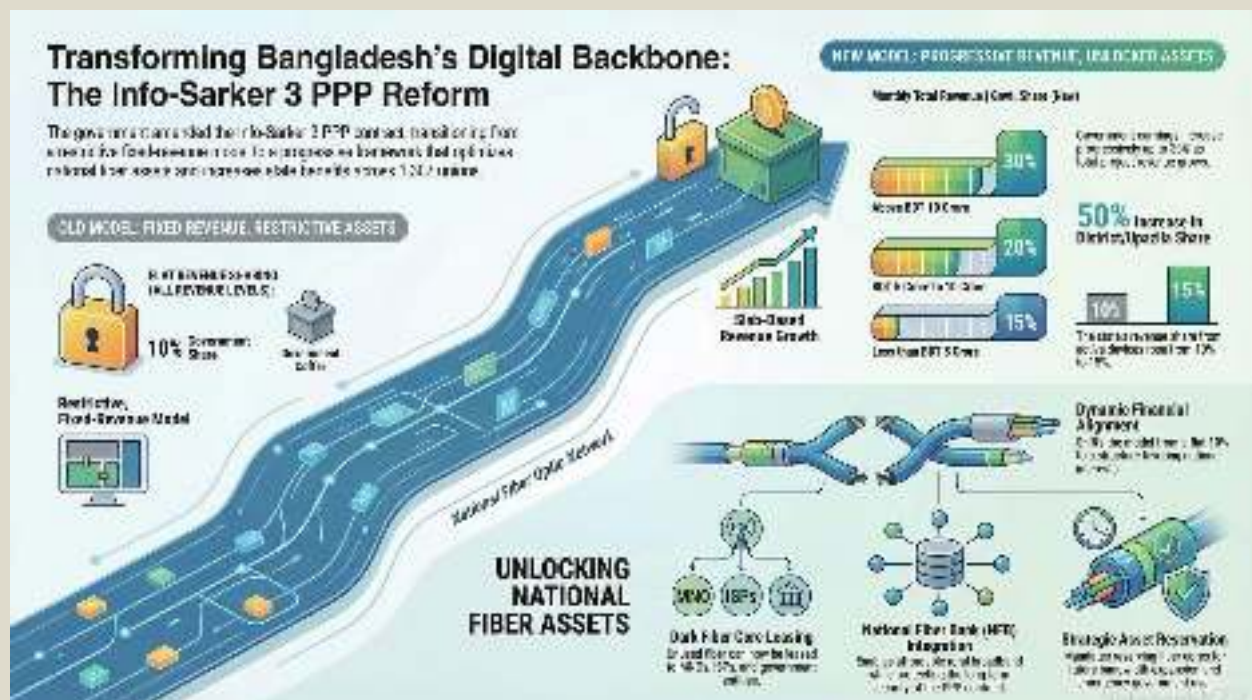
Info-Sarker-3 Network Upgrade: The Info-Sarker network, which connects government offices nationwide down to the union level, is undergoing a comprehensive reform to enhance resilience, redundancy, and nationwide coverage. While the network serves as the country's critical digital artery, historical limitations in monitoring and bandwidth allocation are being addressed through the deployment of **SNMP-ready infrastructure** and modern **Network Management Systems (NMS)**. These technical upgrades are critical to preventing data manipulation and ensuring real-time visibility into network utilization.

Strategic PPP Contract Reform (The February 2026 Amendment): In a landmark move to protect national interests, the Interim Government successfully concluded a six-month renegotiation of the Public-Private Partnership (PPP) agreement for Info-Sarker-3 on February 4, 2026. This reform fundamentally restructures the relationship between the Bangladesh Computer Council (BCC) and the private project companies (SPVs formed by Summit Communications Limited and Fiber@Home Limited), transitioning from a restrictive, fixed-revenue model to a transparent, progressive, and state-friendly framework.

- **Financial Restructuring (The Slab-Based Model):** The previous contract limited the government's revenue share to a **flat 10%**, regardless of the project's financial success. This static model failed to reflect the massive public investment (approx. Tk 1,200 crore) and allowed private operators to maximize profits while the state realized near-zero returns due to opaque reporting. The revised contract introduces a **Progressive Revenue Sharing Model** based on monthly gross revenue slabs:
 - **Tier 1:** For monthly revenue up to BDT 5 Crore, the government share is increased to **15%**.
 - **Tier 2:** For revenue between BDT 5 Crore and 10 Crore, the share rises to **20%**.

- **Tier 3:** For revenue exceeding BDT 10 Crore, the government claims **30%**.
 - Additionally, the revenue share from active devices at District and Upazila levels has been increased from **10% to 15%**, ensuring the state benefits proportionally from network utilization.
- **Unlocking National Assets (Dark Fiber & Fiber Bank):** Previously, the state-built fiber infrastructure was contractually "locked" within the project, preventing broader national use. The reform explicitly authorizes the **leasing of unused "Dark Fiber" cores** to Mobile Network Operators (MNOs), ISPs, and the newly established **National Fiber Bank (NFB)**.
 - **Strategic Reservation:** The new agreement mandates the reservation of specific fiber cores for future government bandwidth expansion, emergency national use, and connecting priority public projects, ensuring that national sovereignty over the infrastructure is never compromised.
 - **Investment Security:** A clear legal provision has been added ensuring that the operationalization of the National Fiber Bank cannot be used as grounds by private partners to cancel or unilaterally modify the PPP contract, thereby securing long-term stability for both state assets and private investors.

This structural overhaul transforms Info-Sarker-3 from a fiscal liability into a self-sustaining national asset, ensuring that the project contributes equitably to the public exchequer while facilitating affordable broadband expansion.



6.1.3 Human Resource Development

Frontier Technology Training Programs: BCC will launch comprehensive training and certification programs covering AI, blockchain, IoT, cloud computing, and quantum computing fundamentals. These programs will be developed in collaboration with international universities and global certification bodies to ensure recognition and relevance in the global ICT market.

6.1.4 Decentralization & Regional Empowerment

BCC's regional offices will be empowered as innovation hubs and service delivery centers, providing localized digital solutions, training, and monitoring of ICT service quality. Their mandate will expand to include certification, skill-building, and entrepreneurship support to strengthen grassroots-level digital ecosystems. This decentralization is essential for achieving the vision of inclusive digital transformation that reaches all divisions of Bangladesh.

6.1.5 National Data & Documentation Ecosystem

National Document Wallet / Digital Locker: BCC will establish a secure digital storage and authentication system for citizen documents. This Digital Locker will integrate with the National ID system and enable citizens to access verified copies of their certificates, licenses, and official documents from any location.

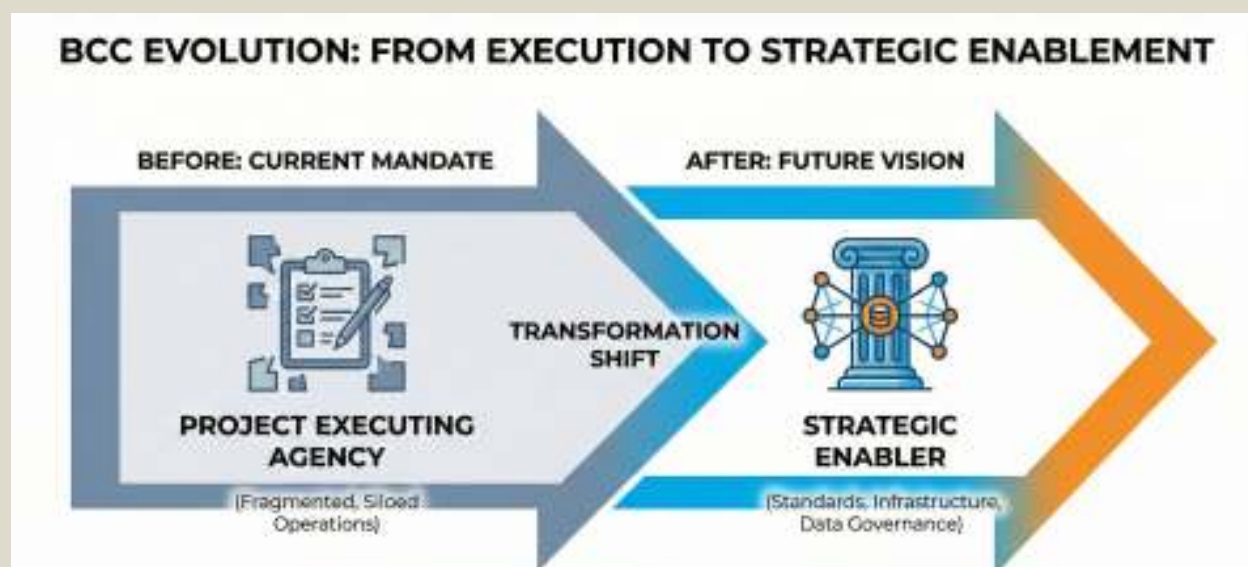
Government Source Code Repository: To ensure government ownership of mission-critical software and reduce vendor lock-in, BCC will create and manage a centralized Government Source Code Repository. This repository will enforce compliance with national digital architecture guidelines, cybersecurity standards, and data protection protocols.

National Data Analytics Platform (NDAP): BCC will build a National Data Analytics Platform to support evidence-based policymaking and sectoral planning. This platform will aggregate anonymized data from across government systems to enable predictive analytics, trend identification, and resource optimization.

6.1.6 Enterprise & Governance Readiness

Government Resource Planning (GRP): BCC will ensure nationwide adoption of Government Resource Planning across all ministries and agencies, enabling unified financial, HR, and asset management systems. This standardization is critical for reducing duplication, improving accountability, and enabling cross-ministry coordination.

Interoperability Standards: BCC will develop and enforce policies and standards promoting interoperability, data sovereignty, and efficiency in all government digital systems. These standards will align with NDGA guidelines and international best practices.



6.1.6 Strategic Infrastructure & Resource Optimization

BCC is undertaking a critical "Hard Reset" of the National Data Center (NDC) to address legacy resource constraints and establish a sovereign cloud backbone.

- **Resource Optimization:** Immediate deployment of 10 new computing nodes/servers into the NDC cloud platform, alongside a rigorous audit to reclaim unused resources from client institutions. This optimization has resolved initial capacity crises.
- **Infrastructure Modernization:** A comprehensive review of hardware/software lifecycles is underway. The "National Data Center Capacity Augmentation" project is being formulated to replace obsolete platforms and meet future demand.
- **Cloud Sovereignty:** Transitioning from reliance on foreign cloud providers to a "Sovereign Cloud" model. BCC is leading the "Secure Cloud" initiative to host all critical government data domestically, ensuring data privacy and national security.
- **Disaster Recovery (DR):** Upgrading the Jashore Disaster Recovery Center with active-active replication capabilities to ensure 99.99% uptime for critical national services like NID and Passport.

6.1.7 Cyber Defense & Security Operations

BCC is operationalizing the **National Security Operations Center (NSOC)** to move from reactive defense to proactive threat intelligence.

- **24/7 Monitoring:** Full-scale monitoring of Critical Information Infrastructure (CII) has commenced.

- **Open Source Security Tools:** Deployment of cost-effective, open-source security visibility tools (SIEM/SOAR) to detect and mitigate threats in real-time without expensive proprietary vendor lock-in.
- **Rohingya Database Handover:** Secure transfer of the Rohingya database management to the Ministry of Home Affairs to streamline national security operations.

6.1.8 Emerging Technology & Innovation Initiatives

BCC is evolving from a support agency to an R&D powerhouse, leveraging frontier technologies for public good.

- **National GPU Cloud:** Repurposing unused GPU resources from Hi-Tech Parks into a centralized **"National GPU Cluster."** This facility will be accessible to researchers and startups for AI model training and high-performance computing.
- **AI-Driven Solutions:**
 - **"Kagoj.AI":** Development of an AI-powered document analysis tool to automate government file processing.
 - **"July Font":** Creation of a standardized, open-source Bangla font to ensure digital inclusivity.
- **Battery-Rickshaw Management System:** Successful piloting of a digital registration system for battery-operated rickshaws in Dhaka North/South City Corporations, registering over **23,900 drivers** to bring order to urban transport.
- **Unified Ticketing System:** Developing a single interoperable ticketing platform integrating Railway, BRTC buses, and Biman Bangladesh Airlines to facilitate seamless multimodal travel.

6.1.9 Institutional Capacity & Governance

- **Recruitment Reform:** Overhauling recruitment rules to allow lateral entry of industry experts (CTOs, Architects) into BCC, breaking the bureaucratic ceiling for technical talent.
- **"July Corner":** Establishing a dedicated memorial corner and training programs for the "July Warriors" to honor the spirit of the student-led revolution.

6.1.10 Infrastructure Expansion

National Data Center (NDC) Expansion: BCC will coordinate the allocation of suitable land in Dhaka Metro for NDC expansion, ensuring scalability for future cloud, AI, and big data needs. Current capacity constraints in BCC's Nutanix and Huawei data centers will be addressed through this expansion.

Green ICT Practices: All data center upgrades will incorporate green ICT practices including renewable energy integration, efficient cooling systems, and carbon-neutral operations.

Bangladesh commits to ensuring at least 30% renewable energy sourcing in public sector ICT infrastructure by 2028.

6.1.11 BCC Reform KPI Framework (2025-2030)

KPI	2025	2026	2027	2028	2030
Government Systems on NSB (%)	30%	50%	70%	85%	100%
Professionals Certified (Cumulative)	10,000	25,000	50,000	80,000	150,000
Regional SHQTCs Operational	2	4	6	8	All Divisions
Agencies on GRP (%)	20%	40%	60%	80%	100%
Renewable Energy in Data Centers	10%	15%	20%	30%	50%

6.2 Department of ICT (DoICT) Reform

6.2.1 Rationale for Reform

The Department of ICT maintains the most extensive grassroots presence of any ICT institution, with officers stationed at every district and upazila. However, this network has been underutilized for systematic digital capacity building. The reform aims to transform DoICT into the primary training and capacity-building arm of the ICT Division, ensuring that digital literacy and skills development reach every corner of Bangladesh.

6.2.2 Strategic Initiatives & Progress Updates

To achieve this transformation, DoICT has executed a series of structural and operational reforms ("Hard Reset"):

1. Specialization via Subject-Matter Pools

- **Action:** Moving away from generalist roles, DoICT has restructured its technical workforce into **04 Specialized Pools** based on expertise:
 1. **Cyber Security Pool**
 2. **Software Development Pool**
 3. **Operation & Maintenance Pool**
 4. **Human Resource Development (HRD) Pool**
- **Impact:** This ensures that the right officer is deployed for the right task—e.g., a cyber-expert handling security audits, while a developer manages software projects. This specialization is now central to all postings, deputations, and training decisions. This initiative fosters a coordinated and supportive work environment at both policy and implementation levels, significantly increasing efficiency and accountability.

2. Field-Level Digital Governance (D-Nothi & Portals)

- **Action:** Upazila-level ICT Officers have been formally designated as "**Office Admins**" for the D-Nothi (Digital Filing) system and National Portal management.
- **Progress:** They are now responsible for overseeing the updating of 25,000+ government web portals and ensuring the smooth operation of digital filing in local administration. This includes regular updates to portals and monitoring of digital file management.
- **Impact:** This decentralization has drastically reduced administrative delays and ensured accountability at the grassroots level. It has increased the speed and efficiency of administrative activities through digital file management.

3. Cyber Security Helpdesks

- **Action:** In collaboration with the **National Cybersecurity Agency (NCSA)**, DoICT has established "**Cyber Security Helpdesks**" at District and Upazila levels based on an MOU signed on November 09, 2025.
- **Function:** These helpdesks provide immediate support to citizens and local government offices for cyber incidents (hacking, harassment, fraud) and implement standard operating procedures (SOPs) for digital safety prepared by NCSA.
- **Impact:** Taking cyber defense from the capital to the community. This ensures faster incident detection, reporting, and management, increasing public awareness about cybersecurity.

4. Freelancer & Entrepreneur Support

- **Action:** Creation of a dynamic database of freelancers and IT entrepreneurs at the district level. DoICT is organizing mentorship programs, workshops, and "Freelancer Summits" to connect local talent with global markets. The **National Freelancers Registration Platform (freelancers.gov.bd)** has been launched (January 13, 2026), and ID cards are being issued (3708 cards issued as of Jan 25, 2026).
- **Synergy:** Integration with the **Freelancer ID** system to ensure legitimate professionals receive banking and social recognition. This includes synergy with **BHTPA** to provide co-working spaces or incubation facilities for LEDP-trained freelancers and entrepreneurs in Software Technology Parks.
- **LEDP Project:** The proposed **Learning, Earning & Entrepreneurship Development (LEDP)** project (2026-2028) aims to train 1,07,320 youth in ICT and freelancing to create entrepreneurship and income opportunities.

5. Administrative Strengthening

- **Action:** Recruitment process for 497 Office Support Staff positions has been completed to strengthen the administrative structure, especially at the field level. Appointment letters have been sent.
- **Impact:** Addressing long-standing manpower shortages in field offices, increasing the speed and efficiency of administrative activities.

6. Surokkha Platform & Office Management

- **Surokkha:** The **surokkha.gov.bd** platform is live, enabling vaccination card downloads for compliance in foreign employment.
- **Smart Office Management System (SOMS):** Implementation of SOMS (soms.gov.bd) across DoICT and other ICT Division agencies. New modules for transfer applications, meeting nominations, and lab/office inspections have been developed. Training has been provided to officials.

7. Specialized Training for Officials

- **Action:** Providing specialized training to officials on Cyber Security (VAPT), Freelancing ToT (Graphics Design & Digital Marketing with AI), and Operation & Maintenance.
- **Progress:** 120 officials trained in Cyber Security (VAPT), 100 in SOMS, and 100 planned for Operation & Maintenance.

6.2.3 Accelerating the EDC Project (2nd Revised DPP)

DoICT is the implementing agency for the **Establishing Digital Connectivity (EDC)** project. Following the cancellation of the Chinese loan component due to high interest rates, the project has been restructured (2nd Revised DPP approved by PIC on Jan 25, 2026) to focus on verifiable outputs:

- **Connectivity:** Providing **1,09,244** broadband connections to end-users (schools, colleges, land offices, etc.) nationwide.
- **Infrastructure:** Construction of **400 Upazila Seva & ICT Training Centers**.
 - *Status:* **100 Centers** were inaugurated on **January 26, 2026** by the Special Assistant to the Chief Advisor. Construction is ongoing in 189 upazilas and is expected to be completed within the next 6 months. Tenders are being processed for the remaining 97 (24 invited, rest in process).
- **Specialized Labs:** Establishment of **20 High-Tech Labs** (17 VLSI & AI, 1 HPC, 1 PKI, 1 Forensic) to build deep-tech capacity.
- **ICT Tower:** Construction of a 12-story Information Technology Building.
- **Network Monitoring System (NMS):** A Central NMS is being procured to monitor all connections in real-time. Already, **49,000** existing connections have been integrated into the monitoring framework to stop billing fraud. A technical committee has been formed to oversee this.
- **School of Future & Digital Labs:**
 - **Python Training:** 27,000 students from School of Future institutions trained in Python programming.
 - **Robotics Corners:** 300 Robotics Corners established in School of Future institutions with drones, VR headsets, 3D printers, etc.
 - **LMS Training:** 8,300 teachers from Digital Lab institutions trained on Learning Management Systems.
 - **Frontier Technology ToT:** Ongoing ToT for 1200 teachers on Frontier Technologies.

6.2.4 Expanded Training Mandate

Selected ICT-related courses will be standardized and approved by the National Skills Development Authority (NSDA), ensuring national recognition and quality assurance. DoICT officers at district and upazila levels will deliver these programs in schools, colleges, and madrasas, reaching populations currently excluded from formal ICT training pathways.

Advanced certifications from global providers including Microsoft, AWS, and CISCO will be integrated gradually. DoICT will facilitate partnerships with these global certifying bodies to ensure that Bangladeshi citizens can obtain internationally recognized credentials without traveling abroad or paying prohibitive fees.

6.2.5 Institutional Strengthening

Grassroots Capacity Building: DoICT will be formally designated as the primary grassroots training and capacity-building institution of the ICT Division. This mandate clarifies roles and prevents duplication with other agencies.

Staff Development: Continuous capacity-building will be provided for existing DoICT staff to ensure they can deliver modern curricula effectively. Training of Trainers (ToT) programs will be mandatory for all field officers.

Training Resource Hubs: Training Resource Hubs will be established at divisional and district levels to support curriculum delivery, provide digital content, and enable teacher upskilling.

In-House Technical Cadre: DoICT will actively cultivate a specialized pool of technical experts within its own ranks to reduce dependency on external vendors. Resource people and officers with technical aptitude will be trained and deployed as software developers, system administrators, and cybersecurity experts. This internal workforce will handle routine software maintenance, minor feature developments, and first-line cyber defense, ensuring faster response times and significantly lowering long-term operational costs previously spent on vendor support contracts.

6.2.6 Curriculum Innovation & Local Integration

DoICT will introduce a tiered learning model to meet learners at their current level and provide clear progression pathways:

Foundation Level: Digital literacy for students and community members covering basic computer operations, internet safety, and online services.

Intermediate Level: E-commerce, freelancing, coding fundamentals, and data entry skills to enable economic participation in the digital economy.

Advanced Level: Cybersecurity, AI basics, cloud computing, and IoT fundamentals for future workforce readiness and specialized career paths.

DoICT DIGITAL LITERACY PYRAMID: TIERED LEARNING MODEL



All curricula will be available in Bangla with accessibility features for rural and marginalized communities. Content will be optimized for low-bandwidth environments common in remote areas.

6.2.7 Future Roadmap & Strategic Priorities

To sustain this momentum, DoICT is focusing on:

- **NMS Implementation:** Completing the procurement of the Central NMS to ensure 100% uptime visibility for all government internet connections and ensuring service quality.
- **School of Future (SoF):** Shifting focus from mere equipment supply to **Sustainable Maintenance** and continuous **Training of Trainers (ToT)** for teachers in Robotics, IoT, and Frontier Technologies to make the program more dynamic.
- **Cloud Sovereignty:** Migrating all local government applications to the **National Data Center (NDC)** cloud infrastructure to ensure data sovereignty and security.
- **Alternative Financing:** Securing funds to complete the remaining EDC components (Upazila Centers, Specialized Labs) through alternative financing following the exit of foreign loan partners (China).
- **Skill Development Synergy:** Coordinating training modules of the EDC project with the Directorate's own training programs.

6.2.8 Monitoring & Accountability

National Digital Learning Dashboard: DoICT will implement a real-time dashboard tracking enrollment, completion, and job-placement outcomes across all training programs. This transparency enables evidence-based program improvement and accountability.

Performance-Based Incentives: DoICT trainers and officers will receive performance-based incentives tied to verified training outcomes, including learner certification rates and employment outcomes.

6.2.9 Digital Service Integration

Beyond training, DoICT offices will be equipped to support local digital service delivery including assistance with e-services, online applications, and grievance systems. Every DoICT office will function as a community digital help desk, ensuring that citizens who face barriers to online services can receive in-person support.

6.2.10 Public-Private Partnerships

DoICT will establish partnerships with tech companies, industry associations, and freelancing platforms to align training with actual market demand. Internship and apprenticeship programs developed through industry collaboration will provide practical experience and employment pathways for program graduates.

6.2.11 Sustainability & Future-Readiness

DoICT reforms will be integrated into the national digital vision as a permanent institution, not a time-bound project. Blended learning models combining offline instruction with online platforms and mobile-based microlearning will ensure scalability and resilience against disruptions.

6.2.12 DoICT Reform KPI Framework (2025-2030)

KPI	2025	2026	2027	2028	2030
Citizens Trained (Annual)	100,000	150,000	200,000	200,000	180,000
Training Hubs Established	8	32	64	All Districts	All Districts
Certification Pass Rate	60%	65%	70%	75%	80%
Women Participants (%)	30%	35%	38%	40%	40%
Employment Rate (6 months)	40%	50%	55%	60%	70%

6.3 National Cybersecurity Agency (NCSA) Reform

6.3.1 Rationale for Reform

As Bangladesh's digital footprint expands through increased e-governance, digital payments, and online services, the attack surface for cyber threats grows proportionally. The National Cybersecurity Agency must evolve from a reactive incident-response entity to a proactive defender capable of anticipating threats, protecting critical infrastructure, and building national cyber resilience. This reform is essential for the effective implementation of the Cyber Safety Ordinance, 2025.

6.3.2 Strategic Initiatives & Progress Updates

1. Legislative and Administrative Reformation

To restore public trust and ensure a transparent legal framework, the NCSA has executed significant legislative overhauls:

- **Repeal and Replacement:** The controversial "Cyber Security Act, 2023" was repealed. Following extensive stakeholder consultations, the new **"Cyber Security Ordinance, 2025"** was gazetted on May 21, 2025.
- **Regulatory Framework:**
 - **Rules 2026:** The "Cyber Security Rules, 2026" have been drafted following workshops (Jan 19–20, 2026) and are being processed for approval.
 - **Strategy 2026-30:** The "Cyber Security Strategy 2026-30" draft is ready for stakeholder consultation (Target: March 2026).
- **Administrative Status:** On September 21, 2025, the NCSA was officially gazetted as an **"Attached Department"** of the ICT Division to streamline administrative decision-making.
- **Recruitment:** Recruitment rules have been sent to the Public Service Commission (PSC), and 70 temporary revenue-sector posts have been approved to build a dedicated workforce (Target: June 2026).

2. Infrastructure and Technical Capacity Building

The agency is establishing state-of-the-art infrastructure to monitor and neutralize threats in real-time:

- **National Security Operation Center (NSOC):** Tender evaluation is ongoing to establish the NSOC, which will serve as the central nerve center for nationwide threat monitoring (Target: June 2026).
- **NCERT Enhancement:** A project is underway to significantly upgrade the **National Computer Emergency Response Team (NCERT)**, including the establishment of a

National Digital Forensic Lab and the procurement of advanced OSINT and Fact-Checking tools.

- **Digital Forensic Standardization:** To ensure international standards (ISO/IEC 17025), a Standard Operating Procedure (SOP) for accrediting forensic labs has been drafted. A technical team comprising experts from BUET, Dhaka University, and MIST is being formed to assess labs operated by CID, PBI, and ACC.

3. Operational Monitoring and Incident Response

The NCSA has operationalized rigorous monitoring mechanisms to protect national assets:

- **CIRRS Deployment:** The **Cyber Incident Reporting & Response System (CIRRS)** has been developed to centralize complaint handling and reporting. It is currently under testing at SQTC (Target: Jan 2026).
- **Threat Intelligence:** In FY 2025, the agency (via BGD e-GOV CIRT) issued:
 - **283** Cyber Threat Intelligence Reports.
 - **147** National Data Center Netflow Alerts.
 - **23** Cyber Sensor Reports regarding risks to Bangladesh Bank and BCC.
 - **16** Security Advisories and 3 Threat Landscape reports.
- **Audits:** IT Audits for three (3) critical systems have been completed, with three more ongoing.

4. Critical Information Infrastructure (CII) Protection

Recognizing the vulnerability of strategic assets, the NCSA has intensified its focus on CII:

- **Assessment Framework:** As per the decision of the **National Cyber Security Council's first meeting** (held on Dec 28, 2025, chaired by the Chief Adviser), a framework for assessing the cyber maturity of government and private institutions has been finalized. Automated assessment software is scheduled for release by January 31, 2026.
- **CII List Update:** The list of CII is being updated under Section 16(5) of the Ordinance, with the Directorate of Posts being processed for inclusion.
- **Maturity Assessment:** A "Cyber Maturity Model Assessment" has been conducted for all existing CII, and five dedicated teams have been formed to handle incident responses within these institutions.

5. Content Control and Financial Sector Security

A zero-tolerance policy has been adopted against digital financial crime and online gambling:

- **Online Gambling:** The agency has referred **1,266 web portals, 194 social media links, and 14,435 Bank/MFS accounts** to BTRC/law enforcement for blocking. Notably,

a legal notice to **ESPNcrinfo** successfully halted the broadcasting of betting advertisements targeting Bangladesh.

- **Disinformation Control:** 147 links related to rumors, fake news, and ethnic violence were processed for removal.
- **Financial Sector:** A seminar on "Cybersecurity in Financial Sector of Bangladesh" was held in August 2025 to align banking stakeholders with the new Ordinance.

6. Awareness and Decentralization

To build a "Cyber Hygiene" culture, operations are being expanded beyond the capital:

- **Student Awareness:** Sessions were held at 16 universities (3,000+ students) and 40 Upazilas (800+ students) to train the youth on cyber safety.
- **Field Operations:** An MoU has been signed with the Department of ICT (DoICT) to establish **Cyber Security Helpdesks** in District and Upazila ICT offices, ensuring service delivery at the grassroots level.
- **Professional Training:**
 - **Law Enforcement:** ~240 officials trained on Advanced Cybersecurity, Fact-Checking, and Sentiment Analysis over two fiscal years.
 - **Govt Officials:** ~140 officials trained on cyber safety protocols.

6.3.3 National Security Operations Center (NSOC)

World Bank Funding: A \$5 million proposal has been submitted to the World Bank through the EDGE project for financial assistance to establish the National Security Operations Center (NSOC). This funding will enable procurement of advanced monitoring equipment, threat intelligence platforms, and forensic laboratory capabilities.

Full Deployment with CIIs: After initial setup, NSOC will be fully deployed with all Critical Information Infrastructures (CIIs). The center will provide 24/7 monitoring, incident response coordination, and threat intelligence sharing.

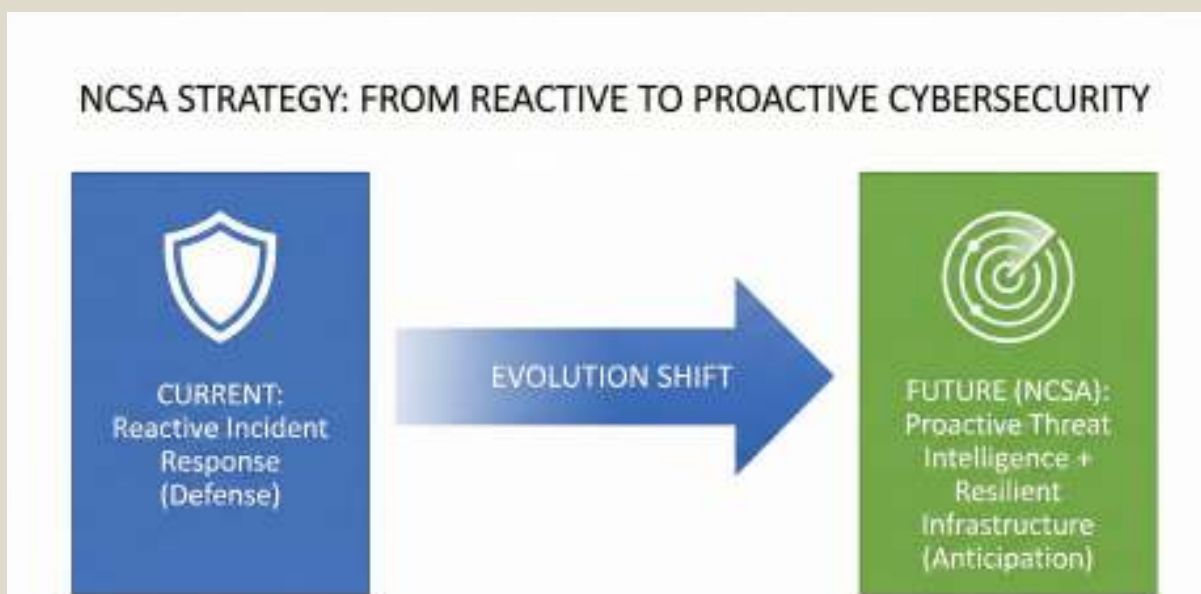
Enhanced Capabilities: NSOC enhancement will include digital forensics laboratory, threat intelligence platform, and centralized incident reporting system. This integrated approach significantly enhances national cyber readiness and enables rapid response to emerging threats.

6.3.4 Strategic Framework Development

Bangladesh Cybersecurity Strategy 2026-2030: NCSA will prepare and finalize a comprehensive national cybersecurity strategy serving as the framework for building a resilient and secure digital ecosystem. This strategy will guide effective implementation of mandates under the Cyber Safety Ordinance, 2025.

Cyber Safety Policy: As a supplementary document to the Ordinance, NCSA will formulate a Cyber Safety Policy providing clearer, actionable guidance for public and private sectors to maintain secure digital operations.

Updated Guidelines and SOPs: NCSA will update and formulate relevant cybersecurity guidelines including revision of the Critical Information Infrastructure Protection Guideline (2022), Digital Forensics Lab Guideline (2023), and development of new guidelines based on current needs and international standards.



6.3.5 Capacity Building & Awareness

Specialized Training for Law Enforcement: NCSA will design and deliver technical training programs tailored to cybercrime investigation, digital forensics, and legal handling of cyber incidents. Judiciary and law enforcement agencies will receive specialized instruction ensuring effective enforcement of the Cyber Safety Ordinance.

Public Awareness Campaigns: NCSA will launch targeted campaigns for government, business, and citizen sectors focusing on digital hygiene, social engineering awareness, and safe online behavior. Content will be delivered through multimedia in both Bengali and English for maximum reach.

Cybersecurity Skills Development: NCSA will support national-level cybersecurity certification tracks and encourage academic-industry-government collaboration. Target: 20,000 certified cybersecurity professionals by 2027, scaling to 50,000 by 2030.

6.3.6 Proactive Threat Monitoring & Victim Support

Proactive Cyber Threat Intelligence: NCSA will implement active threat hunting capabilities to identify evolving cybercrimes (e.g., deepfake fraud, ransomware trends) before they scale. This includes monitoring dark web forums for data leaks and analyzing malware trends targeting Bangladesh.

Protection for Vulnerable Communities: Dedicated units will focus on the cyber safety of women, children, and marginalized groups, addressing issues like non-consensual image sharing ("revenge porn"), sextortion, and cyberbullying with high priority and confidentiality.

Victim Assistance & Legal Navigation: NCSA will establish a dedicated Cyber Victim Support Cell. This unit will not only provide technical remediation advice but also guide victims through the legal process, helping them understand their rights under the Cyber Safety Ordinance and assisting them in filing actionable reports with law enforcement. This ensures that the gap between a "technical incident" and "legal justice" is bridged for the average citizen.

Strategic Challenges and Directives (2026-2030)

While significant progress has been made, the following emerging challenges are identified for strategic review within the digital transformation strategy:

1. **Ransomware & Phishing:** Persistent threats to government and financial archives.
2. **Institutional Inequality:** The gap in cyber resilience between different government agencies creates weak links.
3. **AI & Deepfakes:** The rapid evolution of AI-driven threats requires dynamic countermeasures.

Future Directives: To address these, the NCSA will implement:

- **Automated Cyber Audits:** Full deployment of the assessment software to ensure regular, automated audits of all CIIs.
- **Cyber Resilience Hub:** Establishing a robust "Incident Response Network" for rapid post-attack recovery.
- **International Cooperation:** Strengthening data sharing and legal cooperation with global cybersecurity bodies to tackle cross-border cybercrime.
- **ISO Accreditation:** Ensuring all forensic labs achieve international accreditation for legal validity.

6.3.7 Sustainable Operations Model

While NSOC Phase 1 is supported under World Bank funding, NCSA will establish a sustainable operation model for post-deployment operations. This model incorporates skilled personnel from both government and private sectors for continuous operations, ensuring the center remains effective beyond the initial project period.

6.3.8 NCSA Reform KPI Framework (2025-2030)

KPI	2025	2026	2027	2028	2030
CII's Connected to NSOC	25%	50%	75%	90%	100%
Certified Cyber Professionals	5,000	10,000	20,000	35,000	50,000
Mean Incident Response Time	72 hrs	48 hrs	24 hrs	12 hrs	4 hrs
Law Enforcement Trained	500	1,500	3,000	5,000	8,000

6.4 Bangladesh Hi-Tech Park Authority (BHTPA) Reform

6.4.1 Rationale for Reform

The Bangladesh Hi-Tech Park Authority (BHTPA) was established in 2010 with the visionary mandate to transition the nation toward a knowledge-based economy through specialized infrastructure, investment-friendly policies, and human capital development. However, a comprehensive diagnostic review reveals that over the past 16 years, the authority has largely failed to meet its core objectives due to systemic mismanagement, shortsighted policy-making, unplanned project execution, and deep-seated corruption by syndicates of unskilled officials.

Instead of functioning as a dynamic ecosystem enabler, BHTPA drifted into becoming a mere civil construction agency.

- Infrastructure Decay & Developer Negligence:** The flagship Kaliakoir Hi-Tech Park (355 acres) exemplifies this failure. Despite spending BDT 14 crore (2010-13) and BDT 400 crore (2013-2021) on basic utilities (roads, sewerage, power, gas, fiber), the private developers—**Summit Technopolis** and **Bangladesh Technocity (Fiber@Home)**—failed to build the required superstructure on their 130-acre lease. Critical security infrastructure like CCTV and surveillance systems remains absent. The park's potential has been stifled by incompetent and dishonest project directors who prioritized spending over functionality.
- "Ghost Parks" & Political Planning:** Hi-tech parks and Software Technology Parks (STPs) were frequently established in remote, non-viable locations based on political patronage rather than economic logic. These sites lack fiber connectivity, university proximity, or a talent pool, resulting in "Ghost Parks"—empty buildings in rural areas with zero industry uptake.
- Land Speculation:** Prime industrial plots were allocated to politically connected conglomerates with no track record in tech, leading to land hoarding. Genuine investors faced artificial scarcity and bureaucratic harassment, while valuable land lay fallow.
- Politically Motivated Planning:** High-tech parks and Software Technology Parks (STPs) have frequently been established in remote, non-viable locations based on political considerations rather than economic feasibility, fiber connectivity, or talent availability. This has resulted in "Ghost Parks"—empty buildings in rural areas with zero industry uptake.

- **Infrastructure Corruption & Waste:** Massive public resources have been wasted on substandard construction and luxury amenities in administrative buildings, while critical industrial utilities (stable power, high-speed redundant internet, waste treatment) remain inadequate.
- **Land Speculation:** Industrial plots in prime parks (e.g., Kaliakoir) were allocated to politically connected conglomerates with no track record in tech, leading to land hoarding. Genuine investors often face artificial scarcity and bureaucratic harassment.

6.4.2 Reform Progress Update & Strategic Directives

As of February 2026, the Bangladesh Hi-Tech Park Authority (BHTPA) has transitioned into an active implementation phase of its reform agenda. Moving beyond planning, the authority is now executing specific operational directives aimed at infrastructure optimization, legal modernization, and administrative transparency.

The following outlines the core pillars of this reform progress:

1. Optimization of Infrastructure and Decentralization

To address underutilization issues, BHTPA has initiated a rigorous "De-scoping and Re-scoping" process for all ongoing projects. The primary objective is to ensure 100% utilization of existing Hi-Tech Parks, Software Technology Parks (STPs), and IT Training & Incubation Centers.

- **Action Taken:** Active engagement with investors, associations, and higher education institutions is underway to resolve bottlenecks. Two projects have already been revised, and three are currently under review by the Planning Commission.
- **New Expansion:** A major initiative has been launched to establish **five (5) new STPs** in the satellite townships surrounding Dhaka—specifically **Purbachal, Uttara, Mirpur, Mohammadpur, and Badda**.
- **Strategic Impact:** This move aims to decentralize the IT industry, reducing the pressure on Dhaka's city center while creating "Decent Jobs" in accessible residential zones.

2. Legislative Modernization (BHTPA Ordinance)

A critical overhaul of the legal framework is in the final stages of approval to streamline authority operations and enhance the investment climate.

- **Current Status:** The draft Ordinance has been presented twice to the Advisory Council. Following the second meeting, it is being refined in consultation with the Honorable Finance Advisor regarding VAT and Tax implications.
- **Next Steps:** The Secretary of the ICT Division will facilitate the final presentation for approval by December 2025.
- **Expected Outcome:** Once enacted, this will clarify BHTPA's scope of work and permanently establish an investor-friendly regulatory environment.

3. Administrative Restructuring of Jashore Software Technology Park

In a decisive move to improve governance, BHTPA has canceled the contract with the third-party Property Management Company (PMC) for the Sheikh Hasina Software Technology Park, Jashore.

- **Current Status:** The park is now operated under BHTPA's direct management (effective May 13, 2025).
- **Results:** This transition has allowed for the rationalization of rent rates and has significantly improved the relationship between the authorities and investors, creating a positive work environment in the South-Western IT hub.

4. Land Recovery and Asset Management at Kaliakoir

To ensure the highest utility of national assets, BHTPA has taken a zero-tolerance approach toward non-performance at the flagship Kaliakoir Hi-Tech Park.

- **Action Taken:** Approximately **50 acres of land** have been recovered from two developer companies that failed to utilize the allocation effectively. A firm has been appointed to assess the liabilities of these outgoing developers.
- **Future Plan:** This recovered land is now being re-allocated to capable investors, specifically targeting high-tech manufacturing industries (including semiconductors and electronics), thereby pivoting Bangladesh toward a hardware manufacturing hub.

5. Establishment of the Institute of Frontier Technology (IFT)

To prepare for the Fourth Industrial Revolution (4IR), BHTPA has solidified a partnership with the Bangladesh University of Engineering and Technology (BUET).

- **Collaboration:** An MoU has been signed to establish the **Institute of Frontier Technology (IFT)** directly on the BUET campus.
- **Infrastructure Plan:** The project includes academic and administrative infrastructure at BUET for undergraduate and postgraduate research. Additionally, a specialized center for "Lifelong Learning" is being developed in **Shibchar, Madaripur**, to offer certification courses and PG diplomas.
- **Timeline:** The Revised Development Project Proposal (RDPP) has been submitted, with a Project Evaluation Committee (PEC) meeting scheduled for February 1, 2026.

6. Industry-Academia Collaboration Model

A new operational model has been successfully piloted to bridge the gap between higher education and industry requirements.

- **Pilot Project:** An MoU was signed with **Begum Rokeya University, Rangpur (BRUR)** to manage the Rangpur IT Training & Incubation Center.
- **Status:** The center is currently operational under university management (effective November 24, 2025), serving as a blueprint for future university-led park management.

7. Standardization of Operating Procedures (SOPs)

To eliminate ambiguity in operations and maintenance, five specific Standard Operating Procedures (SOPs) and Guidelines have been drafted and are currently being finalized:

- Park Location Selection Guideline 2026.
- Civil Infrastructure Repair & Maintenance SOP 2026.
- E/M & Networking Repair & Maintenance SOP 2026.
- Kaliakoir Hi-Tech Park Operation SOP (Administrative & Investor Duties) 2026.
- University Innovation Hub Guideline 2026.

6.4.3 Infrastructure Rationalization: The "Viability Audit"

To stop the bleeding of public funds and restore order, the Interim Government has initiated a strict rationalization of all infrastructure assets.

- **Mandatory Feasibility & "De-scoping" Audit:** A third-party technical audit is assessing all ongoing projects. Projects found economically unviable are being **"De-scoped" or "Re-scoped."** For example, the costly "District Level IT/Hi-Tech Park (12 Districts)" project is being radically revised to prevent further waste.
- **Repurposing Non-Viable Assets:** Remote, unoccupied STPs will not be abandoned. They are being repurposed into **"Sheikh Kamal IT Training & Incubation Centers"** or **DolCT Technical Training Hubs**, transferring management to BCC/DolCT for grassroots skills development.
- **Focus on Economic Corridors:** Future investment is being consolidated into high-potential economic corridors (Dhaka, Gazipur, Chattogram, Sylhet, Jashore) and 5 new suburban STPs around Dhaka (Purbachal, Uttara, Mirpur, Mohammadpur, Badda) to create "Decent Jobs" near talent hubs.

6.4.4 Operational Reform: From Bureaucracy to Business

- **Policy Modernization & SOPs:** To eliminate arbitrary decision-making, BHTPA is finalizing a new suite of governance instruments for 2026:
 - **Park Location Selection Guideline 2026:** Ensuring future parks are built only where economically viable.
 - **Infrastructure Maintenance SOP 2026:** Standardizing civil and electrical maintenance to prevent degradation.
 - **Kaliakoir Operation SOP 2026:** Clearly defining the administrative and maintenance responsibilities of investors versus the authority.
 - **University Innovation Hub Guideline 2026:** Formalizing the link between academia and industry.
- **Land Lease "Use-it-or-Lose-it" Policy:** A strict audit of allocated plots is underway. Allottees who failed to commence construction within the stipulated contract period face lease cancellation. A new **"Performance-Based Allocation Policy"** ensures land goes to active tech companies, not real estate speculators.
- **One-Stop Service (OSS) Digitization:** The current manual OSS is being fully digitized to integrate with BIDA, NBR, and RJSC. The goal is a **"Zero-Visit"** investor onboarding experience, transparently delivering licenses, utility connections, and bonded warehouse approvals.

6.4.5 Ecosystem Focus: Soft Infrastructure Over Hard Concrete

Moving beyond concrete, BHTPA is pivoting to "Soft Infrastructure" to create a thriving, innovation-led economy.

- **Institute of Frontier Technology (IFT):** In a landmark move, the "Bangladesh Institute of Frontier Technology" project has been fundamentally restructured. Originally conceived as a construction-heavy project, it is now established as a premier academic and research partnership through a Memorandum of Understanding (MoU) with the **Bangladesh University of Engineering and Technology (BUET)**, signed in September 2025. This collaboration establishes the IFT directly on the BUET campus, shifting focus from erecting empty buildings to fostering cutting-edge research in **Artificial Intelligence (AI), Robotics, and Semiconductor design**. This strategic pivot ensures immediate access to top-tier academic talent and research capabilities, accelerating Bangladesh's entry into deep-tech industries.
- **University-Industry Collaboration:** BHTPA is actively forging strong linkages between academia and industry to ensure a steady pipeline of skilled talent. A prime example is the operationalization of the **Rangpur IT Training & Incubation Center** through an MoU with **Begum Rokeya University (BRUR)** (signed November 24, 2025). This **"Triple Helix"** model (Government-Academia-Industry) mandates that universities manage incubation centers to produce graduates with market-ready skills directly aligned with industry needs, rather than generic training. This approach is being scaled to other universities to maximize the utilization of established infrastructure.
- **Startup & Investor Connect:** To foster a vibrant investment climate, BHTPA has institutionalized regular **investor matchmaking sessions**. These sessions connect park tenants and incubated startups with domestic and foreign capital, shifting the authority's role from passive "tenant management" to active "business success facilitation." The goal is to create a seamless pathway for startups to access funding, mentorship, and global markets.
- **Innovation Hubs & R&D:** The authority is finalizing the **"University Innovation Hub Guideline 2026"** to standardize the operation of innovation hubs across universities. These hubs will serve as centers for research, prototyping, and commercialization, bridging the gap between academic research and industrial application.

6.4.6 Strategic Impact Summary

These reforms are expected to yield three major national outcomes:

1. **Macro-Economic Contribution:** Transparency in infrastructure allocation and the new legal framework will directly boost Foreign Direct Investment (FDI) and strengthen foreign currency reserves.

2. **Brain Drain Prevention:** By creating world-class research facilities (like the IFT) and global-standard work environments locally, Bangladesh will retain top-tier engineering talent.
3. **Hardware Ecosystem:** The strategic recovery of land at Kaliakoir allows Bangladesh to expand beyond software into hardware and semiconductor manufacturing, diversifying the export basket.

6.4.7 BHTPA Reform KPI Framework (2025-2030)

KPI	2025 (Baseline)	2026	2027	2028	2030
Non-Viable Projects Cancelled/Repurposed	0	5	8	-	-
Land Plots Repossessed (Non-Compliant)	0%	30%	60%	100%	-
Park Occupancy Rate (Viable Parks)	~40%	55%	70%	85%	95%
Private Park Operators Onboarded	0	2	4	6	All Major Parks
Actual Employment in Parks	25,000	35,000	50,000	75,000	150,000
Export Revenue from Parks (\$M)	\$100M	\$250M	\$500M	\$1B	\$2.5B

6.5 a2i Transformation: Evolving into a GovTech Agency

6.5.1 Foundation and Achievements

Access to Information (a2i), established in 2006 and later embedded within the ICT Division, has emerged as a pioneering entity in Bangladesh's digital landscape. Through systematic initiatives, a2i has tackled entrenched issues including isolated data systems, incomplete datasets, restricted access for vulnerable populations, and reliance on outdated manual processes.

6.5.2 Key Accomplishments

National Portal: A unified online gateway consolidating information and services from over 35,000 government websites, enabling citizens to access more than 1,000 e-services. Processing times reduced from days to minutes through backend integration and workflow automation.

Union Digital Centers (UDCs): Over 8,500 centers nationwide transforming rural post offices and community spaces into hubs for digital services, skill development, and local entrepreneurship. Serving 30 million users annually and generating thousands of local jobs.

Shared Health Record (SHR): Aggregating patient data to support telemedicine and electronic health records. Enabled 340+ Upazila Health Complexes to deliver over 1 million remote consultations annually.

Teachers' Portal: Digital repository serving 15,000+ schools with educational resources, lesson plans, and collaborative tools promoting blended learning models.

e-Filing System: Digitized administrative processes eliminating 70% of paperwork through automated approvals and document tracking.

MyGov App: 5+ million downloads providing citizen engagement, real-time feedback, service requests, and government notifications.

Impact Metrics: Over 2 billion digital transactions processed, saving citizens approximately 1.5 billion days in administrative tasks, and elevating Bangladesh's UN e-Government ranking from 150th in 2012 to 111th in 2022.

6.5.3 Strategic Initiative Updates & Roadmap

To accelerate the transition to a GovTech agency, a2i is actively executing the following high-priority initiatives:

1. Nagorik Sheba Bangladesh Implementation

- **Current Status:** A total of **1,122 Nagorik Sheba Centers** have been launched. The platform has integrated over **41 services**, and **453 services** are available through MyGov. More than **2,000 entrepreneurs** nationwide have been trained.
- **Target:** Convert all Digital Centers into Nagorik Sheba Centers by **March 2026**.

- **Impact:** Citizens can access end-to-end online services from a single, convenient location, ensuring "zero-visit" delivery for verified services.

2. Digital Toll (D-Toll) Management System

- **Current Status:** Launched on Padma Bridge on September 15, 2025. Over **4,000 vehicles** have used the system seamlessly. Seven financial institutions are integrated.
- **Roadmap:** Implementation on Jamuna Bridge by **February 1, 2026**; Meghna-Gomti Bridge by **February 5, 2026**; and subsequently across all bridges, tunnels (Karnaphuli), and elevated expressways under the Bridge Authority and Roads Division.
- **Impact:** Non-stop vehicle passage in D-Toll lanes, significantly reducing travel time and congestion.

3. E-Apostille System

- **Current Status:** The system is ready for inauguration.
- **Impact:** Digitizes the verification of documents (certificates, affidavits) for students and professionals going abroad, eliminating the need for physical visits to ministries and foreign missions.

4. ekPay Transformation (PLC Formation)

- **Current Status:** The draft Memorandum and Articles of Association (MoA & AoA) for establishing ekPay as a Public Limited Company (PLC) are under review by the ICT Division.
- **Roadmap:** Formation of the company by **December 2026**.
- **Impact:** Establishes a sustainable, autonomous financial ecosystem for government payments, reducing dependency on external funding.

5. Smart D-Nothi & AI Integration

- **Current Status:** D-Nothi is currently active in **12,600+ offices** with over **140,000 users**.
- **Roadmap:**
 - Full implementation in all government offices by **December 2026**.
 - Integration of **AI-assisted file processing** to speed up decision-making.
 - Introduction of a "Tracker" system for real-time file monitoring.
- **Impact:** Complete elimination of paper-based filing (Paperless Office) and drastic reduction in administrative delays.

6. Innovation Ecosystem & Policy

- **National Innovation Policy:** Finalize by **December 2026** to provide a legal and financial framework for innovation.

- **Innovation Repository:** A unified platform to protect intellectual property (IP) and support commercialization for public and private innovations.
- **Challenge Fund:** Launching an "Innovation Challenge Fund" to support 30 new citizen-centric innovations by **October 2026**.

7. Citizen Service Workbook (Standardization)

- **Initiative:** Development of a "**Citizen Service Workbook**" to standardize service delivery parameters across all ministries.
- **Objective:** Mandate adherence to specific service standards to overcome institutional inertia and resistance to manual-to-digital migration.

6.5.4 Operational Challenges & Mitigation Strategies

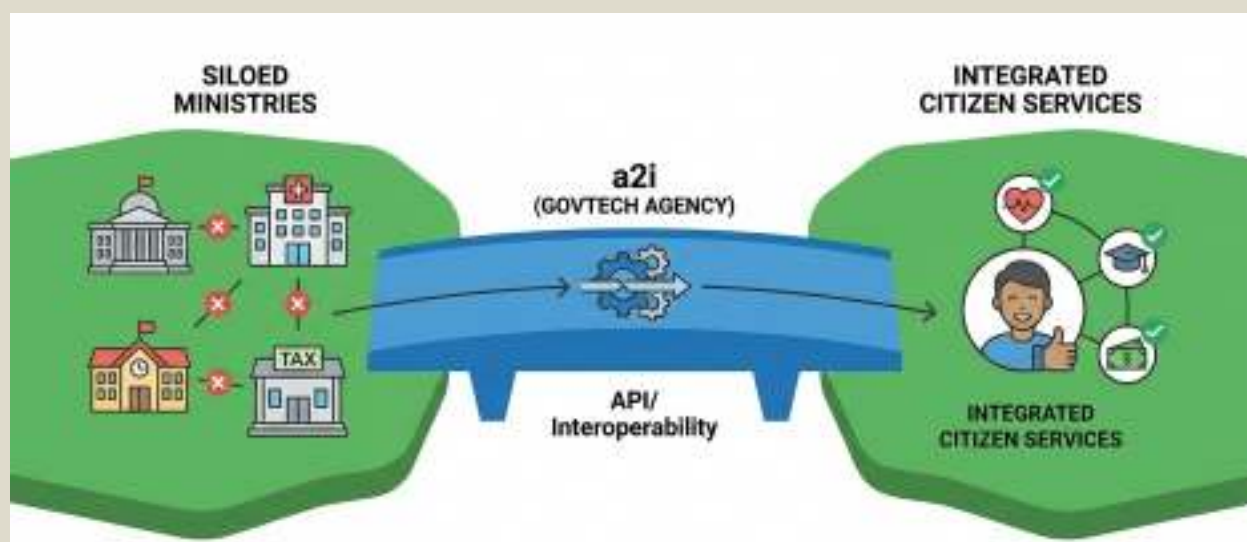
To ensure the success of these initiatives, a2i will address critical challenges:

- **Cybersecurity:** Implement mandatory **Quarterly Security Audits** and establish a dedicated Cyber Incident Response Team (CIRT) for a2i platforms.
- **Financial Compliance:** Form a specialized **Fintech Compliance Team** to navigate Bangladesh Bank regulations for the ekPay PLC formation.
- **Infrastructure:** Leverage **BDCCL's high-performance cloud infrastructure** to handle massive traffic and data loads, ensuring uninterrupted service delivery.

6.5.5 Transformation Roadmap to GovTech Agency

Transforming a2i into a full-fledged GovTech agency requires institutionalizing its role as a hub for digital innovation and cross-governmental coordination. This evolution mirrors global exemplars including Singapore's GovTech Agency, Estonia's X-Road system, and the UK's Government Digital Service (GDS).

Structural Reforms: a2i should gain semi-autonomous status within the ICT Division with independent budget and authority to enforce digital standards across ministries. Dedicated divisions for emerging technologies, data governance, and ethical AI oversight will be established, supported by multi-stakeholder committees.



Scalable DPI Framework: Development of integrated Digital Public Infrastructure incorporating modular components for digital identity, payments, and data sharing. Inclusive design principles will ensure accessibility through multilingual interfaces and low-bandwidth optimizations.

Policy Alignment: Comprehensive GovTech policy framework incorporating robust data protection measures, ethical AI guidelines, and PPP incentives to facilitate collaborations with tech firms.

Monitoring & Evaluation: Advanced monitoring systems with KPIs including service availability rates exceeding 99%, user satisfaction scores above 80%, and quantifiable cost reductions. Real-time analytics dashboards will enable proactive adjustments.

6.5.6 Engineering Talent Strategy

To build sophisticated digital systems, a2i must adopt specialized hiring practices aiming to expand the national ICT workforce to 7-8 million by 2030:

Software Engineers (40%): Experts in cloud computing, microservices, and scalable architectures.

Data Engineers and AI Specialists (30%): Professionals skilled in big data processing and machine learning with ethical AI training.

Cybersecurity Engineers (20%): Specialists in advanced threat detection and zero-trust frameworks.

DevOps Engineers (10%): Individuals proficient in automation tools and containerization.

Hiring will emphasize diversity with 40% representation of women and underrepresented groups, skill-based recruitment through coding challenges and hackathons, continuous upskilling through mandatory certifications, and competitive compensation packages.

6.5.7 Public Procurement Reform for Software

Bangladesh's existing Public Procurement Rules (PPR) often prioritize cost over quality, resulting in vendor dependencies and integration issues. Reforms will include:

PPR Reforms: Transition to balanced evaluation where technical merits (innovation, scalability) account for at least 40% of criteria. Mandatory clauses for open APIs and data interoperability.

Agile Procurement: Segmenting large projects into phased deliverables, favoring open-source solutions and cloud-based deployments with mandatory third-party security audits.

Enhanced e-GP: AI-assisted bid analysis for fairness, provisions for joint ventures, and streamlined approvals. Target: pilot reforms in 2025, full rollout by 2026.

6.5.8 Emerging Technology Adoption

Initiative/Domain	Technologies	Application Examples
Service Delivery (National Portal, MyGov)	AI/ML, Cloud Native, APIs	AI-powered chatbots for 24/7 citizen queries; cloud-native architecture for auto-scaling during peak demand
Education (Shikkhok Batayon, MuktoPaath)	AI/ML, VR/AR, Adaptive Learning	Personalized learning paths; virtual labs for science experiments in remote schools
Healthcare (Telemedicine, Surokkha)	IoT, Blockchain, AI/ML	Remote patient monitoring via IoT; blockchain for secure, patient-controlled health records
Social Safety Nets	AI/ML, Data Analytics	Predictive targeting to identify vulnerable communities before crises
Citizen Engagement (Grievance Redressal)	Natural Language Processing	Automated triage and routing of complaints to relevant agencies
Core Infrastructure (BNDAs)	API-First, Blockchain, Zero-Trust	National consent layer for citizen-controlled data sharing permissions

6.5.9 Interoperability Leadership

a2i is ideally suited to lead Bangladesh's interoperability agenda through:

General Strategy: Champion unified standards for data formats and APIs, establishing governance bodies for data exchanges. Enable 'once-only' data submission reducing redundancy.

Financial Sector: Drive platforms like Binimoy for cross-provider transactions, incorporating blockchain for transparent payments linking mobile wallets with banking systems.

Health Sector: Facilitate integrated pipelines connecting shared records with geo-located facilities, using AI for disease pattern analytics and resource allocation.

Education Sector: Link student profiles with national IDs for automated enrollment and personalized learning paths.

Timeline: Expanded pilots in 2025, full sectoral integration by 2027, potentially yielding efficiencies comparable to Estonia's annual savings equivalent to 2% of GDP.

6.5.10 a2i Initiative Mapping

Initiative	Sector	Impact	Status	Users/Year
National Portal	Governance	2B+ transactions processed	Active	Nationwide
UDCs	Inclusion	Rural digital access & jobs	8,500+ Active	30M
Telemedicine	Health	Remote healthcare access	Active	1M+
Teachers' Portal	Education	Digital resources & collaboration	15,000+ schools	Millions
MyGov App	Engagement	Citizen feedback & services	Active	5M+ downloads
e-Filing	Administration	70% paperwork reduction	Active	All agencies
Innovation Labs	R&D	100+ prototypes developed	Ongoing	Multi-sector

6.5.11 a2i GovTech Transformation KPI Framework

KPI	2025	2026	2027	2028	2030
Service Availability Rate	95%	97%	98%	99%	99.5%
User Satisfaction Score	65%	70%	75%	80%	85%
Ministries Using DPI Standards	20%	40%	60%	80%	100%
Cost Savings (Annual US\$)	\$50M	\$120M	\$250M	\$400M	\$850M

6.6 Data Center Infrastructure Reform (NDC/BDCCL)

6.6.1 Current Challenges

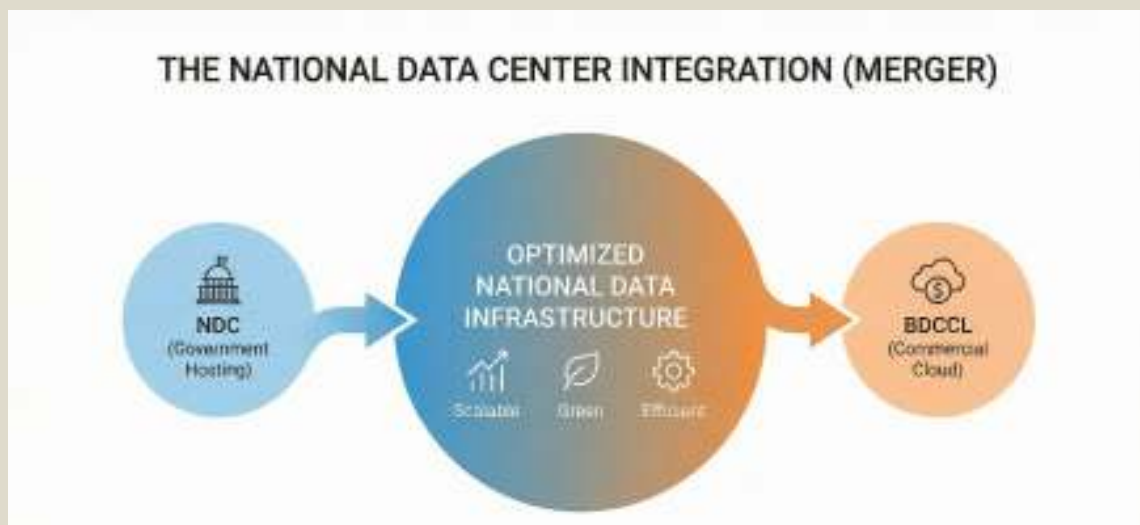
Bangladesh's growing demand for digital services has pushed existing government data centers to full capacity. The Bangladesh Computer Council's (BCC) Nutanix and Huawei data centers face critical constraints in memory, storage, and computing power. Without strategic reform, these limitations will impede the scaling of e-governance, fintech, and digital entrepreneurship services.

Key Challenges Identified:

- **Infrastructure Obsolescence:** A significant portion of the Native Cloud environment was running on legacy hardware that had reached end-of-life status. These assets were consuming excessive power and rack space without delivering commensurate computational value, driving up operational costs (OpEx) while offering poor performance.
- **Cooling & Power Inefficiency:** The data center faced challenges with cooling optimization, leading to a high Power Usage Effectiveness (PUE) ratio (around 1.8). The lack of modern containment strategies meant that energy was being wasted on cooling non-critical or empty rack spaces.
- **Vendor Lock-in & Financial Bleeding:** The previous Oracle Cloud contract was financially unsustainable, with the project consuming **\$23.93 million** in costs while generating less than **\$6 million** in revenue. The terms effectively trapped BDCCL in a "Use-it-or-Lose-it" credit model without ownership of the hardware, forcing the government to pay duties on assets it did not own.
- **Underutilized Capacity:** Despite high demand for cloud services in the country, BDCCL's pricing models and service portfolio were not competitive with private sector or international providers, leading to low occupancy rates.

6.6.2 Reform Measures

1. **Integration of BDCCL and NDC:** The Bangladesh Data Centre Company Limited (BDCCL) and the National Data Centre (NDC) will be integrated for capacity and resource optimization, eliminating duplication and enabling economies of scale.



2. **Capacity Expansion:** Expansion plans are being pursued to significantly increase storage and computing power. Land allocation in Dhaka Metro for NDC expansion will ensure scalability for future cloud, AI, and big data needs.

3. **Service Model Transformation:** A strategic shift from an infrastructure-only model to a product-oriented service platform, enabling value-added services for freelancers, the gaming industry, and the wider digital services ecosystem.
4. **Disaster Recovery Infrastructure:** A national disaster recovery infrastructure expansion plan to safeguard against data loss, ensure business continuity, and strengthen cyber resilience.
5. **Green ICT Practices:** Upgrade data centers with renewable energy, efficient cooling systems, and carbon-neutral operations aligned with the National Green ICT Policy.

6.6.3 Strategic Initiatives & Progress Updates

To transform BDCCL from a passive colocation facility into a dynamic "**National Cloud Service Provider**," a rigorous reform agenda ("Hard Reset") is being executed.

1. Infrastructure Optimization & Decommissioning

- **Action:** A complete audit of the Native Cloud environment was conducted to identify "zombie servers" and obsolete hardware.
- **Progress:** Decommissioning of legacy hardware is in the final stages (Target: **January 2026**). This includes the migration of critical services for the Bangladesh Hi-Tech Park Authority (BHTPA) and DNS systems to optimized platforms.
- **Impact:** This initiative significantly reduces electricity consumption, frees up valuable rack space for high-density modern computing, and lowers the overall carbon footprint of the facility.

2. New Cloud Service Expansion (Revenue Sharing Model)

- **Strategy:** Recognizing that BDCCL cannot solely rely on government capex for rapid expansion, a **Revenue Sharing Model** has been adopted to attract private investment and technical expertise.
- **Progress:** An Expression of Interest (EOI) was floated for a Joint Venture to deploy a new "Sovereign Cloud." As of **Jan 20, 2026**, 5 out of 7 shortlisted firms have submitted detailed proposals. Evaluation is currently underway.
- **Objective:** To deploy a state-of-the-art cloud platform capable of hosting critical government and private sector workloads, reducing reliance on foreign hyperscalers (AWS, Azure) and saving foreign currency.

3. Tariff Rationalization & Market Competitiveness

- **Action:** To make BDCCL competitive against private operators, the 40th Board Meeting approved a strategic reduction in Co-Location service charges.
- **Progress:** The new tariff structure is effective from **January 2026**. Existing contracts with clients are being amended to reflect these lower rates.

- **Impact:** This move aims to attract a broader client base, including private sector enterprises, fintechs, and ISPs, thereby increasing overall revenue through higher volume utilization.

4. Strategic Contract Renegotiation (Oracle Cloud)

- **Context:** Rectifying the financially unbalanced Oracle contract was a top priority.
- **Reform Actions:** High-level renegotiations have been initiated with the following demands:
 - **Duty Reimbursement:** Oracle must reimburse the **\$3.68 million** in import duties paid by BDCCL for hardware owned by Oracle.
 - **Credit Rollover:** Transitioning from a rigid annual forfeiture model to a flexible system where unused credits carry forward to subsequent years.
 - **Market Expansion:** Amending the contract to allow BDCCL to sell Oracle Cloud services to the **Private Sector**, monetizing the idle capacity.
 - **Term Extension:** Extending the service period from 3 to **5 years** to allow sufficient time for market uptake and ROI generation.

5. Hybrid Cloud & Future Architecture

- **Roadmap:** BDCCL is developing a **Hybrid Cloud** architecture. This involves creating a strictly isolated "**Private Cloud**" (potentially air-gapped) for highly sensitive government data (National Security, NID, Health) and a "**Public Cloud**" partition for general citizen services and commercial use. This tiered approach balances supreme security with scalability.

6.6.4 Industry-Academia Collaboration

- **Talent Pipeline:** Addressing the shortage of skilled Data Center Engineers, BDCCL is launching joint training programs with top engineering universities. These programs will focus on **Data Center Operations (DCO)**, **Cloud Architecture**, and **Green IT**, creating a specialized workforce to support the national digital infrastructure.

6.6.5 Expected Impact & Vision

- **Data Sovereignty:** By repatriating critical national data from foreign servers to BDCCL, Bangladesh strengthens its national security and ensures compliance with the **Data Localization** mandates of the NDGO.
- **Economic Efficiency:** A successful Sovereign Cloud will save millions in foreign currency currently paid to international providers. Lower hosting costs will also lower the barrier to entry for local startups.
- **Innovation Enabler:** Affordable, sovereign cloud resources (including the planned GPU cluster) will empower local AI researchers and deep-tech startups to innovate without the high infrastructure costs associated with global providers.

6.6.6 BDCCL Reform: Oracle Cloud Contract Negotiation

The Bangladesh Data Center Company Limited (BDCCL) entered into a strategic agreement with Oracle for the **Dedicated Region Cloud@Customer (DRCC)** service to enhance national cloud sovereignty. However, a comprehensive review by the Interim Government in 2025 revealed critical imbalances in the contractual terms that rendered the project financially unsustainable.

An operational audit indicated that the project was consuming approximately **USD 23.93 million** in costs while generating less than **USD 6 million** in revenue. The existing contract contained provisions that deviated from international best practices, placing disproportionate financial burdens on the state while restricting market flexibility.

Key Issues Identified

- **Unfair Financial Liability:** Despite Oracle retaining ownership of the hardware, BDCCL was designated as the 'Importer of Record,' forcing the government to bear an additional **USD 3.68 million** in import duties and taxes—a burden typically borne by the asset owner.
- **Restrictive Credit Model:** The "Annual Universal Credit (AUC)" structure (USD 3M, 6M, and 9M scaling over 3 years) functioned as a rigid capacity charge with **no rollover** option. Unused credits were forfeited annually, pushing BDCCL toward significant losses due to initial underutilization.
- **Market Restrictions:** The contract limited DRCC usage strictly to government entities, preventing BDCCL from selling excess capacity to the private sector, thereby capping potential revenue.
- **Sovereignty Risks:** The agreement was governed by **English Law** with arbitration in London, and Oracle retained unilateral rights to remove hardware if unused for six months, posing a risk to national data infrastructure stability.

Strategic Reform Measures To rectify these issues and ensure the project's viability, the government has initiated a high-level renegotiation with Oracle, mandated by the Special Assistant to the Chief Advisor. The reform agenda focuses on ten critical amendments to restore balance:

Financial Restructuring

- **Duty Reimbursement:** Demand for Oracle to assume ownership liabilities, including the reimbursement of the **USD 3.68 million** in taxes paid by BDCCL.

- **Credit Rollover & Optimization:** Transitioning from a "Use-it-or-Lose-it" model to a flexible credit system where unused funds **carry forward**, protecting public money from forfeiture.
- **Service Extension:** Proposing a term extension from 3 to **5 years** with a rationalized total value (e.g., USD 6 million inclusive of taxes), aligning costs with actual market uptake.

Market Liberalization

- **Private Sector Access:** Amending the "Usage Rights" clause to allow BDCCL to extend sovereign cloud services to **private sector clients** (e.g., banks, MNCs), subject to security compliance. This is essential to utilize the idle capacity and generate revenue to cover operational costs.

Legal & Operational Sovereignty

- **Governing Law:** Shifting the legal framework to the **Laws of Bangladesh**, with arbitration seated in Dhaka under the Bangladesh International Arbitration Centre (BIAC), ensuring national legal sovereignty.
- **Asset Control:** Revoking Oracle's unilateral right to remove hardware. Any infrastructure changes must now require BDCCL's explicit written consent to prevent service disruption.
- **Data Retention:** Extending the post-termination data retrieval period beyond the restrictive 60-day window to a mutually agreed timeframe, safeguarding critical national data.

Outcome of Renegotiation Efforts

To rectify these issues, the government initiated high-level renegotiations aimed at financial restructuring, market liberalization, and legal sovereignty (shifting arbitration to Dhaka). **However, these negotiations have effectively reached an impasse.** The vendor remained unwilling to align the contract with the financial realities of the local market or the sovereignty requirements of the state. Consequently, continuing the project under the current terms would constitute a significant and recurring loss of public funds.

The Way Forward: Exit Strategy and Mitigation In light of the failed negotiations, BDCCL is triggering a **Strategic Exit and Rationalization Plan** to protect state assets and minimize fiscal impact. The reform agenda now focuses on four immediate actions:

1. Phased Decommissioning of Hardware

- **Capacity Downsizing:** BDCCL will immediately halt the intake of new hardware capacity under the "scaling" clauses of the contract.
- **Asset Return:** A formal process will be initiated to decommission and return unused Oracle hardware racks currently occupying space in the National Data Center. This aims to stop the accrual of maintenance and capacity charges for infrastructure that is effectively idle.

2. Sovereign Cloud Provisioning & Migration

- **Workload Migration:** A "Lift and Shift" roadmap is being executed to migrate existing government workloads from the Oracle DRCC environment to BDCCL's alternative sovereign infrastructure or open-standard private clouds (e.g., OpenStack-based solutions).
- **Service Continuity:** Ensure zero downtime for critical government services during the migration phase by maintaining a minimum viable footprint until the transition is complete.

3. Financial Recovery & Dispute Resolution

- **Duty Reclamation:** The government will pursue legal and administrative channels to reclaim or adjust the **USD 3.68 million** paid in import duties for assets that are being returned to the vendor.
- **Contract Termination:** Legal notice will be served to terminate the agreement based on "Force Majeure" or "Economic Hardship" clauses, shifting dispute resolution to a localized framework to minimize exit penalties.

4. Policy Correction: Open Standards Mandate

- **Vendor Lock-in Prevention:** Moving forward, BDCCL should refrain from signing proprietary, closed-loop cloud contracts. Future procurement must prioritize open-source, interoperable cloud technologies to ensure that the state never again faces data hostage situations or rigid capacity payments.

6.7 Software and Hardware Quality Testing and Certification

Software and Hardware Quality Testing and Certification (SHQTC) Center, accredited by BAB, is a national authority responsible for the mandatory testing, certification, and compliance validation of all government-developed software, hardware, and digital systems. This strategic initiative is designed to enhance the security, reliability, and interoperability of public sector digital services, while also promoting industry-wide quality assurance practices.

The SHQTC Center will play a pivotal role in the reform agenda, certifying the functionality, security, and performance of all ICT solutions procured or developed by government entities. No digital system will be operationalized without undergoing formal certification, ensuring full compliance with national digital architecture standards, cybersecurity frameworks, and data protection policies. To foster a secure, transparent, and efficient digital public service ecosystem, SHQTC will also maintain a centralized Government Software Repository, supporting software reuse, interoperability, and quality standardization across ministries and agencies.

6.7.1 Strategic Empowerment and Institutional Strengthening

To ensure SHQTC can effectively execute its mandate, the government will implement a comprehensive empowerment strategy:

- **Operational Autonomy & Promotion:** SHQTC will operate as an autonomous body with the authority to enforce quality standards. A clear career progression path and promotion policy will be established for technical staff to retain top talent and ensure institutional memory.
- **Dedicated Financial Codes:** The Ministry of Finance will introduce specific economic codes in the national budget for "Software and Hardware Quality Testing." This ensures that every digital project has a mandatory, ring-fenced budget allocation for quality assurance, preventing funds from being diverted to other operational costs.
- **Increased Funding & Staffing:** A significant budget increase will be allocated to recruit specialized engineers, security analysts, and compliance auditors. Staffing levels will be scaled up to meet the growing demand of certifying all government digital assets.
- **Capacity Building:** Continuous professional development programs will be institutionalized, offering certifications in advanced testing methodologies (e.g., ISTQB, CISA) and emerging technologies to keep the workforce at the cutting edge.

6.7.2 Dedicated SHQTC R&D Lab

A state-of-the-art Research and Development (R&D) Lab will be established within SHQTC. This facility will focus on:

- **Automated Testing Frameworks:** Developing AI-driven tools for rapid vulnerability scanning and code quality analysis.
- **Hardware Stress Testing:** Simulating extreme environmental conditions to ensure hardware resilience in diverse Bangladeshi climates.
- **Security Research:** Proactively identifying zero-day vulnerabilities in locally used software stacks.
- **Interoperability Standards:** creating testbeds to validate API integrations against the Bangladesh National Digital Architecture (BNDA).

6.7.3 Market Expansion

In addition to public sector services, the SHQTC will extend its certification services to private sector ICT firms on a voluntary basis. This will enable local software and hardware developers to secure official certification for their products, enhancing credibility, market trust, and competitiveness. By opening these services to the private sector, Bangladesh positions itself to raise national ICT standards and establish itself as a regional hub for trusted, secure, and certified digital solutions.

6.7.4 Key Responsibilities Reform of SHQTC

- **Mandatory Testing:** Certification for all government-developed and procured ICT systems.
- **Voluntary Certification:** Optional, standards-based certification services for private software and hardware firms.
- **Repository Management:** Managing a Government Software Repository for certified and reusable software assets.
- **Compliance Enforcement:** Ensuring adherence to national digital interoperability, cybersecurity, and data governance frameworks.
- **Lifecycle Assurance:** Delivery of structured quality assurance, security, and performance testing throughout system development life cycles.

6.7.5 Implementation Roadmap

- **Mid 2026:** Certification authority operations to commence.
- **September 2026:** Phase 1 launch of the Government Software Repository.
- **2026 onwards:** Optional certification services available for private ICT firms.
- **By 2030:** Full certification compliance expected for all government ICT systems.

To successfully deliver on its mandate, a dedicated capacity development program will be introduced, focusing on secure software development, cybersecurity testing, compliance auditing, and interoperability management. This will be supported by collaborative partnerships with universities, R&D centers, and international digital governance experts to ensure adoption of international best practices and to strengthen institutional capability.

6.8 Reform of the Office of the Controller of Certifying Authorities (CCA)

6.8.1 Background and Context

The Office of the Controller of Certifying Authorities (CCA) is the national trust anchor for electronic signatures and public key infrastructure (PKI) in Bangladesh. To enable end-to-end paperless citizen and government services, the CCA must evolve from a

primarily oversight and compliance function into a modern trust-service regulator and platform operator. This requires legal and policy modernization; resilient 24/7 trust infrastructure (Root CA, OCSP/CRL, signature verification); stronger audits and forensics; a developer-first integration layer (APIs, SDKs, client tools); and systematic cross-ministry adoption that reduces service silos.

Electronic signatures and PKI are foundational enablers for secure digital transactions, providing identity binding, integrity, authentication, and non-repudiation for documents and online workflows. In a government service-delivery context, they enable

- (i) legally enforceable digital approvals;
- (ii) tamper-evident records;
- (iii) auditable decision trails; and
- (iv) faster, less corruptible processes.

The CCA functions as the apex authority that regulates and supervises licensed Certifying Authorities (CAs) and operates national Root CA trust services. Future reform functions include expanding scope beyond licensing into ecosystem enablement, e.g., e-sign API specifications for online e-KYC, interoperability guidance for certificates, and capacity-building programs.

6.8.2 Current Scope of Work and Services

The CCA currently provides or coordinates the following functions:

5. Regulation and supervision of licensed Certifying Authorities (CAs), including review and update of CA Certificate Policy (CP) and Certificate Practice Statement (CPS).
6. Operation and maintenance of national Root CA trust services and trust governance artifacts (Root CA CP/CPS).
7. 24/7 validation services: Online Certificate Status Protocol (OCSP), Certificate Revocation List (CRL), and public digital signature verification services.
8. Support services for subscriber verification, including National Identity Card (NID) verification support for CA operations.
9. Development and periodic update of sector-wide instruments (e.g., PKI Auditing Guideline, CA Licensing Guideline, PKI Enabled Application Guideline, Digital Signature Certificate Interoperability Guidelines, and e-KYC / e-Sign API specifications).
10. Ecosystem development and capacity building: trainings, workshops, PKI summits/hackathons/quizzes, and recognition programs for digital signature adopters.

Operational Trust Services

CCA operates public-facing services that underpin nationwide verification and trust. These should be treated as national critical digital infrastructure and managed with explicit SLAs:

- OCSP service (real-time certificate status).
- CRL publication service (revocation list distribution).
- Public digital signature verification service (document signature checking).

6.8.3 Recent Accomplishments

Strategic Initiatives & Progress Updates

CCA is executing a comprehensive roadmap to integrate digital signatures across all pillars of the government.

1. e-GP System Integration (Transparency in Procurement)

- **Action:** The *Bangladesh e-Government Procurement (e-GP) Guidelines (Revised), 2025* (enacted March 12, 2025) now mandates the use of digital signatures at various stages of the tender process.
- **Progress:** Technical integration is ongoing to ensure full compliance.
- **Target:** Complete integration by **June 2026**.
- **Impact:** This will eliminate procurement fraud, ensure non-repudiation of bids, and guarantee 100% transparency in public spending.

2. Land Service Modernization (Secure Property Rights)

- **Action:** The Ministry of Land decided (August 18, 2025) to introduce digital signatures for critical citizen documents: **Record of Rights (Khatian)**, **Duplicate Carbon Receipts (DCR)**, and **Dakhila** (Rent Receipts).
- **Progress:** CCA is providing technical support for API integration.
- **Target:** Live deployment by **June 2026**.
- **Impact:** This will prevent the forgery of land documents, reduce litigation, and massively increase the legal reliability of online land services.

3. Smart D-Nothi & Office Digitization

- **Current Status:** Digital signatures are currently used for *outgoing letters* (Patra Jari) in 30 offices. A Proof of Concept (POC) has been completed for signing internal *Note Sheets*.
- **Roadmap:** Full-scale implementation of Note Sheet signing across all ministries by **June 2026**.

- **Impact:** Achieving a truly "Paperless Secretariat" where file approval workflows are 100% digital and legally secure.

4. BNDX Platform Integration (Interoperability)

- **Action:** To secure the National Data Exchange Layer, CCA has approved a specialized **Certificate Profile** for the **BNDX (Bangladesh National Digital Exchange)** platform.
- **Status:** Completed (January 2026).
- **Impact:** Ensures that all cross-agency data exchanges (e.g., NID to Passport) are cryptographically signed and trusted.

Other notable accomplishments include:

- **Expansion of Digital Signatures:** Digital signatures were introduced in 20 organizations over the past year, including the National Board of Revenue (NBR), Ministry of Foreign Affairs, Bangladesh Cricket Board, Mutual Trust Bank, Eastern Bank Limited, and BRAC Bank PLC.
- **Service Launch:** Launched the Online Certificate Status Protocol (OCSP) service for the Root CA system.
- **Continuous Availability:** Providing 24/7 Certificate Revocation List (CRL) services via crl.cca.gov.bd.
- **Verification Services:** Maintaining the 'Digital Signature Verification' service (digisigchecker.cca.gov.bd) and providing 24/7 'National ID Verification' services to CA institutions.
- **Regulatory Framework:** Formulated the Office of the Controller of Certifying Authorities (Employee) Recruitment Rules, 2025.
- **Guidelines & Policies:** Updated the Digital Certificate Interoperability Guideline, PKI Enabled Application Guideline, Bangladesh Root CA Certificate Policy (CP), and Root CA Certificate Practice Statement (CPS).
- **CA Oversight:** Reviewed and updated the CP and CPS of various Certifying Authorities, along with PKI Auditing and CA Licensing guidelines.
- **Training & Engagement:** Provided PKI training and distributed e-Signs to 50 university students and teachers on the occasion of July Revolution Day.
- **Events:** Organized the PKI Summit 2025, PKI Hackathon 2025, and PKI Online Quiz 2025.
- **Recognition:** Honored the Ministry of Foreign Affairs, RJSC, BRAC Bank PLC, and Robi Axiata Limited as the best users of digital signatures.
- **Institutional Strengthening:** Submitted a manpower proposal to the Finance Division for the digital signature implementation project.
- **Future Planning:** Developed the DPP for a feasibility study project titled "Electronic Signature Subscriber Verification Ecosystem, Strengthening the Office of the CCA, and Electronic Record Repository."

- **Capacity Building:** Trained 173 government officials across 5 institutions and organized workshops/seminars in 8 institutions regarding digital signatures over the last year.

6.8.4 Current Challenges and Structural Gaps

Despite progress, the ecosystem remains constrained by structural issues that limit nationwide, citizen-facing scale-up:

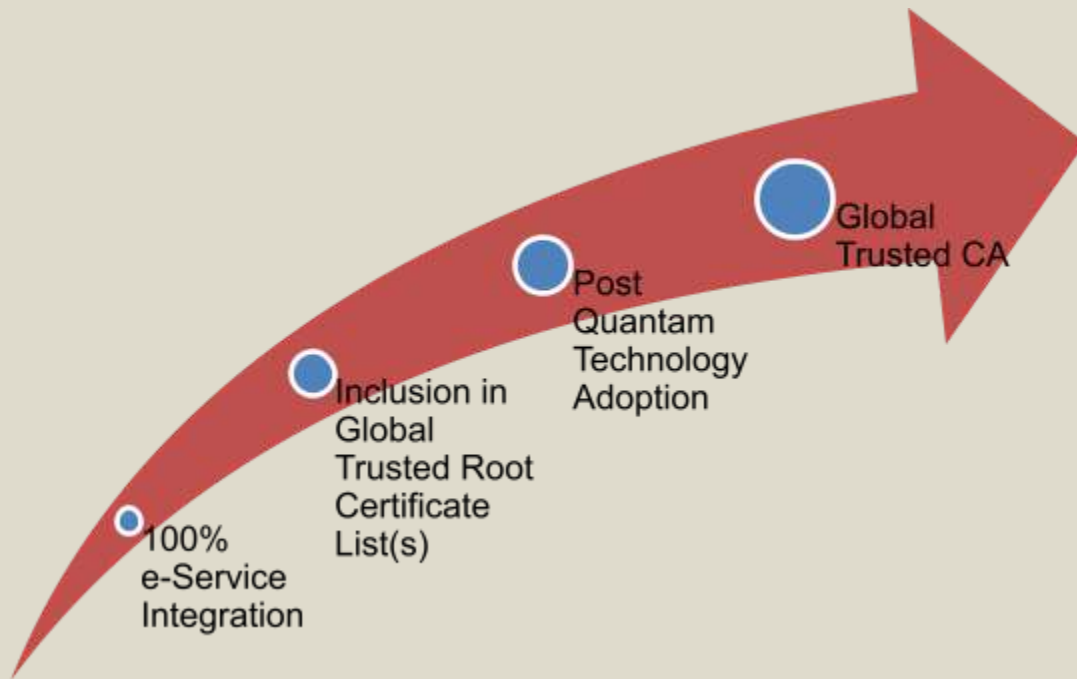
- Siloed ownership of government services across ministries: workflows remain paper-based or hybrid, and electronic signature usage is not consistently mandated in priority e-services.
- Absence of standardized integration patterns: agencies often lack reusable toolkits (signature validation middleware, timestamping, long-term validation, role-based signing policies) and developer enablement for quick onboarding.
- Infrastructure modernization needs: Root CA platform and cryptographic modules require continuous upgrades; OCSP/CRL throughput and latency must scale for higher transaction volumes and stricter availability targets.
- Assurance maturity gaps: independent audits, continuous compliance monitoring, and forensic readiness must be strengthened to protect national trust infrastructure and investigate signature-related cybercrime.
- International recognition barriers: without meeting major root store inclusion and audit transparency requirements, cross-border interoperability and wider private-sector adoption remain limited.
- Human resource and empowerment constraints: the reform proposal highlights gaps in grade/authority, staffing structure, and administrative autonomy expected for 24/7 national services.
- Sustainable financing: recurring OPEX for secure infrastructure, audits, and 24/7 operations requires dedicated budget heads and clear cost recovery mechanisms for certain services.
- Identity privacy and verification risks: evolving NID verification posture requires consent-based, data-minimizing verification mechanisms in e-KYC and signing flows.

6.8.5 Reform Vision, Objectives, and Design Principles

Reform Vision: Establish the CCA as a modern, empowered national Trust Services Authority that enables secure, legally enforceable, paperless transactions for government, citizens, and businesses.

Reform Objectives:

- Scale digital signature usage across priority government workflows (procurement, finance, land, education, health) and progressively expand to citizen-facing services.
- Integrate high-assurance digital signature capabilities with the Citizen ID ecosystem (NID/OneID), enabling remote signing and consent-based identity verification.
- Modernize Root CA infrastructure and operational processes to deliver 24/7 services hosted on government cloud infrastructure with appropriate disaster recovery.
- Strengthen audit, compliance, incident response, and digital forensics capabilities aligned with international best practices.
- Create a developer-first integration layer: standard APIs, SDKs, and reference implementations to reduce onboarding friction for agencies and vendors.
- Progressively achieve international trust and cross-border recognition through conformance, audit transparency, and alignment with global root store policies.



Design Principles:

- Security-by-design and privacy-by-design; least-privilege access; strong key management and HSM governance.
- Interoperability as a policy: standards-based profiles and conformance testing.
- Operational resilience: 24/7 monitoring, defined SLAs, high availability (HA), and disaster recovery (DR).
- Modular enablement: reusable components and APIs usable across sector systems.

- Measurable outcomes: KPI-driven management and transparent reporting.

6.8.6 Roadmap and Priority Actions

1. "Online Fraud Prevention" Project (Secured Repository)

- **Initiative:** A new project titled *"Expansion of Digital Signatures for Cyber Security to Prevent Online Fraud"* was approved in January 2026.
- **Core Component:** Establishment of a **Secured Electronic Repository** to store and verify all digital certificates and signatures centrally.
- **Goal:** To achieve **International Recognition (WebTrust Seal)** for Bangladesh's Root CA, allowing Bangladeshi digital signatures to be recognized globally (e.g., by Adobe, Microsoft).
- **Target:** Architecture design completion by **November 2026**.

2. Judicial Integration (e-Judiciary)

- **Vision:** Extending digital signatures to the judiciary to enable digitally signed court verdicts and bail orders, reducing tampering and delays.

Institutional Synergy & Challenges

To succeed, CCA is building synergies with other agencies:

- **With BDCCL:** Hosting the new Secured Electronic Repository at the **Tier-4 National Data Center** for maximum security.
- **With NCSA:** Establishing a real-time **Threat Intelligence Sharing** mechanism to detect and prevent digital signature forgery.
- **With DoICT:** Launching grassroots awareness campaigns through Upazila-level ICT officers.

Near-term outputs (12–24 months)

- Upgradation of the administrative status of the Controller of the CCA to Grade-2. This elevation is critical for ensuring the CCA has the requisite authority to lead inter-ministerial coordination and enforce compliance across the whole-of-government.
- Procurement integration roadmap with BPPA/e-GP for tender and contract signing.
- iBAS++ integration design and piloting across key approval stages (bills, budgets, funds).
- Land document signing pilots (Namjari, Khatian, and related services) with training and implementation support.

- Academic and health record pilots: digitally-signed certificates/transcripts; EMR/e-prescription authenticity and non-repudiation.
- Onboard at least 10 additional priority government agencies with customized adoption plans.
- Root CA system upgrade planning and execution (hardware, software, cryptographic modules) and CP/CPS modernization.
- CRL/OCSP performance uplift: increased capacity and improved validation speed for real-time use cases.
- To overcome procurement hurdles, the Ministry of Finance will introduce a specific finance code for "Digital Signature & PKI Services" in the national budget. This reform will allow government offices to allocate dedicated funds for acquiring digital signatures, tokens, and related services, eliminating the ambiguity that currently hinders adoption.
- Quarterly third-party cyber security audits of Root CA; strengthening of digital forensics lab operations and SOPs.
- Nationwide awareness program (≥ 20 seminars), user guidelines publication, and quarterly high-level stakeholder review meetings.
- Collaborate with the Ministry of Finance to open specific Economic Revenue Codes for electronic signature services. This will ensure a sustainable, recurring budget for 24/7 Root CA system upgrades, WebTrust audits, and the maintenance of national critical trust infrastructure, reducing dependency on ad-hoc project funding.
- The CCA will actively pursue inclusion in the global trusted stores of Mozilla, Microsoft, and Adobe through rigorous WebTrust audits. This is essential for the international acceptance of Bangladeshi digital certificates and to facilitate Cross-Border Mutual Recognition of digital signatures for international trade and legal cooperation.

Medium-term outcomes (2028–2030)

- Citizen ID-linked signing capability available as a national service, supporting citizen-facing digital approvals and transactions (remote signing with consent-based verification).
- Standard trust-service components (signing gateway, verification middleware, timestamping, long-term validation) reusable across sector systems.
- Government cloud-hosted trust infrastructure with HA/DR, integrated SOC/SRE operations, and mature incident handling.
- International trust milestones: completed audits and transparency requirements aligned with major root store policies; phased mutual recognition pathways initiated.
- Continuous policy modernization: adoption of new cryptographic practices and readiness for post-quantum cryptography transition planning.

- Partner with the Ministry of Education to integrate Digital Signature and PKI concepts into the national ICT curriculum. This will build a foundational 'culture of trust' and ensure a steady pipeline of domestic PKI experts.
- Post-Quantum Cryptography (PQC) Readiness: Establish a dedicated research and development cell within the CCA to explore Post-Quantum Cryptography and its implications for national security. Budgetary priority will be given to PKI-related innovation ideas to ensure Bangladesh's digital trust infrastructure remains resilient against emerging computing threats.

6.8.7 KPI Framework (Outcome and Service KPIs)

The KPI framework should balance (i) adoption outcomes, (ii) service reliability, (iii) assurance and compliance, and (iv) ecosystem enablement. Indicative KPIs and target levels are proposed below; baselines should be confirmed at program inception.

KPI Category	KPI	Target (Indicative)	Notes
Adoption	New ministries/agencies onboarded (production e-sign workflows)	≥10 per year	Includes customized adoption plans
Adoption	Priority workflow coverage (procurement, finance, land, education, health)	≥1 major workflow/sector live by 2027; scale by 2030	Sector penetration
Operations	OCSP availability	≥99.9% monthly uptime	Critical validation
Operations	CRL publication freshness	Defined CRL update SLA met	Revocation safety
Operations	Median OCSP response latency	Target after baseline (e.g., <300ms)	User experience
Security	Independent security audits completed	Quarterly (Root CA) + annual (ecosystem)	Assurance maturity
Security	Mean time to detect/respond critical incidents	Target after baseline; continuous improvement	SOC/SRE maturity

Governance	Root CA CP/CPS review cycle	Annual review completed and published	Policy hygiene
Enablement	Developer platform adoption (API calls/SDK downloads/sandbox users)	Targets per service; YoY growth	Integration at scale
Capacity	Officials trained (government + academia)	≥1,000/year (role-based)	Skill diffusion
Trust	International trust milestones (root store readiness / audits / transparency)	Milestones completed per roadmap	International credibility

6.9 Summary: Integrated Institutional Framework

The institutional reforms outlined in this chapter form an interconnected framework where each agency plays a distinct yet complementary role in achieving Bangladesh's digital transformation objectives:

BCC (Bangladesh Computer Council) functions as the primary standards and infrastructure body, managing critical assets like the National Data Center and National Digital Architecture (BNDA) while ensuring technical standardization across government systems.

DoICT (Department of ICT) serves as the grassroots implementation arm, extending digital services and literacy to the upazila level through its field offices, technical support for local administration, and connectivity maintenance.

NCSA (National Cyber Security Agency) anchors the nation's digital defense, operating the National Security Operations Center (NSOC), conducting critical infrastructure audits, and enforcing the Cyber Safety Ordinance to protect citizen data.

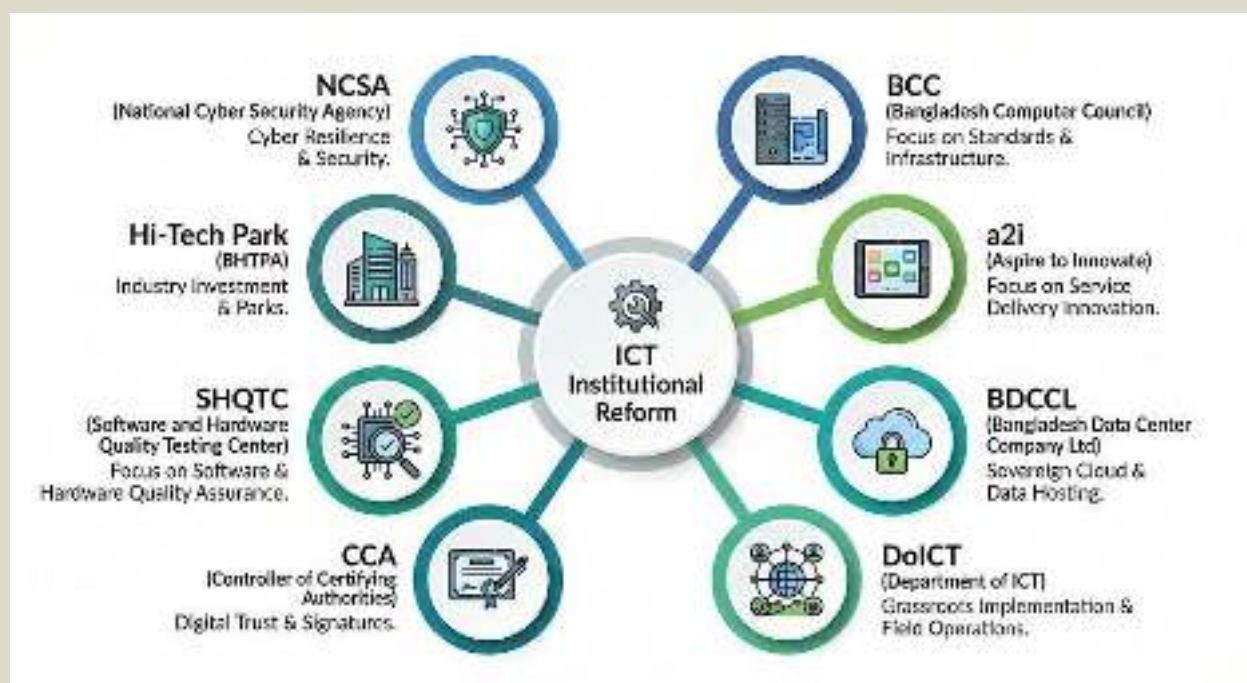
a2i (Aspire to Innovate) operates as the central GovTech innovation hub, designing citizen-centric services, simplifying public service delivery (SPS), and driving the "Zero-Visit" mandate through policy advocacy and pilot initiatives.

BHTPA (Bangladesh Hi-Tech Park Authority) acts as the engine for industry growth, managing Hi-Tech Parks and Software Technology Parks to attract foreign direct investment (FDI) and foster a thriving startup ecosystem.

CCA (Controller of Certifying Authorities) establishes the foundation for digital trust, regulating electronic signatures (CA) and ensuring the legal validity of digital transactions essential for the smart economy.

BDCCL (Bangladesh Data Center Company Ltd) ensures data sovereignty and secure hosting, managing the Tier-IV National Data Center to provide cloud services and colocation for critical government and private sector data.

SHQTC (Software and Hardware Quality Testing Center) provides the essential quality assurance layer, offering standardized testing and certification for government software and hardware procurement to ensure reliability, security, and interoperability.



6.9.1 Coordination Mechanisms

Successful implementation requires coordination mechanisms including quarterly inter-agency coordination meetings, shared KPI dashboards for transparency, joint capacity building programs, and unified standards enforcement through NDGA. The ICT Division will maintain oversight responsibility with clear escalation pathways for resolving inter-institutional challenges.

6.9.2 Success Factors

The reforms will succeed when: government systems achieve full interoperability through the National Service Bus; every citizen can access digital skills training within their district; cyber threats are detected and responded to within hours; public services achieve 99%+ availability with 80%+ user satisfaction; and ICT projects consistently deliver verified outcomes within budget.

6.9.3 Institutional Reform Summary

Institution	Primary Role	Key 2030 Target	Integration Point
BCC	Digital infrastructure backbone & certification	100% NSB integration	BNDA/NDGA standards
DoICT	Grassroots training & digital service support	1M+ citizens trained	NSDA certification
NCSA	Cybersecurity operations & strategy	50K cyber professionals	CSO 2025 enforcement

a2i	GovTech innovation & interoperability	\$850M cost savings	DPI framework lead
Projects	Verified outcome delivery	100% compliance	PPR reform

Through these coordinated institutional reforms, Bangladesh will establish the governance foundations necessary to achieve its vision of becoming a digitally empowered, innovative, and inclusive society by 2030.

End of Chapter 6

7. Project reform

7.1 Introduction and Reform Imperative

Bangladesh's digital transformation journey has been marked by significant ICT project investments over the past decade. However, independent reviews, audit findings, and operational assessments have revealed systemic governance, financial, and technical challenges across several flagship projects. These deficiencies have resulted in cost overruns, suboptimal infrastructure deployment, vendor dependencies, and inadequate accountability mechanisms.

The National Digital Transformation Strategy 2025-2030 recognizes that achieving a digitally empowered Bangladesh requires not only new initiatives but also comprehensive reform of existing projects. This chapter outlines the reform framework for flagship ICT projects, addressing identified weaknesses while establishing scalable models for future digital infrastructure development. The overarching objective is to restore trust, ensure value-for-money, and align all projects with NDGA-led interoperability and cybersecurity standards.

7.2 Expanding Digital Connectivity (EDC) Project Reform

7.2.1 Background and Identified Issues

The Expanding Digital Connectivity (EDC) project, envisioned as a cornerstone for rural digital access, was found to be riddled with systemic malpractice that compromised its integrity and financial viability. Initial audits and independent reviews by the Interim Government uncovered a pattern of falsified reporting and funding mismanagement by service providers and project officials.

1. **Loan Uncertainty & Component Cancellation:** Due to complications with the Chinese loan agreement, major components like the Digital Village (Scope-7) and CRVS Software (Scope-6, BDT 502.09 Crore) were suspended. The Revised Development Project Proposal (RDPP) is now restructuring the project to be completed primarily with GoB funds (BDT 2545.15 Crore).
2. **Financial Irregularities:** Independent reviews revealed inflated connection costs averaging nearly three times international benchmarks and questionable billing practices.
3. **Implementation & Reporting Fraud:** Audit findings flagged unrealistic reporting (e.g., claims of 765+ connections/day) and fake reporting by contractors without third-party verification.
4. **Technical Non-Compliance:** Deployed network devices were not SNMP-ready, and the Network Management System (NMS) was non-functional, leaving infrastructure unmonitored and vulnerable.

5. **Falsified NMS Reporting:** Contractors were found to be submitting fabricated reports regarding network uptime and connectivity status. The Network Management Systems (NMS), which are critical for real-time monitoring, were often non-functional or manipulated to show "active" connections for sites that were physically offline or non-existent. This deception allowed providers to claim payments for services never rendered.
6. **Funding Mismanagement:** Significant portions of the project budget were allocated to non-essential components or overpriced procurement contracts. For instance, the Civil Registration and Vital Statistics (CRVS) software component was allocated BDT 502.09 crore, heavily reliant on a loan agreement with the Chinese government that never materialized. Despite this, administrative overheads continued to drain public resources.
7. **Unrealistic Targets:** The project reported installing over 765 connections per day—a logistical impossibility given the available manpower and equipment—raising red flags about the authenticity of the entire deployment data.

Reform Actions & Restructured Scope

To rescue the project and protect public funds, a comprehensive reform plan was executed, focusing on strict verification, component rationalization, and rebranding for relevance.

The reform plan re-aligns the project to ensure verifiable connectivity and practical skill development, while formally acknowledging the cancellation of unviable components.

1. Expanded End-User Connectivity (Revised Scope-1) In addition to the original targets, the project is expanding its reach to include critical public service points. The 1,09,244 broadband connections will now specifically include:

- **District Level Courts** under the Ministry of Law.
- **All Post Offices** under the Postal Department.
- **Bangladesh Diabetic Association** centers.
- **National University Affiliated Colleges.**
- **Textile Department Institutes.**
- **Compliance:** A functional **Network Monitoring System (NMS)** software is now a mandatory procurement deliverable to monitor these connections in real-time.

2. Upazila Seva & ICT Training Centers (Revised Scope-2) Instead of generic labs, the project will construct **400 "Upazila Seva & ICT Training Centers"** via vertical extension of existing Upazila administration buildings (through LGED deposit work). These centers will serve as local hubs for:

- Advanced IT skill courses for Youth and Women.
- Startup idea incubation and mentorship.

- Specialized training for government officials on e-filing, cybersecurity, and digital governance.

3. Establishment of Specialized Labs (Revised Scope-3) To boost national R&D capacity, the project will establish targeted high-tech labs in universities and the ICT Bhaban:

- **VLSI & AI Labs (17 Units):** Setup in 17 government/private universities and ICT Bhaban to train students in IC design, FPGA, Machine Learning, and Generative AI, directly supporting the semiconductor engineer pipeline.
- **HPC Cloud GPU-Based Lab (1 Unit):** A High-Performance Computing cloud lab at ICT Bhaban (initially hosted at the National Data Center) to provide multi-tenant supercomputing access for university researchers and industry stakeholders.
- **Digital Forensic Lab (1 Unit):** Established at DoICT to handle computer, mobile, and network forensics, supporting law enforcement in cybercrime investigations across jurisdictions.
- **PKI Lab (1 Unit):** A Public Key Infrastructure lab at DoICT to support the Certification Authority, developing software for Digital Certificate Management and E-Sign systems linked to NID.

4. ICT Tower (Tathya Projukti Bhaban) Construction The project includes the construction of a **12-storied ICT Tower** (on a 21-story foundation) at Purbachal. This facility will house the Digital Leadership Academy, specialized research units, the **Network Operations Center (NOC)** for nationwide government network monitoring, and the DoICT head office.

5. Financial Rationalization and Component Cancellation

- **Saving Public Money:** The reform successfully identified and cancelled non-viable components, saving hundreds of crores in public funds. The CRVS Software component (BDT 502.09 crore) and the Digital Village initiative (BDT 251.16 crore) were scrapped due to the failure of the Chinese G2G loan agreement and lack of feasibility. These funds were reallocated or returned to the exchequer.
- **Cost Re-baselining:** The cost per connection was re-evaluated against international benchmarks. Inflated procurement costs for IT equipment were slashed, and a new "value-for-money" standard was enforced for all future tenders.

6. Technical Verification and Accountability

- **End-User Connectivity Verification:** A new Network Monitoring System was introduced to replace the compromised legacy tools. This system mandates real-time, tamper-proof reporting of bandwidth usage. The definition of

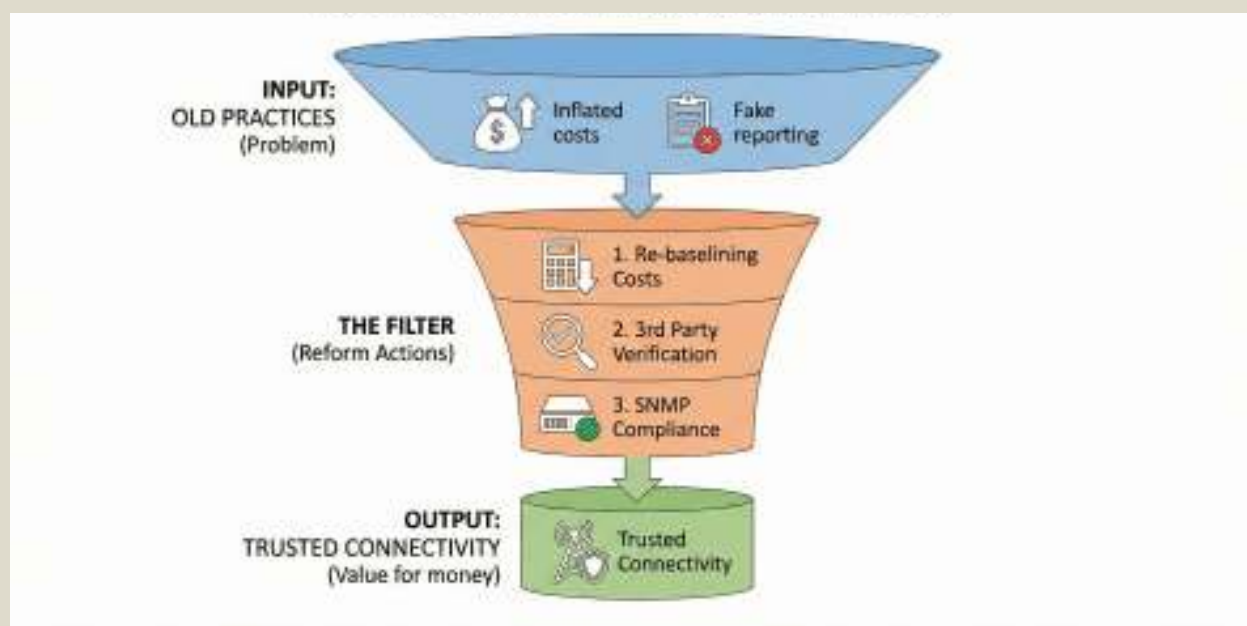
"connectivity" was shifted from simple physical installation to verified active usage, ensuring that payments are only released for functional connections.

- **Third-Party Audits:** To counter falsified reporting, a mandatory third-party verification mechanism was established. Independent technical auditors now physically inspect random sites to validate the data submitted by contractors before any bill is processed.

7. Rebranding and Strategic Realignment

- **Joy Set Center to "Smart Service & Employment Training Centre":** The project component previously named "Joy Set Center" was rebranded to "Smart Service & Employment Training Centre" to better reflect its core objective of youth employment and skill development, removing political branding.
- **Sheikh Russel Digital Lab Reform:** Similarly, the "Sheikh Russel Digital Lab" initiative was restructured. Instead of merely supplying hardware that often went unused, the focus shifted to establishing **20 Specialized Labs** (VLSI, AI, Robotics, Cyber Security) at universities to drive high-value skills, rather than generic computer literacy.

Impact of Reform: These decisive actions have transformed the EDC project from a source of financial leakage into a transparent infrastructure initiative. By stopping payments for fake connections and cancelling unfeasible components, the government has saved significant public capital while ensuring that the digital connectivity actually reaches the intended rural beneficiaries. The project now operates under a "Zero Tolerance" policy for data manipulation, setting a new standard for accountability in public ICT procurement.



7.2.3 Expected Outcomes

- Transparent, corruption-free project implementation with full audit trails
- Functioning, monitored, and secure digital infrastructure across all deployment sites
- Greater taxpayer value with reduced cost per connection aligned to international benchmarks
- Scalable model for future connectivity projects with embedded accountability mechanisms

7.2.4 Key Performance Indicators

KPI	Baseline (2024)	Target (2027)
Cost per connection vs. benchmark	3x international avg.	≤1.2x international avg.
Third-party verification rate	0%	100%
SNMP-compliant devices deployed	<30%	100%
NMS operational coverage	Non-functional	100% real-time monitoring
Payment release without verification	>BDT 99 crore	Zero tolerance

7.3 Enhancing Digital Government and Economy (EDGE) Project Reform

7.3.1 Project Overview

The EDGE Project, supported by the World Bank, represents a significant investment in Bangladesh's digital government infrastructure and economic enablement. The project encompasses multiple components including cybersecurity capacity building, digital service delivery enhancement, and foundational system upgrades.

7.3.2 Strategic Infrastructure Overhaul

The Critical Capacity Crisis At the onset of the Interim Government, a diagnostic review of the Bangladesh Computer Council (BCC) revealed a critical bottleneck in the National Data Center (NDC). Resource utilization had reached dangerous levels—with CPU occupancy at ~82% and storage at ~78%. This "capacity crunch" posed an immediate threat to digital governance continuity, as expanding data center capacity via traditional procurement is typically a complex, time-consuming process.

Historically, World Bank-funded projects in this sector have suffered from slow implementation cycles (averaging 7 years for completion) and low fund utilization (averaging ~18%), often leaving significant allocations unused. With the EDGE project approaching the end of its lifecycle, addressing this crisis through standard procedures was deemed nearly impossible.

The Reform Intervention Recognizing the urgency, the government engaged in rapid high-level negotiations with the World Bank and relevant ministries to restructure specific components of the EDGE project. The objective was threefold: immediate capacity enhancement, service simplification, and qualitative modernization. Through agile project management and the dedicated efforts of BCC as the implementing agency, a comprehensive infrastructure overhaul was executed within strict timelines—a feat previously considered unachievable in the public sector.

Modernization of Monitoring and Security

The review identified severe lapses in the legacy monitoring framework. Virtual Machines (VMs) often lacked active monitoring agents, meaning service failures went undetected until user complaints were filed, leading to prolonged diagnoses and service interruptions. Furthermore, the absence of robust uplink monitoring meant that capacity crunches caused by DDoS attacks or BGP protocol anomalies could not be proactively detected.

Under the EDGE reform, these vulnerabilities have been addressed. BCC has deployed a proactive, real-time monitoring architecture that ensures comprehensive visibility from the network layer to the application layer, significantly enhancing service reliability and security posture.

Deployment of Nutanix Cloud Platform (NCP)

To break the cycle of fragmented infrastructure, the project successfully launched the Nutanix Cloud Platform (NCP). This Hybrid Multi-Cloud infrastructure integrates computation, storage, virtualization, and networking into a single unified stack.

- Impact: Government institutions can now seamlessly manage applications and data across on-premise private clouds, edge locations, and public clouds, ensuring flexibility and scalability.

Establishing Platform as a Service (PaaS) & Microservices

A critical barrier to Bangladesh's digital evolution has been the "Silo Culture" of government data centers. Most existing systems lack essential features like Disaster Recovery (DR), standardized Cyber Security, or DevOps-friendly environments. Integration with the National Data Center was historically difficult, resulting in a near-zero adoption of Microservices Architecture in public services.

To align with global standards of cost optimization, performance, and scalability, the EDGE project has deployed a Platform as a Service (PaaS) layer.

- Technology: A CNCF-certified Kubernetes platform.
- Capabilities: Features Source-to-Image (S2I), built-in CI/CD pipelines, and seamless GitOps integration.
- DevSecOps: The platform includes integrated container registries with vulnerability scanning, ensuring security at build, deploy, and run stages. It supports High Availability (HA), Disaster Recovery (DR), and provides built-in storage services for block, object, and file types.

Enabling Legal Compliance for Critical Information Infrastructure (CII)

The deployed NCP and PaaS infrastructures act as the technical backbone for enforcing three landmark regulations: the Personal Data Protection Ordinance 2025, the National Data Governance Ordinance 2025, and the Cyber Safety Ordinance 2025.

- **CII Protection:** For the first time, the 35 agencies designated as Critical Information Infrastructure (CII) have a platform capable of meeting strict compliance guidelines—a milestone for national digital sovereignty.
- **Data Classification:** The infrastructure supports the 4-tier data classification model, facilitating secure cross-border trade and alignment with global data protection standards.
- **Cloud First Policy:** This ecosystem enables a transition to "Cloud Native" capacity, Containerized Clouds, and CDNs, paving the way for a citizen-friendly digital ecosystem. Negotiations have also been initiated with major traffic providers like Meta to establish local data centers, aimed at reducing internet costs and improving latency.

Resource Optimization and the National GPU Cloud

Amidst a global chip supply crisis and currency devaluation, the project focused heavily on resource optimization.

- **CPU Optimization:** Through technical re-engineering, BCC successfully reclaimed and added over 1,000 vCPUs to the national pool.
- **National GPU Cloud:** Leveraging underutilized assets from the BDSET project, the Bangladesh Hi-Tech Park Authority (BHTPA) and BCC have integrated 20+ NVIDIA Volta Architecture Tensor Core GPUs into a sharable Cloud Resource Management Platform.
 - **Capacity:** Delivers ~2,240 teraFLOPS of Deep Learning capacity, equivalent to the computational power of over 900 CPUs.
 - **Access:** This is the first public sector GPU cloud open to academia, the gaming industry, and researchers for High-Performance Computing (HPC) tasks such as Generative AI, Threat Intelligence, and Geo-science modeling.

This infrastructure reform is the engine powering the shift from fragmented portals to a true Digital Public Infrastructure (DPI).

- **The Connectivity Highway:** The government is constructing a digital highway based on interoperability (mandated by NDGO 2025), centralized source code repositories (BCC/GitHub), and enhanced DevOps capacity at a2i.
- **One ID & Data Reliability:** Addressing the crisis of low-accuracy identity data (errors in names, DOBs across agencies), the reform prioritizes the **"One ID"** foundation. By moving towards Blockchain or Federated Learning-based data management, the system ensures that citizen data is updated automatically

across all 17-42 connected departments, eliminating the harassment of repetitive corrections.

7.3.3 National Security Operations Center (N-SOC) Establishment

A \$5 million proposal has been submitted to the World Bank through the EDGE project for financial assistance to establish the National Security Operations Center (N-SOC). This critical infrastructure will serve as the nerve center for national cybersecurity monitoring and incident response.



Key N-SOC Components:

- 24/7 security monitoring capabilities for Critical Information Infrastructure (CII)
- Digital forensics laboratory for cybercrime investigation support
- Threat intelligence platform for proactive defense
- Integration with N-CERT for coordinated incident response

7.3.4 Revitalizing the Research and Innovation Centre (RIC)

The Research and Innovation Centre (RIC) initiative was originally conceived as the "brain" of the EDGE project; designed to establish 10 cutting-edge R&D hubs across public universities to foster Industry-Academia collaboration. However, a diagnostic review by the Interim Government revealed severe planning mismanagement and financial irregularities that had rendered the initiative largely ineffective.

Instead of functioning as incubators for "Deep Tech" and national problem-solving, the RICs had drifted into administrative lethargy. A significant portion of the funds allocated for research was diverted towards non-essential "soft" activities such as lavish opening ceremonies, repetitive seminars, foreign tours, and general training programs; rather

than actual hardware procurement, prototyping, or patent development. A review of the RIC projects to disburse the rest of the funds found severe irregularities in project selection and evaluation mechanisms both from the government and the donor organization's end. Promising projects suffered from poor procurement planning, hindering successful implementation.

Consequently, despite heavy investment, the initiative failed to produce tangible intellectual property or commercially viable solutions, leading to critical observations from development partners regarding the "misuse of public funds." At the same time, the project planning did not incorporate any scope for professional expertise in identifying potential IP and negotiating IP and revenue share with grantees.

The Reform Intervention: Strategic Realignment To salvage this vital component, ICTD has reformed the RIC operational framework. The focus has been shifted from "Event Management" to "Product Development."

Financial Discipline and Allocation Reform

- **Strict "R&D First" Allocation:** The budget structure has been overhauled to cap "administrative and event" expenses at a strict minimum. 80% of RIC grants are now mandatorily allocated to direct research costs—including lab equipment (GPUs, specialized sensors), prototyping materials, and researcher stipends.
- **Audit and Accountability:** A retrospective audit has been initiated to investigate previous funding anomalies. Going forward, a "Milestone-Based Tranche System" has been introduced. Funds are no longer released as a lump sum; they are unlocked only upon the verification of tangible research outputs (e.g., successful prototype, published peer-reviewed paper, or industry partnership agreement).

Planning and Governance Restructuring

- **Centralized Problem Bank:** Previously, research topics were chosen arbitrarily, leading to disconnected and low-impact projects. The reform introduces a curated list of challenges sourced from industry and government agencies (e.g., AI for traffic management, agricultural IoT, Bangla NLP). RICs will now select projects solely from this strategic list to ensure national relevance.
- **Industry-Led Steering Committees:** The governance structure has been democratized. The academic-heavy committees have been reconstituted to include industry CTOs and Deep-Tech founders. This ensures that research proposals are vetted for commercial viability and market readiness, bridging the gap between university labs and the industrial ecosystem.

Operational Focus: From General IT to Deep Tech

- **Specialized Clusters:** To prevent duplication of effort, each of the 10 RICs should be focused towards becoming a specific domain based on their faculty strength (e.g., BUET for Robotics, SUST for AI/NLP, KUET for Mechatronics).

This specialization ensures focused resource utilization and fosters domain expertise.

- **Intellectual Property (IP) Mandate:** The reform will mandate that RICs must target Patent Filing and University Spin-offs as their primary KPIs. The success of a center is now measured not by the number of students trained, but by the number of startups incubated and patents registered.

These corrective measures are rapidly transforming RICs from passive "training centers" into active Innovation Engines. By stopping the leakage of funds into non-productive channels and enforcing a rigorous, output-oriented planning matrix, the EDGE project ensures that the RICs finally deliver on their promise of creating a self-reliant, knowledge-based economy.

7.3.5 Reform Priorities

1. **Sustainable Operation Model:** Establish a sustainable operation model post-World Bank support, incorporating skilled personnel from both government and private sectors for continuous operations.
2. **Capacity Building Integration:** Align with national cybersecurity certification tracks and encourage academic-industry-government collaboration to ensure an ongoing supply of skilled cybersecurity professionals.
3. **Technology Transfer:** Ensure that financing models prioritize capacity transfer clauses so that local agencies retain skills and systems after donor project completion.

7.4 Digital Economy and Innovation Ecosystem Development (DEIED) Reform

7.4.1 Strategic Context

The Digital Entrepreneurship and Innovation Ecosystem Development (DEIED) project serves as the primary "soft-infrastructure" engine of the Bangladesh Hi-Tech Park Authority (BHTPA). While physical parks provide the shell, DEIED provides the soul—nurturing human capital, fostering startup growth, and creating a decentralized innovation network across eight divisions.

However, previous iterations of the project suffered from disconnected planning, where equipment was procured without user consultation, and startup funding decisions were driven by bureaucratic procedure rather than market viability. The reformed strategy shifts DEIED from a procurement-heavy initiative to a Data-Driven Ecosystem Enabler, anchored by a robust digital backbone and industry-led governance.

7.4.2 Reform Focus Areas & Key Components

The reform focuses on optimizing four key pillars: **Participatory Innovation Hubs, The Integrated Digital Backbone (LMS/MIS), Lifecycle Accelerator Support, and Expert-Led Governance.**

A. University Innovation Hubs (UIHP): Optimization & Resource Sharing

Objective: Transform university campuses into active "Co-Creation Spaces" rather than just equipment storage rooms.

Reform Action: The operationalization of UIHPs in public and private universities will now follow a "**Demand-Driven**" model to avoid the "Ghost Lab" syndrome (where expensive equipment gathers dust).

- **Participatory Needs Assessment:** Before equipping a hub, a mandatory consultation phase involving **Student Bodies and Faculty Researchers** must be conducted. This ensures that the fabrication labs (FabLabs) are stocked with tools relevant to that university's specific research strengths (e.g., IoT sensors for an Agritech-focused university vs. GPU clusters for an AI-focused one).
- **Resource Rationalization (Shared Economy Model):** To prevent duplication of expenditure, hubs will operate on a "Shared Resource" policy. Expensive assets (e.g., high-end 3D metal printers, PCB assembly lines) will be cataloged centrally. A student from University A can book time on a machine at University B via the central management system, maximizing asset utilization.
- **Active Mentorship:** Hubs will not just be physical spaces but will house "Entrepreneurs-in-Residence"—industry leaders who provide direct guidance to student teams, bridging the gap between academic theory and commercial reality.

B. The Digital Backbone: UIHP Integrated LMS/MIS

Objective: Replace manual, fragmented tracking with a unified **Learning Management System (LMS) and Management Information System (MIS)** that tracks a startup's journey from ideation to unicorn status.

Reform Action: Implementation of the **UIHP LMS/MIS**, a secure, role-based platform that acts as the operating system for the national innovation ecosystem.

1. Lifecycle Tracking & Stage-Gated Support:

- The system will digitize the entire startup lifecycle: **Application → Ideation → Prototyping → Pre-Seed Grant → Seed Grant → Commercialization.**
- It utilizes "Stage-Gate" workflows where funding is released only upon the verification of milestones (e.g., prototype completion), replacing arbitrary disbursements.

2. Intellectual Property (IP) & Data Protection:

- **RBAC & PDPO Compliance:** To protect sensitive startup ideas and research data, the platform implements strict **Role-Based Access Control (RBAC)**.
- **Privacy First:** Aligned with the **Personal Data Protection Ordinance (PDPO)**, the system manages consent for data sharing. Access to intellectual property (IP) is restricted to authorized mentors and jury members only, with comprehensive audit logs to prevent IP theft.

3. Advanced Mentoring & Learning Analytics:

- **Mentor Matching:** The system uses domain tagging to match student teams with the right industry mentors (e.g., matching a fintech team with a banking CTO).
- **Learning Analytics:** An AI-driven dashboard tracks student progress, engagement, and curriculum effectiveness, allowing BHTPA to measure the *quality* of innovation, not just the quantity of participants.

4. National Innovation Repository:

- The system feeds into a **Central Digital Hub**, creating an institutional memory of all projects. This prevents "reinventing the wheel" by allowing new cohorts to learn from the case studies and technical outcomes of previous teams (where IP sharing is permitted).

C. Structured Accelerator Programs: The "Unicorn Pipeline"

Objective: Provide continuous support for startups *between* phases, ensuring they survive the "Valley of Death."

Reform Action: Moving beyond generic training, the project will implement targeted tracks with continuous support:

- **Women-Led Digital Enterprise Accelerator:** Dedicated support for female founders to increase gender diversity, featuring specific mentorship on overcoming systemic funding barriers.
- **Deep-Tech & Growth Tracks:** Specialized support for high-potential ventures (Series A readiness). The LMS/MIS will flag high-performing startups for "Fast-Track" support, connecting them directly with Venture Capitalists even after they graduate from the university program.
- **Post-Graduation Tracking:** The system will continue to monitor alumni startups for 2–5 years post-graduation, tracking revenue, job creation, and IP filings to measure long-term economic impact.

D. Expert-Led Governance & Triple-Helix Coordination

Objective: Remove bureaucratic bottlenecks in funding selection and policy planning.

Reform Action: Radical restructuring of the decision-making committees.

- **Decentralized Selection Committees:** Startup selection and funding decisions will no longer be dominated by generalist bureaucrats. Selection committees must comprise **70% Industry Specialists**, including:
 - Active Venture Capitalists (VCs).
 - Domain Experts (e.g., Doctors for Healthtech, Engineers for Robotics).
 - Successful Founders.
- **Triple-Helix Coordination Council:** A standing body representing **Industry, Academia, and Regulators** will meet quarterly to align DEIED activities with market needs. This ensures that:
 - **Academia** produces relevant research.
 - **Industry** provides the problem statements and funding.
 - **Regulators** (Central Bank, BTRC) clear policy hurdles for new innovations (Sandboxes).

7.4.3 Strategic Impact

By combining participatory planning, a secure digital backbone (LMS/MIS), and expert-led governance, the reformed DEIED project will transition from a grant-giving mechanism to a true National Innovation Engine, capable of producing sustainable, investable, and globally competitive startups.

7.5 Institute of Frontier Technology (IFT) Establishment (Formerly BIFT)

7.5.1 Strategic Context

The **Institute of Frontier Technology (IFT)** represents a strategic evolution of the project formerly known as the *Bangladesh Institute of Frontier Technology (BIFT)*. Shifting from a standalone infrastructure project, IFT is now being established as a premier academic and research institution through a **Memorandum of Understanding (MoU)** between the **Bangladesh Hi-Tech Park Authority (BHTPA)** and **BUET**, signed on September 10, 2025.

7.5.2 Reform Focus Areas

A. Strategic Partnership with BUET

- **Model:** The institute will operate under a "Double Affiliation" model where BUET faculty members serve as directors and adjunct faculty, ensuring academic rigor and global standards.

- **Governance:** governed by a high-level **Board of Governors (BoG)** chaired by the Vice Chancellor of BUET, ensuring autonomy and agility.

B. Dual-Campus Structure To balance academic depth with regional development, IFT will operate across two campuses:

1. **BUET Campus, Dhaka:** Focusing on high-end research, postgraduate education, and deep-tech innovation.
2. **Shibchar, Madaripur:** Serving as an extended campus for specialized training, large-scale infrastructure, and applied technology labs.

C. Core Mandate & Facilities

- **Frontier Technologies:** Dedicated research wings for AI, Robotics, Quantum Computing, and Semiconductor design.
- **Zero-Waste Campus:** The infrastructure will be designed as a green, carbon-neutral campus featuring at least **four advanced innovation labs** and space for **eight resident startups**.
- **Sustainability:** A self-sustaining operational model will be developed, leveraging revenue from specialized training, patents, and industry collaborations.

D. Governance Structure

To ensure autonomy and academic rigor, the IFT will be governed by a **14-member Board of Governors (BoG)** chaired by the **Vice Chancellor of BUET**. The Board includes representatives from UGC, BHTPA, line ministries, and industry experts.
+2

E. Sustainability Model

BHTPA will provide capital funding for construction and labs , as well as operating budgets for the initial two years. A sustainable operational framework, including revenue sharing from training and patents, will be developed jointly.

7.6 Enhancement of Bangla Language in ICT (EBLICT) Reform

7.6.1 Background and Context

The “Enhancement of Bangla Language in ICT through Research & Development (EBLICT)” initiative is a national R&D and service delivery programme anchored in Bangladesh Computer Council (BCC) under the ICT Division. Its core mandate is to strengthen Bangla language enablement across government and citizen-facing digital services by developing reusable language technologies (NLP, speech, and translation),

building high-quality datasets, and operationalizing these assets as platforms (e.g., bangla.gov.bd) and public-facing tools.

The attached brochures illustrate how EBLICT has already moved beyond research prototypes to deploy multiple production services such as Bangla machine translation, spell/grammar correction (‘‘সঠিক’’), sentiment analysis, a Bangla virtual assistant, and speech-to-text/text-to-speech. The diverse services offered positions the programme as a foundational layer for Digital Public Infrastructure (DPI) and service transformation.

In 2025, the government also launched ‘‘kagoj.ai’’, a Bangla-centric AI platform aimed at accelerating official documentation and digitization workflows through Bangla OCR and writing assistance, with integrated speech capabilities. This expands the national product portfolio and creates a clear opportunity to rationalize, standardize, and scale Bangla language AI services under a unified operating model.

7.6.2 Current Status and Accomplishments

The EBLICT project has made significant strides in digitizing the Bangla language and creating foundational NLP tools. Recent launches, including the **Kagoj.ai** platform, demonstrate the project's evolving capability to deliver consumer-facing AI products.

Key Achievements:

- **Integrated Service Platform (bangla.gov.bd):** A centralized portal hosting AI and language technologies, including a corpus platform, resource repository, and developer control dashboard.
- **Kagoj.ai:** A recently launched, state-of-the-art AI platform dedicated to document processing. It features advanced OCR (Optical Character Recognition) for digitization, text-to-speech, and speech-to-text capabilities, serving as a critical tool for digitizing government archives and legal documents.
- **Bangla Machine Translator:** A neural machine translation system capable of translating between Bangla and 10 major world languages (including English, Arabic, French, Spanish, and Japanese), facilitating cross-border communication and commerce.
- **Essential NLP Tools:** Development of 16 core software components, including:
 - **‘Dhwani’ (IPA Converter):** For phonetic standardization.
 - **Screen Reader:** Enhancing digital accessibility for the visually impaired.
 - **Sentiment Analysis:** Tools for analyzing public opinion and social media data.

- **Bangla Spell & Grammar Checker:** Ensuring linguistic accuracy in digital content.
- **Crowdsourcing Platform:** A dedicated initiative (collection.bangla.gov.bd) to collect diverse linguistic data from the public to train robust machine learning models.

Asset / Service	Channel	Public endpoint	Primary users	Notes
Bangla language tech portal	Web	bangla.gov.bd	Citizens, agencies, developers	Unified access to tools/resources.
Spell & grammar checker (“সঠিক”)	Web/App	spell.bangla.gov.bd	Citizens, editors, govt staff	Grammar + spelling suggestions; privacy policy indicates encrypted transmission.
Sentiment analysis	Web/API	sentiment.bangla.gov.bd	Service owners, analysts	Monitoring feedback and social content classification.
Bangla virtual assistant	Web/Chat	(via bangla.gov.bd)	Citizens, service desks	Conversational interface for service guidance.
Speech-to-text (STT)	Web/API	stt.bangla.gov.bd	Govt offices, accessibility use-cases	Dictation and transcription; enables faster documentation.

Text-to-speech (TTS)	Web/API	tts.bangla.gov.bd	Accessibility, voice UX	Screen reading and voice responses in Bangla.
Kagoj.ai (Bangla AI platform)	Web	kagoj.ai	Govt officials, enterprises, citizens	Bangla AI platform for writing, official documentation, and OCR-based digitization.

7.6.3 Rationale for Reform and Current Challenges

Despite these successes, the project operates within the constraints of a development project cycle, which poses risks to long-term sustainability and scalability.

1. **Sustainability Risk:** As a "project," EBLICT faces funding cycles that threaten the 24/7 availability and maintenance of critical APIs and services once the project timeline concludes.
2. **Closed Ecosystem:** Currently, many tools operate as web-based interfaces or closed APIs. While many datasets have been made public and some services have public facing APIs, and There is a lack of open access to model weights and raw datasets, which stifles innovation in the private sector and academia.
3. **Technological Shift:** The global AI landscape has shifted towards Large Language Models (LLMs), Vision Language Models (VLM), and edge computing. The current infrastructure needs to evolve from basic NLP tools to advanced, multimodal Generative AI systems.
4. **Developer Experience:** The lack of standardized, production-ready Client Tools (SDKs) and public APIs limits the integration of these technologies into third-party applications by startups and software firms.
5. **Fragmented delivery:** multiple tools/endpoints without a single, versioned API catalogue and developer onboarding path.
6. **Production reliability and capacity:** limited 24/7 operations, monitoring, and incident response for citizen-scale workloads.

7. **Compute constraints for modern models:** GPU availability, cost, and scheduling; difficulty supporting large models, VLM, and speech at scale.
8. **Data governance and licensing:** ensuring datasets are auditable, rights-cleared, and privacy-preserving while still usable for training and open release.
9. **Quality evaluation gaps:** Benchmark regimes across MT, STT, TTS, OCR, and LLM-style tasks; missing acceptance criteria for releases.
10. **Talent and research pipeline:** sustaining top-tier publications and conference participation requires structured research time, partnerships, and incentives.
11. **Interoperability:** weak integration patterns with other DPI layers and with government cloud hosting standards.

7.6.4 Reform vision and strategic objective

The reform goal is to evolve EBLICT from a collection of useful tools into a unified national platform for Bangla language AI—operated with production-grade reliability, aligned to government digital architecture, and designed for rapid reuse by ministries, private innovators, and academia.

Strategic objectives:

- API-first platformization: expose each core capability (MT, OCR, STT, TTS, spell/grammar, sentiment, assistant) through documented, versioned APIs and SDKs.
- Open assets where feasible: publish model weights (including modular weights for large models), training/evaluation datasets, and documentation under clear licenses and governance.
- Next-generation capability development: establish VLM (vision-language) R&D for OCR+understanding, form understanding, and multimodal assistants.
- Sustainable data flywheel: deploy a crowdsourced data collection and labeling toolchain (parser–pause tagger, human-in-the-loop review) to continuously improve models.
- Operational excellence: transition selected services to 24/7 operation with defined SLAs, hosted on government cloud infrastructure (NDC and/or national cloud options).
- Advanced efficiency: prune and compress models to reduce inference cost and enable edge deployments (offline/low-bandwidth use cases).

- Global research visibility: increase submissions to top AI/NLP/VLM venues and Q1 journal publications; strengthen international collaborations.

7.6.5 Strategic Future Outlooks

To address these challenges and position Bangladesh as a leader in AI R&D, the EBLICT initiative will undergo a comprehensive reform focusing on eight strategic pillars:

1. API-First Architecture and Client Tools

The reform will transition EBLICT from a portal-based service to an **API-first platform**. All core technologies (OCR, TTS, Translation, Kagoj.ai services) will be exposed via standardized, high-performance APIs.

- **Action:** Release official Client SDKs (Software Development Kits) for Python, JavaScript, and mobile platforms (iOS/Android) to allow developers to integrate Bangla AI capabilities into their apps seamlessly.

2. Open Science Initiative: Public Model Weights and Datasets

Moving away from a "black box" approach, the project will embrace Open Science principles.

- **Action:** Publish pre-trained model weights (e.g., BERT, GPT-style models, translation engines) and anonymized datasets on national repositories and global hubs like HuggingFace.
- **Modular Release:** Release large models in modular structures to allow researchers to fine-tune specific components without high compute costs.

3. Vision Language Models (VLM) Development

The next frontier of AI involves understanding images and text together.

- **Action:** Develop native VLMs capable of describing images in Bangla, performing visual question answering (VQA), and extracting information from unstructured documents (OCR in the wild). This is critical for accessibility and digitizing archives.

4. Advanced Crowdsourcing Tools

To build high-quality datasets for prosody (speech rhythm) and advanced grammar, simple text collection is insufficient.

- **Action:** Develop sophisticated annotation tools, specifically a **Parser-Pause Tagger**, to collect granular data on speech pauses and syntactic parsing. This will significantly improve the naturalness of TTS engines and the accuracy of grammar checkers.

5. Institutional Merger with BCC for 24/7 Availability

To ensure the permanence of these digital assets, the EBLICT project will be institutionalized.

- **Action:** Merge the project's infrastructure and personnel into the **Bangladesh Computer Council (BCC)** as a permanent "National Language Technology Wing."
- **Service Level:** Transition to government cloud hosting with guaranteed 24/7 uptime, redundant architecture, and dedicated DevOps support to serve critical national applications (e.g., judicial systems, emergency services).

6. Robotics and IoT in Native Languages

AI must move from the cloud to the physical world.

- **Action:** Research and develop lightweight models for Robotics and Internet of Things (IoT) devices that support voice commands and interaction in **Bangla and the ethnic languages of Bangladesh**. This ensures that smart home devices, agricultural IoT, and service robots remain accessible to the entire population.

7. Model Pruning and Edge AI

To reduce inference costs and enable AI on low-cost smartphones, model efficiency is paramount.

- **Action:** Establish a dedicated R&D track for **Model Pruning, Quantization, and Distillation**. The goal is to create "TinyML" versions of Bangla models that run entirely on-device, preserving privacy and reducing latency.

8. Research Excellence and Global Presence

The reform aims to elevate Bangladesh's status in the global AI research community.

- **Action:** Shift focus from software development to scientific contribution. Mandate and incentivize the publication of research findings in **AAA-ranked conferences** (ACL, NeurIPS, ICLR, CVPR) and **Q1 journals**. This peer-reviewed validation ensures our technology meets global state-of-the-art standards.

7.6.6 Reform KPI Framework (2025-2030)

Metric	Baseline (2024)	Target (2027)	Target (2030)
Service Availability	Project-based (Variable)	99.5% Uptime (24/7)	99.9% Uptime (Mission Critical)
Open Model Weights	0 Public Models	10+ Major Models Published	50+ Models (LLM, VLM, Audio)
API Integration	Internal Use Only	Used by 200+ Startups/Gov Apps	Used by 1,000+ External Apps
Research Output	Local Conferences	5 Papers in AAA Conf/Q1 Journals	20+ Papers in AAA Conf/Q1 Journals
Language Coverage	Standard Bangla	Bangla + 2 Ethnic Languages	Bangla + 10 Ethnic Languages
Edge AI Capable	Cloud Only	3 Models Pruned for Mobile	Full Suite of On-Device Models
Dataset Size	<100 GB	1 TB High-Quality Text/Audio	10 TB Multimodal Data

7.6.7 Expected Outcomes

By 2030, the reformed EBLICT initiative will transform Bangla into a "first-class citizen" in the AI world. The transition to an API-first, open-science model will spawn a local ecosystem of AI startups building on top of national models. The institutional merger with BCC will guarantee that the digital voice of Bangladesh remains active, resilient,

and accessible to future generations, preserving our linguistic heritage while powering our digital future.

7.7 Public Procurement Reform for ICT Projects

7.7.1 Role of PPA and PPR in Digital Transformation

The Public Procurement Act (PPA) 2006 and the Public Procurement Rules (PPR) 2008 laid the legal and institutional foundation for transparent and technology-driven procurement. The implementation of the Electronic Government Procurement (e-GP) system represents one of the most significant milestones achieved under this legal framework.

e-GP System Achievements (as of April 2025):

Metric	Value
Agencies on e-GP	1,483
Procuring Entities (PEs)	12,975
Registered Tenderers	130,640
Tenders Invited	935,576
Contract Awards	704,404
Total Tender Value	USD 113.55 Billion
Total Contract Value	USD 72.80 Billion
Estimated Savings	USD 1.1 Billion

7.7.2 Current Challenges in e-GP Implementation

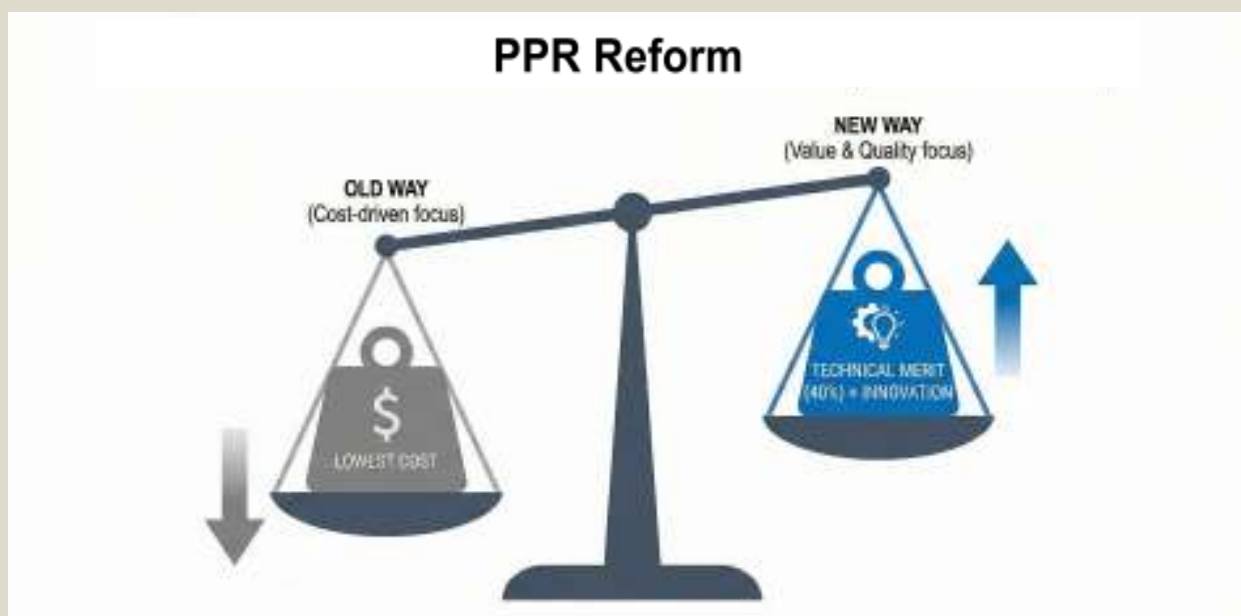
- **Digital Literacy Gap:** Many tenderers, contractors, and consultants lack digital equipment and skills for effective participation.
- **Capacity Constraints:** Procuring entities, especially at local levels, lack trained personnel to operate the system effectively.
- **Connectivity Issues:** Contractors in rural areas face internet access challenges, limiting participation in digital procurement.
- **System Performance:** Users experience slow response times during peak usage and limited mobile device support.

- **Regulatory Gaps:** Current legal framework does not fully address cloud hosting, digital signatures, and e-document validation requirements.
- **Underutilized Data:** Extensive procurement data is not adequately used for performance monitoring, analytics, or risk mitigation.

7.7.3 PPR Reforms for Software Procurement

Bangladesh's existing Public Procurement Rules often prioritize cost over quality, resulting in suboptimal outcomes including vendor dependencies and integration issues. Drawing from global best practices including the UK's agile procurement frameworks, the EU's value-driven evaluations, and Australia's digital marketplaces, the following reforms are recommended:

1. **Balanced Evaluation Model:** Transition to evaluation criteria where technical merits, including innovation and long-term scalability, account for at least 40% of assessment, while mandating clauses for open APIs and data interoperability to prevent vendor lock-in.



2. **Agile Procurement Practices:** Introduce agile methodologies by segmenting large projects into phased deliverables, favoring open-source solutions and cloud-based deployments with mandatory third-party security audits.
3. **Enhanced Tender Processes:** Upgrade the e-GP platform with AI-assisted bid analysis for fairness, incorporate provisions for joint ventures with private entities, and streamline approvals to reduce procurement timelines.
4. **Citizen-Centric Outcome Requirements:** Revise procurement rules to require mandatory user testing with specific groups (women, rural residents, persons with disabilities), service-level KPIs (completion rate, user satisfaction, processing time), and clear government ownership of data and systems.

7.7.4 Implementation Timeline

Pilot reforms will launch in 2025 for select projects, achieving full rollout by 2026 with ongoing evaluations. The recent amendment (4th May 2025) of Section 65 of the PPA made it mandatory for all Procuring Entities to use the e-GP portal where applicable, establishing the legal foundation for comprehensive digital procurement reform.

7.8 Consolidated Project Reform KPI Framework

The following framework establishes measurable indicators to track the progress and impact of project reforms across all flagship ICT initiatives:

Reform Area	KPI	2025 Target	2030 Target
EDC Project	Third-party verification compliance	100%	100%
EDC Project	Cost per connection vs. benchmark	≤1.5x	≤1.1x
EDGE Project	N-SOC operational status	Phase 1 deployed	Full operation
EDGE Project	CII coverage by N-SOC	50%	100%
NDC/BDCCL	Data center capacity utilization	Optimized	Scalable platform
NDC/BDCCL	Disaster recovery readiness	Plan approved	Fully operational
PPR Reform	Agile procurement pilots	5 projects	All ICT projects
PPR Reform	Technical merit weight in evaluation	≥40%	≥40%
Overall	Projects with NDGA compliance	50%	100%

7.9 Strategic Project Planning & Design Improvement

7.9.1 The Crisis of Poor Planning

Historically, ICT projects in Bangladesh have suffered from a "Plan-as-you-go" approach, leading to severe disconnects between objectives and outcomes. Projects are often designed in silos by individual ministries without a holistic view of the national digital architecture. This has resulted in:

- **Duplication of Expenditure:** Multiple agencies procuring identical software (ERPs, Inventory Systems) or infrastructure, wasting hundreds of crores in public funds.
- **Disconnect from Reality:** Projects designed by generalist bureaucrats or retired officials often lack technical depth, leading to procurement specifications that are either obsolete or vendor-locked from day one.
- **Implementation Delays:** The absence of mapped-out phases means critical dependencies (e.g., waiting for fiber optics before buying servers) are missed, causing years of delay and cost overruns.

7.9.2 Reform Framework

The "Design-First" Mandate To rectify this, the government is instituting a rigorous **Project Design & Planning Framework** that must be cleared before any Development Project Proposal (DPP) is approved.

A. Expert-Led Planning Committees

- **Domain Expertise Over Political Appointments:** Project steering committees must now include at least 40% **technical domain experts** (industry CTOs, solution architects, academic researchers) rather than relying solely on generalist officers or retired bureaucrats.
- **Role of the Chief Architect:** No large-scale digital project (>50 Crore) will be approved without a technical vetting by the **National Data Governance Authority (NDGA)** to ensure compliance with the Bangladesh National Digital Architecture (BNDA).

B. Elimination of Silos & Duplication

- **The "Check-Before-Build" Rule:** Before procuring new software, agencies must certify that no similar asset exists in the **National Source Code Repository**. Reusing existing government code is now mandatory where feasible.
- **Unified Feasibility Studies:** For infrastructure projects, a centralized feasibility study by the Planning Commission's digital wing will replace fragmented agency-level studies, ensuring a sector-wide view of requirements.

C. Coordination with Development Partners

- **Donor Harmonization:** A new **Digital Development Coordination Cell (DDCC)** will be established to synchronize efforts with the World Bank, ADB, JICA, and other partners. This ensures that foreign-funded projects do not overlap but

rather complement national priorities (e.g., one donor funds the backbone, another funds the last-mile access).

- **Data-Driven Policy Making:** Project planning must be based on real-time data from the **National Digital Dashboard**, ensuring that investments target actual gaps (e.g., regions with low internet penetration) rather than politically selected constituencies.

7.9.3 Impact

This reform shifts the focus from "spending budget" to "delivering value." By ensuring that projects are professionally designed, technically vetted, and strategically aligned *before* a single Taka is spent, the government aims to eliminate the "Digital Waste" that has plagued previous administrations and ensure that public money truly benefits the citizen.

7.10 Accountability and Monitoring Framework

7.10.1 Governance Structure

Effective project reform requires robust accountability mechanisms and continuous monitoring. The following governance structure ensures oversight and timely intervention:

1. **Project Reform Steering Committee:** A high-level committee chaired by the Secretary of ICT Division with representation from BCC, a2i, NCSA, and relevant ministries to provide strategic oversight and resolve inter-agency issues.
2. **Independent Technical Review Board:** External experts from academia and industry to conduct quarterly technical assessments and provide recommendations for course corrections.
3. **Real-Time Monitoring Dashboard:** A National Digital Dashboard consolidating data from all flagship projects, enabling real-time progress monitoring accessible to policymakers and stakeholders.
4. **Public Accountability Mechanisms:** Annual Digital Transformation Scorecards published to benchmark Bangladesh against regional and global peers, with citizen feedback mechanisms including surveys, grievance redressal portals, and social media analytics.



7.10.2 Enforcement Mechanisms

- Performance reviews linked to budgetary allocations for ministries and agencies
- Divisional investigations under PPR for identified irregularities with clear penalty provisions
- Mandatory disclosure of project status, expenditure, and deviation reports
- Integration of reform compliance into annual civil service performance evaluations

7.11 Conclusion and Future Outlook

Project reform is not merely a corrective measure but a strategic imperative for Bangladesh's digital transformation. The reforms outlined in this chapter address systemic weaknesses while establishing governance frameworks that will prevent recurrence of past challenges. By re-baselining costs, enforcing technical compliance, mandating third-party verification, and aligning all projects with NDGA standards, Bangladesh is building a foundation for sustainable, transparent, and citizen-centric digital infrastructure.

The success of these reforms will be measured not only by improved project outcomes but also by enhanced public trust in government ICT initiatives. As Bangladesh progresses toward its vision of becoming a regional digital leader by 2030, the lessons learned from project reform will inform future initiatives, ensuring that every public investment in digital infrastructure delivers maximum value to citizens and contributes to inclusive national development.

The integration of reformed projects with the broader Digital Public Infrastructure ecosystem will enable Bangladesh to achieve its ambitious targets: \$5 billion in ICT

exports, 7-8 million ICT professionals, and a position among the top 15 nations in the UN E-Government Development Index. These outcomes depend on maintaining reform momentum, institutionalizing accountability mechanisms, and continuously adapting to technological and governance challenges.

End of Chapter 7

8: Startup Ecosystem and Innovation

8.1 Introduction

Over the last decade, Bangladesh's startup ecosystem has shown promising growth, driven by digital adoption, a young population, and government-led innovation initiatives. The ecosystem has evolved from fewer than 100 companies to a dynamic landscape where digital startups across fintech, e-commerce, logistics, healthtech, and education are addressing real-world challenges and creating employment opportunities. Success stories like bKash—which attracted a US\$250 million investment from SoftBank—have demonstrated that Bangladeshi startups can achieve venture-scale growth and global recognition.

However, the ecosystem remains fragmented, with multiple government agencies working in silos, overlapping mandates, and inconsistent policy execution. Access to early-stage finance remains constrained, links between universities and industry are weak, and regulatory bottlenecks impede growth. Compared to regional peers like India and Vietnam, Bangladesh's venture capital flows remain small, and startups face high barriers to scaling both domestically and internationally.

To address these structural inefficiencies and transform the digital economy into a driver of inclusive growth and exports, this chapter proposes the establishment of a Startup Authority / Commission—a centralized, empowered body responsible for policy formulation, implementation, coordination, enforcement, and approval of new initiatives under the government system. The Authority will act as the single coordinating backbone for the startup ecosystem, overseeing fund-of-funds (FoF), VC and accelerator enablement, incubation, infrastructure, R&D, government service digitization, regulatory facilitation, and innovation infrastructure such as makers labs.

8.2 Current Ecosystem Landscape

8.2.1 Ecosystem Strengths

Despite coordination challenges, Bangladesh's startup ecosystem has several inherent strengths that provide a foundation for accelerated growth:

Government Commitment to Innovation: Dedicated programs such as innovation funds, iDEA, ICT innovation grants, and digitization initiatives signal strong policy intent. State-backed venture capital initiatives like Startup Bangladesh Limited (SBL), established in March 2020 with a US\$65 million flagship fund, provide early market confidence and demonstrate government commitment to the startup sector.

Growing Entrepreneurial Talent Pool: Bangladesh has a large youth demographic with increasing exposure to technology, freelancing, and digital entrepreneurship. According to the Oxford Internet Institute's Online Labour Index, Bangladesh accounts for approximately 15% of global freelance labour supply, ranking in the top three

countries globally. BASIS estimates over 650,000 active freelancers, with more than 500,000 registered on international platforms generating approximately US\$400 million annually in foreign exchange. This freelance ecosystem feeds directly into the startup talent pipeline.

Emerging Investment Landscape: The presence of global and local VCs, angel networks, and co-investment mechanisms has grown steadily. Initial success stories—bKash, Pathao, ShopUp, Chaldal, ShareTrip—have created proof-of-concept for venture-scale businesses, attracting international investor attention to the Bangladesh market.

Digital Public Infrastructure Foundation: National ID systems, mobile financial services (MFS), and expanding broadband coverage provide a strong base for civic tech and govtech startups. With over 100 million mobile financial service users and growing internet penetration, Bangladesh offers a substantial addressable market for digital solutions.

8.2.2 Critical Challenges

Despite progress, startups continue to face systemic barriers that constrain ecosystem growth:

Fragmented Institutional Landscape: Multiple ministries and agencies operate independently, leading to duplication, delays, and policy inconsistency. IDEA, SBL, BHTPA, a2i, and various ministry-level initiatives often work in silos without coordinated strategy or shared objectives. This fragmentation creates confusion for entrepreneurs and investors alike, and results in inefficient allocation of public resources.

Weak Regulatory Navigation: Startups struggle to understand and comply with regulations across sectors—finance, health, telecom, logistics. There is no single-window mechanism for approvals or guidance. A fintech startup, for instance, may need to navigate Bangladesh Bank regulations, ICT Division policies, and sector-specific compliance requirements without coordinated support. This regulatory friction diverts entrepreneurial energy from innovation to bureaucratic navigation.

Inefficient Capital Deployment: Public innovation funds often focus on grants rather than market-linked outcomes. While grants serve a purpose in early ideation, they do not build the discipline required for venture-scale growth. Additionally, there is limited capacity building for VCs, accelerators, and angels to operate sustainably. The absence of a vibrant domestic VC ecosystem means Bangladesh lacks the institutional investor base to fuel early-stage growth—unlike India and Vietnam where local VCs and corporates anchor early funding rounds.

Limited R&D and Commercialization Pathways: Research funding is largely disconnected from startup commercialization pathways. Universities produce research papers, but these rarely convert into commercial products, patents, or startups. The

weak linkage between academia, industry, and startups represents a significant missed opportunity for technology transfer and innovation-driven economic growth.

Governance Deficit: Allegations of inefficiency and lack of transparency reduce ecosystem trust. Without performance-based accountability mechanisms, it is difficult to assess which interventions work and which resources are being deployed effectively. This governance gap undermines both domestic and international investor confidence.

Limited Exit Pathways: Rare successful exits prevent the recycling of capital and experience. Without exits, investor confidence remains low and the founder-to-founder investment cycle—critical for ecosystem maturation—cannot develop. The absence of a startup-friendly IPO framework at the Dhaka Stock Exchange further constrains exit options.

Geographic Concentration: Innovation activity remains concentrated in Dhaka and, to a lesser extent, Chattogram. Regional entrepreneurs face limited access to investors, mentors, co-working spaces, and support infrastructure. This concentration not only limits inclusive growth but also leaves regional market opportunities underexplored.

8.3 Strategic Initiatives Updates

A. Investment, Funding & Exit Strategy

1. Direct Startup Investment SBL has accelerated capital deployment to fuel high-potential ventures.

- **Progress:** A total of **BDT 17.15 Crore** has been invested in five key startups: **10 Minute School, Shikho, Chaldal, Bookshunari, and i-Farmer.**
- **Pipeline:** Follow-on investment for one startup is currently in process, while two new investment proposals are pending Board approval.

2. Launch of "Fund of Funds" (FoF) To leverage private capital, SBL is operationalizing a "Fund of Funds" structure.

- **Status:** Policy documents are finalized. **BDT 50 Crore** has been received from the Finance Division, with a request for an additional **BDT 25 Crore** submitted.
- **Implementation:** A "Legal Due Diligence Partner" has been onboarded to ensure compliance. A Request for Expressions of Interest (EOI) for fund managers is ready for release.

3. Portfolio Exit Milestone Demonstrating the maturity of the ecosystem, SBL is executing its first exit.

- **Status:** Exit negotiations with one portfolio startup began in December 2025 and have reached the final stage.
- **Target:** An official exit announcement is expected by the **first week of February 2026.**

B. Policy Formulation & Governance

4. National Startup Policy SBL is leading the formulation of a unified national policy to streamline the startup landscape.

- **Timeline:** The drafting committee held its first meeting on January 6, 2026. Two additional meetings and a workshop are scheduled for Feb-March.
- **Goal:** The final draft will be submitted to the ICT Division by **April 2026**.

5. Formation of "Startup Board" To facilitate public listing and exit routes, SBL is working with the capital market regulator.

- **Action:** A Concept Note has been submitted to the **Bangladesh Securities and Exchange Commission (BSEC)** to form a dedicated "Startup Board."
- **Objective:** This board will provide strategic direction, ensure transparent management, and protect investor interests in the public market.

C. Portfolio Value Creation & Monitoring

6. Portfolio Monitoring & Impact Measurement Monitoring has been strengthened to provide "Smart Capital."

- **Value Addition:** SBL is actively assisting startups with fundraising guidance, business development, and client acquisition.
- **Impact Framework:** An integrated **Impact Measurement & Monitoring (IMM)** framework is being developed to quantify socio-economic contributions. This framework will be incorporated into SBL's future Investment Guidelines.
- **Global Exposure:** "Startup Bangladesh Portfolio Night" was organized to connect portfolio companies directly with foreign investors.

7. Communication & Branding

- **Activity:** Continuous social media promotion and branding activities are being conducted to highlight the achievements of SBL, its portfolio startups, and the broader national ecosystem.

D. Ecosystem Infrastructure & Enablement

8. Ecosystem Data & Knowledge Infrastructure SBL is building the digital backbone for the ecosystem.

- **Startup Universe:** A comprehensive database/dashboard has been created mapping over **1,250 active startups**.
- **Expansion:** Work is ongoing to develop an "Organization Universe" and "People Universe" to map the entire network of stakeholders. The initial dashboard is live and undergoing further development.

9. Advisory, Partnership & Enablement SBL is acting as a strategic advisor to key national and international partners.

- **Partnerships:** Advisory and enablement support provided to **Stanford Seed, Accelerating Bangladesh, EDGE RIC, NSU Startup Next, Build Bangladesh, and Needle Incubator.**
- **Strategic Support:** SBL provided design support for Business Incubation Centers (BIC), structured the "Narayanganj Business Innovation Summit," and advised on the "National Startup Platform."

10. Local & International Events

- **Executed:** SBL successfully hosted "**Bangladesh Startup Connect 2025**", "**Youth Startup Summit 2025**", and two international webinars between March and April 2025 to foster networking and knowledge exchange.

E. Global Connectivity & Capacity Building

11. Global Membership & Integration To internationalize the ecosystem, SBL is building bridges with global venture hubs.

- **Action:** SBL has initiated membership with the **Singapore Venture & Private Capital Association (SVCA).**
- **Benefits:** This membership provides access to a global network of Fund Managers and Limited Partners (LPs), as well as discounted access to capacity-building training and roadshows in Singapore.

12. New Strategic Programs Three new strategic initiatives have been launched to address specific ecosystem needs:

- **Impact Measurement & Management (IMM):** To track sustainable development goals.
- **SBL NRB Connect Program:** To engage Non-Resident Bangladeshi experts and investors.
- **SBL Young Ambassador Program:** To develop future leaders and campus advocates for the startup ecosystem.

13. Team Development Preparing the SBL team for the new "Fund of Funds" mandate.

- **Training:** Initiatives have been taken to equip the team with advanced investment skills. This includes **FMVA certification** from the CFI Institute and weekly internal training sessions to ensure the team is ready to manage larger, more complex capital structures.

8.4 The Case for a Startup Authority / Commission

The challenges outlined above share a common root cause: the absence of centralized coordination, clear mandate allocation, and performance-based governance. Individual agencies—iDEA, SBL, BHTPA, a2i—have achieved results within their domains, but the ecosystem as a whole suffers from fragmentation, overlap, and inconsistent policy execution.

To address this, a Startup Authority / Commission is proposed as a structural shift from fragmented program management to coordinated ecosystem governance. This body would be responsible for:

- **Policy Formulation:** Developing coherent, ecosystem-wide policies that align capital deployment, infrastructure provision, regulatory reform, and talent development
- **Implementation Oversight:** Ensuring that policies translate into action across all stakeholder agencies
- **Coordination:** Serving as the single backbone connecting all ecosystem players—startups, investors, incubators, universities, regulators, and government agencies
- **Enforcement:** Establishing and monitoring performance metrics, ensuring accountability, and course-correcting where necessary
- **Approval Authority:** Streamlining approvals for new initiatives, funding mechanisms, and regulatory relaxations

Critically, this model does not replace existing agencies. Rather, it redefines their roles, improves accountability, and ensures that public resources translate into scalable, sustainable startups. Each agency retains operational responsibility in its area of expertise, but operates within a coordinated framework with clear mandates and performance expectations.

8.4.1 International Benchmarks

The proposed Startup Authority draws on international best practices:

- **Startup India (India):** A flagship initiative providing a single-window clearance system, tax exemptions, simplified compliance, and fund of funds mechanism that has catalyzed one of the world's fastest-growing startup ecosystems
- **Enterprise Singapore:** Combines trade promotion, enterprise development, and startup support under one agency with clear KPIs and accountability mechanisms
- **Startup Chile:** Government-backed accelerator program that attracted global startups through equity-free funding and visa facilitation, demonstrating how coordinated policy can position a country as an innovation hub

- UK's British Business Bank: Operates fund of funds and co-investment programs (including EIS/SEIS tax incentives) that have mobilized billions in private capital for startups

These examples demonstrate that coordinated, empowered institutions with clear mandates can significantly accelerate ecosystem growth while leveraging—rather than crowding out—private capital.

8.5 Central Repository for Innovation, Startup, and Progress (CRISP)

8.5.1 Context: The Need for a Unified Data Backbone

Bangladesh's startup ecosystem has witnessed exponential growth, yet it faces a critical challenge: information asymmetry. Currently, data regarding startups, investors, and support programs is fragmented across siloed databases, making it difficult for policymakers to track growth, for investors to discover investable assets, and for startups to access consolidated resources. To address this, the government will launch **CRISP (Central Repository for Innovation, Startup, and Progress)**—a dynamic, single-source-of-truth platform designed to map and monitor the entire innovation landscape.

8.5.2 Core Objectives and Features

CRISP will serve as the national digital backbone for the startup economy, focusing on three primary objectives:

- **Unified Ecosystem Registry:** Creating a comprehensive, verified directory of all active stakeholders, including startups, investors (VCs, Angels), incubators, accelerators, and mentors.
- **Data-Driven Policymaking:** providing real-time analytics and dashboards to the government (ICT Division, BHTPA) to monitor sector performance, employment generation, and funding inflows, enabling evidence-based policy interventions.
- **Investment Facilitation:** Functioning as a discovery platform where global and local investors can filter startups by sector, stage, and revenue, thereby reducing due diligence friction and accelerating capital deployment.

8.5.3 Functional Architecture

CRISP will be built on a secure, interoperable architecture aligned with the National Digital Architecture (BNDA):

- **Automated Verification:** Integration with **NID** and **RJSC** (Registrar of Joint Stock Companies) APIs to ensure that all registered entities are authentic and compliant.

- **Dynamic Profiles:** Startups will maintain "live" profiles showcasing their traction, fundraising history, and impact metrics, which can be used as a "Digital Passport" to apply for government grants or tax incentives.
- **Interoperability:** The platform will feature open APIs to connect with private sector platforms and international ecosystem trackers, ensuring Bangladesh's data is visible globally.

8.5.4 Strategic Impact

By establishing CRISP, Bangladesh aims to eliminate the "invisible startup" problem. The platform will ensure that support mechanisms—such as grants, tax holidays, and mentorship—reach deserving ventures transparently. Furthermore, it will market the Bangladesh startup ecosystem to the world by presenting a transparent, validated, and growing pipeline of innovation.

8.6 Proposed Institutional Architecture

The following framework assigns clear mandates to each stakeholder within the startup ecosystem, with the Startup Authority / Commission serving as the central coordinating body. This architecture ensures that each agency focuses on its core competency while operating within an integrated, accountable system.

Function	Execution Body	Key Mandate
Policy Oversight	Startup Authority / Commission	Centralized policy formulation, approval, enforcement, and coordination across all ecosystem stakeholders. Sets ecosystem-wide KPIs and monitors performance.
Fund of Funds & VC Enablement	Startup Bangladesh Limited	Developing a market-driven Fund of Funds framework that deploys capital to venture capital firms. Institutionalizes VC Enabler functions to build long-term, profitable fund management capabilities in the domestic market.
Growth Capital Level	Startup Bangladesh Limited	Investment or co-investment in seed, pre-Series A, and Series A rounds. Focus on trust building among ecosystem players, investors, and stakeholders through transparent, market-aligned investment practices.

Accelerator & Angel Enablement	SBL or new organization	Establishing mechanisms to deploy accelerating capital with enabler support. Co-designing inclusive programs including marketplace creation, investment readiness training, and digital academy development to bridge the missing middle.
Incubation	iDEA	Adopting an enablement-first approach by empowering private incubators, accelerators, and venture studios. Fostering a private-sector-driven early-stage support ecosystem rather than government-run incubation.
Innovation Infrastructure	BHTPA	Developing a nationally coordinated infrastructure framework. Institutionalizing Makers Labs to catalyze hardware, deep-tech, and applied innovation. Managing Hi-Tech Park facilities for startup co-location.
R&D Enhancement	ICTD Innovation Fund	Market-aligned R&D funding linked to startups and commercialization pathways. Strengthening university-industry linkages for technology transfer.
Government Services Digitization	a2i	Developing a central design lab (ICTD Playground) as a cross-ministerial innovation platform. Engaging startups in solution design and delivery for national challenges.
Single Point Regulatory Services	All agencies, hosted by BHTPA	Central Regulatory Service Centers with all relevant ministries under one roof for streamlined approvals, compliance guidance, and regulatory navigation.

8.6.1 Rationale for Role Allocation

Each allocation reflects comparative advantage and lessons learned:

SBL for Fund of Funds: As Bangladesh's only government-sponsored VC fund, SBL has the mandate and initial capital base to anchor a Fund of Funds mechanism. Rather than SBL directly picking startup winners—which requires specialized expertise better found in private markets—SBL will select and fund professional VC managers who specialize in startup evaluation, portfolio management, and value addition. This approach leverages market efficiency while deploying government capital to catalyze private investment.

Fund of Funds (FoF) Structure

- Initial FoF Size: BDT 500 Crore, with flexibility to scale further based on performance and ecosystem needs
- Fund Life: 30 years, with option to transition to an evergreen fund thereafter
- Investment Focus: Direct investments in partnership funds established with local and global VC fund managers mandated to invest in Bangladesh
- Ecosystem Support: Portion allocated for government-approved incubators, accelerators, and challenger funds
- Sector Approach: Stage and sector-agnostic, prioritizing high-growth sectors including financial services, e-commerce, logistics, healthcare, education, and agriculture
- ESG Integration: Environmental, Social, and Governance principles embedded to ensure sustainable, long-term value creation

FoF Strategic Objectives

1. Enhance mobilization of foreign capital and promote foreign market access for Bangladesh-focused funds
2. Stimulate local funding to advance domestic VCs and attract funding from local limited partners (LPs) including corporates, family offices, and high-net-worth individuals
3. Drive sustainable development by contributing toward impact goals aligned with national priorities

This represents a fundamental paradigm shift: instead of SBL directly picking startup winners (which requires expertise better found in private markets), SBL will select and fund professional VC managers who specialize in startup evaluation, portfolio management, and value addition. This approach leverages market efficiency while deploying government capital to catalyze private investment.

SBL Recent Investment Activity (September 2024 - January 2025)

During the transition period, SBL invested BDT 11.10 crore across nine startups, supporting both established platforms (Ten Minute School, Sheba Platform, ShareTrip) and early-stage ventures (ParentsCare for elderly healthcare, Dubotech Digital for underwater robotics, Sokrio Technologies for distribution management). This portfolio demonstrates SBL's focus on ventures addressing real-world challenges across healthcare, logistics, education, and technology sectors.

iDEA for Incubation Enablement: iDEA has built infrastructure (co-working spaces, labs) and networks through years of operation. Its future role should shift from direct grant disbursement to enabling a private-sector-driven incubation ecosystem. This means empowering private incubators, accelerators, and venture studios through training, certification, and performance-linked support—rather than competing with them.

BHTPA for Infrastructure: Bangladesh Hi-Tech Park Authority already manages physical infrastructure across multiple locations. Extending this mandate to include Makers Labs, Central Regulatory Service Centers, and regional innovation hubs leverages existing capabilities and real estate assets.

a2i for Govtech: a2i has demonstrated capability in digital service delivery and citizen-centric design through initiatives like the National Portal (over 2 billion transactions processed), Union Digital Centers (8,500+ centers serving 30 million users yearly), and various e-governance platforms. The ICTD Playground concept positions a2i as the bridge between government challenges and startup solutions.

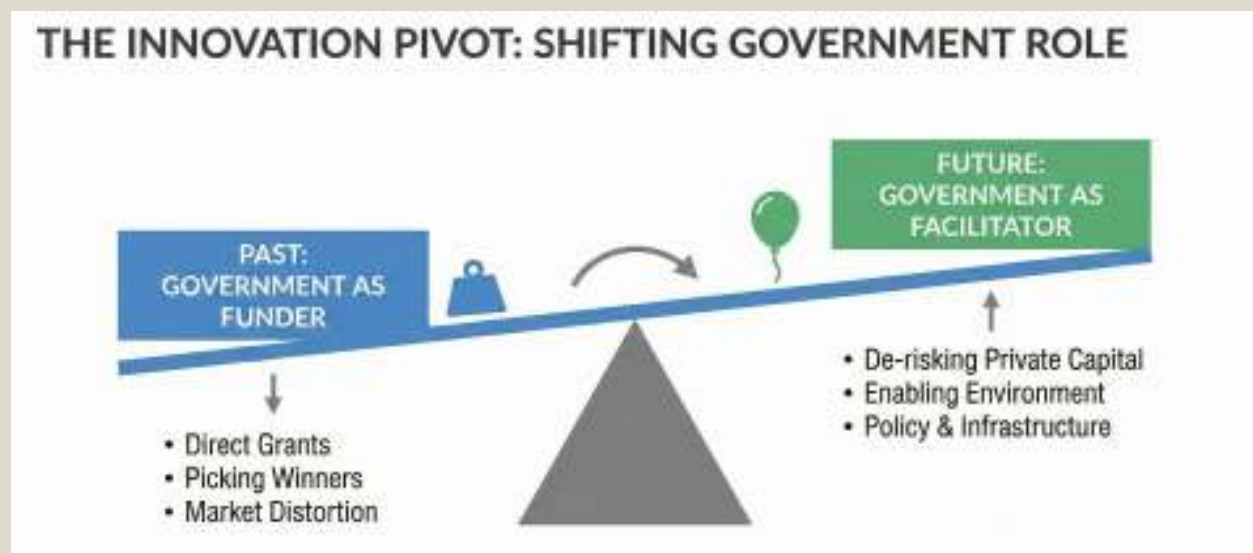
8.7 Core Strategic Principle

The government's reformed approach is anchored in a fundamental principle: sustainable ecosystem growth requires private sector dynamism, with government creating the enabling environment rather than attempting to pick winners directly. This represents a strategic shift from uncoordinated funding to facilitation. To address structural inefficiencies, the government will shift from a fragmented approach to a coordinated model where specific agencies are empowered to deploy capital and resources directly. Unlike previous strategies that limited government participation to facilitation, this framework mandates active state participation in deploying catalytic capital to bridge market gaps.

8.7.1 The Facilitation-Led Model

Under the facilitation-led model, government capital is most effective when used to:

- De-risk private investment through Fund of Funds and co-investment mechanisms, making it attractive for professional VCs to enter the Bangladesh market
- Build ecosystem infrastructure—co-working spaces, labs, regulatory service centers—that private markets may underprovide due to coordination failures or public goods characteristics
- Create market-enabling policies through tax incentives (EIS/SEIS models), regulatory sandboxes, and streamlined approvals that reduce friction for private actors
- Provide training and capacity building for entrepreneurs, investors, and support organizations to strengthen the ecosystem's human capital

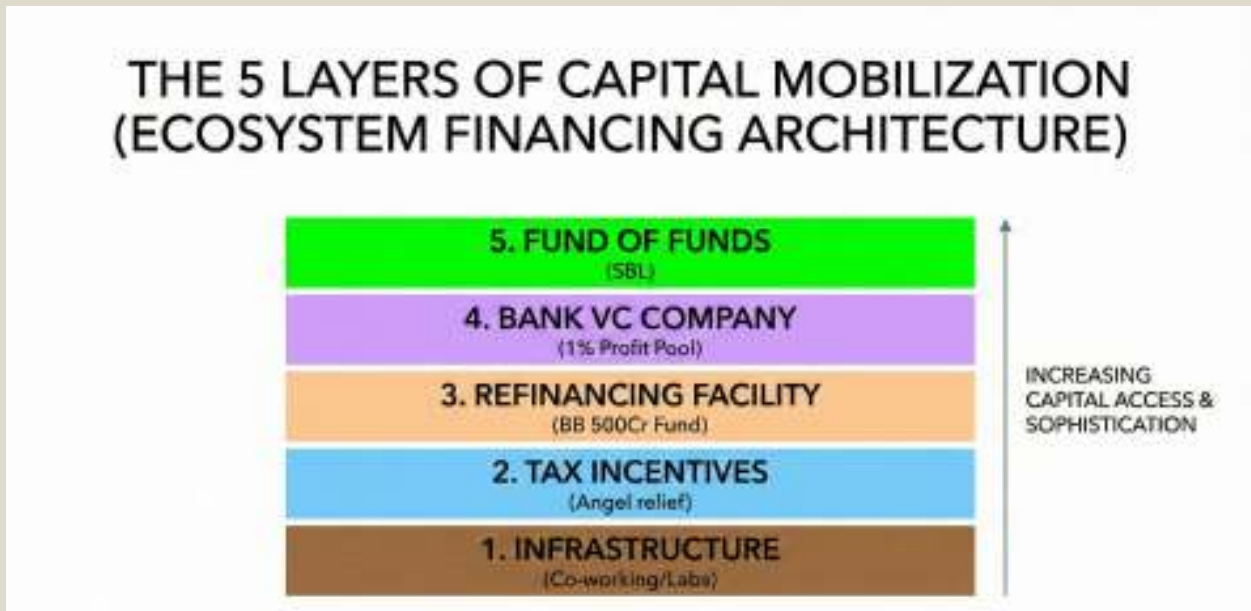


This approach recognizes that the ecosystem needs freedom to operate, compete, and fail—essential elements of entrepreneurial learning that cannot be engineered through government programs alone.

8.7.2 Capital Mobilization Architecture

While the ecosystem matures, the government will play a critical dual role:

Direct Investment: Deploying "Growth Level Capital" (Seed to Series A) via Startup Bangladesh Limited to bridge funding gaps where private risk appetite is low .



Ecosystem Activation: Using a Fund of Funds (FoF) structure to de-risk and attract professional venture capital firms to the Bangladesh market.

The reformed capital mobilization framework operates through five interconnected layers:

Layer	Mechanism	Government Role	Expected Impact
Fund of Funds	SBL deploys capital to professional VC fund managers	Capital provider; strategy oversight; impact alignment	Catalyze US\$300-500M in VC fund formation by 2030
Angel & Accelerator Enablement	Tax incentives and co-investment programs	Policy design via NBR; compliance monitoring	10x increase in domestic angel investment

Debt Financing	Concessional lending through commercial banks	Regulatory framework; refinancing support	US\$100M+ in startup-accessible debt capital
Infrastructure	Co-working spaces, labs, Hi-Tech Park facilities	Direct provision; subsidized access	12 regional hubs by 2030
Regulatory Enablement	Sandboxes, single-window clearance, tax holidays	Policy design; enforcement	50% reduction in compliance burden

8.8 Innovation Infrastructure Framework

Physical infrastructure forms the backbone of startup ecosystem development. While capital is essential, entrepreneurs also need access to facilities, equipment, networks, and regulatory support. BHTPA will lead the development of a nationally coordinated innovation infrastructure framework with three core components:

8.8.1 Makers Labs

Makers Labs address a critical gap in Bangladesh's startup ecosystem: the absence of facilities for hardware, deep-tech, and applied innovation. While software startups can operate from any location with internet access, hardware and deep-tech ventures require specialized equipment and facilities that are prohibitively expensive for individual startups.

- Specialized equipment for prototyping: 3D printers, CNC machines, electronics workbenches, testing facilities
- Technical mentorship and industry partnerships for commercialization pathways
- Shared infrastructure model reducing capital requirements for hardware startups
- Integration with university engineering programs for talent pipeline development

Target: Establish Makers Labs in 6 locations by 2028, co-located with Hi-Tech Parks and university engineering faculties.

8.8.2 Regional Innovation Hubs

To address geographic concentration, sector-focused regional innovation hubs will be established across all divisions. Each hub will specialize based on regional economic strengths and market opportunities:

Location	Sector Focus	Rationale	Target Launch
Dhaka	Fintech, E-commerce	Proximity to financial institutions, largest consumer market, talent concentration	2026
Chattogram	Healthtech, Maritime/Logistics	Port city logistics, pharmaceutical manufacturing base, medical college ecosystem	2026
Rajshahi	Agritech, Food Processing	Agricultural heartland, proximity to farmers, food processing industry	2026
Khulna	Semiconductors, Climate Tech	Industrial base, proximity to Sundarbans for climate research, emerging tech corridor	2027
Sylhet	Diaspora Tech, Remittance Innovation	NRB concentration, remittance flows, international connectivity	2027
Gazipur	Manufacturing IoT, Industry 4.0	Industrial manufacturing hub, factory automation opportunities	2028
Barishal	Blue Economy, Aquaculture Tech	Riverine geography, fisheries industry, water transport innovation	2028
Rangpur	Agricultural Mechanization	Agricultural economy, mechanization needs, cold chain opportunities	2029

8.8.3 Central Regulatory Service Centers

A single-window mechanism for startup approvals will be established through Central Regulatory Service Centers hosted by BHTPA. These centers will co-locate representatives from key regulatory bodies:

- Bangladesh Bank (fintech, payments, lending)
- BTRC (telecom, spectrum, digital services)
- DGDA (healthtech, pharmaceuticals)
- RJSC (company registration, compliance)
- NBR (tax registration, incentives)
- BIDA (foreign investment, work permits)

The goal is to reduce regulatory navigation time by 50% and provide startups with clear, consistent guidance on compliance requirements. Initial pilots will launch at Dhaka and Chattogram Hi-Tech Parks in 2025, with expansion to all regional hubs by 2028.

8.9 ICTD Playground: Govtech Innovation Platform

a2i will develop a central design lab—the ICTD Playground—serving as a cross-ministerial innovation platform. This initiative recognizes that government itself is both a massive market for digital solutions and a source of complex challenges that can drive innovation.

8.9.1 Core Functions

- **Challenge Definition:** Work with ministries to define specific problems suitable for startup solutions—from agricultural extension to healthcare delivery to urban mobility
- **Sandbox Environment:** Provide safe spaces for startups to test solutions with real government data and processes without full compliance burden
- **Pilot Programs:** Run structured pilots with successful startups, measuring impact and refining solutions before scale
- **Procurement Pathways:** Establish streamlined procurement mechanisms for scaling proven solutions into government service delivery

8.9.2 Startup Engagement Model

The ICTD Playground will operate through a structured engagement process:

- **Open Innovation Challenges:** Quarterly challenge announcements with defined problem statements, success criteria, and pilot budgets
- **Technical Support:** API access to government systems, data sharing agreements (PDPO-compliant), technical documentation

- Mentorship: Pairing startups with government domain experts who understand operational realities
- Scaled Procurement: Successful pilots graduate to framework agreements enabling multi-ministry deployment

This model addresses a common complaint from govtech startups: that the government is interested in their solutions but lacks mechanisms to actually buy them. By creating a structured pathway from challenge to pilot to procurement, the ICTD Playground can unlock significant startup revenue while improving government service delivery.

8.10 Strategic Objectives and KPI Framework

8.10.1 Strategic Objectives

#	Strategic Objective	Rationale
1	Position Bangladesh among top 50 global startup ecosystems by 2030	Currently unranked in top 50; requires coordinated improvement across funding, talent, market access, and infrastructure dimensions
2	Grow annual investment to US\$1B with 30% domestic sources	Current local investment is negligible; healthy ecosystems require balanced domestic and foreign capital
3	Enable 5,000 startups with 30% women-led	Scale and inclusion targets; women's participation currently below 15%
4	Establish 10-12 regional innovation hubs	Address geographic concentration; unlock regional market opportunities
5	Facilitate 10+ IPOs/major exits	Exit pathways essential for capital recycling and investor confidence
6	Build 15+ active domestic VC fund managers	Professional VC ecosystem required for sustainable early-stage funding

7	Operationalize Startup Authority with clear governance	Institutional foundation for coordinated ecosystem development
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8.10.2 Startup Ecosystem KPIs (2025-2030)

Metric	2026	2027	2028	2029	2030
New Startups Supported	600	1,200	2,000	3,500	5,000
Total Investment (US\$M)	250	400	600	800	1,000
Local Investment Share (%)	15%	20%	25%	28%	30%
Women-Led Startups (%)	22%	25%	27%	28%	30%
Regional Hubs Operational	4	6	8	10	12
IPOs/Major Exits	2	4	6	8	10+
Active VC Fund Managers	7	9	11	13	15+
Joint R&D Projects	50	100	150	200	300
Patents Commercialized	15	30	50	80	120

8.11 Implementation Roadmap

Phase 1: Foundation (2026-2027)

Establish institutional architecture and launch initial programs:

- Establish Startup Authority / Commission with governance framework, mandate, and staffing
- Redefine agency roles through formal mandate letters: SBL for FoF/VC enablement, iDEA for incubation enablement, BHTPA for infrastructure, a2i for govtech
- Launch Fund of Funds mechanism with US\$33 million initial deployment; issue call for VC fund manager applications
- Establish first two Central Regulatory Service Centers (Dhaka, Chattogram)
- Launch ICTD Playground pilot with 5 challenge areas
- Open first regional innovation hubs (Dhaka fintech, Chattogram healthtech)
- Establish first Makers Lab at BHTPA Kaliakair
- Design and launch angel investor tax incentive scheme modeled on UK EIS/SEIS
- Launch NRB Connect program to engage diaspora investors and mentors

Phase 2: Scaling (2027-2028)

Expand programs based on Phase 1 learnings:

- Expand regional innovation hubs to 6-8 locations with sector specializations
- Scale Fund of Funds deployment; onboard 8-10 professional VC fund managers
- Launch DSE Startup Board with relaxed listing requirements for tech companies
- Expand Central Regulatory Service Centers to all regional hubs
- Scale ICTD Playground to 20+ challenge areas with procurement pathways
- Achieve US\$600 million annual investment with 25%+ domestic participation
- Enable 4-6 IPOs/exits demonstrating viable exit pathways
- Launch women entrepreneurship targeted fund and mentorship program
- Establish regulatory sandboxes for fintech, healthtech, edtech sectors

Phase 3: Maturation (2029-2030)

Achieve scale and ecosystem self-sustainability:

- Complete 10-12 innovation hubs covering all divisions
- Achieve US\$1 billion annual investment target with 30% domestic participation
- Reach 5,000 active startups with 30% women-led
- Enable 10+ IPOs/major exits, establishing DSE as a startup-friendly exchange
- Position Bangladesh in Top 50 global startup ecosystems
- Achieve 15+ active domestic VC fund managers with combined AUM exceeding US\$500 million
- Full integration of ICTD Playground with government procurement systems
- University-industry linkages producing 300 joint R&D projects with 120 commercialized outputs
- Makers Labs operational in 6 locations catalyzing hardware and deep-tech innovation

8.12 Expected Outcomes by 2030

Successful implementation of this strategic framework will deliver transformational outcomes across multiple dimensions:

Institutional Outcomes

- Operational Startup Authority / Commission with clear mandate, accountability mechanisms, and cross-agency coordination
- Streamlined regulatory environment through Central Regulatory Service Centers with 50% reduction in compliance burden
- Performance-based governance with published KPIs and regular public reporting

Investment Outcomes

- Annual US\$1 billion investment inflows with 30% from domestic sources (up from negligible local participation)
- At least 15 active domestic VC fund managers with combined AUM exceeding US\$500 million
- 10+ successful IPOs and major exits, establishing Dhaka Stock Exchange as startup-friendly
- Functional angel investor tax incentive regime catalyzing 10x increase in angel investment

Startup and Innovation Outcomes

- Vibrant ecosystem of 5,000 active startups with 30% led by women entrepreneurs
- Over 500,000 direct and indirect jobs created through startup ecosystem growth
- Strong university-industry linkages with 300 joint R&D projects and 120 commercialized patents/products
- 100,000 entrepreneurs trained through iDEA and partner programs

Infrastructure Outcomes

- 10-12 regional innovation hubs across all divisions with sector specializations
- 6 operational Makers Labs catalyzing hardware and deep-tech innovation
- ICTD Playground integrated with government procurement, enabling startup-delivered public services

National Positioning

- Bangladesh ranked among Top 50 global startup ecosystems
- Recognized regional hub for fintech, agritech, and healthtech innovation
- Globally recognized Bangladesh startup brand contributing to ICT exports, jobs, and national innovation capacity

8.13 Conclusion

The proposed Startup Authority / Commission represents a structural shift from fragmented program management to coordinated ecosystem governance. By centralizing policy, aligning capital with capability, enabling market-driven innovation, and simplifying regulatory engagement, the Authority can significantly accelerate startup growth and economic impact.

This model does not replace existing agencies; rather, it redefines their roles, improves accountability, and ensures that public resources translate into scalable, sustainable startups. SBL focuses on what it does best—capital deployment through market mechanisms. iDEA focuses on what it does best—enabling the broader incubation ecosystem. BHTPA focuses on infrastructure. a2i focuses on govtech. Each agency operates with clear mandate and accountability, coordinated by the Authority.

The core principle of 'Fund Less, Facilitate More' acknowledges that government money can prime the pump, but it cannot replace market forces. Through the Fund of Funds mechanism, VC enablement, infrastructure provision, and regulatory simplification, the government can catalyze private capital without crowding it out.

With the right governance framework, transparency, and performance metrics, the Startup Authority can become the cornerstone of Bangladesh's transition to an innovation-led economy. By 2030, Bangladesh can transform from an ecosystem constrained by fragmentation and capital scarcity to one where domestic investors, professional VCs, and government facilitation work together to nurture world-class startups. The foundations are being laid today.

— End of Chapter 8 —

9. Capacity building and training

Building human capital is the cornerstone of any successful digital transformation. Bangladesh's journey toward becoming a digitally empowered, innovative, and inclusive society hinges critically on developing a workforce capable of creating, implementing, managing, and continuously improving digital systems and services. This chapter presents a comprehensive framework for capacity building and training that addresses the diverse needs of citizens, civil servants, industry professionals, and educators while learning from past initiatives and charting an ambitious roadmap for the future.

9.1 Context and Rationale: The Imperative for Digital Skills Development

Bangladesh's digital economy is expanding rapidly, with ICT exports exceeding US\$1.5 billion and a freelancing sector generating approximately US\$400 million annually. According to the Oxford Internet Institute's Online Labour Index, Bangladesh accounts for approximately 15% of global freelance labour supply, ranking among the top three countries globally. BASIS estimates over 650,000 active freelancers, with more than 500,000 registered on international platforms.

However, this growth has been constrained by critical capacity gaps. Workforce skills remain unevenly distributed across regions, education levels, and sectors. Existing training initiatives have been fragmented, lacking a unified legal framework to ensure continuity, accountability, and alignment with market demand. The Fourth Industrial Revolution (4IR) demands proficiency in technologies such as artificial intelligence, blockchain, IoT, cloud computing, and data analytics — skills that remain scarce in Bangladesh's labour market.

Several structural challenges persist: curricula at educational institutions lag behind industry demand; research rarely converts into commercial products; government policies are designed centrally without continuous feedback from industry and academia; and public sector officials often lack the digital competencies required to implement and manage modern e-government systems. Addressing these challenges requires a multi-track approach that targets different population segments with tailored interventions.

9.2 Review of Past Initiatives: What Has Been Done

9.2.1 Leveraging ICT for Growth, Employment and Governance (LICT) Project

The LICT Project (2013-2019), implemented by Bangladesh Computer Council (BCC) with US\$102.52 million World Bank funding, represents the most significant ICT capacity building initiative to date. The project successfully trained 33,564 individuals — including 10,450 women (31%) — in marketable ICT skills. Training was delivered in two tracks: 10,000 received Top-Up IT training for the IT services segment, while 20,000 received Foundation Skills training for the IT-enabled services (ITES) segment.

The project's approach was notable for its quality assurance mechanisms: pre-assessment and post-assessment of trainees were conducted by the National Association of Software Services Companies (NASSCOM), content was certified by George Washington University (USA), and a Training of Trainers (ToT) pool was established by internationally reputed trainers. Post-training surveys indicated that 90% of trained individuals obtained employment, contributing to the growth of IT/ITES industry employment from 12,000 jobs in 2013 to 47,000 jobs by 2019.

The LICT project also supported institutional capacity by assisting 66% of government ministries and divisions to introduce and maintain at least two or more e-government services, contributing to Bangladesh's improvement in the UN E-Government Development Index from 150th position in 2012 to 115th by 2018.

LICT Project Metrics	Achievement
Project Duration	2013-2019 (Extended)
Total Funding	US\$102.52 million (World Bank)
Total Trained	33,564 individuals
Women Trained	10,450 (31%)
IT Sector Employment Growth	12,000 (2013) to 47,000 (2019)
Employment Rate Post-Training	90%
E-Gov Development Index Improvement	150th (2012) to 115th (2018)

9.2.2 Enhancing Digital Government and Economy (EDGE) Project

Building on LICT's success, the EDGE Project was initiated with US\$300 million World Bank funding to continue digital transformation efforts. The project focuses on creating an enabling environment for digital government and the digital economy through multiple components including IT workforce development, digital infrastructure enhancement, policy reform, and research support.

Under Component 1, EDGE has conducted multifaceted digital skills training programs, training over 51,584 students — including 43% women — in foundational digital skills across more than 25 trades. In January 2024, the ICT Division signed MOUs with 28 public universities to provide digital skills training to 80,000 students over two years. The training is structured in three tiers: 50,000 at foundational level covering basic digital skills; 20,000 at intermediate level addressing more complex technical competencies; and 10,000 at advanced level focusing on frontier technologies including AI, robotics, and blockchain.

The EDGE project has also established partnerships with industry through its Hire & Train (HAT) program, which links training directly to employment guarantees — beneficiary companies must provide employment to at least 80% of trainees for a minimum of six months. The project supports the Digital Leadership Academy (DLA) and Center for 4IR to strengthen institutional capacity for emerging technology adoption.

9.2.3 Bangladesh Korea Institute of Information and Communication Technology (BKIICT)

BKIICT, established in 2005 with technical and financial assistance from the Korean International Cooperation Agency (KOICA), operates as an international-standard ICT training institute under Bangladesh Computer Council. Located at BCC Bhaban, Sher-e-Bangla Nagar, BKIICT offers a comprehensive range of courses including software development, web development, mobile app development, data science, networking, cybersecurity, digital marketing, graphic design, animation, and video editing.

BKIICT has served as a model for specialized ICT training delivery in Bangladesh, demonstrating the value of international partnerships in building advanced technical training capacity. The institute continues to evolve its curriculum to address emerging technology needs and market demands.

9.2.4 Bangladesh-Korea Technical Training Centres (BKTTC)

The Bangladesh-Korea Technical Training Centres represent a long-standing collaboration in technical and vocational education. BKTTC Dhaka, originally established in 1942 as a training center for World War II veterans, was modernized with KOICA funding and renamed in 2012-2013. Similarly, BKTTC Chattogram (established 1962) was remodeled and upgraded with modern equipment and machinery.

These centers operate under the Bureau of Manpower Employment and Training (BMET) and provide a range of services including one-year Skills Certificate courses across six trades, modular courses in 24 trades, pedagogical training for trainers (NTVQF Level 4), mandatory pre-departure orientation for overseas-bound workers, and job placement support. The curriculum is updated by Korea University of Technology (KUT), with trainers receiving professional development through KUT, JOCV, JICA, ILO, and KOICA programs.

9.2.5 Skills and Training Enhancement Project (STEP)

The World Bank-funded STEP project focused on strengthening technical and vocational education and training (TVET) institutions. The project supported 30 polytechnics with implementation grants of up to US\$1 million each, and engaged 50 training providers for short-course delivery. Performance-based incentives rewarded institutions achieving strong pass rates, teacher training outcomes, and graduate employment placement.

STEP supported the development of market-aligned competency standards through Industry Skills Councils (ISCs), including the ICT ISC. The project contributed to

increased enrollment in polytechnics and higher employability of graduates, though challenges remained in coordination between ISCs and the National Skills Development Authority (NSDA).

9.2.6 Union Digital Centers (UDCs) and a2i Capacity Building

The Access to Information (a2i) program has rolled out over 8,500 Union Digital Centers nationwide, transforming rural post offices and community spaces into hubs for digital services, skill development, and local entrepreneurship. These centers provide on-demand access to government services while offering training programs with strong emphasis on women and youth empowerment.

Through initiatives like the Teachers' Portal and multimedia classroom setups in over 15,000 schools, a2i has revolutionized teaching by providing digital resources, lesson plans, and collaborative tools. The Digital Service Accelerator initiative has further expanded service digitization while incorporating user feedback loops for continuous improvement.

9.3 Lessons Learned and Programs Discontinued

The review of past initiatives reveals several critical lessons and identifies programs that were discontinued or scaled back due to various challenges:

9.3.1 Key Success Factors from Past Programs

- **Quality Assurance Mechanisms:** Programs with international certification (NASSCOM, George Washington University) and standardized content achieved higher employment outcomes.
- **Industry Linkages:** Direct partnerships with IT/ITES companies and guaranteed employment clauses (HAT model) significantly improved training-to-employment conversion rates.
- **Training of Trainers (ToT):** Investment in building local training capacity ensured sustainability beyond project timelines.
- **Women's Inclusion:** Targeted outreach and supportive environments enabled significant female participation (31-43% in major programs).
- **Performance-Based Incentives:** Linking institutional funding to outcomes (pass rates, employment placement) drove quality improvements.

9.3.2 Challenges and Discontinued Approaches

- **Fragmented Coordination:** Multiple ministries and agencies operating training programs without unified standards or credential recognition led to duplication and inefficiency.
- **Curriculum-Industry Mismatch:** Static curricula that failed to keep pace with rapidly evolving technology requirements produced graduates with outdated skills.

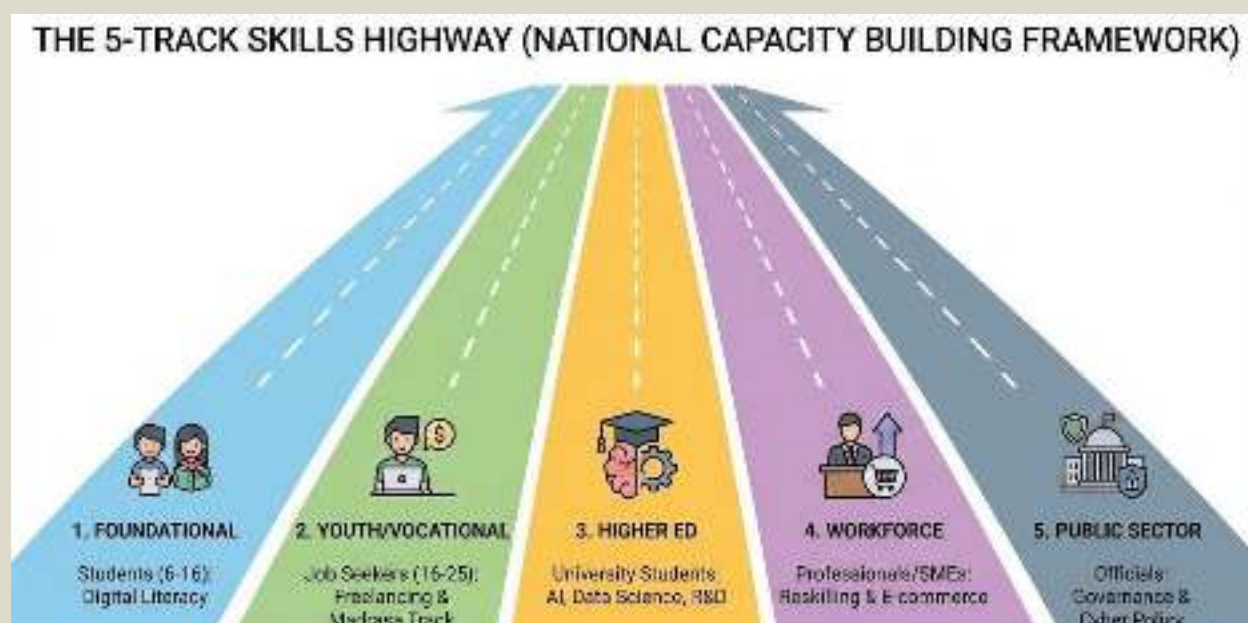
- **Teacher Training Gaps:** Many ICT training activities focused primarily on computer literacy rather than enabling teachers to integrate ICT effectively into teaching and learning.
- **Sustainability Challenges:** Project-based interventions without institutional embedding often lost momentum after donor funding ended.
- **Urban-Rural Divide:** Training infrastructure remained concentrated in metropolitan areas, limiting access for rural populations.
- **ISC Coordination Gaps:** Industry Skills Councils faced challenges in institutional capacity and coordination with NSDC, limiting their effectiveness in curriculum alignment.

9.3.3 Rationale for Reform

These lessons inform the strategic shift toward a more integrated, institutionally embedded, and outcome-focused approach to capacity building. The reformed strategy addresses sustainability through permanent institutional mandates, quality through standardized frameworks and international partnerships, relevance through industry-academia collaboration mechanisms, and inclusion through targeted interventions for underserved populations.

9.4 The Reformed Capacity Building Framework

The National Digital Transformation Strategy introduces a comprehensive, lifecycle-based approach to capacity building that addresses the diverse needs of Bangladesh's population while building sustainable institutional capacity. The framework operates across five interconnected tracks targeting different population segments.



9.4.1 Track 1: Foundational Skills (Schools and Early Learning)

Target Group: Primary and secondary students (ages 6-16)

The foundational track aims to embed digital literacy into the national education curriculum, ensuring that every child develops core competencies in safe device use, online safety, and basic problem-solving through Bangla-first digital tools. The curriculum integrates coding and computational thinking from an early age, while introducing awareness of freelancing and e-commerce concepts.

Strategic Interventions:

- Digital Classrooms Nationwide: Equip schools with smart boards, tablets, and internet access integrated into a National Education Cloud.
- E-Learning Platforms: Expand Bangla-friendly e-learning portals with offline access for rural students.
- Accessible Digital Content: Scale Multimedia Talking Books and interactive learning tools for students with disabilities.
- AI-Enabled Assessment: Deploy AI to track student performance, reduce dropout risks, and provide adaptive learning paths.
- Teacher Training: Mandatory ICT pedagogy certification for teachers; peer learning communities to sustain skills.

Year	Schools with Digital Classrooms	Students Using E-Learning	Teachers ICT-Trained	Accessible Titles
2025	10%	2 million	50,000	5,000
2026	20%	5 million	100,000	10,000
2027	40%	10 million	200,000	20,000
2028	60%	15 million	300,000	30,000
2029	80%	20 million	400,000	40,000
2030	100%	25 million+	500,000+	50,000

Table 9.1: KPI Framework for Foundational Digital Skills (2025-2030)

9.4.2 Track 2: Vocational and Youth Employment Skills

Target Group: School leavers, madrasa students, unemployed youth (ages 16-25)

This track focuses on developing market-ready skills for immediate employment in the digital economy. It includes two specialized pathways: the General Youth Track through the National Outsourcing & Freelancing Accelerator, and the Madrasa-Focused

Arabic/Farsi Employment Track that leverages the linguistic advantages of madrasa graduates for Middle Eastern markets.

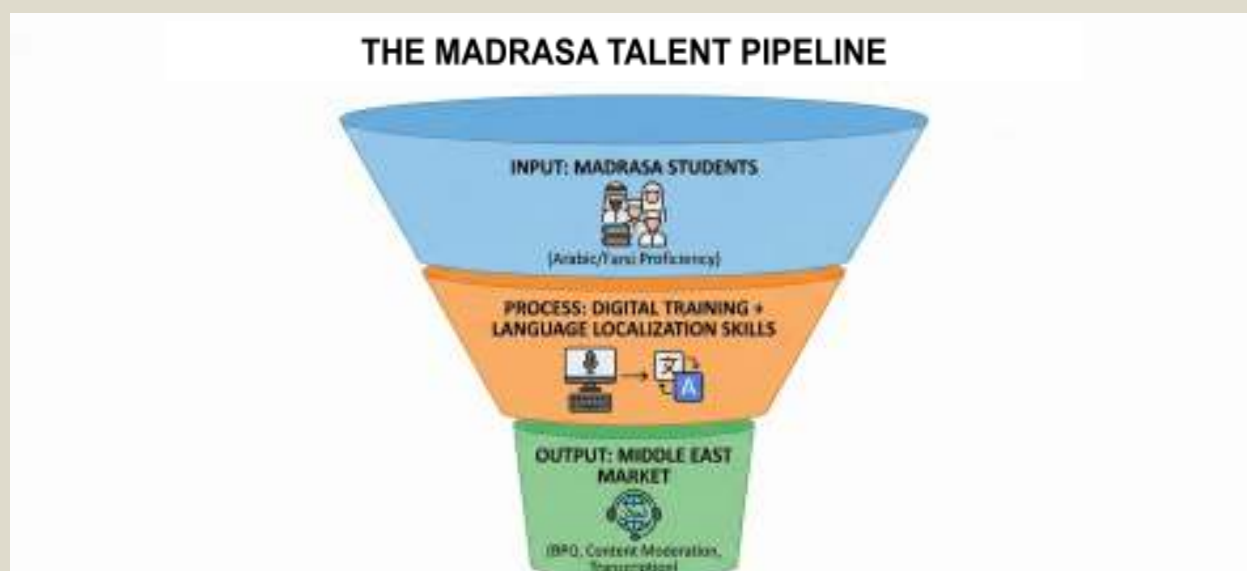
A. National Outsourcing & Freelancing Accelerator (General Youth Track):

- Digital Foundations: ICT literacy, online safety, productivity tools, client-facing English.
- Freelancing Skills: Graphic design, web and app development, digital marketing, transcription, QA, data entry, e-commerce management.
- Emerging Skills: AI data labeling, cloud operations, cybersecurity micro-tasks, fintech back-office, content moderation.
- Platform Skills: Onboarding for Upwork, Fiverr, Freelancer, Amazon MTurk, Remotasks, Clickworker.
- Placement & Market Linkage: Bootcamps, mentorship networks, and verified talent pools with BASIS, BACCO, and global platforms.

B. Madrasa-Focused Arabic/Farsi Employment Track:

Bangladesh's madrasas educate over 2.75 million students in the Alia stream (2023 BANBEIS data), with millions more in the Qawmi stream. Many madrasa learners are proficient in Arabic and Farsi, providing a natural advantage for Middle Eastern outsourcing markets including customer support, transcription, and content moderation.

- Target Niches: Arabic transcription, subtitling, cataloguing, customer support, content moderation, EN ↔ AR localization.
- Curriculum: Modern Arabic, workplace English, freelancing toolsets, specialized software.
- Partners: Alia/Qawmi boards including BEFAQ, GCC BPO firms, freelancing platforms for Arabic talent pools.



Year	Total Trained (General + Madrasa)	Madrasa Track Certified	Active Freelancers (Cumulative)	Export Earnings (US\$)
2025	100,000	15,000	50,000	\$100M
2026	150,000	25,000	120,000	\$250M
2027	200,000	35,000	220,000	\$450M
2028	200,000	30,000	350,000	\$700M
2029	180,000	25,000	500,000	\$950M
2030	170,000	20,000	650,000	\$1B+

Table 9.2: KPI Framework for Vocational and Youth Employment Skills (2025-2030)

9.4.3 Track 3: Higher Education and Advanced Tech Skills

Target Group: University and college students

This track prepares Bangladesh's future technology leaders through advanced training in frontier technologies, close university-industry collaboration, and establishment of Innovation Accelerators at major universities including BUET, DU, CUET, KUET, and RUET.

Focus Areas:

- AI, data science, cloud computing, cybersecurity, blockchain, fintech, and semiconductors.

- University-industry collaboration for R&D and internships with minimum 6 months industry attachment.
- Innovation Accelerators hosted within Hi-Tech Parks with access to labs, co-working spaces, seed grants, and mentorship.
- Joint R&D Grant Facility co-financed by government, private sector, and donors focusing on AI, semiconductors, digital health, fintech, and agriculture tech.

Year	Innovation Hubs	Joint R&D Projects	Patents/Products Commercialized	Students in Industry Placements	% Graduates Industry-Ready
2025	2	20	5	1,000	40%
2026	4	50	15	5,000	50%
2027	6	100	30	15,000	60%
2028	8	150	50	30,000	70%
2029	10	200	80	50,000	80%
2030	12 (all divisions)	300	120	75,000	90%

Table 9.3: KPI Framework for Higher Education and Advanced Tech Skills (2025-2030)

9.4.4 Track 4: Lifelong Learning and Workforce Reskilling

Target Group: Existing workforce, SME owners, returning migrants, women at home

This track addresses the continuous learning needs of the existing workforce, ensuring that professionals remain competitive as technology evolves.

Focus Areas:

- Short online bootcamps and micro-credentials through platforms like Udemy, Coursera, and local portals.
- SME digital adoption training covering finance, logistics, and e-commerce.
- Women's reskilling programmes for home-based freelancing and e-commerce.
- Migrant workers training in digital remittances and GCC outsourcing jobs.

KPI 2030: 500,000 workers reskilled; women's participation in freelancing to reach 40%.

9.4.5 Track 5: Public Sector and Leadership Skills

Target Group: Civil servants, judiciary, law enforcement

Digital transformation of government services requires a public sector workforce equipped with modern digital competencies. This track addresses the unique training needs of government officials at all levels.

Focus Areas:

- Cybersecurity, AI governance, digital ethics, and policy/regulatory skills.
- Judicial and law enforcement training for cybercrime investigation, forensics, and digital evidence handling.
- Senior leadership training on digital economy policymaking.
- Data stewardship and AI ethics training through NDGA.

KPI 2030: 50,000 officials trained in digital governance and cybersecurity.



9.5 Institutional Reforms for Capacity Building

9.5.1 Department of ICT (DoICT) as Grassroots Training Arm

DoICT will be transformed into the primary grassroots training and capacity-building arm of the ICT Division, with expanded mandate and strengthened institutional capacity.

Expanded Training Mandate:

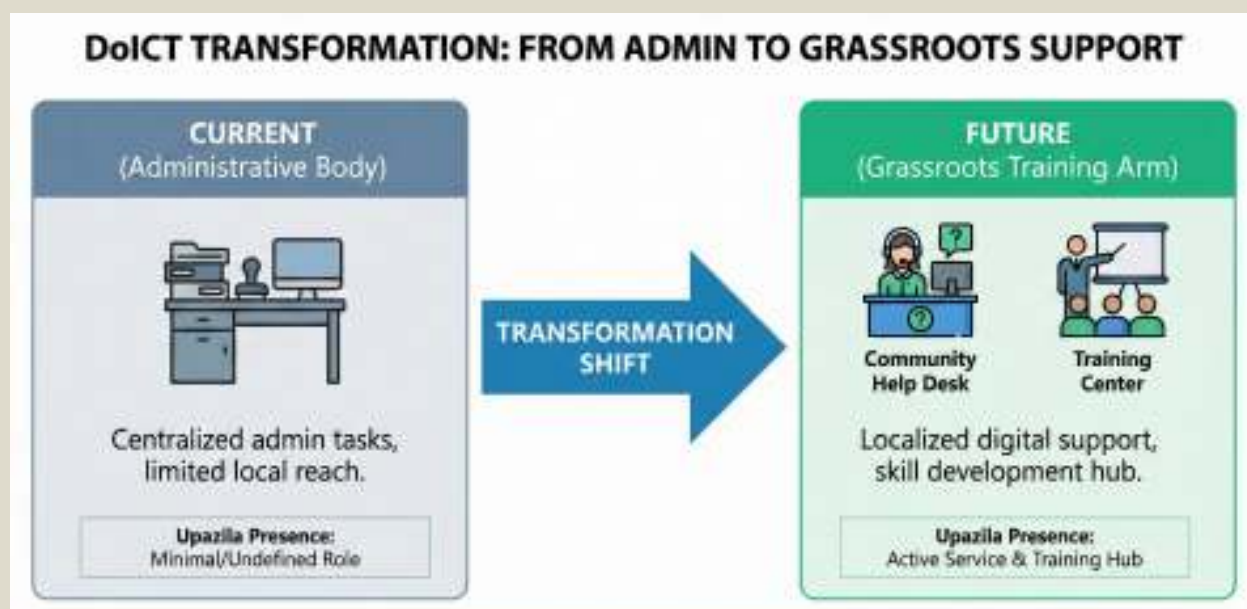
- Selected ICT-related courses (digital literacy, coding fundamentals, office productivity tools, cybersecurity hygiene) standardized and approved by NSDA.
- DoICT officers at district and upazila levels to deliver programs in schools, colleges, and madrasas.
- Advanced certifications (Microsoft, AWS, CISCO) integrated gradually with DoICT facilitating partnerships with global certifying bodies.

Curriculum Innovation and Local Integration:

- Foundation: Digital literacy for students and community members.
- Intermediate: E-commerce, freelancing, coding, and data entry.
- Advanced: Cybersecurity, AI, cloud, and IoT basics for future workforce readiness.
- Local language content and accessibility for rural and marginalized communities.

Monitoring and Accountability:

- National Digital Learning Dashboard to track enrollment, completion, and job-placement outcomes.
- Performance-based incentives for DoICT trainers/officers tied to training outcomes.
- DoICT offices equipped to serve as community digital help desks.



9.5.3 National Data Governance Authority (NDGA) Capacity Building Division

NDGA will establish a dedicated Coordination & Capacity Building Division responsible for training government officials in data governance, interoperability, and AI ethics.

Key Responsibilities:

- Train 5,000 officials in data stewardship and AI ethics by 2030.
- Publish open data handbooks and interoperability toolkits.
- Coordinate sectoral data offices across ministries.
- Conduct awareness programs on data protection and citizen rights under PDPO.

9.6 Digital Skills and Workforce Development Framework

A comprehensive legal and policy framework will underpin the capacity building strategy, providing statutory foundation for training standards, accreditation, and industry participation.

Key Provisions:

- National Skills Registry: Central database of trained professionals accessible to employers and agencies.
- Accreditation Framework: Legal recognition of digital skills certificates, micro-credentials, and global vendor certifications.
- Public Funding Commitment: Annual budgetary allocation to skills development through NSDA and ICT Division.

- **Inclusion Mandate:** At least 40% of training beneficiaries must be women and marginalized groups.
- **Industry Participation:** Private sector co-investment and curriculum advisory roles mandated by law.

9.6.1 Bangladesh Digital Skills Roadmap:

1. Foundational Skills (Schools & Early Learning)

- **Target Group:** Primary and secondary students (ages 6–16).
- **Focus Areas:**
 - Digital literacy (safe device use, online safety, Bangla-first digital tools).
 - Coding and problem-solving basics in school curriculum.
 - Exposure to freelancing/e-commerce concepts (early awareness).
- **KPI 2030:** 20 million students trained in digital literacy and coding basics.

2. Vocational & Youth Employment Skills (TVET, Madrasa, Freelancers)

- **Target Group:** School leavers, madrasa students, unemployed youth (16–25).
- **Focus Areas:**
 - Market-ready freelancing skills: design, coding, e-commerce, transcription, data entry.
 - Specialized *Madrasa Track*: Arabic/Bangla/Farsi-based services (transcription, content moderation, customer support for GCC markets).
 - ICT-enabled vocational trades (repair, IT support, digital agriculture tools).
 - Certification programs with NSDA, BASIS, and global partners (AWS, Cisco, Microsoft).
- **KPI 2030:** 1 million youth certified; 150,000 madrasa learners integrated into freelancing workforce.

3. Higher Education & Advanced Tech Skills

- **Target Group:** University and college students.
- **Focus Areas:**
 - AI, data science, cloud, cybersecurity, blockchain, fintech, semiconductors.
 - University–industry collaboration for R&D and internships.
 - Establish **Innovation Accelerators** at major universities (BUET, DU, CUET, KUET, RUET).
- **KPI 2030:** 100,000 graduates with advanced ICT/AI/data qualifications.

4. Lifelong Learning & Workforce Reskilling

- **Target Group:** Existing workforce, SME owners, returning migrants, women at home.
- **Focus Areas:**
 - Short online bootcamps and micro-credentials (Udemy, Coursera, local platforms).
 - SME digital adoption training (finance, logistics, e-commerce).
 - Women's reskilling programmes for home-based freelancing and e-commerce.
 - Migrant workers trained in digital remittances, GCC outsourcing jobs.
- **KPI 2030:** 500,000 workers reskilled; women's participation in freelancing to reach 40%.

5. Public Sector & Leadership Skills

- **Target Group:** Civil servants, judiciary, law enforcement.
- **Focus Areas:**
 - Cybersecurity, AI governance, digital ethics, policy and regulatory skills.
 - Judicial/law enforcement training for cybercrime, forensics, and digital evidence handling.
 - Senior leadership training on digital economy policymaking.
- **KPI 2030:** 50,000 officials trained in digital governance and cybersecurity.

Expected Outcomes by 2030

- **2 million citizens trained** across all lifecycle stages.
- **1 million active freelancers**, including **150,000 madrasa students**.
- **100,000 advanced ICT graduates** ready for high-value global markets.
- **500,000 reskilled workers** (SMEs, women, migrants).
- Stronger **public sector capacity** in digital governance and cybersecurity.



9.7 Freelancer.gov.bd:

The freelancer.gov.bd platform creates an integrated national infrastructure that:

Strategic Need	Platform Response
Workforce Visibility	National Freelancer Registry with unique ID system enabling real-time tracking of 1M+ freelancers
Credential Trust	Verified digital certificates linked to NID through NDGA interoperability standards
Training Coordination	Single enrollment point connecting to DoICT, BCC, NSDA, and global certification partners
Financial Integration	Remittance tracking, tax compliance, and integration with banking/MFS ecosystem
Market Linkage	Direct connection with BASIS, BACCO, global platforms, and GCC BPO firms

9.7.1 Freelancer.gov.bd Platform Architecture

3.1 Platform Vision

Freelancer.gov.bd will serve as the national digital gateway for Bangladesh's freelancing workforce, providing a single, authoritative platform where freelancers can register, access training, earn certifications, receive the Freelancer ID Card, track their earnings, and connect with domestic and international opportunities.

3.2 Core Platform Modules

Module 1: National Freelancer Registry

The foundational module establishing Bangladesh’s first comprehensive database of digital workers:

Component	Description
Registration Gateway	Online registration with NID verification through oneID/NDGA integration, supporting both web and mobile access with Bangla-first interface design
Profile Management	Comprehensive skill profiles including completed trainings, certifications, portfolio links, platform accounts (Upwork, Fiverr, etc.), and earning history
Unique Freelancer ID	System-generated unique identifier linked to NID for permanent tracking across all training programs, certifications, and earnings
Verification API	Public API enabling employers and platforms to verify freelancer credentials and certification status in real-time
Analytics Dashboard	Administrative interface providing real-time statistics on registered freelancers, active earners, geographic distribution, and skill coverage

Module 2: Training & Certification Hub

A unified gateway connecting freelancers with all available training programs:

Component	Description
Course Catalog	Searchable database of all available courses from DoICT, BCC, a2i, NSDA-accredited providers, and global certification partners
Enrollment System	Single enrollment point with automatic eligibility checking, slot allocation, and waitlist management
Learning Pathways	Structured progression tracks from Foundation (digital literacy) to Intermediate (freelancing skills) to Advanced (AI, cloud, cybersecurity)
Certification Tracker	Digital wallet for all earned certificates including NSDA credentials, global vendor certifications (Microsoft, AWS, Cisco), and platform-specific badges
Progress Dashboard	Individual learning analytics showing completion rates, skill gaps, and recommended next courses

Module 3: Earnings & Payment Integration

Financial tracking and support module addressing payment challenges:

Component	Description
Earnings Tracker	Voluntary self-reporting of earnings from all platforms with aggregated dashboard showing lifetime earnings, monthly trends, and platform breakdown
Remittance Guide	Updated information on legal remittance channels, bank partnerships, and MFS options with comparative cost analysis
Tax Compliance	Integration with NBR API Connectivity Hub for simplified tax registration and compliance reporting for freelance income
Bank Integration	Partnerships with scheduled banks for preferential freelancer accounts with reduced foreign exchange conversion fees

Module 4: Market Linkage & Opportunities

Connecting freelancers with work opportunities:

Component	Description
Job Board	Curated opportunities from verified employers, government projects, and enterprise outsourcing contracts specifically seeking Bangladeshi talent
Platform Onboarding	Step-by-step guides and assisted registration for Upwork, Fiverr, Freelancer, Amazon MTurk, Remotasks, Clickworker, and other global platforms
Talent Pools	Verified talent directories accessible to BASIS members, BACCO companies, and international clients seeking specialized skills
GCC Connect	Dedicated section for Arabic/Farsi-skilled freelancers connecting with Middle Eastern BPO firms and outsourcing contracts
Mentorship Network	Connection with successful freelancers for guidance, bootcamps, and career support through industry partnerships

9.7.2 Freelancer ID Card Initiative

Concept and Purpose

The Freelancer ID Card represents a groundbreaking initiative to formally recognize freelancing as a legitimate profession and provide digital workers with an official

credential that facilitates access to services, builds credibility, and enables government support.

Unlike traditional employment identification, the Freelancer ID is specifically designed to address the unique circumstances of gig economy workers who operate independently across multiple platforms and clients.

Card Features and Specifications

Feature	Description
Physical Card	Standard ISO/IEC 7810 ID-1 format smart card with embedded chip containing digital certificates and NID linkage
Digital Card	Mobile-accessible digital version through freelancer.gov.bd app with QR code for instant verification
Unique ID Number	Permanent identifier format: BD-FLR-XXXXXXX linked to NID and National Skills Registry
Skill Endorsements	Digital attestation of verified skills and certifications with expiry dates and issuing authority
Earning Tier	Voluntary badge system showing lifetime earning ranges: Bronze (<\$5K), Silver (\$5K-25K), Gold (\$25K-100K), Platinum (>\$100K)
Verification API	Real-time verification endpoint for employers/platforms to confirm ID validity and credentials

Eligibility Criteria

To receive the Freelancer ID Card, applicants must meet the following requirements:

Requirement	Specification
Age	Minimum 18 years of age with valid NID
Registration	Active registration on freelancer.gov.bd with verified NID linkage
Training	Completion of at least one NSDA-accredited or government-recognized digital skills course
Platform Activity	Active profile on at least one freelancing platform OR documented freelance work history

Code of Conduct	Agreement to National Freelancer Code of Conduct covering ethical work practices and quality standards
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Benefits of Freelancer ID Card

For Freelancers

Benefit	Description
Official Recognition	Government-backed credential establishing freelancing as a recognized profession
Banking Access	Simplified opening of business accounts with partner banks; serves as income proof for loans
Tax Benefits	Streamlined tax registration and potential future incentives for compliant freelancers
Social Protection	Foundation for future enrollment in social security and health insurance schemes
Client Trust	Verifiable credential increasing credibility with international clients and platforms
Government Services	Priority access to government digital services, subsidized co-working spaces, and training programs

For the Government

Benefit	Description
Workforce Visibility	Accurate data on digital workforce size, skills, earnings, and geographic distribution
Policy Planning	Evidence-based inputs for skills development policy, training investment, and export strategy
Revenue Tracking	Better capture of freelance export earnings data for national accounts and forex planning
Formalization	Pathway for bringing informal digital workers into the formal economy

Training Program Integration

Curriculum Alignment with National Framework

The freelancer.gov.bd platform will offer a structured learning pathway aligned with the Bangladesh Digital Skills Roadmap, delivering courses through integrated partnerships with government training providers:

Level 1: Digital Foundations (0-3 months)

Course	Provider	Certification
ICT Literacy	DoICT District/Upazila Offices	NSDA Digital Literacy Certificate
Online Safety	a2i Online Platform	Cyber Hygiene Badge
Productivity Tools	BCC Regional Offices	Microsoft Office Specialist
Client-Facing English	Partner Language Institutes	Business English Certificate

Level 2: Freelancing Skills (3-9 months)

Track	Skills Covered	Certifications
Graphics & Design	Adobe Suite, Canva, UI/UX basics	Adobe Certified Associate
Web Development	HTML/CSS, JavaScript, WordPress	freeCodeCamp, W3Schools
Digital Marketing	SEO, SEM, Social Media, Analytics	Google Digital Garage
Content & Writing	Copywriting, Transcription, Translation	NSDA Content Certificate
Data Entry & QA	Spreadsheets, Data Validation, Testing	ISTQB Foundation
E-Commerce	Shopify, Amazon Seller, Product Listing	Platform Certifications

Level 3: Emerging Skills (9-18 months)

Track	Skills Covered	Certifications
AI & Data Labeling	ML data prep, annotation, prompt engineering	AWS AI Practitioner

Cloud Operations	AWS, Azure, GCP basics, DevOps intro	AWS Cloud Practitioner
Cybersecurity	Security fundamentals, pen testing basics	CompTIA Security+
Fintech Support	Payment systems, blockchain basics, AML	Fintech Certificate
App Development	React Native, Flutter, mobile UI/UX	Google Associate Developer

Implementation Roadmap

Phase 1: Foundation (2025)

Quarter	Activities	Deliverables
Q1 2025	Platform architecture design; NDGA integration specifications; NID verification API development	Technical blueprint; API specifications
Q2 2025	Core platform development; National Freelancer Registry module; user registration system	Beta platform; registration module
Q3 2025	Training Hub module; NSDA curriculum integration; DoICT office connectivity	Training module; course catalog
Q4 2025	Pilot launch in 10 districts; Freelancer ID Card design finalization; first batch registration	50,000 registered freelancers

Phase 2: Expansion (2026-2027)

Period	Activities	Deliverables
H1 2026	Nationwide rollout to all 64 districts; Freelancer ID Card issuance begins; payment integration module	150,000 freelancers; 50,000 ID cards
H2 2026	Market Linkage module launch; BASIS/BACCO integration; Madrasa Track pilot in 20 madrasas	250,000 freelancers; Job board live

H1 2027	Global certification partnerships (AWS, Microsoft, Cisco); GCC Connect module; bank partnerships	400,000 freelancers; 100,000 ID cards
H2 2027	Madrasa Track expansion to 100+ institutions; women-focused programs; mobile app launch	600,000 freelancers; mobile app live

Phase 3: Consolidation (2028-2030)

Period	Activities	Deliverables
2028	Social protection pilot for ID card holders; advanced AI/4IR tracks; international platform partnerships	800,000 freelancers; social protection pilot
2029	Cross-border payment integration; tax incentive programs; regional hub expansion	950,000 freelancers; payment hub live
2030	Full ecosystem maturity; 1M freelancer milestone; \$1B export earnings verification; global recognition	1M+ freelancers; \$1B+ exports

Key Performance Indicator (KPI) Framework

Platform Performance KPIs

Indicator	2025	2026	2027	2028	2029	2030
Registered Freelancers	50K	250K	500K	750K	900K	1M+
Freelancer ID Cards Issued	5K	75K	200K	400K	600K	800K
Active Earners (Cumulative)	50K	120K	220K	350K	500K	650K
Training Completions	100K	250K	450K	650K	830K	1M
Madrasa Track Certified	15K	40K	75K	105K	130K	150K
Women Freelancers (%)	22%	26%	30%	34%	37%	40%

Economic Impact KPIs

Indicator	2025	2026	2027	2028	2029	2030
Freelance Export Earnings (US\$)	\$100M	\$250M	\$450M	\$700M	\$950M	\$1B+
Average Earning/Freelancer	\$2,000	\$2,100	\$2,050	\$2,000	\$1,900	\$1,540
High-Value Freelancers (>\$10K/yr)	5K	12K	25K	45K	70K	100K
Global Certifications Earned	5K	15K	35K	60K	85K	100K
Districts with Active Freelancers	10	32	48	56	62	64

Service Quality KPIs

Indicator	Target (2030)
Platform Uptime	99.5%
ID Card Issuance Time (from eligibility)	<7 days
Verification API Response Time	<2 seconds
User Satisfaction Score	>80%
Grievance Resolution Time	<15 days
Training-to-Employment Conversion Rate	>65%

9.8 Government-Industry-Academia Collaboration Framework

A structured triple helix collaboration framework will ensure that capacity building efforts are aligned with market demand and leverage the strengths of all stakeholders.

9.8.1 National Council for Digital Innovation (NCDI)

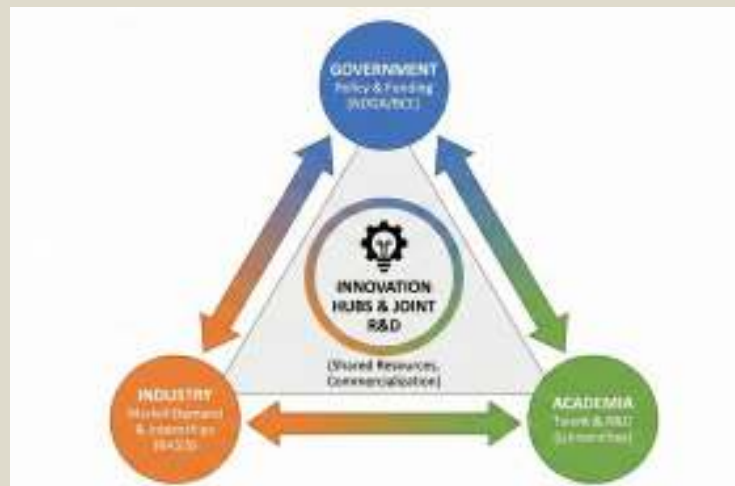
- Chaired by ICT Division, co-chaired by BASIS/BACCO, with Vice-Chancellors of major universities.
- Meets quarterly to align on skills, R&D, and industry challenges.
- Provides recommendations to NDGA, BCC, and Planning Commission.

9.8.2 Curriculum and Talent Alignment

- Industry advisory boards embedded into universities.
- Annual curriculum review aligned with ICT export market trends.
- Internship/apprenticeship pipeline: minimum 6 months industry attachment for ICT students.

9.8.3 Applied Research and Commercialization

- Incentives for faculty to co-publish and co-patent with industry.
- Innovation vouchers for SMEs to access university labs.
- National Startup Incubation Challenge linking student startups to venture funds.



Governance & Institutional Setup

- **National Council for Digital Innovation (NCDI)**
 - Chaired by ICT Division, co-chaired by BASIS/BACCO, with Vice-Chancellors of major universities.
 - Meets quarterly to align on skills, R&D, and industry challenges.
 - Provides recommendations to NDGA, BCC, and Planning Commission.
- **University–Industry Innovation Accelerators**
 - Hubs established at BUET, DU, CUET, KUET, RUET, and leading private universities.
 - Hosted within Hi-Tech Parks, giving access to labs, co-working, seed grants, and mentorship.
- **Joint R&D Grant Facility**

- Co-financed by government, private sector, and donors.
- Grants focused on AI, semiconductors, digital health, fintech, and agriculture tech.
- Outcomes tracked: prototypes, patents, startups.
- **Align Skills with Market Demand:** Ensure curricula, certifications, and training programs respond to real-time industry needs.
 1. **Accelerate Research to Market:** Translate academic R&D into products, patents, and startups.
 2. **Build Innovation Hubs Nationwide:** Foster university–industry accelerators in every region, beyond Dhaka and Chattogram.
 3. **Facilitate Policy Feedback Loops:** Formalize government–industry–academia councils for co-creation of strategies.
 4. **Create Talent & Startup Pipelines:** Leverage students and researchers as a source of entrepreneurship and innovation.

Key Interventions

1. Curriculum & Talent Alignment

- Industry advisory boards embedded into universities.
- Annual curriculum review aligned with ICT export market trends.
- Internship/apprenticeship pipeline: minimum **6 months industry attachment** for ICT students.

2. Applied Research & Commercialization

- Incentives for faculty to **co-publish and co-patent** with industry.
- Innovation vouchers for SMEs to access university labs.
- National Startup Incubation Challenge linking student startups to venture funds.

3. Innovation Hubs & Regional Balance

- **12 hubs by 2030** across all divisions.
- Sector-focused hubs: fintech (Dhaka), agritech (Rajshahi), healthtech (Chattogram), manufacturing IoT (Gazipur), semiconductors (Khulna).
- Annual National Digital Innovation Expo to showcase outcomes.

4. Policy Co-Creation

- Government agencies to co-develop digital standards with universities and industry.
- Regulatory sandboxes tested in collaboration with academia.

- Real-time feedback loop: research insights → policy pilots → scaled solutions.

KPI Framework (2025–2030)

Year	Innovation Hubs	Joint R&D Projects	Patents/Products Commercialized	Students in Industry Placements	% Graduates Industry-Ready
2025	2	20	5	1,000	40%
2026	4	50	15	5,000	50%
2027	6	100	30	15,000	60%
2028	8	150	50	30,000	70%
2029	10	200	80	50,000	80%
2030	12 (all divisions)	300	120	75,000	90%

Expected Outcomes by 2030

- **12 innovation hubs** connected to universities, industry, and hi-tech parks.
- **300 joint R&D projects** with at least **120 commercialized outputs**.
- **75,000 students** gaining direct industry placements.
- **90% ICT graduates** considered industry-ready.
- Stronger **startup pipelines** emerging directly from universities.
- Bangladesh is recognized as a **regional leader in government–industry–academia collaboration**.

9.9 International Partnerships and Knowledge Transfer

International cooperation will be leveraged strategically to build sustainable domestic capacity while ensuring that partnerships are mutually beneficial rather than donor-driven.

Key Principles:

- All major ICT projects with foreign partners will include mandatory knowledge transfer programs, on-site training, and certification of Bangladeshi officials.
- Bilateral programs will focus on co-innovation rather than simple aid, including joint pilots in 5G, health tech, and smart agriculture.
- Joint R&D projects with universities abroad will feed into the Government-Industry-Academia collaboration framework.
- Engagement with E-Government Academy (EGA), ITU, and OECD networks for benchmarking and standards localization.



Year	% Donor Projects with Knowledge Transfer	Bilateral Projects with Local Content	Officials Trained	Startups in Joint Projects	Global Forum Participation
2025	30%	20%	500	50	2
2026	50%	30%	1,500	150	3
2027	70%	50%	3,000	300	5
2028	85%	70%	5,000	500	6
2029	95%	85%	7,000	700	7
2030	100%	100%	10,000+	1,000+	10

Table 9.4: KPI Framework for International Partnerships (2025-2030)

9.10 Expected Outcomes by 2030

The comprehensive capacity building framework is designed to achieve transformative outcomes across all population segments:

9.10.1 Aggregate Training Outcomes

- 2 million citizens trained across all lifecycle stages.
- 1 million active freelancers, including 150,000 madrasa students.
- 100,000 advanced ICT graduates ready for high-value global markets.
- 500,000 reskilled workers (SMEs, women, migrants).
- 50,000 public sector officials trained in digital governance and cybersecurity.
- 500,000+ teachers ICT-trained with 25 million students engaged in digital learning.

9.10.2 Economic Impact

- US\$1 billion+ annual freelance export earnings.
- Bangladesh positioned as #1 freelancing workforce hub globally.
- Women's participation in freelancing increased from 20% to 40%.
- 90% of ICT graduates considered industry-ready.

9.10.3 Institutional Outcomes

- 12 innovation hubs connected to universities, industry, and hi-tech parks across all divisions.
- 300 joint R&D projects with at least 120 commercialized outputs.
- 75,000 students gaining direct industry placements.
- 10,000+ Bangladeshi officials trained in frontier technologies through international partnerships.
- Bangladesh recognized as a regional leader in government-industry-academia collaboration.

9.11 Consolidated Capacity Building Framework Summary

Track	Target Group	Key Focus Areas	2030 Target
1. Foundational Skills	Students ages 6-16	Digital literacy, coding basics, online safety	20M students trained

2. Youth Employment	Youth 16-25, madrasa students	Freelancing, ITES, Arabic services	1M certified, 650K active freelancers
3. Higher Education	University students	AI, data science, cloud, cybersecurity	100K advanced graduates
4. Workforce Reskilling	Workers, SMEs, women, migrants	Micro-credentials, e-commerce, digital adoption	500K reskilled
5. Public Sector	Civil servants, judiciary, law enforcement	Digital governance, cybersecurity, AI ethics	50K officials trained

Table 9.5: Consolidated Capacity Building Framework Summary

The capacity building and training reforms outlined in this chapter represent a paradigm shift from fragmented, project-based interventions to a comprehensive, institutionally embedded, and outcome-focused approach. By addressing the diverse needs of different population segments through tailored tracks, building sustainable institutional capacity, and leveraging international partnerships for knowledge transfer, Bangladesh will develop the human capital foundation essential for realizing its vision of becoming a digitally empowered, innovative, and inclusive society by 2030.

End of Chapter 9

10. Strategic and governance reforms

10.1 Introduction: The Imperative for Strategic Governance Reform

Bangladesh's digital transformation journey requires more than technological adoption—it demands fundamental changes in how government institutions coordinate, make decisions, and deliver services to citizens. The National Digital Transformation Strategy recognizes that sustainable digital progress is impossible without parallel reforms in governance structures, inter-agency coordination, funding mechanisms, and accountability frameworks. This chapter outlines the comprehensive strategic and governance reforms necessary to ensure the successful implementation of Bangladesh's digital vision through 2030 and beyond.

The current state of governance in Bangladesh's ICT ecosystem presents significant challenges. Fragmented digital assets across ministries, siloed decision-making processes, overlapping mandates, and insufficient coordination mechanisms have historically undermined digital initiatives. Major data repositories—including the National ID (NID), Birth and Death Registration System (BDRIS), and National Board of Revenue (NBR) records—lack secure interoperable mechanisms. This fragmentation has resulted in redundant infrastructure, higher operational costs, poor coordination among government agencies, and citizen data vulnerabilities. The reforms outlined in this chapter address these systemic challenges through institutional restructuring, enhanced coordination frameworks, sustainable financing models, and robust monitoring mechanisms.

10.2 Coordination Framework

Effective implementation of the National Digital Transformation Strategy requires a robust coordination framework that aligns diverse institutions, minimizes duplication, and ensures timely delivery of reforms. Bangladesh's digital ecosystem spans ministries, regulators, state-owned enterprises, and private sector actors. Without structured inter-agency mechanisms, overlapping mandates and siloed decision-making risk undermining progress.

10.2.1 Three-Tier Coordination Structure

The proposed coordination framework consists of three interlinked tiers, each serving distinct but complementary functions in ensuring cohesive digital governance:

Tier 1: National Digital Transformation Council (NDTC)

The National Digital Transformation Council serves as the apex strategic oversight body, providing high-level policy direction and ensuring alignment with national development priorities like Sustainable Development Goals. The Council is chaired at the highest level of government with membership including key ministers, secretaries, and heads of regulatory bodies. Its responsibilities include approving major policy frameworks, resolving inter-ministerial conflicts, and ensuring political commitment to digital transformation objectives. The NDTC convenes quarterly to review progress and provide strategic guidance.

Tier 2: Inter-Agency Taskforces and Committees

Issue-specific taskforces form the operational backbone of the coordination framework. These include dedicated taskforces for Cybersecurity, Digital Public Infrastructure, AI & Data Governance, E-Commerce and Logistics, Land Automation, and Digital Financial Services. Each taskforce is chaired by the relevant line ministry with co-chairing by the ICT Division to ensure technical consistency. Taskforces are empowered to draft policies, oversee implementation pilots, and coordinate resource sharing across agencies. This structure ensures that specialized expertise is brought to bear on sector-specific challenges while maintaining alignment with overall digital transformation objectives.

Tier 3: Implementation and Monitoring Secretariat

A permanent secretariat under the ICT Division serves as the central coordination hub, responsible for day-to-day coordination and monitoring activities. The secretariat's core functions include convening taskforces, tracking progress through a National Digital Dashboard, and providing quarterly reports to the NDTC. It ensures accountability through transparent KPIs, citizen feedback loops, and cross-agency performance reviews. The secretariat also serves as the primary interface with development partners, coordinating technical assistance and funding flows.

10.2.2 National Digital Transformation Taskforce (NDTT)

To ensure effective implementation of the strategy's bold goals, the government will establish a National Digital Transformation Taskforce (NDTT). This taskforce will include representatives from the ICT Division, Bangladesh Computer Council (BCC), Bangladesh Bank, the Ministry of Finance, the National Board of Revenue (NBR), the Bangladesh Telecommunication Regulatory Commission (BTRC), the Cabinet Division, and development partners such as the World Bank, JICA, the EU, and UNDP. The inclusion of development partners ensures access to global expertise, financing, and best practices while maintaining national ownership of the transformation agenda.

The NDTT operates under a clear mandate to break down institutional silos, harmonize digital initiatives across ministries, and ensure that all government agencies feel ownership of the transformation process. Unlike previous ICT Division-led initiatives, the NDTT's inclusive structure ensures that digital transformation is not viewed as the responsibility of a single ministry but as a whole-of-government priority. Biannual progress reports will track key performance indicators through global benchmarking and public feedback.

Coordination Framework Structure

Tier	Body	Chair	Frequency	Primary Functions
1	National Digital Transformation Council (NDTC)	Prime Minister / Chief Adviser	Quarterly	Strategic oversight, policy approval, conflict resolution
2	Inter-Agency Taskforces	Relevant Line Ministry + ICT Division	Monthly	Policy drafting, pilot oversight, resource coordination
3	Implementation Secretariat	ICT Division	Continuous	Daily coordination, monitoring, reporting, partner liaison

10.3 Public-Private Partnership (PPP) Model

The success of Bangladesh's digital transformation requires mobilizing both public and private resources. While the government provides the regulatory framework, strategic direction, and foundational infrastructure, the private sector contributes innovation, investment, and operational efficiency. A structured Public-Private Partnership (PPP) model will therefore be central to accelerating priority digital projects.

10.3.1 Strategic Principles of the PPP Framework

Under this model, the government will identify flagship initiatives—such as 5G rollout, National Fibre Bank, Digital Public Infrastructure, and Semiconductor Roadmap—that require joint investment. The framework ensures transparent procurement and risk-sharing mechanisms, creating a predictable environment for private investors. Innovation sandboxes and pilot projects will enable startups and enterprises to co-develop solutions with government agencies. Blended financing through a mix of public funds, concessional loans, foreign direct investment (FDI), and private capital will accelerate implementation.

By leveraging PPPs, Bangladesh can fast-track implementation timelines, reduce fiscal burdens, and ensure that cutting-edge global expertise is embedded in national digital systems. This collaborative approach aligns with the strategy's vision of building a resilient, citizen-centric, and innovation-driven digital ecosystem. Key principles guiding PPP engagements include value for money, risk allocation to the party best able to manage it, transparency in procurement and contracting, and accountability through clearly defined performance metrics.

10.3.2 Priority Sectors for PPP Engagement

Sector	Initiative	Government Role	Private Sector Role	Expected Investment
Telecommunications	5G Network Rollout	Spectrum allocation, regulatory framework	Network deployment, operations	\$2-3 billion
Infrastructure	National Fibre Bank	Asset inventory, policy framework	Leasing operations, maintenance	\$500 million
Digital Services	Digital Public Infrastructure	Standards, governance, core platforms	Service development, integration	\$300 million
Industry	Semiconductor Roadmap	R&D funding, workforce development	Facility investment,	\$1-2 billion

			commercialization	
Finance	Universal Payment Gateway	Regulatory oversight, interoperability standards	Platform development, operations	\$200 million

THE PPP VALUE EXCHANGE (PUBLIC-PRIVATE PARTNERSHIP)



10.4 Role of Public Procurement Act and Rules in Digital Transformation

Bangladesh's journey toward a more efficient and accountable public sector has been strongly supported by digital transformation. A key enabler in this process has been the Public Procurement Act (PPA) 2006 and the Public Procurement Rules (PPR) 2008, which laid the legal and institutional foundation for transparent and technology-driven procurement. The implementation of the Electronic Government Procurement (e-GP) system is one of the most significant milestones achieved under this legal framework, contributing to increased efficiency, reduced corruption, and improved service delivery across public sector entities.

10.4.1 Role of PPA 2006 and PPR 2008 in Digital Transformation

The PPA and PPR provided a forward-thinking legal environment that facilitated the digitalization of procurement operations. Key contributions include:

Legal Authority for e-GP: A recent amendment (4th May 2025) of Section 65 of the PPA made it mandatory for all Procuring Entities (PEs) to use the e-GP portal where applicable.

Process Standardization: The Act and Rules introduced standardized procedures, formats, and thresholds, which were critical to designing an automated, auditable procurement workflow.

Support for ICT Sector Engagement: The legal framework allowed for the procurement of IT-enabled services, facilitating the development of national digital platforms and systems through engagement with local ICT firms.

Decision Capacity for Software Specifications: Procuring Authorities should have the decision capacity to define software technology specifications to ensure appropriate technical requirements are established for digital procurement.

Through these contributions, the procurement law not only ensured transparency and competitiveness but also actively supported Bangladesh's broader digital service delivery goals.

10.4.2 e-GP Maturity Status (as of 12 April 2025)

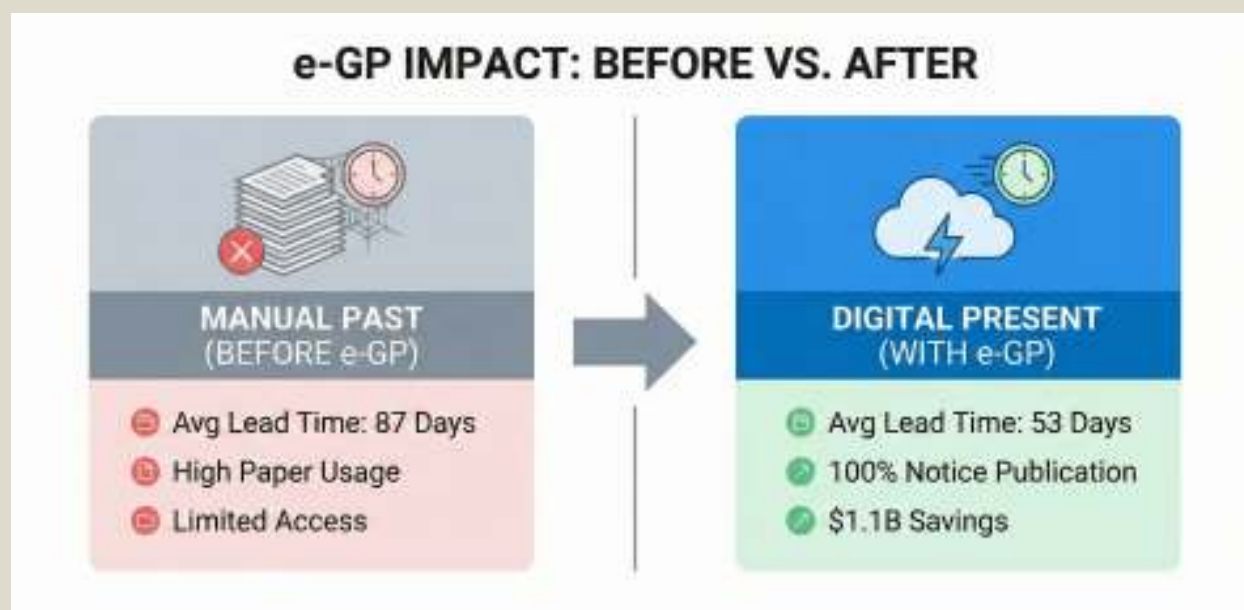
Indicator	Value
Agencies	1,483
PEs in e-GP	12,975
Registered Tenderers	130,640
Tender Invited	935,576

Contract Awards	704,404
Tender Value	USD 113.55 Billion
Contract Value	USD 72.80 Billion
Registered Banks	53

10.4.3 Impact of e-GP

The e-GP system has delivered significant measurable improvements:

Publication of 100% of procurement notices and contract award information has been achieved. Average procurement lead time (from invitation to contract signing) decreased from 86.7 days in FY12 to 53 days in FY23. The percentage share of awarded bids within the original bid validity period increased from 10% in FY07 to 99% in FY23. The percentage share of rejected bids decreased from 8% in FY12 to 3% in FY19. An estimated USD 1.1 billion in savings has been realized. Improved market access and bidding environment compared to manual bidding has been established. An estimated 497 million km of travel distance reduction and decrease of 153,559 tons of CO2 emission has been achieved, along with savings of 1,053 million pages of paper.



10.4.4 Current Challenges in e-GP Implementation

Digital Literacy Issues: Not all tenderers, contractors, and consultants are digitally equipped to effectively use the e-GP system.

Capacity Constraints: Many procuring entities lack trained and experienced personnel to operate the system effectively, especially at the local level.

Connectivity and Access Issues: Contractors in rural areas often face challenges related to internet access, limiting their participation.

System Performance Limitations: Users continue to experience slow response times, technical issues during peak usage, and limited support for mobile devices.

Regulatory Gaps: The current legal framework does not fully reflect the operational needs of digital procurement, such as cloud hosting, digital signatures, and e-document validation.

Limited Use of Procurement Data: While the e-GP system captures extensive data, it is not being adequately used for performance monitoring, analytics, or risk mitigation through IMED and other oversight bodies.

10.4.5 Strategic Recommendations

Legal and Regulatory Update:

Amend the PPA and PPR to explicitly incorporate provisions for e-signatures and secure electronic records, incorporate provisions for Agile procurement practices for software development, enable cloud-based platforms and third-party integrations, and establish simplified procedures for procuring digital services.

Capacity Development:

Scale up mandatory training and certification for procurement practitioners, including a focus on digital procurement tools and ethical compliance. Establish dedicated procurement support cells in key ministries and local government bodies.

System Enhancement and User Support:

Upgrade the infrastructure of the e-GP system to ensure better performance and scalability. Improve the user interface with mobile-responsive design and multilingual access. Strengthen technical support through helpdesks and online troubleshooting tools.

Performance Monitoring and Analytics:

Deploy real-time dashboards and automated performance indicators for procurement oversight bodies including IMED. Utilize procurement data to detect red flags, ensure competition, and improve value-for-money outcomes.

10.5 National Computing Infrastructure: Centralized GPU Clusters

As Bangladesh accelerates its transition into the Fourth Industrial Revolution (4IR), access to High-Performance Computing (HPC) and Artificial Intelligence (AI) resources is no longer a luxury but a necessity for academic and industrial innovation. To maximize resource utilization and ensure equitable access, the government will adopt a **Centralized GPU Cluster** model for University Innovation Hubs, moving away from fragmented, siloed procurement.

10.5.1 The Strategic Pivot: Centralization vs. Fragmentation

Traditionally, computing resources have been procured in silos, with individual universities establishing small, isolated server rooms. This decentralized approach has led to significant inefficiencies:

- **Underutilization:** Academic GPU servers in isolated environments typically see less than **40% average utilization**, wasting valuable computational potential.
- **Operational Inefficiency:** Managing ten separate small clusters requires ten times the cooling, power, and maintenance overhead (OPEX), alongside a duplication of skilled system administrator roles.
- **Inequity:** Smaller universities with lower budgets are left behind, creating a "digital divide" within the academic ecosystem.

The Reform Directive: Bangladesh will establish a **National GPU Cluster** hosted in a central Tier-III/IV data center (e.g., BDREN, BUET, or National Data Center). This consolidated infrastructure will serve all University Innovation Hubs via the high-speed BDREN optical backbone.

10.5.2 Technical Architecture & Governance

The centralized architecture is designed to offer "elastic scaling"—allowing researchers to access massive compute power when needed, which would be impossible with local hardware limits.

Core Infrastructure:

- **Hosting:** Centralized facility with optimized cooling and power redundancy.
- **Compute Nodes:** High-density Multi-GPU servers (utilizing enterprise-grade GPUs like A100/H100 equivalents).
- **Connectivity:** Accessed remotely via the **Bangladesh Research and Education Network (BDREN)**, ensuring low-latency data transfer (99%+ uptime).

Workload Management: To manage demand fairly across institutions, the cluster will utilize industry-standard schedulers:

- **Slurm:** For batch processing of heavy research jobs (e.g., training Large Language Models, genomic sequencing).
- **Kubernetes + GPU Operator:** For containerized, interactive AI/ML development environments.

Security & Isolation: Data sovereignty and security are paramount. The system will employ **Role-Based Access Control (RBAC)** and **Namespaces** to ensure per-project isolation. Sensitive research data remains protected, accessible only to authorized users despite the shared infrastructure.

10.5.3 The "Hybrid Model": Balancing Efficiency with Autonomy

To address the pedagogical need for "hands-on" hardware access, a **Hybrid Model** will be implemented:

1. **The Central Core (Heavy Lifting):** The bulk of the investment will go into the central cluster for heavy model training, scalable workloads, and collaborative research.
2. **The Edge (Teaching & Tinkering):** Innovation Hubs at universities will retain **1-2 edge GPU units** locally.
 - *Purpose:* These units are strictly for teaching basic system administration, robotics interfacing, and offline inference tasks.
 - *Outcome:* Students gain physical hardware experience without compromising the efficiency of national-scale computing.

10.5.4 Strategic Benefits

Adopting this centralized approach aligns with global best practices (e.g., Japan's Fugaku, US National Research Cloud) and delivers four key advantages:

1. **Cost Efficiency:** Bulk procurement and consolidated maintenance significantly reduce Total Cost of Ownership (TCO).
2. **Academic Equity:** A student at a remote university gains access to the same high-end compute power as a researcher at a leading engineering university.
3. **Future-Proofing:** Upgrades are managed centrally, eliminating the "hardware obsolescence" cycle for individual campuses.
4. **Collaboration:** A shared software stack fosters cross-institutional research, preventing the emergence of "software silos."

By unifying our computing power, Bangladesh avoids the risk of falling behind due to fragmentation. This infrastructure will serve as the engine for the next generation of AI researchers, ensuring sustainable and scalable innovation for complete digital transformation.

10.6 Funding and Financial Mechanisms

The long-term success of Bangladesh's digital transformation will depend on the availability of sustainable and diversified financing mechanisms. Given the scale of planned reforms and infrastructure projects, traditional budget allocations alone will not be sufficient. A blended financial strategy is therefore essential, combining national resources, foreign capital, and innovative funding instruments.

10.6.1 Non-Resident Bangladeshi (NRB) Involvement

Mobilizing the global Bangladeshi diaspora represents a strategic opportunity for investment in digital infrastructure and startups. NRB-focused bonds, venture funds, and digital innovation platforms will channel remittances into productive, high-impact projects. The diaspora community possesses not only financial resources but also technical expertise, global market connections, and entrepreneurial experience that can accelerate Bangladesh's digital transformation. Specific mechanisms include Digital Infrastructure Bonds targeting NRB investors, Diaspora Venture Funds focused on technology startups, and Digital Innovation Platforms enabling direct investment in approved projects.

10.6.2 Foreign Loans and Development Assistance

Concessional loans from multilateral institutions and bilateral partners will finance large-scale infrastructure such as the National Fibre Bank, 5G rollout, and Smart Governance systems. Partnerships with global development agencies including the World Bank, Asian Development Bank, JICA, KOICA, EU, and UNDP will bring technical expertise, policy benchmarking, and risk-sharing support. Bangladesh has already secured significant commitments through the World Bank's EDGE project and other initiatives, providing a foundation for continued engagement.

10.6.3 Operational Cost Management

A sustainability model will be introduced for flagship digital projects, where cost recovery is achieved through service fees, transaction charges, and shared savings. Ministries and agencies will adopt performance-based budgeting, ensuring efficiency and value-for-money. Public-private revenue-sharing models will further reduce the government's fiscal burden while maintaining affordability for citizens. Key mechanisms include user fee structures for premium digital services, transaction-based cost recovery for payment systems, and efficiency gain sharing between government and private operators.

Projected Funding Sources (2025-2030)

Source	2025	2026	2027	2028	2029	2030	Total
Government Budget (BDT Cr)	2,500	3,500	4,500	5,500	6,500	7,500	30,000
Development Partners (\$M)	150	250	400	500	600	700	2,600
NRB Investment (\$M)	50	100	200	300	400	500	1,550
Private Sector FDI (\$M)	200	400	600	800	1,000	1,200	4,200

PPP Co-Investment (\$M)	100	200	350	500	700	900	2,750
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10.7 IT Export Strategy & Action Plan:

The Information Technology (IT) sector has emerged as a priority engine for Bangladesh's future export growth. However, despite significant progress, IT export performance remains below potential due to structural gaps in international market access, talent development, and branding. To position Bangladesh as a global IT powerhouse, targeted reforms and enabling policies must be adopted.

10.7.1 Problem Statement

Despite rising global demand for software and IT/ITS services, Bangladeshi exporters increasingly:

- register offices/entities abroad to operate normally in target markets,
- retain export earnings offshore for extended periods to preserve operational solvency,
- route IT/ITS earnings through remittance channels instead of formal IT/ITS export mechanisms.

These practices reduce formal export figures, distort policy data and incentives, create “policy blind spots,” and constrain sectoral scaling—particularly for SaaS/product-led growth and globally distributed delivery models.

10.7.2 Key Structural Constraints to Address

Inadequate foreign-currency flexibility for export operations

IT/ITS exporters routinely incur legitimate foreign expenses, including:

- cloud infrastructure and SaaS subscriptions,
- international marketing and client acquisition,
- foreign consultants/technical professionals,
- cross-border licensing, compliance, and certification costs.

However, outward remittance for business expenses is capped (e.g., USD 40,000 annually) irrespective of firm scale—constraining normal operations for firms earning both USD 100,000 and USD 1 million per year.

Exchange-rate risk and forced value loss

Rapid depreciation of the Taka creates real losses for firms that repatriate early and later need to pay foreign expenses. This rationally encourages offshore retention of earnings to avoid operational disruption.

Mid-level management deficit and founder bottlenecks

Many firms remain founder-dependent for routine decisions, limiting speed, scalability, and export delivery maturity. The ecosystem lacks structured pipelines for mid-level managerial talent (critical for scaling from small teams to export enterprises). *(A solution addressing this has been under development by Adnan Dewan/SBL.)*

Unequal access to overseas subsidiary formation

Existing provisions enabling certain startups to open overseas subsidiaries (including limited setup remittance such as USD 10,000 with post-notification) do not extend equitably to older registered firms—even if they became export-ready later. Registration age is treated as a maturity proxy, which does not reflect sector realities.

Misclassification of IT/ITS exports as remittance

IT/ITS exports are eligible for a cash incentive (e.g., 3%), while remittances receive a comparable benefit (e.g., 2.5%) with significantly simpler procedures. Due to documentation burdens and approvals, many freelancers and firms remit as “remittance” rather than IT/ITS export—undercounting export volumes and weakening sector statistics.

10.7.3 Strategic Priorities and Policy Actions

Policy Support, FX Modernization & Market Access

1.1 Overseas subsidiaries and branches (global footprint enablement)

- Establish a clear regulatory framework enabling Bangladeshi IT/ITS companies to set up foreign branches/subsidiaries across all firm ages, with risk-based safeguards and transparent reporting.
- Standardize allowable purposes (sales offices, customer success, compliance representation, overseas hiring, local contracting) and simplify post-notification workflows.

1.2 Revenue-linked foreign expense caps (replace flat ceilings)

- Replace a flat annual outward remittance cap (e.g., USD 40,000) with **tiered limits linked to export turnover** to reflect export-driven cost structures.

Illustrative model (policy to finalize by regulator):

- Tier 1: Export turnover up to USD 250k → allowable FX business expense cap up to USD 75k (or up to X% of turnover)
- Tier 2: USD 250k–1M → cap up to USD 200k (or up to X% of turnover)
- Tier 3: > USD 1M → cap up to USD 500k (or up to X% of turnover)
- Apply enhanced AML/CFT and audit requirements progressively by tier (risk-based regulation).

1.3 Foreign-currency retention windows and onshore FC accounts

- Permit exporters to retain foreign earnings for defined operational purposes before mandatory conversion (e.g., 90–180 days, renewable based on documented pipeline).
- Alternatively (or additionally), allow exporters to hold earnings in foreign currencies in local banks (special FC retention accounts) to reduce offshore retention incentives while preserving solvency.

1.4 International payment gateways

- Ensure full operational integration of global payment platforms (e.g., PayPal/Stripe or equivalent rails) to facilitate seamless cross-border transactions and reduce informal routing.

1.5 Outward FDI allowance (controlled)

- Permit **controlled outward FDI** by IT/ITS firms to accelerate market entry, acquisitions, and overseas delivery capability—subject to export performance thresholds and reporting.

1.6 Tax and cash incentives (predictable, digitized, performance-linked)

- Sustain and enhance export-linked benefits (e.g., restore cash incentive from 6% toward 10% if policy permits), digitize application workflows, and publish predictable tax exemption/withholding frameworks.
- Implement fast-track incentive processing for compliant firms with verified export receipts.

1.7 Innovation funding aligned to export opportunities

- Expand industry–academia collaboration grants for export-relevant R&D (AI services, cybersecurity, healthtech, fintech, govtech, supply-chain tech), with clear pathways from prototype to international commercialization.

Marketing, Branding & Global Positioning

2.1 Unified “Bangladesh Tech” global presence

- Participate in major global events (GITEX, VivaTech, Web Summit, Japan IT Week, etc.) with a **unified Bangladesh Tech Pavilion** supported by a national brand kit, deal-room, and pitch roster.

2.2 Bangladesh IT Summits abroad

- Organize annual IT summits in priority markets (USA, UK, Japan, UAE, Germany), designed as B2B/B2G deal accelerators (buyers, integrators, venture funds, compliance advisors, diaspora CTO networks).

2.3 Embassy engagement and ICT trade desks

- Embed ICT trade support within embassies; designate honorary ICT consuls and formalize matchmaking and investor outreach KPIs per mission.

2.4 International media and analyst relations

- Launch a structured global branding campaign to place Bangladeshi IT success stories in global platforms and analyst ecosystems (industry publications, research notes, rankings, and market reports), supported by a consistent narrative: quality delivery, cost competitiveness, security compliance, and scaling maturity.

Talent Development & Leadership Growth

3.1 Hire & Train (HAT) expansion

- Scale the EDGE project’s HAT program to produce industry-ready professionals with placement-linked outcomes.

3.2 Mid-level management capacity programs (to break founder bottlenecks)

- Create structured pathways for product managers, delivery managers, engineering managers, QA leads, customer success leads, and finance/ops managers.
- Link participation and subsidies to export-growth benchmarks and scaling milestones.
- Integrate ecosystem solutions being developed by SBL leadership initiatives (including the work referenced under Adnan Dewan/SBL).

3.3 Leadership development

- Strengthen CXO and ACMP programs to develop globally competent leadership and export-ready governance practices (contracts, compliance, pricing, risk management).

3.4 Global certification support

- Provide reimbursement or voucher-based support for internationally recognized certifications (e.g., Microsoft/AWS/Cisco), tied to employment retention and export delivery needs.

3.5 Retention & return incentives

- Adopt milestone-based support to retain local talent and enable NRB professionals to contribute (remote mentorship, short-term placements, co-founding support), with clear deliverables and accountability.

Export Growth Pathways

4.1 Scaling SMEs (3–5 to 50+ employees)

- Provide structured scaling support: export readiness diagnostics, sales enablement, legal/compliance toolkits, offshore partner matching, and tiered FX facilitation.

4.2 Expanding mid-large firms

- Enable global scaling through export-linked tax incentives, R&D rebates, and facilitation for overseas operations and payments.

4.3 Boosting SaaS & product exports

- Offer targeted support for product-led growth: international market access programs, compliance and data protection readiness, marketplace onboarding, and customer acquisition co-funding.

4.4 BPO transformation

- Move BPO up the value chain via AI-driven service innovation (automation, multilingual support, analytics services) and next-gen outsourcing models.

Strategic Learnings & Global Best Practices

- **JETRO model (Japan):** international offices, market facilitation, and brand-building support through coordinated government-industry structures.

DANIDA-style milestone funding (Denmark): performance-linked funding tranches (e.g., USD 100k–800k equivalents) tied to verified export outcomes, accelerating scaling while protecting public funds.

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10.8 Monitoring and Evaluation Framework

A robust Monitoring and Evaluation (M&E) Framework is essential to ensure that the National Digital Transformation Strategy delivers measurable results, maintains accountability, and adapts to emerging challenges. The framework will provide government, private sector, and citizens with transparent insights into progress while enabling timely course corrections.

10.8.1 Key Performance Indicators (KPIs)

A national KPI matrix will be developed, aligning with Sustainable Development Goals (SDGs). Indicators will cover areas such as digital service adoption, infrastructure expansion, cybersecurity resilience, ICT sector growth, and citizen satisfaction. The KPI framework is designed to track both output metrics (what is being delivered) and outcome metrics (what impact is being achieved), ensuring a comprehensive view of transformation progress.

Category	Key Indicators	Baseline (2024)	Target (2030)
Digital Services	Government services digitized end-to-end	30%	95%
Infrastructure	Mobile towers fiber-connected	25%	100%
Cybersecurity	Professionals trained	5,000	50,000
Digital Economy	ICT export revenue (\$B)	1.5	10.0
Citizen Access	Active digital service users (millions)	15	110
E-Government Ranking	UN EGDI position	100th	Top 15
Interoperability	Systems integrated via BNDA	20%	100%

Data Governance	Agencies PDPO-compliant	0%	100%
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10.8.2 Digital Dashboard for Real-Time Tracking

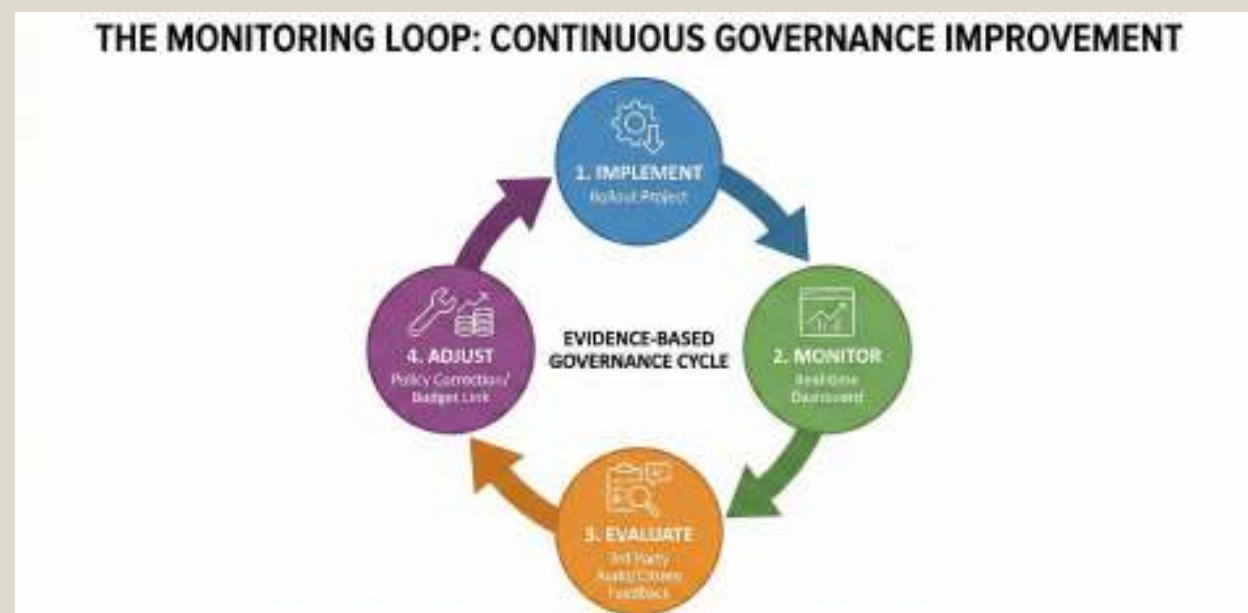
A National Digital Transformation Dashboard will consolidate data from ministries and agencies, enabling real-time progress monitoring. Dashboards will be accessible to policymakers and select stakeholders, with periodic public reports to strengthen transparency. The dashboard will provide visualization of KPI trends, early warning alerts for underperforming initiatives, comparative benchmarking against regional peers, and integration with citizen feedback systems.

10.8.3 Independent Audits and Third-Party Evaluation

External assessments by academic institutions, industry experts, and civil society will validate progress and provide recommendations. Annual Digital Transformation Scorecards will be published to benchmark Bangladesh against regional and global peers including Vietnam, India, and Malaysia. Independent evaluations will focus on value-for-money assessments, technical compliance verification, user experience analysis, and impact assessment on citizen outcomes.

10.8.4 Accountability and Feedback Loops

Ministries and agencies will be held accountable through performance reviews linked to budgetary allocations. Citizen feedback mechanisms—including surveys, grievance redressal portals, and social media analytics—will help evaluate the effectiveness and inclusivity of services. Through this framework, Bangladesh will establish a culture of evidence-based policymaking, ensuring that digital transformation is not only implemented but also continuously refined to maximize impact.



10.9 International Partnerships

Bangladesh's digital transformation requires more than domestic reforms—it must be anchored in sustainable, mutually beneficial international partnerships. Such collaborations should not only bring finance but also ensure technology transfer, institutional strengthening, and skills development so that local capacity grows and dependency is reduced over time.

10.9.1 Development Partners

The World Bank, UNDP, EU and others will support e-governance, cybersecurity, and digital inclusion initiatives. Financing models will prioritize capacity transfer clauses, ensuring that after donor projects end, local agencies retain skills and systems. Digital inclusion projects will include measurable targets for women, rural communities, and persons with disabilities, not just infrastructure rollout. Key ongoing partnerships include the World Bank's EDGE Project supporting digital government and economy, the EU's Effective E-Governance program enhancing public service delivery, and UNDP's technical assistance for policy development and South-South cooperation.

10.9.2 Bilateral Cooperation

Partners like JICA and KOICA will go beyond infrastructure loans to co-develop innovation ecosystems, including research labs and skills academies. Agreements will mandate local content requirements, so a minimum share of hardware/software is sourced or assembled in Bangladesh. Bilateral programs will focus on co-innovation rather than simple aid—for example, joint pilots in 5G, health tech, and smart agriculture. This approach ensures that Bangladesh builds indigenous capability while benefiting from partner expertise.

10.9.3 Global Platforms and Benchmarking

Engagement with the E-Government Academy (EGA), ITU, and OECD networks will help Bangladesh benchmark against peers. Instead of adopting 'copy-paste' solutions, global models will be localized through NDGA for cultural, linguistic, and socio-economic fit. Participation will ensure Bangladesh contributes to global digital norms (AI ethics, cross-border data flows), giving the country a voice in shaping standards. Bangladesh will actively participate in WTO e-commerce negotiations, ITU digital standards development, and AI ethics frameworks.

10.9.4 Technology and Knowledge Transfer

All major ICT projects with foreign partners will include mandatory knowledge transfer programs, on-site training, and certification of Bangladeshi officials. Partnerships will incentivize the growth of local startups, ensuring they are included in procurement and pilot programs. Joint R&D projects with universities abroad will feed into Bangladesh's Government–Industry–Academia collaboration framework, building a long-term skills pipeline.

International Partnership KPI Framework (2025-2030)

Year	Donor Projects w/ Knowledge Transfer	Bilateral Projects w/ Local Content	Officials Trained	Startups in Joint Projects	Global Forums
2025	30%	20%	500	50	2
2026	50%	30%	1,500	150	3
2027	70%	50%	3,000	300	5
2028	85%	70%	5,000	500	6
2029	95%	85%	7,000	700	7
2030	100%	100%	10,000+	1,000+	10

10.10 Bangladesh in the Global Digital Economy

Bangladesh's ICT exports are growing steadily, exceeding US\$1.5 billion, mainly in software, BPO, and freelancing. However, integration into regional platforms (BIMSTEC, SAARC, ASEAN) and participation in global digital governance remain limited. To sustain growth and credibility, Bangladesh must evolve from a cost-driven outsourcing hub to a trusted and globally integrated digital partner.

10.10.1 Strategic Objectives

The global positioning strategy aims to position Bangladesh as a regional digital trade hub by 2030, secure cross-border agreements on data flows, digital trade, and payment systems, actively participate in global forums on AI ethics, data protection, and open-source governance, leverage the diaspora tech community to channel investment, skills, and market access, and build Bangladesh's global reputation as a secure and resilient digital service provider.

10.10.2 Key Interventions

Digital Trade and Regional Cooperation

Bangladesh will negotiate digital trade provisions in FTAs and regional agreements (BIMSTEC, SAARC, ASEAN, GCC), establish cross-border e-commerce corridors with South Asia and the Middle East, and develop regional digital payment systems to reduce transaction costs for freelancers and SMEs.

Global Standards and Governance

The strategy calls for aligning national regulations with global standards (GDPR, ISO, UN Open Source Principles), actively participating in WTO e-commerce negotiations, ITU digital standards, and AI ethics frameworks, and positioning NDGA as Bangladesh's representation node in global data governance forums.

Diaspora Tech Talent and Investment

Key initiatives include launching a Bangladesh Tech Diaspora Network to attract mentorship and venture investment, creating a Reverse Brain Drain programme offering incentives for returning ICT professionals, and facilitating diaspora-led venture funds targeting startups in fintech, healthtech, and agritech.

Bangladesh as a Trusted Digital Hub

The government will market Bangladesh's cybersecurity, IP protection, and service reliability globally, establish 'Trusted Bangladesh' branding for ICT exports similar to global 'Made in...' labels, and build partnerships with GCC and ASEAN countries to position Bangladesh as a preferred outsourcing hub.

Global Digital Economy KPI Framework (2025-2030)

Year	ICT Exports (\$B)	Trade Agreements	Diaspora Investment (\$M)	Global Forums	Trust Recognition
2025	2.0	1	50	2	Strategy launched
2026	2.8	2	100	3	Framework drafted
2027	3.8	3	200	5	50 firms certified
2028	5.0	4	300	6	Top 30 IT hub
2029	7.0	5	400	7	100 firms certified
2030	10.0	6+	500+	10	Top 20 global hub

EXPANDING BANGLADESH'S DIGITAL INFLUENCE



10.11 Phased Implementation Roadmap for Strategic Governance Reforms

The strategic and governance reforms outlined in this chapter will be implemented through a carefully phased approach over six years (2025-2030). This phased structure ensures that foundational reforms are established before more complex initiatives are undertaken, and that lessons learned in early phases inform subsequent activities.

10.11.1 Phase 1: Foundation Building (2025-2026)

The first phase focuses on establishing the institutional foundations for digital transformation governance. Priority activities include establishing the National Digital Transformation Taskforce (NDTT) with full membership and operational mandate, operationalizing the Implementation and Monitoring Secretariat under ICT Division, launching the National Digital Dashboard for real-time monitoring, establishing inter-agency taskforces for priority areas (Cybersecurity, DPI, AI & Data Governance), developing and adopting the initial KPI framework aligned with SDGs, and initiating first-round PPP negotiations for 5G and National Fibre Bank.

10.11.2 Phase 2: Scaling and Optimization (2027-2028)

The second phase focuses on scaling successful pilots and optimizing coordination mechanisms. Key activities include full operationalization of all inter-agency taskforces with clear deliverables, launching performance-based budgeting across all participating ministries, expanding PPP portfolio to include DPI, semiconductor roadmap, and Universal Payment Gateway, establishing diaspora engagement mechanisms including Digital Infrastructure Bonds and Venture Funds, implementing capacity transfer clauses in all major donor-funded projects, and achieving 70% of international projects with knowledge transfer and local content requirements.

10.11.3 Phase 3: Consolidation and Global Integration (2029-2030)

The final phase focuses on consolidating gains and achieving global recognition. Priority activities include achieving 100% compliance with coordination framework requirements across government, reaching US\$10 billion annual ICT exports, securing Bangladesh's position in UN E-Government Index Top 15, signing 6+ regional digital trade agreements, training 10,000+ officials in frontier technologies, engaging 1,000+ local startups in international pilot projects, and achieving active participation in at least 10 global digital governance forums.

Strategic Governance Reform Timeline

Reform Area	Phase 1 (2025-26)	Phase 2 (2027-28)	Phase 3 (2029-30)
Coordination Framework	NDTT established, Secretariat operational	All taskforces operational	Full integration achieved

PPP Model	Framework adopted, first projects launched	Portfolio expanded to 5+ sectors	Mature ecosystem, private sector leadership
Funding Mechanisms	NRB bonds launched, donor commitments secured	Diversified funding mix achieved	Self-sustaining model established
M&E Framework	Dashboard launched, baseline KPIs set	Real-time tracking operational	Evidence-based governance culture
International Partnerships	Initial agreements signed	Technology transfer scaled	Global leadership in digital governance
Global Positioning	Branding strategy launched	Regional integration advanced	Top 20 trusted digital hub

10.12 Expected Outcomes by 2030

The successful implementation of the strategic and governance reforms outlined in this chapter will fundamentally transform how Bangladesh approaches digital development. By 2030, the following outcomes are expected:

Governance Transformation

The three-tier coordination framework will eliminate institutional silos and ensure coherent digital strategy execution across government. All ministries will operate under unified digital standards with clear accountability mechanisms. Performance-based budgeting will ensure efficient resource allocation and value-for-money in all digital investments.

Public-Private Collaboration

Mature PPP mechanisms will have mobilized over \$4 billion in private sector investment in digital infrastructure. Innovation sandboxes and pilot programs will have created sustainable pathways for technology adoption. Risk-sharing frameworks will have reduced government fiscal burden while accelerating implementation timelines.

Sustainable Financing

Diversified funding sources—including government budget, development partners, NRB investment, private sector FDI, and PPP co-investment—will have mobilized over \$11 billion for digital transformation. Self-sustaining revenue models for flagship digital services will reduce dependency on external financing.

Evidence-Based Governance

The National Digital Dashboard will provide real-time visibility into transformation progress. Annual Digital Transformation Scorecards will benchmark Bangladesh against global peers. Citizen feedback mechanisms will ensure that digital services remain responsive to user needs.

Global Integration

All international projects will include built-in knowledge transfer and sustainability clauses. Over 10,000 Bangladeshi officials and professionals will have been trained in frontier technologies. 1,000 local startups will be integrated into international pilot projects, creating export-ready innovation. Bangladesh will have become an active voice in at least 10 global digital forums, shaping international digital governance.

Economic Impact

ICT exports will reach US\$10 billion annually. Bangladesh will be a signatory to at least 6 regional digital trade agreements. Diaspora will contribute US\$500 million+ annually in investment and mentorship. Bangladesh will be recognized among the Top 20 trusted outsourcing destinations globally.

10.13 Risk Management and Mitigation Strategies

The ambitious reforms outlined in this chapter carry inherent risks that must be proactively managed. A comprehensive risk management framework will be integrated into the governance structure to identify, assess, and mitigate potential challenges.

Risk Category	Specific Risks	Likelihood	Impact	Mitigation Strategies
Institutional	Resistance to coordination, turf protection	High	High	Clear mandates, incentive alignment, leadership commitment
Political	Policy discontinuity, changing priorities	Medium	High	Cross-party consensus, institutional embedding
Financial	Funding shortfalls, budget constraints	Medium	High	Diversified funding, PPP expansion, cost recovery
Technical	Integration failures, security breaches	Medium	High	Robust architecture, cybersecurity investment
Capacity	Skills gaps, brain drain	High	Medium	Training programs, diaspora engagement, retention incentives
External	Global economic shocks, partner withdrawal	Low	High	Contingency planning, diversified partnerships

10.14 Conclusion

The strategic and governance reforms outlined in this chapter represent a fundamental shift in how Bangladesh approaches digital transformation. By establishing robust coordination mechanisms, sustainable financing models, transparent accountability frameworks, and strategic international partnerships, Bangladesh positions itself not merely as a consumer of digital technology but as a contributor to global digital governance.

The three-tier coordination structure—comprising the National Digital Transformation Council, Inter-Agency Taskforces, and the Implementation Secretariat—ensures vertical and horizontal integration across all levels of government. The National Digital Transformation Taskforce brings together key ministries, regulators, and development partners under a unified mandate, breaking down the institutional silos that have historically undermined digital initiatives.

The blended financing strategy—combining government resources, diaspora investment, development partner support, and private sector capital—provides the sustainable funding base necessary for long-term success. Performance-based budgeting and cost-recovery mechanisms ensure fiscal sustainability while maintaining service affordability.

The monitoring and evaluation framework, anchored by the National Digital Dashboard and regular third-party assessments, creates a culture of evidence-based governance. Transparent KPIs, citizen feedback mechanisms, and global benchmarking ensure continuous improvement and accountability.

International partnerships structured around knowledge transfer, local content requirements, and co-innovation ensure that Bangladesh builds indigenous capacity while benefiting from global expertise. The goal is not dependency but partnership—positioning Bangladesh as a contributor to global digital norms and standards.

By 2030, these reforms will have transformed Bangladesh into a regional digital leader—not through technological adoption alone, but through the governance innovations that make sustainable digital development possible. This chapter's reforms are not merely administrative adjustments; they are the foundation upon which Bangladesh's digital future will be built.

End of Chapter 10

11. Strategic roadmap to 2030 and beyond

11.1 Introduction

The National Digital Transformation Strategy 2025-2030 represents Bangladesh's most comprehensive effort to transition from a technology consumer to a regional digital leader. This roadmap consolidates all reforms, initiatives, and projects outlined in the preceding chapters into a unified, phased implementation plan. The roadmap is designed to ensure that each initiative builds upon the foundations laid by earlier reforms, creating a cohesive and self-reinforcing digital ecosystem.

The implementation timeline spans six years, organized into three primary phases, with provisions for longer-term initiatives extending to 2035 and 2040 where foundational infrastructure investments require extended development cycles. Each phase has been calibrated to balance immediate priorities with sustainable, long-term growth objectives.

11.2 Strategic Phases Overview

The roadmap is structured around three core implementation phases, each with distinct objectives that progressively build national digital capacity:

Phase	Timeline	Strategic Focus
Phase 1: Foundation	2025-2026	Legal frameworks, institutional reform, data governance establishment, core infrastructure deployment
Phase 2: Scale & Optimize	2027-2028	Nationwide rollout, infrastructure expansion, sectoral integration, private sector engagement
Phase 3: Smart Governance	2029-2030	AI-driven governance, DPI consolidation, global positioning, predictive services
Phase 4: Long-term Vision	2030+	Emerging technology leadership, regional hub status, sustainable digital sovereignty

11.3 Phase 1: Foundation (2025-2026)

Phase 1 establishes the legal, institutional, and technical foundations for Bangladesh's digital transformation. This phase prioritizes governance frameworks, regulatory compliance, and critical infrastructure deployment.

11.3.1 Legal and Governance Framework

Legal & Regulatory Foundations (2025-2026)	
Key Actions	Target Outcomes
Enact NDGA Rules and Regulations under NDGO 2025	2025-2026
Establish NDGA Secretariat and Policy Board	2025-2026

Implement PDPO 2025 compliance framework	2025-2026
Deploy Cyber Safety Ordinance enforcement mechanisms	2025-2026
Appoint Chief Data Officers across 10-20 ministries	End 2026
Begin PDPO compliance audits government-wide	Q4 2026
Establish National Cybersecurity Task Force (N-CERT)	2026
Introduce Digital Service Act for public digital services	2026

11.3.2 Institutional Reform

Institutional Restructuring (2025-2026)	
Key Actions	Target Outcomes
BCC restructuring: Strengthen NSB integration with BNDA	2026
Reform Info-Sarker-3 network for enhanced resilience	2025-2026
Transform DoICT into primary grassroots training arm	2026
Launch SHQTC certification authority operations	Dec 2025
Phase 1 Government Software Repository launch	2026
Establish Training Resource Hubs at divisional levels	2026
a2i structural reforms: Semi-autonomous status within ICT Division	2025-2026
NCSA: Submit NSOC proposal, finalize Cybersecurity Strategy 2026-2030	2025

11.3.3 Digital Public Infrastructure

DPI Foundation - STAR Phase (2025-2026)	
Key Actions	Target Outcomes
Deploy SITA project across 5-10 ministries	2026
Establish Bangladesh National Digital Architecture (BNDA) under NDGA	2026
Launch National Data Exchange pilot (1-3 cross-sector APIs)	2026
Begin Unified Identity Layer integration (NID, BRN, Passport)	2026
Certify 0-10 AI systems under new framework	2026
Train 500-1,000 officials in data and AI governance	2025-2026

11.3.4 E-Government Acceleration

E-Government Foundation (2025-2026)	
Key Actions	Target Outcomes
Digitize 800+ government services end-to-end	2026
Achieve 30-50% services fully digital	2026
Launch National e-Service Dashboard	2025
Implement Universal Employee Payment System with iBAS++	2025
Launch Police Automation Program (FIRs, case tracking)	2025-2026
Deploy 20-40 services in Bangla	2025-2026
Achieve 15-30 million active e-service users	2026

11.3.5 Infrastructure Development

Infrastructure Foundations (2025-2026)	
Key Actions	Target Outcomes
National Fiber Bank: Complete 25-50% fiber inventory	2025-2026
Achieve 35-50% mobile towers fiber-connected	2025-2026
Fiber Bank platform operational in 3-6 divisions	2025-2026
National Drone Policy: NDCC setup, CAAB regulations finalized	2025
Drone pilots in agriculture, land, and disaster sectors	2025
Train 200-500 licensed drone pilots	2025-2026
Complete 5-20% Upazila drone survey coverage	2026

11.3.6 Economic Initiatives

Digital Economy Launch (2025-2026)	
Key Actions	Target Outcomes
Introduce Universal Payment Gateway (UPG)	2025
Launch Startup Bangladesh 2.0 program	2025
Establish National Innovation & Venture Fund	2025
Support 300-600 new startups	2025-2026

Achieve \$100-250M startup investment inflows	2025-2026
Launch 2-3 regional innovation hubs	2025-2026
ICT exports reach \$2-2.8B annually	2025-2026
Sign first regional digital trade agreement (BIMSTEC MoU)	2025

11.3.7 Capacity Building

Workforce Development (2025-2026)	
Key Actions	Target Outcomes
Establish National AI & 4IR Hub	2026
Launch nationwide digital skills and AI literacy programs	2025-2026
Standardize training curricula through NSDA	2026
Launch National Outsourcing & Freelancing Accelerator	2025-2026
Train initial cohort of madrasa students for GCC markets	2025-2026
Establish 2-4 University-Industry Innovation Accelerators	2025-2026
Train 500-1,500 Bangladeshi officials in frontier technologies	2025-2026

11.4 Phase 2: Scale & Optimize (2027-2028)

Phase 2 expands the foundational work into nationwide deployment, deepens cross-sectoral integration, and accelerates private sector engagement. This phase builds Bangladesh's digital backbone through infrastructure scaling and regulatory optimization.

11.4.1 Data Governance & Interoperability - DSTAR Phase

DSTAR Integration (2027-2028)	
Key Actions	Target Outcomes
Operationalize NRDEX nationwide	2027
Integrate 20-40 ministries and regulators	2027-2028
Launch AI registry and certification regime	2027
Conduct first cross-sector data-exchange pilot	2027
Introduce national interoperability dashboard	2027
Certify 25-50 AI systems	2027-2028
Deploy 5-8 cross-sector APIs under NRDEX	2027-2028
Appoint 40-60 Chief Data Officers	2027-2028
Train 2,000-3,500 officials in data/AI governance	2027-2028

11.4.2 Infrastructure Expansion

National Infrastructure Scale-up (2027-2028)	
Key Actions	Target Outcomes
Roll out nationwide 5G and fiber-optic broadband	2027-2028
Enact National Cloud Act for government/enterprise data	2027
Deploy AI-driven digital identity verification (e-KYC)	2027
Complete 70-85% fiber inventory	2027-2028
Achieve 65-80% mobile towers fiber-connected	2027-2028
Utilize 30-50% unused fiber cores	2027-2028
Fiber Bank platform operational in 8-10 divisions	2027-2028
Integrate BDCCL and NDC for capacity optimization	2027
Launch National Code Repository Initiative	2027

11.4.3 E-Services Expansion

E-Government Expansion (2027-2028)	
Key Actions	Target Outcomes
Achieve 65-75% of all services fully digital	2027-2028
Integrate all services into MyGov super-app	2028
Introduce real-time grievance redressal across ministries	2027
Implement AI-based chatbots for citizen interaction	2027
Deploy 60-80 services in Bangla	2027-2028
Achieve 50-75 million active e-service users	2027-2028
Realize \$250-400M in cost savings	2027-2028

11.4.4 Sectoral Digital Transformation

Education Sector (2027-2028)	
Key Actions	Target Outcomes
40-60% schools with digital classrooms	2027-2028
10-15 million students using e-learning platforms	2027-2028
200,000-300,000 teachers ICT-trained	2027-2028
20,000-30,000 accessible digital titles available	2027-2028
Health Sector (2027-2028)	
Key Actions	Target Outcomes
50-75% health facilities digitized	2027-2028
25-35 million citizens with unique Health ID	2027-2028
5-10 million referrals managed digitally	2027-2028
8-15 million teleconsultations conducted annually	2027-2028
50-70% MCCoD coverage for registered deaths	2027-2028
Agriculture Sector (2027-2028)	
Key Actions	Target Outcomes
5-10 million farmers using agri apps	2027-2028

30-50% villages with agricultural info hubs	2027-2028
2,000-5,000 drone-assisted crop surveys	2027-2028
\$500M-\$1B in digital agricultural transactions	2027-2028

11.4.5 Economic & Industrial Growth

Digital Economy Scale-up (2027-2028)	
Key Actions	Target Outcomes
Support 1,200-2,000 new startups	2027-2028
Achieve \$400-600M startup investment inflows	2027-2028
Enable 4-6 startup IPOs/exits	2027-2028
Launch 4-6 regional innovation hubs	2027-2028
ICT exports reach \$3.8-5B annually	2027-2028
Sign 3-4 regional digital trade agreements	2027-2028
Onboard 400,000-800,000 rural entrepreneurs on e-commerce	2027-2028
Achieve 45-70% GPS/geotag address coverage	2027-2028

11.4.6 Drone & Emerging Technology

Drone & Advanced Tech (2027-2028)	
Key Actions	Target Outcomes
Drone units operational in all divisions	2027
National Drone Repository and UTM operational	2027
1,000-2,000 licensed drone pilots	2027-2028
50-75% Upazila coverage for drone surveys	2027-2028
1,500-3,000 datasets in national repository	2027-2028
8-10 government agencies using drones	2027-2028

11.5 Phase 3: Smart Governance & Digital Economy (2029-2030)

Phase 3 realizes Bangladesh's vision of a fully functional Digital Public Infrastructure ecosystem, AI-driven predictive governance, and regional leadership in digital innovation. This phase consolidates all prior investments into citizen-centric smart services.

11.5.1 DPI Consolidation

DPI Consolidation (2029-2030)	
Key Actions	Target Outcomes
Full NDGA autonomy with independent funding	2029
Integrate private-sector APIs (banks, telcos, logistics)	2029-2030
Establish AI innovation sandbox and cross-border pilots	2029
Achieve zero manual data exchange in government	2030
Integrate 60-100+ ministries/agencies via NDGA	2029-2030
Deploy 12-15+ cross-sector APIs under NRDEX	2029-2030
Certify 75-100+ AI systems	2029-2030
Appoint 80-100+ Chief Data Officers	2029-2030
Reduce reported government data breaches to near zero	2030

11.5.2 Smart E-Government

Smart Service Delivery (2029-2030)	
Key Actions	Target Outcomes
Reach 85-95% end-to-end digital service coverage	2029-2030
Enable predictive governance: AI anticipates citizen needs	2029
Integrate digital ID and document wallet for instant verification	2029
Bangladesh enters UN E-Gov Index Top 15	2030
Deploy 100-120 services in Bangla	2029-2030
Achieve 95-110 million active e-service users	2029-2030
Realize \$600-850M in cost savings	2029-2030
Achieve 80-85% citizen satisfaction	2030

11.5.3 Complete Infrastructure Deployment

Infrastructure Completion (2029-2030)	
Key Actions	Target Outcomes
Complete 100% fiber inventory with GIS visibility	2030
Achieve 100% mobile towers fiber-connected	2030
Utilize 70-90% unused fiber cores	2029-2030
Fiber Bank generating BDT 15,000-25,000M+ in lease revenue	2029-2030
Cross-border fiber nodes operational	2030
Full disaster recovery infrastructure expansion	2029

11.5.4 Sectoral Transformation Completion

Education Sector Completion (2029-2030)	
Key Actions	Target Outcomes
80-100% schools with digital classrooms	2029-2030
20-25+ million students using e-learning platforms	2029-2030
400,000-500,000+ teachers ICT-trained	2029-2030
40,000-50,000 accessible digital titles available	2030
Health Sector Completion (2029-2030)	
Key Actions	Target Outcomes
90-100% health facilities digitized	2029-2030
45-50+ million citizens with unique Health ID	2029-2030
15-20+ million referrals managed digitally	2029-2030
25-40+ million teleconsultations annually	2029-2030
85-100% MCCoD coverage	2029-2030
Agriculture Sector Completion (2029-2030)	
Key Actions	Target Outcomes
15-20+ million farmers using agri apps	2029-2030
75-100% villages with agricultural info hubs	2029-2030
10,000-20,000 drone-assisted crop surveys annually	2029-2030

\$2-3+ billion in digital agricultural transactions	2029-2030
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11.5.5 Digital Economy Maturity

Digital Economy Targets (2029-2030)	
Key Actions	Target Outcomes
Support 3,500-5,000 new startups	2029-2030
Achieve \$800M-\$1B annual startup investment	2029-2030
Enable 8-10+ startup IPOs/exits	2029-2030
Launch 8-10 regional innovation hubs across all divisions	2029-2030
30% of startups women-led	2030
ICT exports reach \$7-10B annually	2029-2030
Sign 5-6+ regional digital trade agreements	2029-2030
Diaspora tech investment reaches \$400-500M+ annually	2029-2030
Bangladesh recognized as Top 20 global outsourcing hub	2030
1 million active freelancers including 150,000 madrasa students	2030
\$1B+ annual freelance export earnings	2030

11.5.6 National Security & Safety

Security Infrastructure (2029-2030)	
Key Actions	Target Outcomes
AI-enabled CCTV in 100% major cities	2030
100% pipeline/power grid segments with IoT sensors	2030
Satellite/drone coverage for all districts	2030
1,500 km smart highway segments operational	2030
50% reduction in emergency response times	2030

11.6 Phase 4: Long-term Vision (2030 and Beyond)

Certain strategic initiatives require investment horizons extending beyond 2030. These long-term programs ensure Bangladesh's sustained competitiveness and leadership in emerging technologies.

11.6.1 Semiconductor Industry Development

Semiconductor Roadmap (2030-2040)	
Key Actions	Target Outcomes
Train 10,000+ semiconductor engineers under BASICS initiative	2030
\$1B+ annual semiconductor export revenue	2030
Establish design, testing, and OSAT hubs	2030
\$500M public investment leveraged into \$2B+ return	2030
\$1B+ private and foreign co-investment mobilized	2030
Green Foundry: 30% renewable energy adoption	2030
Green Foundry: 100% renewable energy adoption	2040

11.6.2 National Drone Coverage

Complete Drone Infrastructure (2030+)	
Key Actions	Target Outcomes
All ministries using drones for operations	2030
5,000+ licensed drone pilots nationwide	2030
100% district coverage for drone surveys	2030
10,000+ datasets in national repository	2030
National ortho-maps updated on 3-year cycle	2030+
Commercial drone delivery ecosystem established	2030+

11.6.3 Deep-Tech Identity (BEAR)

Bangladesh's long-term deep-tech identity will be anchored in the BEAR framework:

- Biotech: Health tech innovations, genomics, and precision medicine
- Electronics: Semiconductor design, IoT manufacturing, and smart devices
- AI: Ethical AI governance, NLP for Bangla, and predictive systems
- Robotics: Industrial automation, agricultural robotics, and smart manufacturing

11.7 Master KPI Dashboard (2025-2030)

The following consolidated KPI framework tracks progress across all strategic pillars:

11.7.1 Data Governance & Interoperability

Indicator	2025	2026	2027	2028	2029	2030
Ministries/Agencies integrated via NDGA	5	10	25	40	60	100+
Chief Data Officers appointed	10	20	40	60	80	100+
AI systems certified	0	10	25	50	75	100+
Cross-sector APIs live under NRDEX	1	3	5	8	12	15+
Officials trained in data/AI governance	500	1,000	2,000	3,500	4,500	5,000+

11.7.2 E-Services & Citizen Engagement

Indicator	2025	2026	2027	2028	2029	2030
% Services End-to-End Digital	30%	50%	65%	75%	85%	95%
Services in Bangla	20	40	60	80	100	120
Active Users (millions)	15M	30M	50M	75M	95M	110M
Citizen Satisfaction (%)	60%	65%	70%	75%	80%	85%
Cost Savings (US\$)	\$50M	\$120M	\$250M	\$400M	\$600M	\$850M

11.7.3 Digital Infrastructure

Indicator	2025	2026	2027	2028	2029	2030
Fiber Inventory Completed (%)	25%	50%	70%	85%	95%	100%
Mobile Towers Fiber-Connected (%)	35%	50%	65%	80%	90%	100%
Unused Fiber Cores Utilized (%)	5%	15%	30%	50%	70%	90%
Fiber Bank Revenue (BDT M)	500	2,000	4,500	8,000	15,000	25,000+

11.7.4 Digital Economy & Startup Ecosystem

Indicator	2025	2026	2027	2028	2029	2030
New Startups Supported	300	600	1,200	2,000	3,500	5,000
Startup Investment (US\$)	\$100M	\$250M	\$400M	\$600M	\$800M	\$1B

Women-Led Startups (%)	20%	22%	25%	27%	28%	30%
ICT Export Earnings (US\$)	\$2B	\$2.8B	\$3.8B	\$5B	\$7B	\$10B
Regional Innovation Hubs	2	3	4	6	8	10

11.8 Critical Success Factors

The successful implementation of this roadmap depends on several interdependent factors that must be addressed consistently throughout the six-year timeline:

11.8.1 Governance & Coordination

- **Strong Inter-Ministerial Coordination:** The ICT Division must maintain leadership in coordinating cross-agency initiatives, with NDGA serving as the technical authority for data governance and interoperability standards.
- **Political Commitment:** Sustained high-level political support is essential to navigate institutional resistance and ensure continuity across electoral cycles.
- **Accountability Mechanisms:** Real-time monitoring dashboards, regular progress reviews, and transparent KPI reporting will ensure accountability at all levels.

11.8.2 Financial Sustainability

- **Blended Financing:** National budget allocations must be supplemented by foreign loans, development assistance, PPP investments, and NRB diaspora contributions.
- **Revenue Generation:** The Fiber Bank, SHQTC certification services, and other infrastructure assets should generate sustainable revenue streams to reduce fiscal dependence.
- **Cost Efficiency:** The target of \$850M in e-service cost savings by 2030 will partially offset implementation costs while demonstrating return on investment.

11.8.3 Human Capital Development

- **Workforce Pipeline:** The target of 7-8 million ICT workforce by 2030 requires sustained investment in education, vocational training, and reskilling programs.
- **Public Sector Capacity:** Training 5,000+ officials in data governance and AI ethics is critical to ensuring competent implementation across government.
- **Inclusive Growth:** Specific targets for women (40% freelancing participation, 30% startup leadership) and marginalized groups (madrasa students, rural entrepreneurs) ensure equitable benefits distribution.

11.8.4 Technology & Security

- **Vendor Neutrality:** Strict enforcement of open standards, government source code ownership, and interoperability requirements will protect digital sovereignty.
- **Cybersecurity Resilience:** The N-CERT, NSOC, and PDPO compliance framework must be fully operational to protect critical infrastructure and citizen data.
- **Technology Transfer:** All international partnerships must include mandatory knowledge transfer clauses to build sustainable local capacity.

11.9 Risk Mitigation Framework

Risk Category	Potential Impact	Mitigation Strategy
Funding Shortfalls	Delayed infrastructure deployment, reduced service quality	Diversified funding sources; PPP models; phased implementation with priority sequencing
Institutional Resistance	Slow adoption, siloed operations, duplicated systems	Change management programs; performance incentives; mandatory interoperability compliance
Cybersecurity Threats	Data breaches, service disruptions, citizen trust erosion	NSOC deployment; PDPO enforcement; regular VAPT audits; incident response protocols
Skills Gap	Implementation delays, quality issues, vendor dependence	National training programs; university partnerships; international certifications; competitive compensation
Technology Obsolescence	Stranded investments, incompatible systems	Open standards mandate; modular architecture; regular technology reviews; sunset provisions
Political Discontinuity	Policy reversals, budget cuts, institutional restructuring	Legal frameworks (NDGO, PDPO); independent NDGA; broad stakeholder coalitions
Digital Divide	Exclusion of rural/marginalized populations, inequality	Bangla-first interfaces; mobile optimization; community digital hubs; accessibility standards

Conclusion: Pathway to 2030 and Beyond

This comprehensive roadmap transforms Bangladesh's digital vision into actionable milestones spanning 2025-2030 and beyond. By 2030, Bangladesh will have achieved:

- A fully independent NDGA regulating data, privacy, and AI across 100+ ministries and agencies
- Zero manual data transfers in government through secure API-based exchanges
- 95% of government services delivered digitally end-to-end with 110 million active users
- \$10 billion in annual ICT exports, positioning Bangladesh among the Top 20 global outsourcing hubs
- A vibrant ecosystem of 5,000 startups with \$1 billion in annual investment
- Complete fiber infrastructure with 100% mobile tower connectivity
- Transformed education (25M+ digital learners), health (50M+ citizens with Health IDs), and agriculture (20M+ digitally connected farmers) sectors
- Bangladesh's recognition as a regional leader in digital governance, entering the UN E-Government Index Top 15

The journey from MESH to STAR to DSTAR represents more than technological evolution; it embodies Bangladesh's commitment to inclusive, citizen-centric digital governance. Success hinges on sustained political will, cross-institutional coordination, public-private collaboration, and unwavering focus on transparency and accountability.

With bold reforms and sustained investment, Bangladesh will not merely participate in the global digital economy but will lead the region in smart governance, digital inclusion, and technological innovation by 2030 and beyond.

Annex

Note to Successor

আমার আন্তরিক সালাম এবং মোবারকবাদ গ্রহণ করুন।

০৫ মার্চ ২০২৫ থেকে ১৪ ই ফেব্রুয়ারি ২০২৬ সময়কালে তথ্য প্রযুক্তি বিভাগের দায়িত্ব পালন করি। এই সময়ে গৃহীত গুরুত্বপূর্ণ উদ্যোগ এবং এর সাথে সংশ্লিষ্ট পাইপলাইন এক্টিভিটি গুলো সম্পর্কে আপনাকে অবহিত করার অভিপ্রায় থেকে লিখছি।

১। মানবাধিকার সংস্থা, ডিজিটাল প্ল্যাটফর্ম, জাতিসংঘ, ইউরোপীয় ইউনিয়ন সহ আন্তঃমন্ত্রণালয় পর্যায়ে পর্যাপ্ত স্টেকহোল্ডার কন্সাল্টেশন করে আমরা সাইবার সুরক্ষা অধ্যাদেশ ২০২৫ প্রণয়ন করেছি। এই আইন বিগত ফ্যাসিস্ট রেজিমের ঘৃণ্য ডিজিটাল সিকিউরিটি আইন ও সাইবার সিকিউরিটি আইনকে প্রতিস্থাপন করেছে, নাগরিকদের সুরক্ষাকে প্রাধান্য দেয়ার পাশাপাশি রাষ্ট্রের গুরুত্বপূর্ণ স্থাপনা এবং নারী ও শিশুদের জন্য বিশেষ সুরক্ষার জায়গা তৈরী করেছে। বিগত সরকারের অপব্যবহারের কথা মাথায় রেখে যতটা সম্ভব এই আইনের অপব্যবহার কমানোর ফ্রেমওয়ার্ক তৈরী করা হয়েছে। এতে একইসাথে ন্যাশনাল সাইবার সেফটি এজেন্সি'কে কারিগরিভাবে শক্তিশালী করার মেন্ডেট রাখা হয়েছে। বর্তমানে বিশ্বব্যাংকের সাথে একটি NSoC প্রজেক্ট চলমান আছে, কাজটি স্বচ্ছতার সাথে এগিয়ে নেওয়ার জন্য, এবং প্রয়োজনে আরো বৃহদাকারে National Cyber Operation Center নির্মাণ করার জন্য আপনার দৃষ্টি আকর্ষণ করছি।

২। একইভাবে বাংলাদেশে অপারেশনে থাকা প্রায় সব আন্তর্জাতিক ডিজিটাল প্ল্যাটফর্ম, জাতিসংঘ, ইউরোপীয় ইউনিয়ন, মানবাধিকার সংস্থা এবং বিশেষভাবে ওয়ার্ল্ডব্যাংক আইনজীবীদের (মার্ক রোরি-ওয়াশিংটন ডিসি, রিলিফ-ঢাকা) সহ আন্তঃমন্ত্রণালয় পর্যায়ে বিশদ স্টেকহোল্ডার কন্সাল্টেশন শেষে আমরা আইসিটি ডিভিশন থেকে 'ব্যক্তিগত উপাত্ত সুরক্ষা অধ্যাদেশ ২০২৫' এবং 'জাতীয় উপাত্ত ব্যবস্থাপনা অধ্যাদেশ ২০২৫' প্রণয়ন করেছি। এটা বাংলাদেশের ইতিহাসে পার্সোনাল ডেটা প্রটেকশন এবং ন্যাশনাল ডেটা গভর্নেন্স সংক্রান্ত প্রথম আইন।

এই আইনে প্রধানমন্ত্রীর কার্যালয়ের অধীনে একটি 'ন্যাশনাল ডেটা গভর্নেন্স অথরিটি' তৈরির ম্যান্ডেট দেয়া হয়েছে। ডেটা গভর্নেন্স অথরিটির অর্গানোগ্রাম তৈরিতে কমিটি প্রতিবেদন তৈরি করেছে যা ক্যাবিনেট ডিভিশনে পাঠানো হয়েছে। আমার আন্তরিক নিবেদন এই যে, দেশের বিভিন্ন সরকারি প্রতিষ্ঠান এবং বিশেষভাবে নাগরিকদের ব্যক্তিগত উপাত্ত ঝুঁকির ব্যাপকতার বিষয়টি আমলে নিয়ে অতি দ্রুততার সাথে আপনি এই অথরিটি গঠনের ব্যাপারে সর্বোচ্চ চেষ্টা করবেন। নতুন আইনে ডেটা অথরিটির ফাংশনালিটি সবিস্তারে উঠে এসেছে।

এই আইনে ব্যক্তিগত উপাত্তকে দরকারি শ্রেণীবিন্যাস করা হয়েছে, ডেটা লোকালাইজেশন'কে উঠিয়ে দেওয়া হয়েছে (দুটি কৌশলগত শর্তসাপেক্ষে) -

ক। ক্রিটিক্যাল ইনফরমেশন ইনফ্রাস্ট্রাকচার (CII, Referebce- CII Guideline, NCSA) গুলোর হার্ডওয়্যার/সফটওয়্যার সরবরাহকারীরা কোনো আঞ্চলিক কিংবা আন্তর্জাতিক নিষেধাজ্ঞায় পড়লে প্যারалаইজ না হয়ে যায়- সেজন্য এই কৌশলগত এবং জাতীয় গুরুত্বপূর্ণ খাতগুলোর সাথে সংশ্লিষ্ট বাল্ক অ্যামাউন্ট ক্রস বর্ডার পার্সোনাল ডেটা ট্রান্সফারের ক্ষেত্রে একটা সিস্কো-নাস লোকাল ব্যাকআপ রাখার কথা বলা হয়েছে।

খ। ডিজিটাল ইকোনমি প্লেনার এবং প্ল্যাটফর্ম গুলোর সুবিধা বিবেচনায়, বাল্ক এমাউন্ট পার্সোনাল ডেটার ক্রস বর্ডার ট্রান্সজেকশন এ কর্তৃপক্ষের অনুমতির বাধ্যবাধকতা উঠিয়ে দেওয়া হয়েছে। তবে কেবলমাত্র বাল্ক এমাউন্ট পার্সোনাল ডেটা বা সংবেদনশীল ডেটা বা PII (Personal Identifiable information) Data cross border ট্রান্সজেকশন এর ক্ষেত্রে National Data Governance authority বা কর্তৃপক্ষকে ডেটাডিকশনারি জানানোর কথা বলা হয়েছে যাতে করে সিভিয়ার ডেটা ব্রেক হলে তার রিকভারিতে দ্রুত কারিগরি উদ্যোগ নেয়া যায়।

আমার দায়িত্ব পালনকালীন সময়ে প্ল্যাটফর্ম গুলো থেকে ডেটা লোকালাইজেশন প্রশ্ন বিভিন্ন ধরনের যৌক্তিক অযৌক্তিক এবং অযাচিত চাপ এসেছে। বিষয়টিকে ইউএসটিআর (যুক্তরাষ্ট্রের সাথে বাণিজ্য ও রিসিপ্রোকাল ট্যারিফ নেগোসিয়েশন) এ এসকালেশন করা হয়েছে। তথাপি আমরা একটি যৌক্তিক অবস্থানে পৌঁছে দেশের ডেটা সত্রেন্টি রক্ষার নেগোসিয়েশন করেছি।

ব্যক্তিগত উপাত্ত সুরক্ষা অধ্যাদেশ ২০২৫ এ ২টি সংশোধনী আনা হয়েছে। ডেটা ক্লাসিফিকেশন, সিক্রোনাস লোকাল ব্যাপক, সংবেদনশীল ব্যক্তিগত উপাত্ত ডেটার cross border ট্রান্সফার, এবং সাইবার আক্রমণ বিপরীতে ক্রিমিনাল অফেন্স- এই তিনটি বিষয়ে দেশি-বিদেশি বিশেষজ্ঞদের সাথে সমন্বয় করে আমরা দেশের স্বার্থে যৌক্তিক সিদ্ধান্ত নিয়েছি। আমার মতে, প্ল্যাটফর্মগুলোকে এর বেশি ছাড় দিলে দেশের সার্বিক ডেটা সার্বভৌমত্ব নষ্ট হবে, এবং ডিজিটাল ফিউচার বিঘ্নিত হবে।

৩। দায়িত্ব পালনকালে আমি ন্যাশনাল ইন্টার-অপারেবিলিটি এবং সিটিজেন আইডি অথেন্টিকেশন লেয়ার তৈরির লিটারেসি এবং এতদসংক্রান্ত ডিপ্লোম্যা প্রণয়ন গুরুত্বপূর্ণ ফোকাসে রেখেছি। বিশ্বব্যাংক এখানে ফান্ডিং করবে। এর সত্যিকার বাস্তব রূপ লাভ আপনার সময়ে সম্ভব হবে বলে আমি আশা রাখি।

এজন্য একটি বিশদ লিগ্যাল এন্ড ইনফ্রা ফ্রেইমওয়ার্ক তৈরি করেছি। বাংলাদেশের ডিজিটাল ইকোনোমিকে সত্যিকারভাবে বাস্তবায়ন করতে হলে আমাদের দরকার একটি সক্ষম DPI বা ডিজিটাল পাবলিক ইনফ্রাস্ট্রাকচার।

DPI এর ফাউন্ডিং পিলার তিনটি-

ক। সাইবার সুরক্ষা, ব্যক্তিগত উপাত্ত সুরক্ষা, ডেটা গভর্নেন্স এর আইনি ভিত্তি। জাতীয় ক্লাউড পলিসি, এন্টারপ্রাইজ সফটওয়্যার পলিসি, ইন্টার অপারেবিলিটি পলিসি, ইত্যাদি পলিসি পরিকাঠামো যার জন্য ডেটা গভর্নেন্স আইনে ম্যান্ডেট দেয়া আছে।

খ। প্রজেক্ট ভিত্তিক আইসোলেটেড কাজ করতে করতে দেশের প্রতিটি মন্ত্রণালয় এক একটি ডিজিটাল আইল্যান্ড, এখানে আন্তঃসংযোগ এবং আন্তঃপরিবাহিতা ব্যাপকভাবে অনুপস্থিত। বিগত সরকারের আমলে আন্তঃপরিবাহিতা, আন্তঃসংযোগ, ন্যাশনাল এপিআই এক্সচেঞ্জ, ন্যাশনাল রেস্পন্সিবল ডেটা শেয়ারিং এক্সচেঞ্জ এবং সত্যিকারের ইলেকট্রনিক আইডি অথেন্টিকেশন লেয়ার তৈরি না করে, সরাসরি সার্ভিস লেয়ারে এ গিয়ে কিছু অ্যাপ তৈরি চেষ্টা হয়েছে, যা ব্যর্থ হয়েছে, মানহীন আনসিকিউর এপ গুলো সব অকেজো হয়ে গেছে। ইন্টারপারেবিলিটি এবং এক্সেস টু ফার্স্ট হ্যান্ড ডেটার সমস্যা সমাধান না করে এপ তৈরির চেষ্টা চিরতরে বন্ধ করতে হবে।

ফলে বিগত সরকারের ডিজিটাল বাংলাদেশ প্রপাগান্ডাই ফেইল করেছে এবং স্মার্ট বাংলাদেশ মিশনও ফেইল করেছে। বিপরীতে দরকার ডিজিটাল ট্রান্সফরমেশন এবং ইন্টারঅপারেবল ডেটা গভর্নেন্স যার জন্য ডিজিটাল পাবলিক ইনফ্রাস্ট্রাকচার পরিকাঠামো জরুরি- এখানে থাকবে

ক। লিগ্যাল ফাউন্ডেশন

খ। ইন্টার কানেক্টিভিটি, ন্যাশনাল কানেক্টিভিটি হাইওয়ে

গ। সিটিজেন আইডি ম্যানেজমেন্ট, আইডি ম্যাপিং, ওয়ান আইডি অথেন্টিকেশন লেয়ার

ঘ। সাইবার সিকিউরিটি, ডেটা রেগুলেশন, ডেটা গভর্নেন্স, কমপ্লায়েন্স, এনফোর্সমেন্ট এবং সর্বোপরি ডিজিটাল সার্ভিস লেয়ার।

ডিজিটাল সাইলো হিসেবে গড়ে ওঠায় প্রত্যেকটা মন্ত্রণালয়ের হার্ডওয়্যার এবং সফটওয়্যার সিস্টেমকে ইন্টার কানেক্ট এবং ইন্টার অপারেবল করে ন্যাশনাল কানেক্টিভিটি হাইওয়ে তৈরি করা এখন সময়ের সবচেয়ে জরুরি কাজ। এই ইন্টার-অপারেবিলিটি কানেক্টিভিটি লেয়ারের উপরে ইলেকট্রনিক আইডি ম্যানেজমেন্ট লেয়ার কাজ করবে, বর্তমানে এই লেয়ারগুলো নেই বলে বাংলাদেশের সত্যিকারের ডিজিটাল পাবলিক ইনফ্রাস্ট্রাকচার গড়ে ওঠেনি।

ন্যাশনাল রেস্পন্সিবল ডিজিটাল আর্কিটেকচার এন্ড ন্যাশনাল এক্সচেঞ্জ (NRDEx) হাব তৈরির জন্য আমরা পরিকল্পনা হাতে নিয়েছি। এলিমেন্টারি পর্যায়ে এপিআই ভিত্তিক ইন্টার-কানেক্টিভিটির মাধ্যমে আমরা ন্যাশনাল ইন্টারঅপারেবিলিটির যাত্রা শুরু করেছি। রেস্পন্সিবল শব্দটার মাধ্যমে রেস্পন্সিবল ডাটা শেয়ারিংকে গুরুত্ব দেয়া হয়েছে যাতে আগের মত নাগরিকদের সেন্সিটিভ এবং ক্রিটিকাল তথ্যের অপব্যবহার না হয়। এর সার্ভিস লেয়ারও নাগরিক সেবার মাধ্যমে শুরু হয়েছে, নাগরিক সেবার মাধ্যমে সব মন্ত্রণালয়ের সব সেবাকে একই এপিআই এক্সচেঞ্জ এনে দেশের জন্য ওয়াল ডিজিটাল সার্ভিস উইন্ড খোলা হচ্ছে। ইতিমধ্যে ১১ মন্ত্রণালয়ের ৪১ সেবা এখানে এপিআই দিয়ে সংযুক্ত। কিন্তু এটাকে সফল রূপ দেওয়ার জন্য দরকার একটি ন্যাশনাল কানেক্টিভিটি

এক্সচেঞ্জ, যা সাইবার নিরাপত্তা ও ডেটা গভর্নেন্স এবং ডেটা রেগুলেশন এনফোর্সমেন্ট একসাথে করবে। ডি-স্টার (DSTAR) প্রজেক্টের ডিপিপি সেভাবে প্রণয়ন করেছি। আমার আন্তরিক বিশ্বাস, আমার লিখিত ডিপিআই সংক্রান্ত ডিপিপি-টি বিশ্বব্যাপকের অর্থায়নে আপনি বাস্তবায়নের সর্বোচ্চ গুরুত্ব আরোপ করবেন।

৪। বাংলাদেশ ডেটা সেন্টার কোম্পানির সাথে ওরাকলের সাথে যে যুক্তিটি আছে তা দেশের স্বার্থ বিরোধী। এই চুক্তির অন্যান্য শর্তগুলো নিয়ে আমরা প্রতিবেদন প্রকাশ করেছি, সে মতে ওরাকলকে চুক্তি সংশোধনের অনুরোধ করা হয়েছে যাতে তারা সায় দেয়নি।

প্রতিবেদন মতে, বাজার মূল্যে অনুধ্ব ৬ মিলিয়ন ডলার রেভিনিউ আর্ন করতে সক্ষম এরকম একটা ক্লাউড ফেসিলিটি বিদ্যুৎ খাতের ক্যাপাসিটি চার্জের আদলে ২৪ মিলিয়ন ডলার দিয়ে কেনা হয়েছে, এখানে মূল ক্রেডিট ১৮ মিলিয়নের পাশাপাশি আড়াই মিলিয়ন ডলার শুদ্ধ রয়েছে। হার্ডওয়্যারের মালিকানা ওরাকলের কাছে থাকায় শুদ্ধ তাদের দেয়ার কথা থাকলেও এই শুদ্ধ বাংলাদেশের উপরে চাপিয়ে দেয়া হয়েছে। পাশাপাশি হার্ডওয়্যারের ক্যাপাসিটির সম্পূর্ণ ব্যবহার নিশ্চিত না হওয়া সত্ত্বেও সরকারকে সাড়ে তিন মিলিয়ন ডলারের বেশি ভ্যাট পরিশোধের মত ন্যাকারজনক ঘটনা ঘটেছে। যুক্তরাষ্ট্রের একটি আন্তর্জাতিক কোম্পানির সাথে চুক্তি স্বাক্ষরিত হয়ে যাওয়ায় সেখান থেকে ফিরে আসা যায়নি।

তবে ডি-নথির জন্য কন্টেনারাইজড নেটিভ ক্লাউড ফ্যাসিলিটি তৈরির উদ্যোগ নিয়েছি। এর জন্য ডি-নথি কে কন্টেনার বেজড আর্কিটেকচারে কমপ্লায়েন্ট করার উদ্দেশ্য নেয়া হয়েছে যা এ সংক্রান্ত ToR এ রিকমেন্ডেড রয়েছে। এটি হোস্ট করার জন্য ন্যাশনাল ডাটা সেন্টারে Platform-as-a-Service (PaaS) প্ল্যাটফর্মটি এক্সপ্যান্ড করে ডি-নথি ব্যবস্থার Processing Capacity আনা যেতে পারে। বর্তমানে ডি-নথি অফিসগুলো মার্জ করা, আলাদা আলাদা মন্ত্রণালয়ের অফিসগুলোকে ভার্চুয়ালি সেপারেট নেয়া হয়েছে। বাংলাদেশ ডাটা সেন্টার কোম্পানিকে বেসরকারি বিনিয়োগে আনার চেষ্টাটি আরেকটি পর্যায়ে আছে, রেভিনিউ শেয়ারিং এর মাধ্যমে বেসরকারি বিনিয়োগে BDCCL কে লোকসানি প্রতিষ্ঠান হওয়া থেকে বাঁচানো না গেলে কোম্পানি বন্ধ করে ন্যাশনাল ডেটা সেন্টারের সাথে মার্জ করা উত্তম বলে মনে করি। বিষয়টি আপনার সদয় বিবেচনার জন্য দেখে যাচ্ছি, সময় স্বল্পতায় আমার পক্ষে কাজটি শতভাগ সম্পাদন করা সম্ভব হয়নি।

৫। আমার দায়িত্ব পালনের শুরুতে এজ (EDGE) প্রজেক্টের বাস্তবায়ন সক্ষমতা ছিল মাত্র ১৭ শতাংশ। এটাকে আমি ৮১ শতাংশে উন্নীত করেছি। এর মাধ্যমে চারটি data center component রয়েছে যার দুটি সম্পন্ন হয়েছে, যশোরে গুরুত্বপূর্ণ ডিসি/ডিআর ডেটা সেন্টার ও ক্লাউড ফ্যাসিলিটি নির্মাণ কর্মকান্ডটি চলমান আছে যা দ্রুত কমপ্লিশন করতে হবে। World Bank এর পক্ষ থেকে এই প্রজেক্ট টি আগামী জুন ২০২৬ পর্যন্ত ডেডলাইন রয়েছে। আপনি জেনে অবাক হবেন যে, ন্যাশনাল ডেটা সেন্টারের কার্যকর কোন ডিজাস্টার রিকভারি (DR) সেন্টার এই মুহূর্তে নেই। ২০০ টি ক্যাবিনেটের বিপরীতে মাত্র ১৬ টি ক্যাবিনেট, ৮ কিলোওয়াট এর পাওয়ার এর বিপরীতে মাত্র ৪ কিলোওয়াট ব্যাক করে ডিআর করার মতো ভয়ঙ্কর অবস্থা রেখে গিয়েছে বিগত সরকার, যার ফলে বেশিরভাগ ক্যাপাসিটিই অব্যবহারযোগ্য ছিল।

ডাটা সেন্টারগুলোতে ভি এম মনিটরিং, এনভায়রমেন্টাল মনিটরিং ড্যাশবোর্ড, ডিডস (DDoS), লোড ডিস্ট্রিবিউশন, থ্রোপুট ডিস্ট্রিবিউশন ইত্যাদি মৌলিক কম্পোনেন্ট ছিল। আপনার দায়িত্ব পালনকালে এই বিষয়গুলো পর্যাপ্ত গুরুত্ব পাবে বলে আমি বিশ্বাস করি।

৬। ইনোভেশন টিম হিসেবে এটুআইকে অত্যন্ত জরাজীর্ণ অবস্থায় পেয়েছি। দুর্নীতিগ্রস্ত অ্যাপ অ্যাপ্লিকেশন তৈরিতে ব্যস্ত রাখা ছিল প্রতিষ্ঠানটিকে। বাজেটে নাগরিক সেবা দিয়ে এই প্রতিবন্ধকতা কাটিয়ে উঠার চেষ্টা করেছি।

ডি-নথির সেবার মান বাড়িয়েছি, এখন সেখানে দরকার ভার্চুয়াল অফিস সেপারেশন, সাবস্কাইবেবল ক্লাউড সলিউশন, এবং এআই ইন্টিগ্রেশন এই বিষয়গুলো কীভাবে করতে হবে তার বিস্তারিত ধারণা পত্র তৈরী করা হয়েছে।

এর বাইরে মাইগভ প্ল্যাটফর্মকে আরো সমৃদ্ধ করা হয়েছে সিটিজেন সেন্দ্রিক সার্ভিস দিয়ে, মাই গভ কে নাগরিক সেবায় মার্জ করা হয়েছে। বেসরকারি ইউটিলিটি পেমেন্ট গ্রিগেটের গুলো যাতে মনোপলি হয়ে উঠতে না পারে সেজন্য আমার মতে একটি ন্যাশনাল পিএসএস থাকা

ভালো। সেই লক্ষ্যে একপে ইনিশিয়েটিভ টি রেখে তার সাথে এবং নাগরিক সেবার ইন্টিগ্রেশন করা হয়েছে, আমার স্বপ্ন সময়ে একপের ইন্টিগ্রেশন বহু গুণে বেড়েছে, ডিটো ই টেকেটিং ইত্যাদি সম্প্রসারিত হয়েছে। আমি আশা করি এই কাজ টার্গেট ভিত্তিক এপ্রচে দ্রুততার সাথে সব সেক্টরে এগিয়ে যাবে। বাংলাদেশ ব্যাংকের সাথে একপের কোম্পানি ঘটনার বিষয়টি আপনি সুরাহা করবেন বলে আমি বিশ্বাস করি।

৭। বাংলাদেশের ডকুমেন্ট জালিয়াতি বৃহৎ পর্যায়ে পৌঁছেছে। CCA শক্তিশালী করার জন্য একটি ডিপিপি প্রণয়ন করা হয়েছে। আমরা এটির সাথে পোস্টাল সার্ভিস টা চালু করেছি। আমি আশা করি আপনার সময়ে সকল স্কুল কলেজ বিশ্ববিদ্যালয় সার্টিফিকেট, সকল ট্রেড লাইসেন্স, ট্রেডমার্ক, কাবিননামা, ব্যবসায়িক সার্টিফিকেট, বিদেশগামীদের জন্য বিএমইটি সার্টিফিকেট মেডিকেল ক্লিনারেন্স সহ দেশের যাবতীয় দলিল দস্তা পেয়েছে, ভূমির দলিল আয়করনামা তিন টিম সার্টিফিকেট, ইত্যাদি সবকিছুতে রুট কি PKI ভিত্তিক ডিজিটাল সিগনেচার ব্যবস্থা প্রবর্তিত হবে। এই বিষয়ে যথাযথ গুরুত্ব না দিলে বাংলাদেশী প্রবাসীরা বিদেশে জাল ডকুমেন্টের পাশাপাশি ডিজিটাল সিস্টেমে অনুপযুক্ত ডকুমেন্টের শিকার হয়ে বিভিন্নভাবে বঞ্চিত হবেন।

৮। ডিওআইসিটির মাধ্যমে ফ্রিল্যান্সারদের জন্য ফ্রি ডিজিটাল ভেরিফিকেশন এবং আইডি কার্ড প্রণয়নের ব্যবস্থা করেছি। বাংলাদেশ ব্যাংকের সাথে দেখা সেট করে এই ব্যবস্থা বাংলাদেশের সকল বাণিজ্যিক ব্যাংকে পরিচিত করে ফ্রিল্যান্সারদের কাজটি সহজ করবে। Pay-pal এর সাথে একটা কার্যকর আলোচনা আমরা শুরু করেছে, রুস বর্ডার ট্রানজেকশন এর ক্ষেত্রে পেপাল বাংলাদেশে লঞ্চ করার বিষয়টি বিবেচনাধীন এবং বর্তমানে, এর ফলো-আপ ডিজিটাল ইকোনমি'র বিকাশে জরুরি।

৯। ইডিসি প্রকল্পের ইনপুট গুলো অনেক পূরনো, ২% সুদে চিনা ঋণ এ চাইনিজ এক্সিম ব্যাংক রাজি হয়নি বলে এই প্রকল্পটি সংকুচিত করার প্রস্তাবনা নিয়েছি। এখানকার, হঠকারী এবং পলিটিক্যালি মোটিভেটেড কম্পনেন্ট গুলো বন্ধ করার দরকার বিধায় ইডিসিকে সংকুচিত করে নতুন আর ডি পি প্রণয়ন করা হয়েছে, যাতে আর কোন অপচয় না হয়।

এই প্রকল্পের এন্ড ইউজার কানেক্টিভিটি ভেন্ডরগুলো রাজনৈতিক সংশ্লিষ্ট ছিল, কানেক্টিভিটি জানাতে অনেকভাবে দিয়ে যাচাই না করে তাদেরকে পেমেন্ট দেওয়ার যে চর্চা ছিল সেটা আমরা বন্ধ করেছি, সেজন্য একটি কার্যকর NMS এবং ডেটা ভিত্তিক Work Order Bill Payment চালু করা হয়েছে।

১০। বাংলাদেশ হাইটেক পার্কের চল্লিশ একরের বেশি বেদখলকৃত জমি উদ্ধার করা হয়েছে, আমি বিশ্বাস করি আপনার আন্তরিকতার মাধ্যমে বিগত সরকারের আমলে বরাদ্দকৃত অন্যান্য জমিগুলো উদ্ধার করবেন, যেসব প্লট ডেভেলপমেন্ট করে শিল্প স্থাপন করা হয়নি সেগুলো Reallocate করাই সমীচীন। জনতা টাওয়ার রেনোভেশনের ব্যবস্থা নিয়েছি, জনতা টাওয়ার রেনোভেশন এর পাশাপাশি কাওরান বাজারে সফটওয়্যার টেকনোলজি পার্ক নির্মাণাধীন প্রকল্পটিও শেষ করা দরকার। বাংলাদেশের সফটওয়্যার কোম্পানিগুলোর জন্য সহজ শর্তে এবং সহনীয় ভাড়া অফিস স্পেস তৈরি করতে না পারলে দেশের আইসিটি রপ্তানি ২০৩০ সালের মধ্যে পাঁচ বিলিয়নে পৌঁছানো যাবে না।

পাশাপাশি দেশের বিভিন্ন নিউ সিটি ডেভেলপমেন্ট প্রকল্প বিশেষ করে ঢাকার পূর্বাচল জলসিঁড়ি কিংবা ঝিলিমিল এসব এলাকার যেখানে মধ্যবিত্ত কিংবা উচ্চ মধ্যবিত্তের অনুকূলে আপওয়ার্ড মবিলিটি তৈরি হচ্ছে, সেখানে আমাদের শিক্ষার্থীদের ভবিষ্যৎ কর্মসংস্থানের জন্য পর্যাপ্ত অফিসের স্পেস এবং সফটওয়্যার পার্ক দরকার যেখানে কম ভাড়া কোম্পানিগুলো অফিস নিতে পারবে এবং মানসম্পন্ন কাজের পরিবেশ সেখানে থাকবে। এ উদ্দেশ্যে জমি বরাদ্দের জন্য আমি রাজউক এবং গণপূর্তকে চিঠি দিয়েছি, এর ফলোআপ করতে অনুরোধ করছি।

পাশাপাশি বিটিসিএল এর জমি থেকে মহাখালী বনানী এলাকার জমির পাশে কড়াইল বস্তি এলাকায় হাইটেক পার্কের জমি রয়েছে। তেজগাঁও এ বিটিসিএল এর পরিত্যক্ত জমি রয়েছে, হাইটেক পার্ক কর্তৃপক্ষ বিটিসিএল এর সাথে যৌথ উদ্যোগে গুলশান এবং বনানীর কাছাকাছি অবস্থিত এই এলাকায় একাধিক সফটওয়্যার পার্ক তৈরি করতে পারে যেখানে সফটওয়্যার কোম্পানিগুলোর জন্য পর্যাপ্ত অফিস স্পেস তৈরি করা দরকার।

১১। বাংলাদেশ হাইটেক পার্কের আট আইটি ১২ আইটি ইত্যাদি নামে একাধিক প্রকল্প রয়েছে যেখানে সারাদেশে আইটি ইনকিউবেটর নির্মাণ করা হয়েছে এবং হচ্ছে। অত্যন্ত দুঃখের বিষয় এই যে এখানে শুধু ভবন কেন্দ্রিক ইনফ্রাস্ট্রাকচার ফ্যাসিলিটি আছে কিন্তু সফট স্কিল ডেভেলপমেন্টের কোন প্রোগ্রাম সুস্থানে রাখা হয়নি। যেহেতু অধিকাংশ ক্ষেত্রেই ভবনগুলোর কাজ শেষ হয়নি এবং হ্যান্ডওভার হয়নি, তাই দুর্নীতির প্রশ্ন তৈরি না করতে আমি এখানে স্কিল ডেভেলপমেন্ট প্রোগ্রাম চালু করিনি, তবে DoICT, DEIED প্রজেক্ট এবং BCC'র সমন্বয়ে আমরা বেশ কিছু নতুন ট্রেনিং কারিকুলাম এবং স্কিল ডেভেলপমেন্ট কারিকুলাম প্রণয়ন করেছি।

১২। আপনি জেনে খুশি হবেন যে আইসিটি থেকে যেই মানহীন ট্রেনিংগুলো বিগত সরকারের আমলে দেওয়া হতো তার সবগুলো বন্ধ করা হয়েছে। প্রতিটি ট্রেনিংকে ক্যারিয়ার সার্টিফিকেট এর সাথে সংযুক্ত করার নতুন পলিসি গ্রহণ করা হয়েছে।

এই পলিসির দুটো দিক আছে -

ফ্রিল্যান্সার লার্নিং আর্নিং কিংবা স্টার্টআপ সংক্রান্ত ট্রেনিং গুলোর আধুনিক কারিকুলাম তৈরি করা হয়েছে, এই কারিকুলাম ন্যাশনাল স্কিল ডেভেলপমেন্ট অথরিটির (NSDA) সাথে নেগোসিয়েট করে স্বীকৃতি দেওয়া হয়েছে। EDGE প্রকল্প থেকে যেখানে পাইথন, সাইবার সিকিউরিটি ট্রেনিং, কিংবা এডভান্স স্কিল ডেভেলপমেন্ট ট্রেনিং যার সাথে ইন্টারন্যাশনাল রিকগনিশন দরকার সেখানে আন্তর্জাতিক সার্টিফিকেট এর বাধ্যবাদকতা নিশ্চিত করা হয়েছে।

বিগত সরকারের আমলে হার পাওয়ার, লার্নিং আর্নিং, এবং স্টার্টআপ ট্রেনিং গুলো যেনতেন ভাবে একই NID এর বিপরীতে দেয়ার মাধ্যমে অর্থলোপাটের ঘটনা ঘটেছে। উপরন্তু লোকাল ভেন্ডর সার্টিফিকেট ভিত্তিক ট্রেনিং গুলো শিক্ষার্থীদের কোন কাজে আসে না, অধিকাংশ ক্ষেত্রেই এসব ভেন্ডর সার্টিফিকেট জাল এবং ট্রেনিং না করেই বা যেনোতেনো ভাবে ট্রেনিং করেই দেয়া হতো, ফলে সরকারি অর্থের নিদারুণ অপচয় হতো ট্রেনিংয়ের নামে।

আপনি আমার সাথে একমত হবেন যে, আমাদেরকে পরীক্ষাভিত্তিক ট্রেনিংয়ে যেতে হবে যেখানে সঠিক জাতীয় এবং আন্তর্জাতিক মূল্যায়ন এর ভিত্তিতে যে সার্টিফিকেট দেয়া হবে তা ব্যক্তি শিক্ষার্থী তার কর্মজীবনে কাজে লাগাতে পারেন।

১৩। IDEA প্রকল্পের এসএমই এবং সিট Allocation এ ব্যাপক দুর্নীতি ছিল। আমরা এসব প্রতিবেদন তৈরি করেছি। স্টার্টআপ স্পেস গ্র্যান্ট দেয়ার প্রসেসে শৃঙ্খলা এনেছি, ইনোভেশন গ্র্যান্ট বরাদ্দ দিয়ে দুর্নীতি করার রাস্তা বন্ধ করেছি। শুধুমাত্র DEIED প্রজেক্ট, IDEA, ICT Promotion থেকে এবং StartUp Bangladesh থেকে উপযুক্ত ইনিশিয়েটিভগুলোকে ফান্ড/স্পেস বরাদ্দ দেয়া হয়েছে।

আমি সচিব মহোদয়কে সব লার্নিং ম্যানেজমেন্ট, ফান্ড ম্যানেজমেন্ট, প্রজেক্ট ম্যানেজমেন্ট, ইত্যাদি এক জায়গায় এনে একটি সেন্ট্রাল এমআইএস প্রতিষ্ঠার জন্য তাগাদা দিয়েছি যাতে সফটওয়্যারগুলোর ডুপ্লিকেশন এবং সাইলো ডেভেলপমেন্ট না হয়। আপনি এই কাজটি এগিয়ে নিবেন বলে আমি আশা প্রকাশ করি। বর্তমানে মন্ত্রণালয়ের তিনটি জব পোর্টাল আছে, এটিও একটি টেলিটক একটি এবং বিসিসি তার একটি, এগুলোর যে যে ফিচারগুলো ভালো সেগুলো এক জায়গায় করে শনাক্ত করে মার্জ করাই ভালো হবে। কনফারেন্স এবং কমিউনিকেশনের জন্য কমিউনিকেশনের জন্য বিসিসির বিভিন্ন ধরনের প্রোডাক্ট আছে, ডিওআইসিটি এবং এটুআই এর আছে, এগুলো মার্জ করার জন্য নির্দেশনা দেওয়া হয়েছে।

১৪। বিগত সরকারের সময়ে বাংলাদেশের ফান্ড ব্যাপক অপচয় হয়েছে। বাংলাদেশের অরেঞ্জ এবং রেন্ট ক্যাটাগরির কোম্পানির কাছে বিপুল পরিমাণ ফান্ড আটকে আছে। সামান্য কয়েকটা কোম্পানি মাত্র ভালো করছে যা নিবিড় পর্যবেক্ষণ প্রয়োজন। অদক্ষ ও অপেশাদার কোম্পানিগুলোর কাছ থেকে পুনরুদ্ধারে আইনি ব্যবস্থা নেওয়া দরকার। দুর্নীতি এবং সার্বিক অপব্যবস্থাপনার কথা বিবেচনা করে আমরা ভেঞ্চার ক্যাপিটাল ফান্ড ডিরেক্ট ফান্ডিং থেকে সরে আসার সিদ্ধান্ত নিয়েছি। বিপরীতে জাইকার সাথে জয়েন কোলারেশনে ফান্ড অফ ফান্ডস চালুর উদ্যোগ নিয়েছি। এখানে আন্তর্জাতিকভাবে স্বীকৃত ভেঞ্চার ক্যাপিটালগুলোর সাথে ম্যাচিং ফান্ড তৈরি করা হবে, এবং আন্তর্জাতিকভাবে স্বীকৃত প্রতিষ্ঠানের মান সিলেকশন ক্রাইটেরিয়াতে ফান্ডিং করা হবে, আপনার সময়ে এই কাজ এমনভাবে করা হবে যে দেশের স্টার্টার গুলো বিকশিত হয়।

১৫। আমার দায়িত্ব প্রাপ্তির সময় ন্যাশনাল লেটার সেন্টারের ক্যাপাসিটি অত্যন্ত নাজুক পরিস্থিতিতে ছিল। অন্তর্বর্তীকালীন সরকারের সময়ে ন্যাশনাল ডেটা সেন্টার (NDC) এর ক্লাউড ফেসিলিটি গুলোর বিভিন্ন কম্পোনেন্ট এর সক্ষমতা বহুগুণ বৃদ্ধি করা হয়েছে। স্টোরেজ, মেমোরি, এবং প্রসেসিং পাওয়ার প্রতিটি কম্পোনেন্টের সক্ষমতা তিনগুণের বেশি বাড়ানো হয়েছে। আমরা যশোরে তৈরি করছি সম্পূর্ণ নতুন টায়ার-থ্রি ডেটা সেন্টার। যশোরে রাতদিন কাজ চলছে পুরোদমে। ক্লাউড ফ্যাসিলিটির সব সার্ভার, যন্ত্রপাতি, সাপোর্ট ইকুইপমেন্টও আসতে শুরু করেছে। ৩ মাসের মধ্যেই ডাটা সেন্টার এবং ক্লাউড ফ্যাসিলিটি পুরোদমে অপারেশনে যাবে বলে আশা রাখি। দেশের ডিজিটাল ট্রান্সফর্মেশন এর ফিউচার সিকিউর করার জন্য প্রয়োজনীয় অবকাঠামো তৈরি করে যাচ্ছি আমরা, নিরবচ্ছিন্ন পরিশ্রমের মাধ্যমে অল্প সময়েই বড় ধরনের অগ্রগতি সম্ভব হয়েছে। এ সময় আমরা ৫ টা ক্লাউড ফ্যাসিলিটি তৈরি করে যাচ্ছি-

ক। ন্যাশনাল ডেটা সেন্টার এর ক্যাপাসিটি বৃদ্ধি, নিউট্রিনক্স ক্লাউড;

খ। ন্যাশনাল ডেটা সেন্টারের ক্যাপাসিটি বৃদ্ধি, হ্যাওয়ে ক্লাউড;

গ। সম্পূর্ণ নতুন প্ল্যাটফর্ম এজ এ সার্ভিস, পাস প্লাটফর্ম। ভার্সুয়াল কন্টেনারাইড ক্লাউড সার্ভিস;

ঘ। ছোট ছোট এআই মডেল ট্রেন করার জন্য NVIDIA Volta Architecture Tensor Core GPU ভিত্তিক Cloud Resource Management Platform;

ঙ। যশোরে বড় ক্যাপাসিটির নতুন DC/DR ক্লাউড।

এনডিসি এখন থেকে সরকারের বহু মন্ত্রণালয় ও দপ্তর সংস্থার সফটওয়্যার সার্ভিস, পেলোড হোস্ট করতে পারবে। প্রত্যেকটা মন্ত্রণালয় যাতে আলাদা আলাদা ডেটা সেন্টার না করে সেজন্য আন্তঃ মন্ত্রণালয়ে যোগাযোগ শক্ত করা দরকার।

১৬। আমরা বাংলাদেশের প্রথম সোর্স কোড রিপোজিটরি তৈরি করেছি। এর মাধ্যমে ভেদুর লক পরিস্থিতি থেকে দেশের সব মন্ত্রণালয়কে মুক্তি দিতে হবে। বর্তমানে সোর্স কোড নিয়ে কোনো সচেতনতা না থাকায় প্রতিটি মন্ত্রণালয়ে একাধিক ভেদুর সফটওয়্যার বন্টন করে কিন্তু সোর্স কোড না দেয়ায় সেসবে কন্টিনিউয়াস ডেভেলপমেন্ট ও কন্টিনিউয়াস ইন্টিগ্রেশন সম্ভব নয়। বরং একটি সফটওয়্যার চলে গেলে নতুন কোম্পানিকে দিয়ে সব কিছু আবার স্ক্র্যাচ থেকে তৈরি করতে হয়। এই অবস্থা থেকে মুক্তিতে বিসিসি থেকে একটি জাতীয় সোর্স কোড রিপজিটরি তৈরি প্রয়োজন। এ বিষয়ে সচেতনতা তৈরি করে মন্ত্রণালয় ভিত্তিক গিট হাব এবং ন্যাশনাল গিট হাব ভিত্তিক সোর্স কোড রিপজিটরি পলিসি নেয়া হয়েছে। আমার আন্তরিক বিশ্বাস আপনি এটা বাস্তবায়নে ভূমিকা রাখবেন, এ বিষয়ে আমরা পর্যাপ্ত সচেতনতা এবং আন্তঃ মন্ত্রণালয় সভা করেছে। পাশাপাশি ন্যাশনাল এআই ও ক্লাউড পলিসি তৈরি করা হয়েছে। এসব পলিসি ক্যাবিনেটের মাধ্যমে পাশ করা দরকার যাতে এতে সকল মন্ত্রণালয় অংশ নেয়। এই দেশে যাতে আবারও নতুন নতুন ডিজিটাল সাইলো তৈরি না হয়।

১৭। আইসিটি থেকে আমরা বিভিন্ন মন্ত্রণালয়ে ডিজিটাল ট্রান্সফর্মেশন ইনপুট দিয়েছি -

ভূমি ডিজিটাইজেশনে, আইন মন্ত্রণালয়ের ই-রেজিস্ট্রেশন, ই জুডিশিয়ারি, ই বেইল বন্ড ইত্যাদির সমন্বয় করা দরকার, যাতে সেখানে আবারও নতুন করে অর্থ অপচয় না হয়। এজন্য ডিপিসি গুলোর সংশোধন করা দরকার। বর্তমান ডিপিসি গুলো কিছু অযোগ্য এবং দুর্নীতিগ্রস্ত কম্পালেন্ট দিয়ে প্রণয়ন করা। এমনকি ওয়ার্ল্ড ব্যাংক, এডিবি, জাইকার কম্পালেন্টদেরকেও অত্যন্ত অযোগ্য হিসেবে পেয়েছি।

SITA প্রকল্পে বিবিএস, এনবিআর, প্ল্যানিং কমিশন, অডিট এবং পাবলিক প্রকিউরমেন্ট এই ৫টি অর্থনৈতিকভাবে গুরুত্বপূর্ণ ডিপার্টমেন্ট এ সেক্টরাল ইন্টারকানেক্টিভিটি তৈরি করা দরকার যাতে ডেটা প্রাপ্তি, পেমেন্ট প্রাপ্তি, ভ্যাট ও ট্যাক্স প্রাপ্তি ডিজিটাল হয় এবং এফিশিয়েন্ট হয়। সেকেন্ডারি ডেটা এবং সেকেন্ডারি ট্রানজেকশন ডেটা নির্ভরতা দুর্নীতি বাড়ানো এবং রেভিনিউ লিকেজ হচ্ছে। ফলে এনবিআরকে যেমন দেশের সকল ইআরপির সাথে সংযুক্ত করতে হবে, অন্যদিকে বিবিএস- বিপিপিএ - ওডিটকে ও সোর্স থেকে প্রাপ্ত মূল ডেটার ভিত্তিতে কাজ করতে হবে। এতে দুর্নীতির সমস্যা কমবে, এক্সেস টু ডেটার সমস্যা যাবে এবং কৃত্রিম বুদ্ধিমত্তার প্রয়োগ ঘটানো যাবে। এর বাইরেও বিবিএস, এনবিআর, প্ল্যানিং কমিশন, অডিট এবং পাবলিক প্রকিউরমেন্ট, বাংলাদেশ ব্যাংক সহ সকল ভূমি আইন, শিক্ষা ও

স্বাস্থ্যের মত মন্ত্রণালয়ে সেক্টরাল এপিআই ইন্টারকানেক্টিভিটি হাব দরকার যা ধীরে ধীরে ন্যাশনাল ডেটা গভর্নেন্স এবং ইন্টার অপারেবিলিটি এক্সচেঞ্জ হয়ে উঠবে। তবে একইভাবে গুরুত্বপূর্ণ বিষয় হচ্ছে, বিভিন্ন মন্ত্রণালয় যাতে একই রকম সফটওয়্যার আলাদা আলাদা করে সফটওয়্যার না করে বিদ্যমান সফটওয়্যারকে CI/CD করে মান উন্নয়ন করে সব মন্ত্রণালয়ের এবং দরকারে মডিউলার ব্যবহার নিশ্চিত করা দরকার। পাশাপাশি সরকারের সকল মন্ত্রণালয় কিভাবে ভেন্টর লকড, সফটওয়্যার লকড এবং সোর্স কোড লকড সিচুয়েশন থেকে বের হতে পারে সেই লক্ষ্যে আইসিটিকে কাজ করতে হবে।

আমি SHQTC'কে সক্ষম করতে আপনার মনোযোগ আকর্ষণ করছি। এই টিম আগে বছরে ৪০-৫০টি সফটওয়্যার টেস্ট সম্পন্ন করতো, গত এক বছরে প্রায় ২৪০টি সফটওয়্যার টেস্ট করেছে। সার্ট/বিজিডি ই-গভ সার্ট মিলে এমন একটা কালচার তৈরি করতে হবে যেখানে প্রতিটি মন্ত্রণালয়ের প্রতিটি সফটওয়্যার এবং Applications লাইভ প্রডাকশনে আনার আগে সফটওয়্যার কোয়ালিটি টেস্ট এবং থার্ড পার্টি VAPT বাধ্যতামূলক করা হয়। এবিষয়ে মন্ত্রণালয়গুলোর সচেতনতা একেবারেই নগণ্য।

১৮। বিগত সময়ে আমরা একটি ফ্রাঞ্চেইজড ডিজিটাল সার্ভিস ভিত্তিক পরিকাঠামোর উপর ভিত্তি করে বাংলাদেশের যে ডিজিটাল অলিকম্বাট্রা রচিত হয়েছিলো, তাতে দেশের আপামর জনসাধারণের নাগরিক সেবা সহজীকরণের স্থায়ী রূপরেখা ছিলো অনুপস্থিত ও বিচ্ছিন্ন। কারণ নাগরিক সেবা দেয়ার ক্ষেত্রে যে ডিজিটাল পাবলিক ইনফ্রাস্ট্রাকচার (DPI) থাকা প্রয়োজন তা ন্যূনতম পরিমাণে বিদ্যমান ছিল না। উপরন্তু এসবই ছিল প্রকল্প ভিত্তিক এবং এক একটি এক এক রকম সাইলো (Silo)। Interoperability Concept না থাকার ফলে এক এজেন্সি আরেক এজেন্সির সাথে উপাত্ত শেয়ারিং এ বাধাগ্রস্ত হত। একই সাথে পয়েন্ট টু পয়েন্ট Mesh Connectivity বৃদ্ধির ফলে নাগরিকদের ব্যক্তিগত উপাত্ত চুরি থেকে ব্যাংকের ড্রাজেকশন ডার্ক ওয়েবে সহজলভ্য ছিল। ডাটা সেন্টারের ক্যাপাসিটি ইউটিলাইজেশন বিপদজনক পর্যায়ে চলে এসেছিল এবং দেশে হাই ক্যাপাসিটির কোন ডিজেন্স্টার রিকোভারি সেন্টার ছিল না। তদুপরি সরকারি এজেন্সি সমূহ বিশেষত CII এজেন্সিরাও উপাত্ত সুরক্ষার ব্যাপারে উদাসীন ছিল।

এমতাবস্থায় বিগত সময়ে নিম্নোক্ত বিষয়গুলোর বর্তমান অগ্রগতি ও ভবিষ্যৎ বাস্তবায়নের দিকে আপনার বিশেষ দৃষ্টি আকর্ষণ করছি:

- বিভিন্ন মন্ত্রণালয় ও দপ্তর সংস্থায় প্রাতিষ্ঠানিক সক্ষমতা বৃদ্ধির লক্ষ্যে আমরা কেবল জনবল বৃদ্ধি নয়, কার্যমোগত ও গুণগত রূপান্তরের ওপর গুরুত্ব দিয়েছি। ভূমি মন্ত্রণালয়ের বিদ্যমান সিস্টেমসমূহ পর্যালোচনা করে সমস্যা নিরসন, Land Single Gateway প্রস্তাব, উপাত্ত সংরক্ষণ কার্যক্রম শক্তিশালীকরণ ইত্যাদি উদ্যোগের ফলে স্বল্প সময়ের মধ্যে দৈনিক রাজস্ব প্রায় চারগুণ বৃদ্ধি পায়। চট্টগ্রাম পোর্ট অথরিটির অনুরোধে তাদের ডিজিটাল সিস্টেম আপগ্রেডেশনের পাশাপাশি একটি ত্রিমাত্রিক Interoperable Port Single Window (PSW) ধারণা প্রদান করা হয় যা (ক) পোর্টের অভ্যন্তরীণ সিস্টেম, (খ) আন্তঃএজেন্সি সংযোগ যেমন NBR NSW, এবং (গ) আন্তর্জাতিক মেরিটাইম ইন্ডাস্ট্রি, IMO, শিপিং ও লজিস্টিকস সেক্টরের সাথে আইনসম্মত সংযোগ নিশ্চিত করবে। একইভাবে BRTA-কে ভেন্টর-লকড অবস্থা থেকে একটি উন্মুক্ত ও ইন্টারঅপারেবল আর্কিটেকচারে রূপান্তরের পরামর্শ দেওয়া হয়। eJudiciary, eRegistration, SITA (Strengthening Institutions for Transparency and Accountability) প্রভৃতি উদ্যোগে সমন্বিত ডিজিটাল রূপান্তরের জন্য নীতিগত গাইডলাইন প্রণয়ন করা হয়েছে।
- বাংলাদেশে Interoperability বহুমাত্রিকভাবে বাস্তবায়ন অপরিহার্য। প্রথমত, এজেন্সির অভ্যন্তরে সমন্বিত তথ্যপ্রবাহ নিশ্চিত করে দক্ষতা বৃদ্ধি করতে হবে। দ্বিতীয়ত, আন্তঃএজেন্সি পর্যায়ে ব্যক্তিগত উপাত্ত সুরক্ষা আইন, জাতীয় উপাত্ত গভর্নেন্স আইন ২০২৫ এবং সাইবার সুরক্ষা আইন অনুসরণ করে সুরক্ষিত ডাটা এক্সচেঞ্জ অবকাঠামো গড়ে তুলতে হবে। তৃতীয়ত, আন্তর্জাতিক সিস্টেমের সাথে সংযোগ স্থাপন করে বৈশ্বিক সুবিধা গ্রহণ করতে হবে যেমন ই-এপোস্টিল ইতোমধ্যে আন্তর্জাতিক পর্যায়ে কার্যকরতার প্রমাণ দিয়েছে। শিক্ষা, বন্দর, মেরিটাইম ও শিপিং খাতে একটি Interoperable National Responsible Data Exchange (NRDEx) প্রতিষ্ঠা গুরুত্বপূর্ণ ভূমিকা রাখতে পারে; এ বিষয়ে জাতীয় উপাত্ত ব্যবস্থাপনা কর্তৃপক্ষকে আইনি ম্যান্ডেট প্রদান করা হয়েছে।
- এ প্রেক্ষিতে বিশ্বব্যাংক অর্থায়িত DSTAR (Digital Service Transformation for Access and Resilience) প্রকল্প প্রস্তাব করা হয়েছে, যার মাধ্যমে NRDEx স্থাপন ও Identity Management-এর জন্য OneID বাস্তবায়নের পরিকল্পনা রয়েছে। এর ফলে নাগরিকদের জীবনঘটিত সেবা (Life Event Services) সহজ, সমন্বিত ও কার্যকরভাবে ডিজাইন করা সম্ভব হবে। পাশাপাশি NagorikSheba প্ল্যাটফর্ম প্রান্তিক পর্যায়ে সেবা-অ্যাক্সেস পরীক্ষামূলকভাবে সক্ষমতা প্রদর্শন করেছে। ভবিষ্যতে Digital Public Infrastructure (DPI), Digital Certificate, Universal Interoperable Payment Ecosystem এবং সমন্বিত Data Sharing কার্যক্রম আরও শক্তিশালী ও প্রাতিষ্ঠানিক রূপ পাবে এমন প্রত্যাশা রইল।

১৯। আইসিটি বিভাগের বিভিন্ন অধিদপ্তরে নিচের কাজগুলো হাতে নেয়া হয়েছে যায় কিছু সুসম্পন্ন হয়েছে, বাকীগুলো চলমান যা পরবর্তী সরকার আরো ভালোভাবে চালিয়ে নিবেন বলে আমরা আশাবাদ ব্যক্ত করছি –

তথ্য ও যোগাযোগ প্রযুক্তি বিভাগ

- ০ সাইবার সুরক্ষা (সংশোধন) অধ্যাদেশ ২০২৫, ব্যক্তিগত সুরক্ষা আইন ২০২৫, জাতীয় উপাত্ত ব্যবস্থাপনা অধ্যাদেশ ২০২৫ অনুমোদন
- ০ বাংলাদেশ হাই-টেক পার্ক কর্তৃপক্ষ আইন (সংশোধন) খসড়া অধ্যাদেশ, ২০২৫ অনুমোদন প্রক্রিয়াধীন
- ০ ন্যাশনাল এআই পলিসি, ক্লাউড পলিসি, সোর্স কোড পলিসি, আইসিটি রডম্যাপ নিয়ে কার্যক্রম চলমান

বাংলাদেশ কম্পিউটার কাউন্সিল

- ০ ডাটা সেন্টার ও ক্লাউড সক্ষমতা বহুগুণ বৃদ্ধি (৪০,০০০ ভার্সুয়াল কোর, ১৭০ TB মেমোরি, ৮ PB স্টোরেজ)
- ০ ক্লাউড সেবা ক্যাটালগ ১১ থেকে সম্প্রসারিত হয়ে মোট ৩০+ সেবা (Cyber Security, AI, Database, Monitoring ইত্যাদি)
- ০ যশোরে টায়ার-৩ DR ডাটা সেন্টার নির্মাণাধীন; AI ওয়ার্কলোডসহ পূর্ণ অপারেশন প্রত্যাশিত
- ০ Nutanix Private Cloud, GPU ও OpenShift ভিত্তিক নতুন ক্লাউড মডেল চালু
- ০ GPU অবকাঠামোতে ৪০ পেটালপ সক্ষমতা; HPC ও Deep Learning সেবা
- ০ মনিটরিং, টেস্টিং ও VAPT সক্ষমতা উল্লেখযোগ্যভাবে জোরদার
- ০ National Cyber Security Operation Center ও RIC-C4IR স্থাপন প্রক্রিয়াধীন
- ০ DSTAR প্রকল্পটির DPP প্রণয়ন শেষে পরিকল্পনা কমিশনে যাচাই বাছাইয়ের জন্য রয়েছে
- ০ EDGE প্রকল্পটি RDPP পরিকল্পনা কমিশন কর্তৃক অনুমোদিত হয়েছে

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- ০ ভেন্টুর-ডিপেন্ডেন্ট মডেল থেকে Competency Driven Organization এ রূপান্তর
- ০ ই-এপোস্টিল: ৮ মাসে ১৮ লক্ষ+ সার্টিফিকেট অটোমেটেড সত্যায়ন
- ০ ডিজিটাল টোল: পদ্মা সেতুতে চালু; ৪,০০০+ যানবাহন, ৭টি আর্থিক প্রতিষ্ঠান সংযুক্ত
- ০ এক-পে: বাংলাদেশ ব্যাংকের অনুমোদনপ্রাপ্ত; জাতীয় ই-ওয়ালেটের পথে
- ০ নাগরিক সেবা বাংলাদেশ: ১,১২২ কেন্দ্র, ৪১+ API সেবা, ২,০০০+ উদ্যোক্তা প্রশিক্ষণ
- ০ ই-টিকেটিং: প্রধান পর্যটন স্পটে বাস্তবায়ন
- ০ NPF 2.0: সকল মন্ত্রণালয় ও অধিদপ্তরে চালু
- ০ D-Nothi তে AI সংযোজন ও Oracle থেকে নিজস্ব ক্লাউডে মাইগ্রেশন প্রক্রিয়াধীন
- ০ ন্যাশনাল ইনোভেশন রিপোজিটরি সিস্টেম: টেন্ডার সম্পন্ন

BHTPA

- ০ বাংলাদেশ হাই-টেক পার্ক কর্তৃপক্ষ আইন উপদেষ্টা পরিষদে উত্থাপিত; পুনরায় উপস্থাপন প্রক্রিয়াধীন
- ০ SOP ও গাইডলাইন (২০২৬) চূড়ান্ত; অনুমোদন অপেক্ষমাণ
- ০ যশোর পার্কে নিজস্ব ব্যবস্থাপনা; কালিয়াকৈরে ৫০ একর জমি পুনরুদ্ধার
- ০ রংপুর IT ইনকিউবেশনে একাডেমিয়া-ইন্ডাস্ট্রি MoU
- ০ বাংলাদেশ ফ্রন্টিয়ার টেকনোলজি প্রকল্পের DPP পরিকল্পনা কমিশনে দাখিল

DoICT

- ০ উপজেলা ICT সেন্টার: ১০০টি উদ্বোধন, বাকিগুলো চলমান
- ০ স্কুল অব ফিউচার: ৩০০ স্কুলে রোবোটিক্স, ২৭,০০০ শিক্ষার্থী প্রশিক্ষিত
- ০ ফ্রিল্যান্সার স্বীকৃতি: ৩,৭০৮ আইডি কার্ড; LEDP-এর আওতায় ১,০৭,৩২০ প্রশিক্ষণ লক্ষ্য
- ০ জেলা-উপজেলায় সাইবার নিরাপত্তা হেল্পডেস্ক

- o মার্চ পর্যায়ে ডি-নথি ও তথ্য বাতায়ন তদারকি জোরদার
- o দক্ষতা-ভিত্তিক ৪টি টেকনিক্যাল পুল গঠন
- o EDC প্রকল্পে ৩৬,০০০ প্রাথমিক বিদ্যালয়ে ইন্টারনেট; বিকল্প অর্থায়নে অবশিষ্ট কাজ সম্পন্ন প্রয়োজন

BDCCL

- o Native Cloud এর অপ্রয়োজনীয় হার্ডওয়্যার ডিকমিশনিং
- o নতুন ক্লাউড সেবার জন্য RFP চলমান
- o Oracle Cloud চুক্তি পুনঃনেগোশিয়েশন অসম্পন্ন; ভবিষ্যৎ সিদ্ধান্ত অপেক্ষমাণ

NCSA

- o সাইবার সুরক্ষা আইন ২০২৬ গেজেট প্রকাশ; গাইডলাইন প্রণয়ন চলমান
- o সরকারি-বেসরকারি IT ও সাইবার সক্ষমতা যাচাই ফ্রেমওয়ার্ক উন্নয়ন
- o BTRC-এর সহায়তায় ২৫০+ অনলাইন জুয়া সাইট ব্লক

CCA

- o eGP, NBR, D-Nothi ও ভূমি মন্ত্রণালয়ে ডিজিটাল স্বাক্ষর বাধ্যতামূলক
- o Secured Electronic Repository প্রকল্প অনুমোদিত
- o আন্তর্জাতিক স্বীকৃতির লক্ষ্যে ডিজিটাল সিগনেচার সম্প্রসারণ কার্যক্রম চলমান

SBL

- o ৫ টি স্টার্টআপে বিনিয়োগ করা হয়েছে, আরো ৩ টি প্রক্রিয়াধীন
- o Fund of Fund চালুর লক্ষ্যে সংশ্লিষ্ট পলিসি ডকুমেন্টস তৈরী ও প্রয়োজনীয় অনুমোদন গ্রহণ করা হয়েছে। অর্থবিভাগ থেকে অনুদান পাওয়া গেছে
- o সকল স্টার্টআপের মনিটরিং জোরদার করা হয়েছে

এসবের সংক্ষিপ্ত পরিচিতি নিম্নে দেয়া হলো -

বাংলাদেশ কম্পিউটার কাউন্সিল (BCC)

- ডাটা সেন্টার সক্ষমতা বৃদ্ধি – ভৌত অবকাঠামো, স্টোরেজ, মেমোরি, প্রসেসিং, সিকিউরিটি ও মনিটরিংসহ ক্লাউড ফ্যাসিলিটির মূল ৫টি কম্পোনেন্ট বহুগুণ সম্প্রসারণ করা হয়েছে; এতে প্রায় ৪০,০০০ ভার্সুয়াল কোর, ১৭০ টেরাবাইট মেমোরি ও ৮ পেটাবাইট স্টোরেজ সক্ষমতা অর্জিত হবে।
- ক্লাউড সেবা ক্যাটালগ সম্প্রসারণ – বিদ্যমান ১১টির সঙ্গে সাইবার সিকিউরিটি, ডেটা সুরক্ষা, পারফরমেন্স মনিটরিং, AI ও ক্লাউড ডাটাবেইজসহ মোট ১৯টি নতুন সেবা যুক্ত হয়েছে; উল্লেখযোগ্য সেবার মধ্যে রয়েছে CCE, OBS 3.0, CFW 2.0, DBAS, NDR, AOM, APM ও Database।
- ডিজাস্টার রিকভারি ডাটা সেন্টার উন্নয়ন – যশোর সফটওয়্যার টেকনোলজি পার্কে নতুন টায়ার-৩ সার্টিফায়েড DR ডাটা সেন্টার নির্মাণাধীন; ২০০+ উচ্চ ক্ষমতাসম্পন্ন রেকসহ AI ওয়ার্কলোড হোস্টিং সক্ষমতা যুক্ত হবে এবং আগামী ছয় মাসে পূর্ণ অপারেশন প্রত্যাশিত।
- নতুন ক্লাউড মডেল চালু – Nutanix Private Cloud Infrastructure স্থাপন করে ৬০০+ সরকারি সংস্থার ই-মেইল ও স্টোরেজ সেবা চালু; AI ও LLM ওয়ার্কলোড সমর্থনে GPU অন্তর্ভুক্ত এবং Application Modernization এর জন্য PaaS হিসেবে Red Hat OpenShift চালু।
- ক্লাউডভিত্তিক GPU অবকাঠামো – EBLICT ও EDGE প্রকল্পে ৪০ পেটাক্লপ GPU সক্ষমতা তৈরি; NVIDIA Volta GPU ইন্টিগ্রেশনের মাধ্যমে সর্বোচ্চ ২,২৪০ teraFLOPS ডিপ লার্নিং ক্যাপাসিটি ও HPC সেবা প্রদান সম্ভব।
- মনিটরিং ব্যবস্থা শক্তিশালীকরণ – নিজস্ব ও ওপেন সোর্স টুল ব্যবহার করে নেটওয়ার্ক, সিকিউরিটি ও সার্ভিস মনিটরিং আধুনিকায়ন ও সমন্বিত করা হয়েছে।

- সফটওয়্যার ও হার্ডওয়্যার টেস্টিং সক্ষমতা বৃদ্ধি – TMMi Level-5 ও ISO/IEC 17025:2017 অর্জনের মাধ্যমে টেস্টিং ল্যাব স্থাপন করে প্রায় ৩৮৫টি টেস্ট সক্ষমতা তৈরি; CIRT কর্তৃক ১,০৭৮টি সরকারি আইটি অ্যাসেস্টের VAPT, ৩৭৯টি থ্রেট ইন্টেলিজেন্স রিপোর্ট ও ১০টি ডিজিটাল ফরেনসিক কেইস সম্পন্ন। EDGE প্রকল্পের আওতায় Tools as a Service (TaaS) কার্যক্রম চলমান এবং SHQTC’র জন্য প্রি-টেস্টিং ল্যাব স্থাপনের নীতিগত পর্যালোচনা প্রক্রিয়াধীন।
- জাতীয় সাইবার অপারেশন সেন্টার প্রতিষ্ঠাকরণ – National Cyber Security Operation Center প্রতিষ্ঠা প্রকিউরমেন্ট ফেজে রয়েছে। এর মাধ্যমে NDC এর ক্লাউড কম্পোনেন্টস গুলো এবং প্রাথমিক পর্যায়ে পাইলট প্রোগ্রামের আয়তায় একটি CII কে SEIM, SOAR, PAM, ইত্যাদি সাইবার সিকিউরিটি এলিমেন্টস স্থাপন করা সম্ভব হবে।
- RIC এবং C4IR প্রতিষ্ঠা - চলমান কাজের অংশ হিসেবে আগামী ফেব্রুয়ারীতে ইনোভেশন প্রডাক্টগুলোর কাজ সম্পন্ন হবে।

এসপায়ার টু ইনোভেট A2i

ভেন্ডর ডিপেন্ডেন্ট প্রতিষ্ঠান থেকে এটিকে কম্পিউক্সি ডিভেন অর্গানাইজেশনে রূপান্তরের মাধ্যমে সোর্স কোড আল্লিকরণ, ট্রান্সপারেন্সি, ও একাউন্টেবিলিটি নিশ্চিত করা হয়েছে। ফলত স্বল্প সময়ে নিচের ইনোভেটিভ এবং কার্যকরী সেবাগুলো প্রদান করা সম্ভবপর হয়েছে-

- ই-এপোস্টিল - বিদেশগামী অথবা বিদেশে অবস্থানকারী শিক্ষার্থী, পেশাজীবীসহ বিভিন্ন ব্যক্তির নানা ধরনের সনদ সত্যায়ন করা যায় আটোমেটিক ভাবে। মাইগভের মাধ্যমে বর্তমানে প্রচলিত ম্যানুয়াল (সনাতনী) প্রথার বদলে সমন্বিত অনলাইন সত্যায়ন ব্যবস্থাপনা যুক্ত করা হয়েছে। যার মাধ্যমে ৮ মাসের মধ্যে ১৮ লক্ষেরও অধিক সার্টিফিকেট সত্যায়ন করা হয়েছে। সিস্টেমটি যথাযথভাবে মেইন্টেন করা হবে (ITSM)।
- ডিজিটাল টোল (ডি-টোল) ম্যানেজমেন্ট সিস্টেম বাস্তবায়ন - বাংলাদেশ সেতু কর্তৃপক্ষের সহযোগিতায় গত ১৫ সেপ্টেম্বর পদ্মা সেতুতে ডি-টোল সেবা চালু করা হয়। এ কার্যক্রমের আওতায় ইতোমধ্যে প্রায় ৪,০০০টি যানবাহন নির্বিঘ্নে টোল প্লাজা অতিক্রম করেছে। এছাড়াও, ৭টি আর্থিক প্রতিষ্ঠান সংযুক্ত রয়েছে। যমুনা সেতু, মেঘনা-গোমতী সেতু, এবং পরবর্তীতে সড়ক বিভাগের আওতাধীন ১৭ টি সেতুতে সিস্টেমটি পর্যায়ক্রমে বাস্তবায়ন করা হবে যার টেকনিকেল সলুশন সম্পূর্ণ প্রস্তুত।
- এক-পে - কার্যক্রম পরিচালনার জন্য বাংলাদেশ ব্যাপক থেকে অনুমোদন গ্রহণ করা হয়েছে। এটিকে একটি কোম্পানি হিসেবে প্রতিষ্ঠার লক্ষ্যে বাংলাদেশ ব্যাংক ও সংশ্লিষ্ট প্রতিষ্ঠানের অনুমোদন প্রক্রিয়া চলমান রয়েছে, যা পরবর্তীতে জাতীয় ই-ওয়ালেট হিসেবে কাজ করতে পারবে।
- নাগরিক সেবা বাংলাদেশ - বর্তমানে ১,১২২টি নাগরিক সেবা কেন্দ্র রূপান্তর করা হয়েছে। প্ল্যাটফর্মটিতে ইতোমধ্যে ৪১টির বেশি সেবা API ইন্টিগ্রেশন সম্পন্ন হয়েছে এবং মাইগভের মাধ্যমে আরোও ৪৫৩টি সেবা উন্মুক্ত রয়েছে। এ পর্যন্ত সারাদেশের প্রায় ২,০০০ (দুই হাজার) এর অধিক নাগরিক সেবা কেন্দ্রের উদ্যোক্তাকে প্রশিক্ষণ প্রদান করা হয়েছে। নাগরিক সেবা বাংলাদেশ নীতিমালাটি স্থানীয় সরকার বিভাগ হতে পরিপত্র জারির বিষয়টি প্রক্রিয়াধীন রয়েছে।
- ই-টিকেটিং - পর্যটন খাতে ডিজিটাল সেবা সম্প্রসারণের অংশ হিসেবে দেশের সকল দর্শনীয় ও পর্যটনস্থলের টিকেট ব্যবস্থাপনা একটি কেন্দ্রীয় অনলাইন প্ল্যাটফর্মে আনার লক্ষ্যে মাইগভ প্ল্যাটফর্মে ‘ই-টিকেটিং’ সেবা লালবাগ কেল্লা, জাতীয় উদ্ভিদ উদ্যান, বাগেরহাটের ষাটগম্বুজ মসজিদ ও বাগেরহাট জাদুঘরে ইমপ্লিমেন্ট করা হয়েছে।
- জাতীয় তথ্য বাতায়নের নতুন ভার্সন চালুকরণ (NPF2.0) - সকল মন্ত্রণালয় (৫৮টি), ৯৭০টি অধিদপ্তর এবং ৪টি (সিলেট, ময়মনসিংহ, খুলনা, বরিশাল) বিভাগীয় কার্যালয় ও তার অধীনস্থ দপ্তর/সংস্থায় ন্যাশনাল পোর্টাল ফ্রেমওয়ার্ক (NPF) এর ২য় ভার্সন আনুষ্ঠানিকভাবে চালু করা হয়েছে। ২০২৬ এর ৪টি বিভাগীয় কার্যালয় ও তার অধীনস্থ দপ্তর/সংস্থা সংযুক্ত হবে।
- জাতীয় বিশ্ববিদ্যালয়ের শিক্ষার্থীদের জন্য বাধ্যতামূলক আইসিটি কোর্স চালুকরণ
- ডিজিটাল নথি (D-Nothi) সিস্টেমে কৃত্রিম বুদ্ধিমত্তা (AI) যুক্তকরণ - ডি-নথিতে কৃত্রিম বুদ্ধিমত্তা (AI) যুক্ত করতে প্রকিউরমেন্ট চলমান আছে। Oracle কে এ পর্যন্ত এই সিস্টেমের জন্য বকেয়া প্রায় ৩০০ কোটি টাকা অর্থবিভাগের মাধ্যমে পরিশোধের উদ্যোগ গ্রহণ করা হয়েছে। একই সাথে আর্কিটেকচার রিভিউ করে ডাটা সেন্টারের রিসোর্স অপ্টিমাইজ করা হয়েছে। পরবর্তীতে D-Nothi System টি ওরাকল থেকে NDC বা BDCCL এর নিজস্ব সক্ষমতায় মাইগ্রেশনের সিদ্ধান্ত হয়েছে, মাইগ্রেশনের কাজ চলমান রয়েছে।
- ন্যাশনাল ইনোভেশন রিপোজিটরি সিস্টেম - একই উদ্ভাবনী কার্যক্রমে একাধিক উৎসের অর্থায়ন রোধে এটুআই কর্তৃক ন্যাশনাল ইনোভেশন রিপোজিটরি সিস্টেম প্রস্তুতের উদ্যোগ গ্রহণ করা হয়েছে। টেন্ডার সম্পন্ন হয়েছে, কার্যক্রম বাস্তবায়ন করতে হবে বাধ্যতামূলকভাবে। এছাড়াও ইনোভেশন নীতিমালা প্রণয়নের কাজ চলমান রয়েছে।

বাংলাদেশ হাইটেক পার্ক অথরিটি BHTPA

- বাংলাদেশ হাই-টেক পার্ক কর্তৃপক্ষ আইন: উপদেষ্টা পরিষদের দুইটি সভায় উপস্থাপিত; ভ্যাট-ট্যাক্স বিষয়ে অর্থ উপদেষ্টার সাথে আলোচনার পর পুনরায় উপস্থাপনের সিদ্ধান্ত।
- এসওপি, গাইডলাইন ও নীতিমালা: পার্ক প্রতিষ্ঠার জমি/স্থান নির্বাচন গাইডলাইন ২০২৬, SOP এবং বিশ্ববিদ্যালয় ইনোভেশন হাব পরিচালনা গাইডলাইন ২০২৬ চূড়ান্ত; অনুমোদন প্রক্রিয়াধীন।
- টেকনোলজি পার্ক ব্যবস্থাপনা: যশোর সফটওয়্যার টেকনোলজি পার্কে পিএমসি চুক্তি বাতিল করে নিজস্ব ব্যবস্থাপনা চালু; কালিয়াকৈর হাই-টেক পার্কে ৫০ একর জমি পুনরুদ্ধার ও বিনিয়োগকারীদের বরাদ্দ শুরু; রংপুর আইটি ট্রেনিং ও ইনকিউবেশন সেন্টারে একাডেমিয়া-ইন্ডাস্ট্রি সহযোগিতায় বেগম রোকেয়া বিশ্ববিদ্যালয়ের সাথে MoU স্বাক্ষর।
- বাংলাদেশ ফ্রন্টিয়ার টেকনোলজি প্রকল্প: DPP প্রস্তুত করে অনুমোদনের জন্য পরিকল্পনা কমিশনে দাখিল।

তথ্য ও যোগাযোগ প্রযুক্তি অধিদপ্তর DoICT

- উপজেলা আইসিটি সেন্টার: ৪০০টির মধ্যে ১০০টি উপজেলা প্রশিক্ষণ কেন্দ্র উদ্বোধন সম্পন্ন; অবশিষ্ট কেন্দ্রগুলোর কাজ চলমান।
- স্কুল অব ফিউচার ও রোবোটিক্স: ৩০০টি স্কুলে রোবোটিক্স কর্নার স্থাপন এবং ২৭,০০০ শিক্ষার্থীকে পাইথন প্রোগ্রামিং প্রশিক্ষণ প্রদান সম্পন্ন।
- ফ্রিল্যান্সার স্বীকৃতি ও LEDP: freelancers.gov.bd-এর মাধ্যমে ৩,৭০৮ জন ফ্রিল্যান্সারকে আইডি কার্ড প্রদান; ১,০৭,৩২০ জনকে প্রশিক্ষণের লক্ষ্যে LEDP প্রকল্প বাস্তবায়ন।
- সাইবার নিরাপত্তা হেল্পডেস্ক: NCSA-এর সাথে MOU অনুযায়ী জেলা ও উপজেলা পর্যায়ে সাইবার নিরাপত্তা হেল্পডেস্ক ও SOP বাস্তবায়ন।
- ডি-নথি ও তথ্য বাতায়ন: উপজেলা পর্যায়ে অফিস অ্যাডমিন নিয়োগের মাধ্যমে ডি-নথি ও জাতীয় তথ্য বাতায়ন হালনাগাদ ও তদারকি জোরদার।
- বিষয়ভিত্তিক পুল গঠন: সাইবার নিরাপত্তা, সফটওয়্যার, অপারেশন ও HRD - এই চারটি ক্যাটাগরিতে দক্ষতা-ভিত্তিক কর্মকর্তা পুল গঠন সম্পন্ন।
- সরকারি প্রতিষ্ঠানসমূহে ইন্টারনেট সেবা প্রদান - EDC প্রকল্পের মাধ্যমে এ পর্যন্ত প্রায় ৩৬ হাজার প্রাথমিক স্কুলে ইন্টারনেট সেবা প্রদান চালু করা হয়েছে। এই প্রকল্পের NMS Reform এর মাধ্যমে Data Driven Billing ব্যবস্থা চালু করে স্বচ্ছতা ও জবাবদিহিতা নিশ্চিত করা হয়েছে। একটি সেন্ট্রাল ITAP প্রকিউরমেন্ট প্রক্রিয়াধীন রয়েছে। চীনের সাথে ঋণচুক্তি না হওয়ায় ২য় সংশোধিত ডিপিপি-র কাজগুলো (উপজেলা সেন্টার ও স্পেশলাইজড ল্যাব) বিকল্প অর্থায়নের মাধ্যমে দ্রুত শেষ করার ব্যবস্থা নিতে হবে।

বাংলাদেশ ডাটা সেন্টার কোম্পানি লিমিটেড BDCCL

- **Native Cloud** এর অপ্রয়োজনীয় হার্ডওয়্যারসমূহ ডিকমিশনিং - বিদ্যুৎ সাশ্রয় এবং নতুন ক্লাউড স্থাপনে স্থান সংকুলনের লক্ষ্যে ন্যাটিভ ক্লাউড এর অপ্রয়োজনীয় হার্ডওয়্যারসমূহ ডিকমিশনিং করা হচ্ছে।
- নতুন ক্লাউড সেবা সংযোজন - RFP চলমান রয়েছে;
- **Oracle Cloud Negotiation** - বিদ্যমান চুক্তি সরকারের জন্য লাভজনক না হওয়াতে চুক্তির শর্তাবলী এবং শিডিউল নেগোশিয়েশনের জন্য Oracle কে আহ্বান জানানো হলেও তারা অনড় অবস্থান প্রকাশ করেন। পরবর্তী সরকার এ সংক্রান্ত উদ্যোগ নিবেন।

জাতীয় সাইবার সুরক্ষা এজেন্সি NCSA

- সাইবার সুরক্ষা আইন, ২০২৬ প্রণয়ন - নাগরিক অধিকার নিশ্চিতকরণ, সাইবার বুলিং কে অপরাধ হিসেবে গন্য করণ, ৩৬ টি CII সমূহের অবকাঠামোগত সুরক্ষা প্রদানের নিমিত্তে সাইবার সুরক্ষা আইন ২০২৬ প্রণয়ন করা হয়েছে কেবিনেটের সম্মতি স্বাপেক্ষে যা গেজেট আকারে প্রকাশিত হয়েছে। এই আইনের অধীনে গাইডলাইন প্রণয়ন চলমান।
- সরকারি-বেসরকারি প্রতিষ্ঠানসমূহের তথ্য-প্রযুক্তি ও সাইবার নিরাপত্তা সক্ষমতা যাচাই এর নিমিত্ত ফ্রেমওয়ার্ক – চলমান;
- হুমকিস্বরূপ কন্টেন্ট ব্লক/অপসারণ এবং অনলাইন জুয়া প্রতিরোধে কার্যক্রম - BTRC এর সহায়তায় Detection and Blocking প্রায় ২৫০ টি অনলাইন জুয়ার সাইট বন্ধ করা হয়েছে;

কন্ট্রোলার অব সার্টিফাইং অথরিটিজ CCA

- **Digital Signature** ইন্টিগ্রেশন - eGP, NBR, D-Nothi, ভূমি মন্ত্রণালয়ে বিভিন্ন ধাপে ডিজিটাল স্বাক্ষর ব্যবহার বাধ্যতামূলক করা হয়েছে;
- **Secured Electronic Repository** স্থাপন - প্রকল্পটির অনুমোদন পাওয়া গেছে;
- সাইবার সুরক্ষা ও আন্তর্জাতিক স্বীকৃতি - আন্তর্জাতিকভাবে বাংলাদেশি ডিজিটাল সিগনেচারের গ্রহণযোগ্যতা বৃদ্ধিকল্পে অনলাইন জালিয়াতি প্রতিরোধে ডিজিটাল স্বাক্ষর সম্প্রসারণ প্রকল্পের মাধ্যমে রুট সিএ-এর আন্তর্জাতিক স্বীকৃতির লক্ষ্য নির্ধারণ করা হয়েছে।

সব শেষে, আমি আশা প্রকাশ করি আপনার সময়ে এনসিএসএ একটি শক্তিশালী অবস্থানে আসবে। বর্তমানে এখানে পর্যাপ্ত ট্রেইন্ড জনবল অনুপস্থিত, প্রতিষ্ঠানটিকে একটি কারিগরিভাবে সক্ষম প্রতিষ্ঠানে রূপান্তর করতে সিম সোর প্যাম, থ্রেট এনালাইসিস ও প্রিভেনশনের এর জন্য উন্নত ল্যাব, সাইবার ট্রেনিং এর ব্যবস্থা করা দরকার। বর্তমানে এনসিএসএ শুধুমাত্র রিপোর্টিং করছে, প্রতিষ্ঠানটিকে এমনভাবে রূপান্তর করতে হবে যাতে তারা অন্য দপ্তর ও অন্য মন্ত্রণালয়ের সাইবার সমস্যা সমাধান করতে পারে।

আমার আন্তরিক বিশ্বাস এই যে, আইসিটিকে বাংলাদেশের ডিজিটাল ট্রান্সফর্মেশন মিনিস্টি হয়ে উঠা দরকার। দেশের সব মন্ত্রণালয়ের ডিজিটাল ট্রান্সফর্মেশন এমনভাবে করতে হবে যাতে প্রত্যেকে আলাদা আলাদাভাবে কাজ না করে, আলাদা প্রকল্প না নিয়ে পরস্পর পরস্পরের সাথে সফটওয়্যার ও সাইবার সিকিউরিটি রিসোর্স শেয়ার করে, ইনফ্রাস্ট্রাকচার শেয়ার করবে এবং দেশের ডিজিটাল ট্রান্সফর্মেশনের যাত্রা সাশ্রয়ী এবং টেকশই করবে।

আপনার সুস্বাস্থ্য এবং দীর্ঘায়ু কামনা করি।